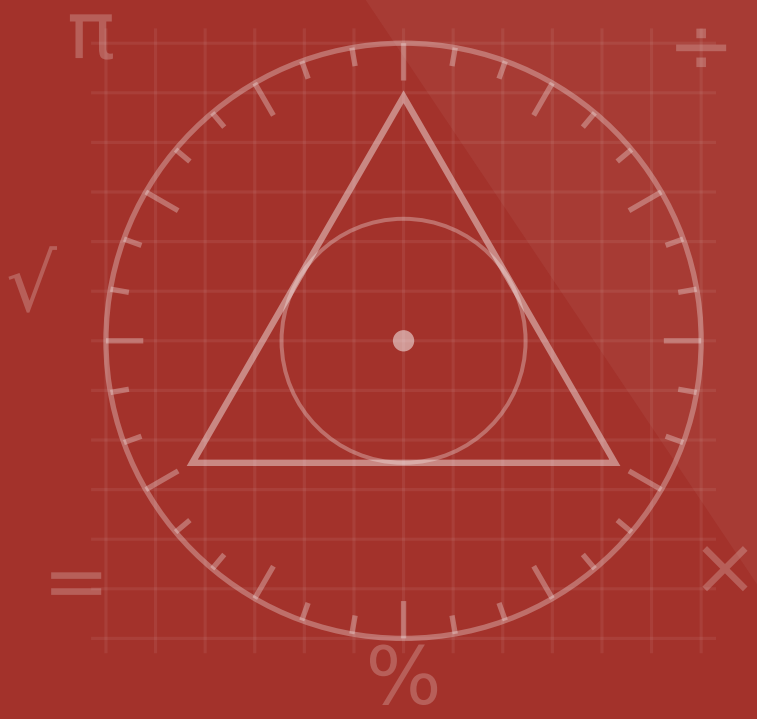
VENVLEARNING

TEACHER EDITION



Mathematics

GRADE 4

Guided lessons, pacing, and complete answer keys

10 units

VENVLEARNING

Grade 4
Mathematics

TEACHER EDITION

Unit 1 Place Value, Rounding, and Whole Numbers
Unit 2 Multiplication
Unit 3 Division
Unit 4 Factors, Multiples, and Patterns
Unit 5 Fraction Equivalence and Comparison
Unit 6 Fraction Operations
Unit 7 Decimals
Unit 8 Geometry, Angles, and Line Plots
Unit 9 Measurement, Area, and Perimeter
Unit 10 Year in Review

A VenvLearning family homeschool curriculum

How to use this Teacher Edition

This book holds everything needed to teach the course: unit overviews, a plan for every lesson with what to say and do, the student lesson with Try It answers, and answer keys for every workbook page, weekly check and unit test.

- Daily routine (about 45–55 minutes): Warm-Up 5 · Teach 15 · Guided Practice 10 · Independent Practice 15 · Exit Ticket 5.
- Say: lines are a suggested script. Use your own words.
- Four lessons a week, then the Weekly Check (about 20 minutes), then corrections or a project.
- Each plan's Pacing row and each unit's pacing guide give the week and lesson order. Set the days to fit your own calendar.

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UNIT 1 · TEACHER OVERVIEW

Place Value, Rounding, and Whole Numbers

How our number system works, and how to read, compare, round, add, and subtract numbers up to one million.

Every number you will work with this year is built from place value. In this unit you will see how each place is worth ten times the place to its right, read and write numbers up to one million, compare and order them, round them to friendly numbers, and add and subtract them with the standard algorithm.

Big ideas

- Explain why each place is worth 10 times the place to its right.
- Read and write numbers to 1,000,000 in standard, word, and expanded form.
- Compare and order multi-digit numbers using >, =, and <.
- Round numbers to any place and use rounding to estimate.
- Add and subtract multi-digit numbers with the standard algorithm and check if answers are reasonable.

Standards

Code	Standard
4.NBT.1	Recognize that in a multi-digit whole number, a digit in one place represents ten times what it represents in the place to its right.
4.NBT.2	Read and write multi-digit whole numbers using base-ten numerals, number names, and expanded form. Compare two multi-digit numbers using >, =, and <.
4.NBT.3	Use place value understanding to round multi-digit whole numbers to any place.
4.NBT.4	Fluently add and subtract multi-digit whole numbers using the standard algorithm.
4.OA.3	Solve multistep word problems; assess the reasonableness of answers using mental computation and estimation strategies including rounding.

What students should already know

- Read, write, and compare numbers to 1,000 (Grade 3).

- Round to the nearest 10 or 100 (Grade 3).
- Add and subtract within 1,000 with regrouping (Grade 3).
- Addition and subtraction facts to 20 from memory.

Materials for the unit

- Base-ten blocks (or paper cut-outs of cubes, flats, rods, and units)
- Place-value chart (six places) in a sheet protector and a dry-erase marker
- Play money: $1, $10, $100, and $1,000 bills
- Painter's tape or a long paper strip for a floor number line
- Grid paper
- Digit cards 0–9 (two sets)

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 4	1.1	How Place Value Works
Week 1	Lesson 2 of 4	1.2	Place Value to One Million
Week 1	Lesson 3 of 4	1.3	The Value of a Digit
Week 1	Lesson 4 of 4	1.4	Standard, Word, and Expanded Form
Week 1	End of week	Check	Weekly Check 1: Place Value
Week 2	Lesson 1 of 4	1.5	Renaming Numbers
Week 2	Lesson 2 of 4	1.6	Comparing Multi-Digit Numbers
Week 2	Lesson 3 of 4	1.7	Ordering Numbers
Week 2	Lesson 4 of 4	1.8	Large Numbers on a Number Line
Week 2	End of week	Check	Weekly Check 2: Comparing and Ordering
Week 3	Lesson 1 of 4	1.9	Rounding with a Number Line
Week 3	Lesson 2 of 4	1.10	Rounding to Any Place
Week 3	Lesson 3 of 4	1.11	Estimating Sums and Differences
Week 3	Lesson 4 of 4	1.12	Is My Answer Reasonable?
Week 3	End of week	Check	Weekly Check 3: Rounding and Estimating
Week 4	Lesson 1 of 4	1.13	Adding Multi-Digit Numbers
Week 4	Lesson 2 of 4	1.14	Subtracting with Regrouping
Week 4	Lesson 3 of 4	1.15	Subtracting Across Zeros

Week	Order	Lesson	Title
Week 4	Lesson 4 of 4	1.16	Solving Multi-Step Word Problems
Week 4	Before the test	Review	Unit 1 Review
Week 4	End of unit	Test	Unit 1 Test

TEACHING THIS UNIT AT HOME

Keep the place-value chart and money handy all unit. Ask your child to read big numbers they see in daily life (gas prices, mileage, store totals) and to round them. A few minutes of this each day builds strong number sense.

UNIT VOCABULARY

place value	The value a digit has because of its place in a number.
period	A group of three places separated by a comma.
expanded form	A number written as the sum of the values of its digits.
round	To change a number to a nearby friendly number.
estimate	A close answer found with rounded numbers.
regroup	To trade 10 of one unit for 1 of the next bigger unit, or the reverse.

LESSON 1.1 · TEACHER PLAN
Week 1 · Lesson 1 of 4
Unit 1: Place Value, Rounding, and Whole Numbers

Lesson 1.1: How Place Value Works

Student goal	I can explain that a digit in one place is worth ten times what it would be worth in the place to its right.
Objective	Students recognize that each place is worth ten times the place to its right and use this to compare digits.
Standard	4.NBT.1 — Recognize that in a multi-digit whole number, a digit in one place represents ten times what it represents in the place to its right.
Pacing	Unit 1, week 1, lesson 1 of 4
Time	About 50 minutes
Materials	Base-ten blocks (or paper drawings), place-value chart, workbook Lesson 1.1
Get ready	If you have no blocks, draw a cube, flat, rod and unit on an index card for reference.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Have students count aloud by tens to 100, then by hundreds to 1,000. Ask: How many tens did we count to get to 100?
Launch	5	Read Let's Explore together. Ask: How would you bundle 1,000 paper clips? Listen for: bundles of 10, then bundles of 10 bundles.
Teach	15	Show 1 unit, 1 rod, 1 flat, 1 cube. Stack 10 units next to a rod to prove they match. Say: “Every time we get ten of something, we can trade it for one of the next bigger size.” Work Worked Example 1 (770) together on the place-value chart. Say: “The digit is the same, but its place changes its value.”
Guided Practice	10	Work Worked Example 2 and the three Try It problems together. Have the student explain each answer using the word ‘place’.
Independent	12	Workbook Lesson 1.1, Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking a digit has the same value wherever it appears.
- Believing ×10 just means “stick a zero on the end” without understanding why (this breaks later with decimals).

SUPPORT

- Use real blocks or bundles of 10 straws.
- Keep the ten-times diagram visible during practice.

EXTEND

- What number is 100 times as much as 40? 1,000 times?
- Explain why 10 × 10 × 10 = 1,000 using blocks.

STUDENT LESSON 1.1 · WITH ANSWERS

Lesson 1.1: How Place Value Works

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.1 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.1, Part A.
- [ ] Complete the rest of Workbook Lesson 1.1 (Parts B–E).

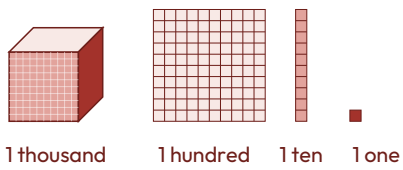
LET'S EXPLORE

Imagine you have a big pile of 1,000 paper clips. Counting them one at a time would take a long time! How could you bundle them to count faster? What size bundles would you make?

We write every number, no matter how big, with just ten digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9. The trick that makes this work is place value. Where a digit sits tells us how much it is worth.

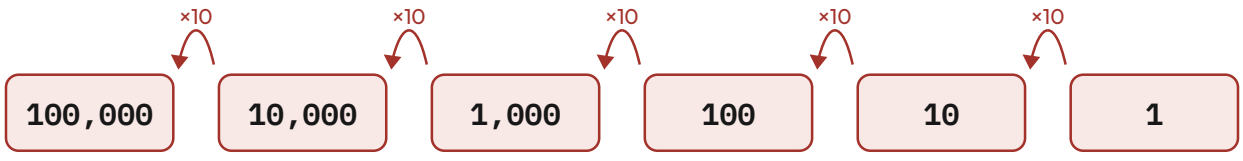
Bundling by tens

Our number system is called base-ten because we group things in tens. Ten ones make one ten. Ten tens make one hundred. Ten hundreds make one thousand. The pattern keeps going forever.



Base-ten blocks: a thousand cube, a hundred flat, a ten rod, and a one cube.

Look at the blocks. A ten rod is 10 one cubes stacked together. A hundred flat is 10 ten rods side by side. A thousand cube is 10 hundred flats stacked up.



Each place is worth 10 times the place to its right.

★ KEY IDEA

In a multi-digit number, a digit in one place is worth 10 times what it would be worth in the place to its right.

WORKED EXAMPLE · COMPARE TWO 7S

In the number 770, how many times greater is the first 7 than the second 7?

Hundreds	Tens	Ones
7	7	0

1 The first 7 is in the hundreds place. Its value is 700.

2 The second 7 is in the tens place. Its value is 70.

3 Ask: 70 times what number equals 700? 70 × 10 = 700.

Answer: The first 7 is worth 10 times as much as the second 7.

WORKED EXAMPLE · TEN TIMES AS MUCH

What number is 10 times as much as 400?

1 Think about place value. 400 is 4 hundreds.

2 Ten times 4 hundreds is 4 thousands, because each place is worth 10 times the place to its right.

3 Write 4 thousands as a number: 4,000.

Answer: 10 × 400 = 4,000

! WATCH OUT

When you multiply a number by 10, the digits move one place to the left. A zero fills the ones place. It is not just “adding a zero”: 10 × 35 = 350 because 3 tens become 3 hundreds and 5 ones become 5 tens.

✎ TRY IT

Answers are shown in color.

1. How many tens are in 100?

Answer: 10

2. What number is 10 times as much as 60?

Answer: 600

3. In 5,550, how does the 5 in the thousands place compare with the 5 in the hundreds place?

Answer: The thousands-place 5 (5,000) is 10 times the hundreds-place 5 (500).

Workbook Answer Key · Lesson 1.1

#	Question	Answer
1	How many tens are in 100?	10
2	What number is 10 times as much as 60?	600
3	In 5,550, how does the 5 in the thousands place compare with the 5 in the hundreds place?	The thousands-place 5 (5,000) is 10 times the hundreds-place 5 (500).
4	Count by tens: 10, 20, 30, ___, ___, ___	40, 50, 60
5	How many ones are in 1 ten?	10
6	How many hundreds are in 1 thousand?	10
7	How many tens are in 1 thousand?	100
8	What number is 10 times as much as 8?	80
9	What number is 10 times as much as 300?	3,000
10	What number is 10 times as much as 2,000?	20,000
11	700 is 10 times as much as what number?	70
12	In 44, the first 4 is worth ___ times the second 4.	10
13	In 3,300, the 3 in the thousands place has a value of _____. The 3 in the hundreds place has a value of _____.	3,000 and 300
14	Maya says the two 6s in 6,600 have the same value because they are the same digit. Do you agree? Explain.	No. The first 6 is worth 6,000 and the second is worth 600. The first is 10 times the second.
15	Draw base-ten blocks to show that 10 hundreds is the same as 1 thousand.	Drawing of 10 hundred flats = 1 thousand cube.
16	What number is 10 times as much as 90? Explain how you know.	900. 9 tens times 10 is 9 hundreds.

LESSON 1.2 · TEACHER PLAN
Week 1 · Lesson 2 of 4
Unit 1: Place Value, Rounding, and Whole Numbers

Lesson 1.2: Place Value to One Million

Student goal	I can read and write numbers up to one million using periods and commas.
Objective	Students read and write numbers to 1,000,000 in standard form and word form using periods.
Standard	4.NBT.2 — Read and write multi-digit whole numbers using base-ten numerals, number names, and expanded form.
Pacing	Unit 1, week 1, lesson 2 of 4
Time	About 50 minutes
Materials	Place-value chart (6 places), workbook Lesson 1.2, a newspaper or website with large numbers (optional)
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Write 325, 3,250 and 32,500. Ask the student to read each one.
Launch	5	Read Let's Explore. Ask: Where do you see big numbers in real life? (population, prices, distances)
Teach	15	Show 452,318 in the chart. Cover the ones period with your hand and read “452 thousand.” Say: “Read the left group, say the comma's name, then read the next group.” Work both Worked Examples. Point out the zero in 60,405.
Guided Practice	10	Dictate three numbers aloud (e.g., 85,020; 400,700; 9,009). Student writes them in standard form. Do the Try It problems together.
Independent	12	Workbook Lesson 1.2.
Exit Ticket	3	Collect the Exit Ticket.

COMMON MISCONCEPTIONS

- Leaving out zeros: writing 60,45 for sixty thousand forty-five.
- Saying “and” in whole numbers (“four hundred and two”). In math, “and” is saved for decimals.

SUPPORT

- Write the period names over the commas on a whiteboard.
- Use a place-value chart for every problem.

EXTEND

- Find the population of two Mississippi cities and write them in words.
- Read 1,000,000 as “one thousand thousands.” Why does that work?

STUDENT LESSON 1.2 · WITH ANSWERS

Lesson 1.2: Place Value to One Million

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.2 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.2, Part A.
- [ ] Complete the rest of Workbook Lesson 1.2 (Parts B–F).

LET'S EXPLORE

The town of Hattiesburg has about 48,000 people. The state of Mississippi has close to 3 million. How do you read numbers this big out loud?

Big numbers are easier to read when we split the digits into groups of three called periods. A comma separates each period. Starting from the right, the periods are the ones period, the thousands period, and the millions period.

THOUSANDS PERIOD			ONES PERIOD		
Hundred Thousands	Ten Thousands	Thousands	Hundreds	Tens	Ones
4	5	2	3	1	8

452,318 in a place-value chart.

Each period has the same three places: hundreds, tens, and ones. In the thousands period they are called hundred thousands, ten thousands, and thousands.

Reading a large number

- Read the number in the left period like a normal three-digit number.
- Say the period name when you reach the comma (“thousand”).
- Read the last period. You do not say “ones.”

WORKED EXAMPLE · READ A SIX-DIGIT NUMBER

Read 452,318 out loud.

1 The thousands period shows 452. Say “four hundred fifty-two thousand.”

2 The ones period shows 318. Say “three hundred eighteen.”

Answer: four hundred fifty-two thousand, three hundred eighteen

WORKED EXAMPLE · WRITE A NUMBER IN STANDARD FORM

Write “sixty thousand, four hundred five” in standard form.

1 The thousands period is sixty, so write 60 and then a comma.

2 The ones period is four hundred five: 405. It needs three digits.

3 Put them together.

Answer: 60,405

! WATCH OUT

Every period after the first one needs exactly three digits. Sixty thousand, forty-five is 60,045, not 60,45. Use zeros to hold empty places.

★ KEY IDEA

One million is written 1,000,000. It is 10 times 100,000. That is the largest number we work with this year.

✎ TRY IT

Answers are shown in color.

1. Read 307,250 out loud. Write the words.

Answer: three hundred seven thousand two hundred fifty

2. Write “ninety-one thousand, six” in standard form.

Answer: 91,006

3. How many digits does one million have?

Answer: 7

Workbook Answer Key · Lesson 1.2

#	Question	Answer
1	Read 307,250 out loud. Write the words.	three hundred seven thousand two hundred fifty
2	Write “ninety-one thousand, six” in standard form.	91,006
3	How many digits does one million have?	7
4	Write the number shown by 3 hundreds, 2 tens, 5 ones.	325
5	Put a comma in the right place: 48296	48,296
6	forty-two thousand, one hundred sixteen	42,116

#	Question	Answer
7	seven hundred thousand, nine hundred	700,900
8	three hundred thirty thousand, thirty	330,030
9	one million	1,000,000
10	five hundred eight thousand, sixty-four	508,064
11	eighteen thousand, two	18,002
12	26,410	twenty-six thousand four hundred ten
13	903,075	nine hundred three thousand seventy-five
14	Jamal wrote “fifty thousand, sixty” as 50,60. What mistake did he make? Write it correctly.	The ones period needs three digits: 50,060.
15	Write 600,047 in words.	six hundred thousand forty-seven

LESSON 1.3 · TEACHER PLAN
Week 1 · Lesson 3 of 4
Unit 1: Place Value, Rounding, and Whole Numbers

Lesson 1.3: The Value of a Digit

Student goal	I can name the place and the value of any digit in a number up to one million.
Objective	Students identify the place and value of digits in numbers to 1,000,000.
Standard	4.NBT.1, 4.NBT.2 — Use place value understanding to name the value of digits in multi-digit whole numbers.
Pacing	Unit 1, week 1, lesson 3 of 4
Time	About 50 minutes
Materials	Place-value chart, workbook Lesson 1.3, highlighter
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Mental math: 10 × 60, 10 × 600, 10 × 6,000.
Launch	5	Read Let's Explore. Ask: Is the 9 in 59 worth the same as the 9 in 95,000? Listen for: no, because the 9 is in a different place.
Teach	15	Model 386,254 on the chart. Highlight one digit at a time and say its place and value. Say: "To find a digit's value, multiply the digit by its place." Work both Worked Examples. In Example 2, connect back to Lesson 1.1 (ten times).
Guided Practice	10	Try It 1–3. Then call out a number and a digit; the student says place and value.
Independent	12	Workbook Lesson 1.3.
Exit Ticket	3	Check the Exit Ticket.

COMMON MISCONCEPTIONS

- Counting places from the left instead of the right.
- Giving the digit (8) instead of its value (80,000).

SUPPORT	EXTEND
• Always start counting at the ones place. • Write the place names above the digits in pencil.	• Make the largest and smallest six-digit numbers using the digits 3, 0, 8, 1, 5, 7. Explain your thinking.

STUDENT LESSON 1.3 · WITH ANSWERS

Lesson 1.3: The Value of a Digit

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.3 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.3, Part A.
- [ ] Complete the rest of Workbook Lesson 1.3 (Parts B–F).

LET'S EXPLORE

Two numbers both have a 9 in them: 59 and 95,000. Is the 9 worth the same amount in both numbers?

Every digit has a place and a value. The place is the name of its column. The value is the digit times the place.

THOUSANDS PERIOD			ONES PERIOD		
Hundred Thousands	Ten Thousands	Thousands	Hundreds	Tens	Ones
3	8	6	2	5	4

In 386,254 the 8 is in the ten thousands place.

In 386,254 the digit 8 is in the ten thousands place. Its value is 8 × 10,000 = 80,000.

WORKED EXAMPLE · FIND PLACE AND VALUE

What is the place and the value of the 5 in 652,019?

THOUSANDS PERIOD			ONES PERIOD		
Hundred Thousands	Ten Thousands	Thousands	Hundreds	Tens	Ones
6	5	2	0	1	9

1. Count places from the right: ones (9), tens (1), hundreds (0), thousands (2), ten thousands (5).
2. The 5 is in the ten thousands place.
3. Value = 5 × 10,000.

Answer: ten thousands place; value 50,000

WORKED EXAMPLE · COMPARE VALUES

In 444,000, how does the value of the first 4 compare with the second 4?

1 First 4: hundred thousands place, value 400,000.

2 Second 4: ten thousands place, value 40,000.

3 40,000 × 10 = 400,000.

Answer: The first 4 is worth 10 times the second 4.

★ KEY IDEA

Value = digit × place. A zero has a value of 0, but it still holds its place.

✎ TRY IT

Answers are shown in color.

1. What is the value of the 7 in 27,315?

Answer: 7,000

2. In which place is the 3 in 830,642?

Answer: ten thousands

3. Which digit is in the hundred thousands place in 914,208?

Answer: 9

Workbook Answer Key · Lesson 1.3

#	Question	Answer
1	What is the value of the 7 in 27,315?	7,000
2	In which place is the 3 in 830,642?	ten thousands
3	Which digit is in the hundred thousands place in 914,208?	9
4	What is 10 × 600?	6,000
5	How many places are in the thousands period?	3
6	43,921	thousands
7	508,117	hundred thousands
8	62,740	hundreds
9	194,305	ten thousands
10	26,480	6,000

#	Question	Answer
11	813,005	800,000
12	75,312	300
13	409,999	0 The zero holds the ten thousands place.
14	357,000	50,000
15	1,000,000	1,000,000
16	Write a six-digit number that has a 4 in the ten thousands place, a 0 in the hundreds place, and a 9 in the ones place.	Answers vary, e.g. 248,079 (digit 4 in ten thousands, 0 in hundreds, 9 in ones).
17	In 777,777, what is the value of the 7 in the hundred thousands place? How many times greater is it than the 7 in the ten thousands place?	700,000; 10 times greater than 70,000
18	What is the value of the 2 in 520,816?	20,000

LESSON 1.4 · TEACHER PLAN
Week 1 · Lesson 4 of 4
Unit 1: Place Value, Rounding, and Whole Numbers

Lesson 1.4: Standard, Word, and Expanded Form

Student goal	I can write a number in standard form, word form, and expanded form.
Objective	Students write numbers to 1,000,000 in standard, word, and expanded form and move between forms.
Standard	4.NBT.2 — Read and write multi-digit whole numbers using base-ten numerals, number names, and expanded form.
Pacing	Unit 1, week 1, lesson 4 of 4
Time	About 50 minutes
Materials	Place-value chart, workbook Lesson 1.4, index cards (optional matching game)
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Review: value of a digit (Lesson 1.3).
Launch	5	Read Let's Explore. If you have an old check or receipt, show the written amount.
Teach	15	Show the table for 32,605. Read each form aloud. Say: "Expanded form is like taking the number apart into the value of each digit." Work Example 1 and 2. In Example 2, have the student fill the chart before writing the number.
Guided Practice	10	Matching game: write one number in three forms on three cards. Mix two numbers' cards; student sorts them. Try It 1–3.
Independent	12	Workbook Lesson 1.4.
Exit Ticket	3	Collect. Tomorrow is the Weekly Check.

COMMON MISCONCEPTIONS

- Writing 0 values in expanded form (e.g., + 0 + 0). It is not wrong, but it is not needed.
- Dropping zeros when changing back to standard form.

SUPPORT	EXTEND
• Fill a place-value chart first for every expanded-form problem.	• Write 52,340 in expanded form two different ways (e.g., 5 × 10,000 + 2 × 1,000 + 3 × 100 + 4 × 10).

STUDENT LESSON 1.4 · WITH ANSWERS

Lesson 1.4: Standard, Word, and Expanded Form

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.4 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.4, Part A.
- [ ] Complete the rest of Workbook Lesson 1.4 (Parts B–F).

LET'S EXPLORE

Your grandmother writes a check for $1,250 and must write the amount in words. How would she write it?

There are three common ways to show the same number.

Form	Example: 32,605
Standard form	32,605
Word form	thirty-two thousand six hundred five
Expanded form	30,000 + 2,000 + 600 + 5

Expanded form shows the value of each digit. Skip any zeros, because their value is 0.

WORKED EXAMPLE · WRITE IN EXPANDED FORM

Write 470,903 in expanded form.

1 List the value of each digit that is not zero: 4 → 400,000; 7 → 70,000; 9 → 900; 3 → 3.

2 Join the values with plus signs.

Answer: 400,000 + 70,000 + 900 + 3

WORKED EXAMPLE · FROM EXPANDED FORM TO STANDARD FORM

Write 80,000 + 6,000 + 40 + 2 in standard form.

THOUSANDS PERIOD		ONES PERIOD		
Ten Thousands	Thousands	Hundreds	Tens	Ones
8	6	0	4	2

1 Find each place: 8 ten thousands, 6 thousands, 0 hundreds, 4 tens, 2 ones.

2 Write a zero for the missing hundreds.

Answer: 86,042

! WATCH OUT

In expanded form, the hundreds place in 86,042 is empty. When you go back to standard form, put a 0 there or you will get 8,642, which is a much smaller number.

★ KEY IDEA

Standard form, word form, and expanded form are three names for the same number.

✎ TRY IT

Answers are shown in color.

1. Write 25,019 in expanded form.

Answer: 20,000 + 5,000 + 10 + 9

2. Write 300,000 + 50,000 + 700 + 1 in standard form.

Answer: 350,701

3. Write 41,300 in word form.

Answer: forty-one thousand three hundred

Workbook Answer Key · Lesson 1.4

#	Question	Answer
1	Write 25,019 in expanded form.	20,000 + 5,000 + 10 + 9
2	Write 300,000 + 50,000 + 700 + 1 in standard form.	350,701
3	Write 41,300 in word form.	forty-one thousand three hundred
4	What is the value of the 6 in 16,402?	6,000
5	Read 300,500 aloud. Write the words.	three hundred thousand five hundred
6	6,724	6,000 + 700 + 20 + 4
7	53,180	50,000 + 3,000 + 100 + 80
8	907,046	900,000 + 7,000 + 40 + 6
9	20,000 + 4,000 + 300 + 9	24,309
10	700,000 + 6,000 + 50	706,050

#	Question	Answer
11	90,000 + 8	90,008
12	five hundred thousand, two hundred twelve	500,212
13	Which shows 64,050 in expanded form? A. 6,000 + 400 + 50 B. 60,000 + 4,000 + 50 C. 60,000 + 4,000 + 500 D. 64,000 + 500	B
14	Write 318,407 in expanded form and word form.	300,000 + 10,000 + 8,000 + 400 + 7; three hundred eighteen thousand four hundred seven

WEEKLY CHECK · ANSWER KEY
After Lesson 1.4

About 15 minutes

Weekly Check 1: Place Value — Answer Key

Answers

#	Question	Answer
1	What number is 10 times as much as 500?	5,000
2	What is the value of the 3 in 734,120?	30,000
3	Write 208,016 in word form.	two hundred eight thousand sixteen
4	Write 60,000 + 900 + 4 in standard form.	60,904
5	Write 45,370 in expanded form.	40,000 + 5,000 + 300 + 70
6	In 99,000, the first 9 is worth ___ times the second 9.	10
7	Write “one hundred four thousand, ten” in standard form.	104,010

Scoring

Score	What it means / next step
7/7 or 6/7	Ready to move on.
4–5	Review the missed lessons using the Key Idea boxes before the next lesson.
0–3	Reteach Lessons 1.3–1.4 with a place-value chart and blocks.

LESSON 1.5 · TEACHER PLAN
Week 2 · Lesson 1 of 4
Unit 1: Place Value, Rounding, and Whole Numbers

Lesson 1.5: Renaming Numbers

Student goal	I can rename a number using different place-value units, such as 3,400 = 34 hundreds.
Objective	Students rename whole numbers using different place-value units.
Standard	4.NBT.1 — Recognize that a digit in one place represents ten times what it represents in the place to its right (applied to renaming).
Pacing	Unit 1, week 2, lesson 1 of 4
Time	About 50 minutes
Materials	Base-ten blocks or play money ($1, $10, $100 bills), workbook Lesson 1.5
Get ready	Print or make play money: 10 × $100, 10 × $10.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Quick review of Weekly Check 1 errors.
Launch	5	Read Let's Explore. Act it out with play money: trade a "$1,000" card for ten $100 bills.
Teach	15	Model 3,400 with blocks; trade each thousand cube for 10 flats and count the flats. Say: "Same amount, different units." Work the three Worked Examples.
Guided Practice	10	Try It 1–3. Then the student makes up a renaming problem for you.
Independent	12	Workbook Lesson 1.5.
Exit Ticket	3	Check: does the student give two correct names?

COMMON MISCONCEPTIONS

- Saying 3,400 has 4 hundreds (only reading the hundreds digit) instead of 34 hundreds.

SUPPORT

- Use play money for every problem.

EXTEND

- How many hundreds are in 1,000,000? (10,000)

STUDENT LESSON 1.5 · WITH ANSWERS

Lesson 1.5: Renaming Numbers

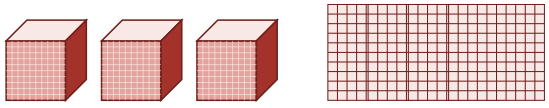
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.5 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.5, Part A.
- [ ] Complete the rest of Workbook Lesson 1.5 (Parts B–E).

LET'S EXPLORE

A bank gives you $3,400 in $100 bills. How many $100 bills do you get?

Because 10 of one unit equals 1 of the next bigger unit, we can rename numbers in different ways. This will help a lot when we add, subtract, multiply, and divide.



3 thousands

4 hundreds

3,400 is 3 thousands and 4 hundreds.

Each thousand is 10 hundreds, so 3 thousands is 30 hundreds. Add the 4 hundreds: 3,400 = 34 hundreds. That means you get 34 bills of $100.

WORKED EXAMPLE · RENAME WITH TENS

How many tens are in 2,560?

1 Cover the ones digit: 2,560 → 256.

2 There are 256 tens in 2,560. Check: 256 × 10 = 2,560.

Answer: 256 tens

WORKED EXAMPLE · RENAME WITH THOUSANDS

Write 45 thousands in standard form.

1 1 thousand = 1,000, so 45 thousands = 45 × 1,000.

2 45 thousands is 4 ten thousands and 5 thousands.

Answer: 45,000

WORKED EXAMPLE · MANY WAYS TO SHOW ONE NUMBER

Show 1,200 in three ways.

1 1 thousand + 2 hundreds

2 12 hundreds

3 120 tens

Answer: All three name 1,200.

★ KEY IDEA

To find how many of a unit are in a number, look at all the digits from the left down to that unit's place. There are 56 hundreds in 5,600 and 560 tens in 5,600.

✎ TRY IT

Answers are shown in color.

1. How many hundreds are in 7,000?

Answer: 70

2. Write 120 tens in standard form.

Answer: 1,200

3. Fill in: 6,300 = ___ hundreds

Answer: 63

Workbook Answer Key · Lesson 1.5

#	Question	Answer
1	How many hundreds are in 7,000?	70
2	Write 120 tens in standard form.	1,200
3	Fill in: 6,300 = ___ hundreds	63
4	10 × 40 = ___	400
5	How many tens make 1 hundred?	10
6	4,000 = ___ hundreds	40
7	4,000 = ___ tens	400
8	2,700 = ___ hundreds	27
9	590 = ___ tens	59
10	38 hundreds = _____	3,800

#	Question	Answer
11	16 thousands = _____	16,000
12	250 tens = _____	2,500
13	70,000 = ___ thousands	70
14	A store packs pencils in boxes of 10. How many boxes does it need for 1,800 pencils?	180 boxes
15	Lena has 52 hundred-dollar bills. How much money is that?	$5,200
16	Show 2,400 in two different ways using place-value units.	Examples: 24 hundreds; 240 tens; 2 thousands 4 hundreds

LESSON 1.6 · TEACHER PLAN
Week 2 · Lesson 2 of 4
Unit 1: Place Value, Rounding, and Whole Numbers

Lesson 1.6: Comparing Multi-Digit Numbers

Student goal	I can compare two multi-digit numbers using >, =, and <.
Objective	Students compare two multi-digit whole numbers using place value and the symbols >, =, <.
Standard	4.NBT.2 — Compare two multi-digit numbers based on meanings of the digits in each place, using >, =, and < symbols.
Pacing	Unit 1, week 2, lesson 2 of 4
Time	About 50 minutes
Materials	Place-value chart, workbook Lesson 1.6, digit cards 0–9 (optional)
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Show 72 and 27. Ask which is greater and why (the tens place).
Launch	5	Read Let's Explore. Ask the student to guess, then prove it.
Teach	15	Write both numbers in a place-value table. Move left to right with your finger. Say: "We stop at the first place where the digits are different." Work both examples.
Guided Practice	10	Digit-card game: each player draws 5 cards, makes a number, and the greater number wins the round. Try It 1–3.
Independent	12	Workbook Lesson 1.6.
Exit Ticket	3	Check the explanation, not just the symbol.

COMMON MISCONCEPTIONS

- Comparing from the right (ones place) first.
- Mixing up > and <. Read the sentence aloud: "382,910 is greater than..."

SUPPORT	EXTEND
• Line numbers up in a chart before comparing.	• Write three numbers between 499,950 and 500,050.

STUDENT LESSON 1.6 · WITH ANSWERS

Lesson 1.6: Comparing Multi-Digit Numbers

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.6 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.6, Part A.
- [ ] Complete the rest of Workbook Lesson 1.6 (Parts B–F).

LET'S EXPLORE

Two video games sold 245,600 copies and 254,600 copies. They have the same digits! Which sold more?

To compare numbers, compare the digits in the same place, starting with the greatest place.

Steps to compare

- Line the numbers up by place value (ones under ones).
- If one number has more digits, it is greater.
- If they have the same number of digits, start at the left. Find the first place where the digits are different.
- The number with the greater digit in that place is greater.

WORKED EXAMPLE · COMPARE

Compare 245,600 and 254,600.

	HTh	TTh	Th	H	T	O
245,600	2	4	5	6	0	0
254,600	2	5	4	6	0	0

1 Both have six digits.

2 Hundred thousands: 2 and 2, the same.

3 Ten thousands: 4 and 5. They are different, and 4 < 5.

Answer: 245,600 < 254,600

WORKED EXAMPLE · DIFFERENT NUMBERS OF DIGITS

Compare 98,999 and 101,000.

1 98,999 has five digits. 101,000 has six digits.

2 A six-digit number is always greater than a five-digit number.

Answer: 98,999 < 101,000

★ KEY IDEA

The symbol opens toward the greater number, like a mouth that wants the bigger meal: 7 > 3 and 3 < 7.

! WATCH OUT

Do not compare the first digits you see if the numbers have different lengths. 98,999 starts with a 9, but it is less than 101,000.

✎ TRY IT

Answers are shown in color.

1. Compare: 63,417 ___ 63,471

Answer: <

2. Compare: 500,010 ___ 499,999

Answer: >

3. Compare: 80,000 + 700 ___ 80,700

Answer: =

Workbook Answer Key · Lesson 1.6

#	Question	Answer
1	Compare: 63,417 ___ 63,471	<
2	Compare: 500,010 ___ 499,999	>
3	Compare: 80,000 + 700 ___ 80,700	=
4	Which is greater, 72 or 27?	72
5	4,000 = ___ hundreds	40
6	3,125 ___ 3,152	<
7	47,900 ___ 47,090	>
8	205,000 ___ 250,000	<
9	99,999 ___ 100,000	<
10	61,308 ___ 61,308	=

#	Question	Answer
11	718,402 ___ 718,399	>
12	8,040 ___ 8,400	<
13	530,215 ___ 530,251	<
14	40,000 + 3,000 + 60 ___ 43,600	<
15	six hundred thousand, nine ___ 600,090	<
16	Use the digits 4, 7, 1, 9, 0 once each. Write the greatest and least five-digit numbers you can. Compare them with a symbol.	97,410 > 10,479 Least must not start with 0.
17	Compare 382,910 and 382,190. Explain which place helped you decide.	382,910 > 382,190; the hundreds place (9 > 1).

LESSON 1.7 · TEACHER PLAN
Week 2 · Lesson 3 of 4
Unit 1: Place Value, Rounding, and Whole Numbers

Lesson 1.7: Ordering Numbers

Student goal	I can order three or more multi-digit numbers from least to greatest or greatest to least.
Objective	Students order three or more multi-digit whole numbers.
Standard	4.NBT.2 — Compare two multi-digit numbers based on meanings of the digits in each place (extended to ordering).
Pacing	Unit 1, week 2, lesson 3 of 4
Time	About 50 minutes
Materials	Index cards, workbook Lesson 1.7, a map of Mississippi (optional)
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Compare two numbers on the whiteboard; student explains.
Launch	5	Read Let's Explore. Find the rivers on a map if you have one.
Teach	15	Write the three numbers from Example 1 on cards. Line them up in a place-value table. Say: "Ordering is just comparing, one place at a time." Work both examples.
Guided Practice	10	Give the student 4 cards with numbers to physically put in order. Try It 1–3.
Independent	12	Workbook Lesson 1.7.
Exit Ticket	3	Check.

COMMON MISCONCEPTIONS

- Ordering by the first digit only, ignoring how many digits each number has.

SUPPORT

- Cross out each number as you place it in the list.

EXTEND

- Find the lengths of three more U.S. rivers and order them.

STUDENT LESSON 1.7 · WITH ANSWERS

Lesson 1.7: Ordering Numbers

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.7 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.7, Part A.
- [ ] Complete the rest of Workbook Lesson 1.7 (Parts B–F).

LET'S EXPLORE

Three Mississippi rivers have these lengths in miles: Pascagoula 81, Big Black 330, Yazoo 188. Which is longest? How would you list them from shortest to longest?

Ordering is comparing again and again. Line up the numbers by place value and compare one place at a time.

WORKED EXAMPLE · ORDER FROM LEAST TO GREATEST

Order 36,812; 36,281; 38,126.

Number	TTh	Th	H	T	O
36,812	3	6	8	1	2
36,281	3	6	2	8	1
38,126	3	8	1	2	6

1 Ten thousands: all are 3.

2 Thousands: 6, 6, 8. So 38,126 is greatest.

3 Compare the other two at the hundreds place: 36,812 and 36,281. 2 < 8.

Answer: 36,281; 36,812; 38,126

WORKED EXAMPLE · ORDER FROM GREATEST TO LEAST

Order 902,415; 92,415; 920,415.

1 92,415 has only five digits, so it is least.

2 Compare 902,415 and 920,415: ten thousands place 0 < 2.

3 Greatest first.

Answer: 920,415; 902,415; 92,415

★ KEY IDEA

Read the directions carefully: least to greatest starts with the smallest number. Greatest to least starts with the largest.

✎ TRY IT

Answers are shown in color.

1. Order from least to greatest: 4,509; 4,950; 4,095

Answer: 4,095; 4,509; 4,950

2. Order from greatest to least: 71,300; 17,300; 73,100

Answer: 73,100; 71,300; 17,300

3. Which number is greatest: 650,001; 605,100; 650,010?

Answer: 650,010

Workbook Answer Key · Lesson 1.7

#	Question	Answer
1	Order from least to greatest: 4,509; 4,950; 4,095	4,095; 4,509; 4,950
2	Order from greatest to least: 71,300; 17,300; 73,100	73,100; 71,300; 17,300
3	Which number is greatest: 650,001; 605,100; 650,010?	650,010
4	Compare: 5,280 ___ 5,820	<
5	What place decides 61,400 vs 61,040?	hundreds
6	2,315; 2,153; 2,531	2,153; 2,315; 2,531
7	48,600; 46,800; 48,060	46,800; 48,060; 48,600
8	300,999; 309,000; 30,999; 390,000	30,999; 300,999; 309,000; 390,000
9	7,402; 7,240; 7,420	7,420; 7,402; 7,240
10	125,000; 152,000; 15,200; 125,500	152,000; 125,500; 125,000; 15,200
11	Mississippi rivers: Pascagoula 81 mi, Big Black 330 mi, Yazoo 188 mi, Pearl 444 mi. List them from longest to shortest.	Pearl 444, Big Black 330, Yazoo 188, Pascagoula 81
12	Order from least to greatest: 61,090; 60,919; 61,009	60,919; 61,009; 61,090

LESSON 1.8 · TEACHER PLAN
Week 2 · Lesson 4 of 4
Unit 1: Place Value, Rounding, and Whole Numbers

Lesson 1.8: Large Numbers on a Number Line

Student goal	I can place and find large numbers on a number line.
Objective	Students locate multi-digit numbers on number lines and identify the nearer of two benchmark numbers.
Standard	4.NBT.2 — Compare multi-digit numbers; builds the number-line model used for rounding (4.NBT.3).
Pacing	Unit 1, week 2, lesson 4 of 4
Time	About 50 minutes
Materials	Painter's tape or a long strip of paper for a floor number line, sticky notes, workbook Lesson 1.8
Get ready	Make a floor number line from 30,000 to 40,000 with 11 marks.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Count by 1,000s and by 10,000s.
Launch	5	Read Let's Explore. Student stands on the floor line where 5,000 would go.
Teach	15	Say: "First find how much each jump is worth." Work Example 1 on the floor line, then Example 2.
Guided Practice	10	Call out numbers; the student walks to where they belong and says which mark they are closer to. Try It 1–3.
Independent	12	Workbook Lesson 1.8.
Exit Ticket	3	Check. Weekly Check 2 is next.

COMMON MISCONCEPTIONS

- Assuming every jump is 1 or 10 without checking the interval.

SUPPORT

- Label every mark before answering.

EXTEND

- Make a number line from 0 to 1,000,000 and place your town's population.

STUDENT LESSON 1.8 · WITH ANSWERS

Lesson 1.8: Large Numbers on a Number Line

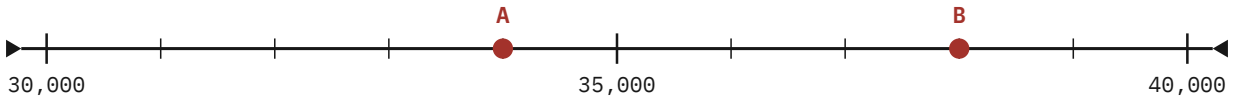
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.8 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.8, Part A.
- [ ] Complete the rest of Workbook Lesson 1.8 (Parts B–E).

LET'S EXPLORE

A number line goes from 0 to 10,000. Where would you put 5,000? Where would 9,900 go?

A number line shows numbers in order. Numbers get greater as you move right. To use a number line with big numbers, first find the interval: how much each jump between marks is worth.



Each mark is 1,000 more than the mark before it.

WORKED EXAMPLE · FIND THE INTERVAL

On the number line above, what numbers are at A and B?

1 From 30,000 to 40,000 there are 10 jumps.

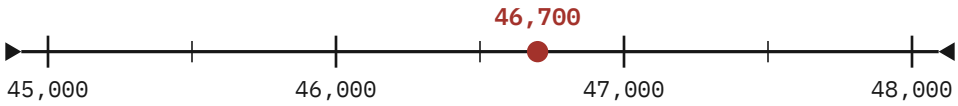
2 10,000 ÷ 10 = 1,000, so each jump is 1,000.

3 A is 4 jumps past 30,000: 34,000. B is 8 jumps past 30,000: 38,000.

Answer: A = 34,000 and B = 38,000

WORKED EXAMPLE · PLACE A NUMBER

Between which two marks does 46,700 go on a number line marked by thousands?



1 46,700 is more than 46,000 and less than 47,000.

2 700 is a little more than halfway (500) between them.

Answer: Between 46,000 and 47,000, a bit past the middle

★ KEY IDEA

Number lines help us see which numbers are close to each other. Tomorrow's skill, rounding, uses this idea.

✎ TRY IT

Answers are shown in color.

1. A number line goes from 0 to 1,000 with 10 equal jumps. How much is each jump?

Answer: 100

2. Which thousand is 52,900 closer to: 52,000 or 53,000?

Answer: 53,000

3. What number is halfway between 200,000 and 300,000?

Answer: 250,000

Workbook Answer Key · Lesson 1.8

#	Question	Answer
1	A number line goes from 0 to 1,000 with 10 equal jumps. How much is each jump?	100
2	Which thousand is 52,900 closer to: 52,000 or 53,000?	53,000
3	What number is halfway between 200,000 and 300,000?	250,000
4	Count by 1,000s from 7,000 to 12,000.	7,000; 8,000; 9,000; 10,000; 11,000; 12,000
5	What number is at point P? (figure)	6,000
6	What number is at point Q? (figure)	75,000
7	What number is at point R? (figure)	430
8	What number is at point S? (figure)	250,000 Labels use k for thousands.
9	Is 3,800 closer to 3,000 or 4,000?	4,000
10	Is 61,200 closer to 61,000 or 62,000?	61,000
11	Is 540 closer to 500 or 600?	500

#	Question	Answer
12	What number is halfway between 7,000 and 8,000?	7,500
13	Draw a number line from 20,000 to 30,000 with marks every 1,000. Put a dot at 27,000.	Number line with 11 marks; dot at the 8th mark (27,000).

WEEKLY CHECK · ANSWER KEY
After Lesson 1.8

About 15 minutes

Weekly Check 2: Comparing and Ordering — Answer Key

Answers

#	Question	Answer
1	3,700 = ___ hundreds	37
2	Write 46 thousands in standard form.	46,000
3	Write >, <, or =: 508,213 ___ 508,312	<
4	Write >, <, or =: 90,000 + 400 + 5 ___ 90,405	=
5	Order from least to greatest: 72,018; 70,281; 72,108	70,281; 72,018; 72,108
6	What number is at point M? (figure)	64,000
7	Is 8,300 closer to 8,000 or 9,000?	8,000

Scoring

Score	What it means / next step
6–7	Ready for rounding.
4–5	Review the number line (Lesson 1.8) before the next lesson: rounding builds on it.
0–3	Reteach comparing with a place-value table.

LESSON 1.9 · TEACHER PLAN
Week 3 · Lesson 1 of 4
Unit 1: Place Value, Rounding, and Whole Numbers

Lesson 1.9: Rounding with a Number Line

Student goal	I can use a number line to round a number to the nearest ten, hundred, or thousand.
Objective	Students round whole numbers to the nearest ten, hundred, and thousand using number lines.
Standard	4.NBT.3 — Use place value understanding to round multi-digit whole numbers to any place.
Pacing	Unit 1, week 3, lesson 1 of 4
Time	About 50 minutes
Materials	Floor number line from Lesson 1.8 (relabel 2,000 to 3,000), sticky notes, workbook Lesson 1.9
Get ready	Relabel the floor number line from 2,000 to 3,000 with marks every 100.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Ask: What number is halfway between 20 and 30? 200 and 300? 2,000 and 3,000?
Launch	5	Read Let's Explore. Ask: Why would a newspaper use 3,000? Listen for: it's easier to read; nobody needs the exact count.
Teach	15	Student stands at 2,846 on the floor line. Ask: Which end is closer? Say: "Rounding means finding the closer friendly number." Work Example 1 on paper with a number line, then Example 2 (the halfway rule).
Guided Practice	10	Try It 1–3. For each one the student first names the two benchmarks out loud.
Independent	12	Workbook Lesson 1.9.
Exit Ticket	3	Check that the number line shows the halfway point.

COMMON MISCONCEPTIONS

- Rounding to the wrong place (e.g., rounding to the nearest ten when asked for hundred).
- Rounding halfway numbers down.

- Changing digits to the left of the rounding place incorrectly, e.g. 4,950 → 4,900 instead of 5,000.

SUPPORT	EXTEND
• Always draw the number line and label the halfway point. • Highlight the rounding place.	• Round 4,950 to the nearest ten, hundred, and thousand. Why are two of the answers the same?

STUDENT LESSON 1.9 · WITH ANSWERS

Lesson 1.9: Rounding with a Number Line

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.9 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.9, Part A.
- [ ] Complete the rest of Workbook Lesson 1.9 (Parts B–F).

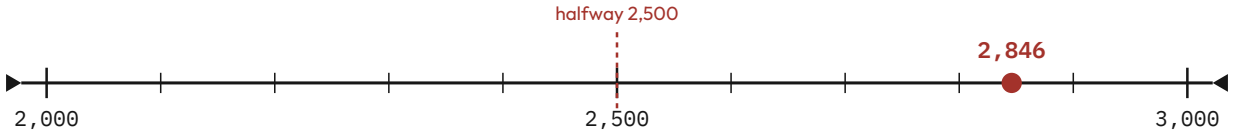
LET'S EXPLORE

A newspaper headline says "About 3,000 people came to the county fair." The real count was 2,846. Why might the newspaper use 3,000 instead of the exact number?

We round numbers when we do not need an exact answer. A rounded number is close to the real number but easier to say, remember, and work with.

Rounding is finding the closer benchmark

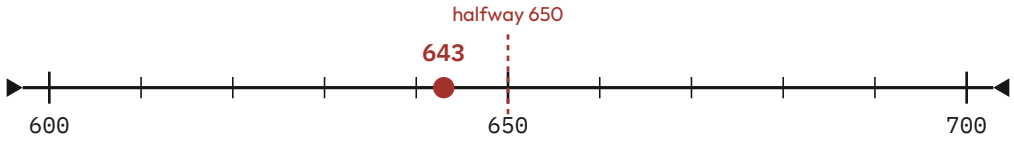
To round 2,846 to the nearest thousand, ask: is it closer to 2,000 or to 3,000? A number line shows the answer.



2,846 is past the halfway point (2,500), so it is closer to 3,000.

WORKED EXAMPLE · ROUND TO THE NEAREST HUNDRED

Round 643 to the nearest hundred.



1 Find the two hundreds that 643 is between: 600 and 700.

2 Find halfway: 650.

3 643 is less than 650, so it is closer to 600.

Answer: 643 rounds to 600

WORKED EXAMPLE · A NUMBER EXACTLY HALFWAY

Round 75 to the nearest ten.

1 75 is between 70 and 80.

2 75 is exactly halfway.

3 The rule: when a number is exactly halfway, we round up.

Answer: 75 rounds to 80

★ KEY IDEA

To round: (1) find the two benchmarks the number is between, (2) find the halfway point, (3) choose the closer benchmark. Halfway numbers round up.

! WATCH OUT

Rounding to the nearest hundred does not mean the answer has to end in one zero. It ends in two zeros, like 600 or 1,200.

✎ TRY IT

Answers are shown in color.

1. Round 38 to the nearest ten.

Answer: 40

2. Round 1,250 to the nearest hundred.

Answer: 1,300

3. Round 7,480 to the nearest thousand.

Answer: 7,000

Workbook Answer Key · Lesson 1.9

#	Question	Answer
1	Round 38 to the nearest ten.	40
2	Round 1,250 to the nearest hundred.	1,300
3	Round 7,480 to the nearest thousand.	7,000
4	What number is halfway between 300 and 400?	350
5	Is 4,700 closer to 4,000 or 5,000?	5,000
6	Round 26 to the nearest ten. (figure)	30
7	Round 340 to the nearest hundred. (figure)	300

#	Question	Answer
8	Round 5,500 to the nearest thousand. (figure)	6,000 Exactly halfway rounds up.
9	Round 8,120 to the nearest thousand. (figure)	8,000
10	Round 84 to the nearest ten.	80
11	Round 615 to the nearest ten.	620 Halfway between 610 and 620: round up.
12	Round 2,391 to the nearest hundred.	2,400
13	Round 4,950 to the nearest hundred.	5,000
14	Round 3,099 to the nearest thousand.	3,000
15	Round 9,602 to the nearest thousand.	10,000
16	Name three different numbers that round to 500 when rounded to the nearest hundred.	Any three numbers from 450 to 549, e.g. 460, 500, 549.
17	The fair report said “about 3,000 people.” What is the least number of people who could have come, if the number was rounded to the nearest thousand?	2,500
18	Round 1,673 to the nearest hundred. Draw a number line to show how you know.	1,700 (between 1,600 and 1,700; past 1,650)

LESSON 1.10 · TEACHER PLAN
Week 3 · Lesson 2 of 4
Unit 1: Place Value, Rounding, and Whole Numbers

Lesson 1.10: Rounding to Any Place

Student goal	I can use place value to round a number to any place, up to the hundred thousands.
Objective	Students round multi-digit numbers to any place using place value reasoning.
Standard	4.NBT.3 — Use place value understanding to round multi-digit whole numbers to any place.
Pacing	Unit 1, week 3, lesson 2 of 4
Time	About 50 minutes
Materials	Place-value chart, highlighter, workbook Lesson 1.10
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Two quick number-line roundings from yesterday.
Launch	5	Read Let's Explore. Ask: How could place value help us skip the number line?
Teach	15	Write the four-step rule on a card the student can keep. Say: "Underline, look right, decide, fill with zeros." Work all three examples. For Example 3, connect to regrouping from Lesson 1.5.
Guided Practice	10	Round one number (527,381) to every place together. Try It 1–3.
Independent	12	Workbook Lesson 1.10.
Exit Ticket	3	Check.

COMMON MISCONCEPTIONS

- Looking at the rounding digit itself instead of the digit to its right.
- Leaving digits after the rounding place instead of changing them to zeros (e.g., 46,281 → 50,281).

SUPPORT	EXTEND
• Use the rule card and a highlighter. • Check with a quick number line when unsure.	• Find the greatest whole number that rounds to 60,000 when rounded to the nearest ten thousand. (64,999)

STUDENT LESSON 1.10 · WITH ANSWERS

Lesson 1.10: Rounding to Any Place

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.10 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.10, Part A.
- [ ] Complete the rest of Workbook Lesson 1.10 (Parts B–G).

LET'S EXPLORE

Drawing a number line every time takes a while. Is there a faster way to round 46,281 to the nearest ten thousand?

Number lines show why rounding works. Place value gives us a quick way to do it.

The place-value rounding rule

- Underline the digit in the rounding place.
- Look at the digit just to its right.
- If that digit is 5 or more, add 1 to the underlined digit. If it is 4 or less, keep the underlined digit the same.
- Change every digit to the right of the rounding place to 0.

WORKED EXAMPLE · ROUND TO THE NEAREST TEN THOUSAND

Round 46,281 to the nearest ten thousand.

Ten Thousands	Thousands	Hundreds	Tens	Ones
4	6	2	8	1

1 Underline the ten thousands digit: 4̲6,281.

2 Look right: the thousands digit is 6.

3 6 is 5 or more, so 4 becomes 5.

4 Change the digits to the right to zeros.

Answer: 50,000

WORKED EXAMPLE · ROUND TO THE NEAREST HUNDRED THOUSAND

Round 734,900 to the nearest hundred thousand.

1 Underline the hundred thousands digit: 7̲34,900.

2 Look right: the ten thousands digit is 3.

3 3 is 4 or less, so 7 stays 7.

4 Change the digits to the right to zeros.

Answer: 700,000

WORKED EXAMPLE · WHEN ROUNDING UP MAKES A 9 BECOME 10

Round 3,970 to the nearest hundred.

1 Underline the hundreds digit: 3,9̲70.

2 Look right: 7 is 5 or more, so round up.

3 9 hundreds + 1 hundred = 10 hundreds = 1 thousand. Regroup: 3 thousands become 4 thousands.

Answer: 4,000

★ KEY IDEA

Look one place to the right. 5 or more: round up. 4 or less: stay the same. Then fill with zeros.

! WATCH OUT

Only the digit in the rounding place can change (unless it regroups). Never change digits to the left for no reason. Round 46,281 to the nearest thousand: 46,000, not 40,000.

✎ TRY IT

Answers are shown in color.

1. Round 58,412 to the nearest thousand.

Answer: 58,000

2. Round 58,412 to the nearest ten thousand.

Answer: 60,000

3. Round 249,999 to the nearest hundred thousand.

Answer: 200,000

Workbook Answer Key · Lesson 1.10

#	Question	Answer
1	Round 58,412 to the nearest thousand.	58,000
2	Round 58,412 to the nearest ten thousand.	60,000
3	Round 249,999 to the nearest hundred thousand.	200,000
4	Round 76 to the nearest ten.	80
5	Round 425 to the nearest hundred.	400
6	6,720	7,000
7	31,455	31,000
8	89,512	90,000
9	409,700	410,000
10	23,900	20,000
11	75,002	80,000
12	148,000	150,000
13	996,000	1,000,000
14	381,000	400,000
15	650,000	700,000
16	129,999	100,000
17	870,500	900,000
18	Round 527,381 to the nearest ten, hundred, thousand, ten thousand, and hundred thousand.	527,380, 527,400, 527,000, 530,000, 500,000
19	Kayla rounded 68,503 to the nearest thousand and got 69,000. Ben got 68,000. Who is correct? Explain.	Kayla. The hundreds digit is 5, so round up.
20	Round 963,400 to the nearest ten thousand.	960,000

LESSON 1.11 · TEACHER PLAN
Week 3 · Lesson 3 of 4
Unit 1: Place Value, Rounding, and Whole Numbers

Lesson 1.11: Estimating Sums and Differences

Student goal	I can round numbers to estimate sums and differences.
Objective	Students estimate sums and differences by rounding.
Standard	4.NBT.3 — Use place value understanding to round multi-digit whole numbers to any place (applied to estimation).
Pacing	Unit 1, week 3, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 1.11, a road map or trip-planning website (optional)
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Round two numbers to the nearest thousand.
Launch	5	Read Let's Explore. Ask the student to guess before calculating.
Teach	15	Work Examples 1 and 2. Say: Round first, then do the easy math. Show the table: different rounding places give different estimates, and both are fine.
Guided Practice	10	Try It 1–3. Ask after each: Is the exact answer more or less than your estimate? How do you know?
Independent	12	Workbook Lesson 1.11.
Exit Ticket	3	Accept any reasonable rounding place if the work is shown.

COMMON MISCONCEPTIONS

- Finding the exact answer and then rounding it (that is not estimating).
- Thinking there is only one correct estimate.

SUPPORT

- Round to the nearest hundred for 3-digit numbers and nearest thousand for 4- and 5-digit numbers.

EXTEND

- Plan a pretend road trip across three cities; estimate the total miles.

STUDENT LESSON 1.11 · WITH ANSWERS

Lesson 1.11: Estimating Sums and Differences

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.11 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.11, Part A.
- [ ] Complete the rest of Workbook Lesson 1.11 (Parts B–F).

LET'S EXPLORE

Your family drives 386 miles on Saturday and 412 miles on Sunday. Did you drive more or less than 1,000 miles? Can you tell without finding the exact answer?

An estimate is a smart guess that is close to the exact answer. Estimating is useful when you only need to know "about how many," and it helps you check your work later.

Round, then compute

WORKED EXAMPLE · ESTIMATE A SUM

Estimate 386 + 412.

1 Round each number to the nearest hundred: 386 → 400 and 412 → 400.

2 Add the rounded numbers: 400 + 400 = 800.

Answer: about 800 miles, so less than 1,000 miles

WORKED EXAMPLE · ESTIMATE A DIFFERENCE

Estimate 7,815 − 2,290.

1 Round to the nearest thousand: 7,815 → 8,000 and 2,290 → 2,000.

2 Subtract: 8,000 − 2,000 = 6,000.

Answer: about 6,000

Choosing a place to round to

Rounding to a greater place makes the math easier. Rounding to a smaller place makes the estimate closer to the exact answer. Usually, round to the greatest place of the smaller number.

Estimate 4,628 + 1,357	Rounded numbers	Estimate
Nearest thousand	5,000 + 1,000	6,000
Nearest hundred	4,600 + 1,400	6,000
Exact answer	4,628 + 1,357	5,985

★ KEY IDEA

To estimate, round each number first, then add or subtract the rounded numbers. Use the word about in your answer.

✎ TRY IT

Answers are shown in color.

1. Estimate 52 + 37 by rounding to the nearest ten.

Answer: 50 + 40 = 90

2. Estimate 6,120 − 3,890 by rounding to the nearest thousand.

Answer: 6,000 − 4,000 = 2,000

3. Estimate 24,610 + 13,250 by rounding to the nearest thousand.

Answer: 25,000 + 13,000 = 38,000

Workbook Answer Key · Lesson 1.11

#	Question	Answer
1	Estimate 52 + 37 by rounding to the nearest ten.	50 + 40 = 90
2	Estimate 6,120 − 3,890 by rounding to the nearest thousand.	6,000 − 4,000 = 2,000
3	Estimate 24,610 + 13,250 by rounding to the nearest thousand.	25,000 + 13,000 = 38,000
4	Round 4,512 to the nearest thousand.	5,000
5	Round 349 to the nearest hundred.	300
6	287 + 512	300 + 500 = 800
7	645 + 238	600 + 200 = 800
8	903 − 471	900 − 500 = 400
9	1,382 − 619	1,400 − 600 = 800
10	3,782 + 5,104	4,000 + 5,000 = 9,000
11	12,450 + 6,701	12,000 + 7,000 = 19,000

#	Question	Answer
12	9,388 – 4,620	9,000 – 5,000 = 4,000
13	40,215 – 18,960	40,000 – 19,000 = 21,000
14	A school library has 2,846 fiction books and 1,512 nonfiction books. About how many books does it have?	about 4,300 (2,800 + 1,500) or about 5,000 (3,000 + 2,000) Nearest hundred: 2,800 + 1,500 = 4,300. Nearest thousand: 3,000 + 2,000 = 5,000. Accept reasoned estimates.
15	A stadium holds 48,200 people. 31,750 tickets are sold. About how many seats are left?	about 16,000 (48,000 – 32,000)
16	Estimate 6,489 + 2,730. Show the rounded numbers.	6,000 + 3,000 = 9,000 (or 6,500 + 2,700 = 9,200)

LESSON 1.12 · TEACHER PLAN
Week 3 · Lesson 4 of 4
Unit 1: Place Value, Rounding, and Whole Numbers

Lesson 1.12: Is My Answer Reasonable?

Student goal	I can use estimation to decide whether an answer makes sense.
Objective	Students use estimation to judge the reasonableness of answers and to solve "is it enough?" problems.
Standard	4.NBT.3, 4.OA.3 — Assess the reasonableness of answers using mental computation and estimation strategies including rounding (4.OA.3).
Pacing	Unit 1, week 3, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 1.12, a store flyer with prices (optional)
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Two quick estimates.
Launch	5	Read Let's Explore. Ask: How did you know so fast?
Teach	15	Work Example 1. Say: An estimate is like a guard dog. It barks when an answer is way off. Work Example 2 and discuss the Watch Out: rounding up vs. down.
Guided Practice	10	Shop from a store flyer: choose two items, estimate the total, decide if $50 (or $500) is enough.
Independent	12	Workbook Lesson 1.12.
Exit Ticket	3	Check. Next: Weekly Check 3.

COMMON MISCONCEPTIONS

- Trusting an exact answer without checking it.
- Thinking an estimate proves you have enough money even when it is very close.

SUPPORT

- Use money amounts the student knows.

EXTEND

- Make up a wrong answer for the teacher to catch.

STUDENT LESSON 1.12 · WITH ANSWERS

Lesson 1.12: Is My Answer Reasonable?

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.12 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.12, Part A.
- [ ] Complete the rest of Workbook Lesson 1.12 (Parts B–E).

LET'S EXPLORE

A classmate says 4,895 + 3,210 = 18,105. Without adding, how can you tell something is wrong?

Even good mathematicians make mistakes. A quick estimate is your safety net. If your exact answer is far from your estimate, check your work.

WORKED EXAMPLE · CHECK AN ANSWER

Is 4,895 + 3,210 = 18,105 reasonable?

1 Estimate: 5,000 + 3,000 = 8,000.

2 18,105 is much more than 8,000.

3 The answer is not reasonable. The exact sum is 8,105.

Answer: No. It should be close to 8,000.

WORKED EXAMPLE · USE AN ESTIMATE IN A STORY

Mom has $500. She wants a $289 tablet and a $178 bike helmet. Does she have enough money?

1 Round to the nearest hundred: 300 + 200 = 500.

2 Both numbers were rounded up, so the exact total is less than $500.

3 Check: 289 + 178 = 467.

Answer: Yes. $467 is less than $500.

! WATCH OUT

When both numbers are rounded up, the estimate is greater than the exact sum. When both are rounded down, the estimate is less. Thinking about this tells you whether you have enough.

★ KEY IDEA

Estimate before or after you solve. If the exact answer and the estimate are far apart, look for a mistake.

✎ TRY IT

Answers are shown in color.

1. Is 612 + 297 = 909 reasonable? Use an estimate.

Answer: Yes. 600 + 300 = 900, close to 909.

2. Is 8,050 − 2,980 = 6,070 reasonable?

Answer: No. 8,000 − 3,000 = 5,000. Exact answer is 5,070.

3. You have $60. Books cost $18, $24, and $15. Do you have enough? Use estimation.

Answer: Yes. 20 + 20 + 20 = 60 and each number was rounded up, so the total is less than $60. (Exact: $57)

Workbook Answer Key · Lesson 1.12

#	Question	Answer
1	Is 612 + 297 = 909 reasonable? Use an estimate.	Yes. 600 + 300 = 900, close to 909.
2	Is 8,050 − 2,980 = 6,070 reasonable?	No. 8,000 − 3,000 = 5,000. Exact answer is 5,070.
3	You have $60. Books cost $18, $24, and $15. Do you have enough? Use estimation.	Yes. 20 + 20 + 20 = 60 and each number was rounded up, so the total is less than $60. (Exact: $57)
4	Estimate 476 + 318.	500 + 300 = 800
5	Estimate 5,820 − 1,950.	6,000 − 2,000 = 4,000
6	315 + 489 = 804 A. Yes B. No	Yes (300 + 500 = 800)
7	2,705 + 4,390 = 9,095 A. Yes B. No	No (3,000 + 4,000 = 7,000; exact 7,095)
8	9,120 − 6,850 = 2,270 A. Yes B. No	Yes (9,000 − 7,000 = 2,000)
9	63,400 − 21,900 = 51,500 A. Yes B. No	No (63,000 − 22,000 = 41,000; exact 41,500)

#	Question	Answer
10	Grandpa wants to buy a $389 lawn mower and a $129 trimmer. He has $500. Is that enough? Explain using estimation.	No. 400 + 100 = 500, but 389 + 129 = 518, which is more than $500. The estimate is too close to tell, so find the exact sum. Great discussion: when the estimate is very close to the limit, find the exact answer.
11	A farm picked 1,240 pounds of pecans in October and 2,860 pounds in November. Lee says they picked about 3,000 pounds in all. Is Lee's estimate reasonable?	No. 1,000 + 3,000 = 4,000. Exact: 4,100.
12	Write an addition problem whose answer would be about 5,000. Show your estimate.	Answers vary, e.g., 2,950 + 2,080 is about 3,000 + 2,000 = 5,000.

WEEKLY CHECK · ANSWER KEY
After Lesson 1.12
About 15 minutes

Weekly Check 3: Rounding and Estimating — Answer Key

Answers

#	Question	Answer
1	Round 467 to the nearest ten.	470
2	Round 3,250 to the nearest hundred.	3,300
3	Round 81,604 to the nearest thousand.	82,000
4	Round 355,000 to the nearest hundred thousand.	400,000
5	Estimate 2,870 + 4,150 by rounding to the nearest thousand.	3,000 + 4,000 = 7,000
6	Estimate 9,412 − 3,780 by rounding to the nearest thousand.	9,000 − 4,000 = 5,000
7	Is 5,120 + 2,990 = 7,110 reasonable? Explain.	No. 5,000 + 3,000 = 8,000. Exact: 8,110.

Scoring

Score	What it means / next step
6–7	Ready for adding and subtracting.
4–5	Practice the rounding rule card for 5 minutes a day this week.
0–3	Reteach with number lines (Lesson 1.9) before Lesson 1.13.

LESSON 1.13 · TEACHER PLAN
Week 4 · Lesson 1 of 4
Unit 1: Place Value, Rounding, and Whole Numbers

Lesson 1.13: Adding Multi-Digit Numbers

Student goal	I can add multi-digit numbers using the standard algorithm.
Objective	Students fluently add multi-digit whole numbers using the standard algorithm.
Standard	4.NBT.4 — Fluently add and subtract multi-digit whole numbers using the standard algorithm.
Pacing	Unit 1, week 4, lesson 1 of 4
Time	About 50 minutes
Materials	Grid paper, base-ten blocks (optional), workbook Lesson 1.13
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Solve 678 + 245 together. Point out each regroup.
Launch	5	Read Let's Explore. Student estimates first: about 3,000 + 3,000.
Teach	15	Work Example 1 on grid paper, one digit per square. Say: "Ones first. If it's 10 or more, trade 10 ones for 1 ten and carry it." If the student struggles, model the ones and tens with blocks to show the trade. Work Example 2 (three addends).
Guided Practice	10	Try It 1–3 on grid paper. Student says each step aloud.
Independent	12	Workbook Lesson 1.13.
Exit Ticket	3	Check alignment and regrouping.

COMMON MISCONCEPTIONS

- Writing the whole two-digit sum in one column (e.g., writing 14 under the ones).
- Forgetting to add the regrouped 1.
- Lining numbers up on the left.

SUPPORT	EXTEND
• Use grid paper so each digit has its own box. • Color-code the regrouped digits.	• Find two 5-digit numbers whose sum is exactly 100,000.

STUDENT LESSON 1.13 · WITH ANSWERS

Lesson 1.13: Adding Multi-Digit Numbers

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.13 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.13, Part A.
- [ ] Complete the rest of Workbook Lesson 1.13 (Parts B–E).

LET'S EXPLORE

On Saturday 2,768 people visited the aquarium. On Sunday 3,456 people visited. How many visitors came in all?

You already know how to add 2- and 3-digit numbers. The same steps work for numbers of any size.

The standard algorithm for addition

- Line up the numbers by place value: ones under ones, tens under tens.
- Add the ones first. If the sum is 10 or more, regroup: write the ones digit and carry 1 ten to the tens column.
- Add each place, moving left, and regroup whenever a sum is 10 or more.
- Estimate to check that your answer is reasonable.

WORKED EXAMPLE · ADD WITH REGROUPING

Find 2,768 + 3,456.

```
  1 1 1
  2,768
+ 3,456
-------
  6,224
```

1 Ones: 8 + 6 = 14. Write 4, regroup 1 ten.

2 Tens: 1 + 6 + 5 = 12. Write 2, regroup 1 hundred.

3 Hundreds: 1 + 7 + 4 = 12. Write 2, regroup 1 thousand.

4 Thousands: 1 + 2 + 3 = 6.

5 Check: 3,000 + 3,000 = 6,000. 6,224 is close.

Answer: 6,224 visitors

WORKED EXAMPLE · ADD THREE NUMBERS

Find 14,205 + 8,390 + 1,076.

1 Line up all three numbers by place value.

2 Ones: 5 + 0 + 6 = 11 → write 1, regroup 1.

3 Tens: 1 + 0 + 9 + 7 = 17 → write 7, regroup 1.

4 Hundreds: 1 + 2 + 3 + 0 = 6. Thousands: 4 + 8 + 1 = 13 → write 3, regroup 1. Ten thousands: 1 + 1 = 2.

Answer: 23,671

! WATCH OUT

Line the numbers up on the right. When numbers have different lengths, the ones must still be in the same column.

★ KEY IDEA

Add from right to left. When a place adds to 10 or more, regroup 10 of that unit as 1 of the next unit.

✎ TRY IT

Answers are shown in color.

1. 4,526 + 3,371

Answer: 7,897

2. 6,859 + 2,764

Answer: 9,623

3. 45,308 + 27,996

Answer: 73,304

Workbook Answer Key · Lesson 1.13

#	Question	Answer
1	4,526 + 3,371	7,897
2	6,859 + 2,764	9,623
3	45,308 + 27,996	73,304
4	678 + 245 (figure)	923
5	Estimate 3,912 + 2,088.	4,000 + 2,000 = 6,000

#	Question	Answer
6	3,214 + 5,463 (figure)	8,677
7	6,758 + 1,347 (figure)	8,105
8	2,905 + 4,698 (figure)	7,603
9	15,623 + 21,479 (figure)	37,102
10	48,097 + 36,815 (figure)	84,912
11	209,354 + 485,976 (figure)	695,330
12	The Mississippi Gulf Coast had 27,480 visitors in June and 31,915 in July. How many visitors came in both months?	59,395 visitors
13	Find the missing digits: 4,_27 + 2,58_ = 7,016 (each _ is one missing digit)	4,427 + 2,589 = 7,016 (missing digits 4 and 9) Ones: 7 + 9 = 16. Tens: 1 + 2 + 8 = 11. Hundreds: 1 + 4 + 5 = 10. Thousands: 1 + 4 + 2 = 7.
14	38,476 + 25,867 (figure)	64,343

LESSON 1.14 · TEACHER PLAN
Week 4 · Lesson 2 of 4
Unit 1: Place Value, Rounding, and Whole Numbers

Lesson 1.14: Subtracting with Regrouping

Student goal	I can subtract multi-digit numbers using the standard algorithm, regrouping when needed.
Objective	Students subtract multi-digit numbers with regrouping and check with addition.
Standard	4.NBT.4 — Fluently add and subtract multi-digit whole numbers using the standard algorithm.
Pacing	Unit 1, week 4, lesson 2 of 4
Time	About 50 minutes
Materials	Grid paper, base-ten blocks or play money, workbook Lesson 1.14
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	83 − 47 with blocks or money: trade a ten for 10 ones.
Launch	5	Read Let's Explore. Estimate: 6,000 − 4,000 = 2,000.
Teach	15	Work the example slowly on grid paper. Cross out and rewrite each regrouped digit neatly. Say: "Not enough ones? Go next door and trade 1 ten for 10 ones." Check by adding.
Guided Practice	10	Try It 1–3. Student checks each with addition.
Independent	12	Workbook Lesson 1.14.
Exit Ticket	3	Check.

COMMON MISCONCEPTIONS

- Subtracting the smaller digit from the larger regardless of position (the "Sam" mistake).
- Forgetting to reduce the digit that was regrouped from.

SUPPORT	EXTEND
• Play money trades for each regroup. • Ask before each column: Is the top digit big enough?	• Write a subtraction problem that needs regrouping in every place.

STUDENT LESSON 1.14 · WITH ANSWERS

Lesson 1.14: Subtracting with Regrouping

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.14 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.14, Part A.
- [ ] Complete the rest of Workbook Lesson 1.14 (Parts B–F).

LET'S EXPLORE

A school raised $6,215 for a new playground. It has spent $3,748. How much money is left?

When you subtract, start in the ones place. If the top digit is smaller than the bottom digit, you need to regroup (some people say “borrow”) from the next place to the left.

WORKED EXAMPLE · SUBTRACT WITH REGROUPING

Find 6,215 − 3,748.

$$\begin{array}{r} {\scriptstyle 5\,1\,11\,0\,15} \\ 6,215 \\ -\,3,748 \\ \hline 2,467 \end{array}$$

1. Ones: 5 − 8 won't work. Regroup 1 ten: the tens become 0 and the ones become 15. 15 − 8 = 7.
2. Tens: 0 − 4 won't work. Regroup 1 hundred: hundreds become 1, tens become 10. 10 − 4 = 6.
3. Hundreds: 1 − 7 won't work. Regroup 1 thousand: thousands become 5, hundreds become 11. 11 − 7 = 4.
4. Thousands: 5 − 3 = 2.
5. Check by adding: 2,467 + 3,748 = 6,215. ✓

Answer: $2,467 is left

Check subtraction with addition

Subtraction and addition are opposites. If 6,215 − 3,748 = 2,467, then 2,467 + 3,748 must equal 6,215. Always check your subtraction this way.

★ KEY IDEA

Subtract from right to left. If the top digit is too small, regroup 1 from the next place to make 10 more. Check with addition.

! WATCH OUT

Do not subtract the smaller digit from the bigger one when the bigger one is on the bottom. 5 – 8 is not 3! Regroup instead.

✎ TRY IT

Answers are shown in color.

1. 8,652 – 3,421

Answer: 5,231

2. 7,314 – 2,856

Answer: 4,458

3. 52,610 – 18,374

Answer: 34,236

Workbook Answer Key · Lesson 1.14

#	Question	Answer
1	8,652 – 3,421	5,231
2	7,314 – 2,856	4,458
3	52,610 – 18,374	34,236
4	83 – 47 (figure)	36
5	Add to check: 36 + 47 = ___	83
6	5,874 – 2,531 (figure)	3,343
7	6,342 – 1,857 (figure)	4,485
8	9,123 – 4,768 (figure)	4,355
9	47,315 – 28,649 (figure)	18,666
10	81,604 – 35,278 (figure)	46,326
11	530,412 – 278,653 (figure)	251,759

#	Question	Answer
12	Sam's work: 4,523 − 1,867 = 3,344. Explain the mistake and find the correct answer.	Sam subtracted the smaller digit from the larger in each place instead of regrouping. Correct: 2,656.
13	The Pearl River is 444 miles long. The Pascagoula River is 81 miles long. How much longer is the Pearl River?	363 miles
14	72,345 − 46,789 (figure)	25,556

LESSON 1.15 · TEACHER PLAN
Week 4 · Lesson 3 of 4
Unit 1: Place Value, Rounding, and Whole Numbers

Lesson 1.15: Subtracting Across Zeros

Student goal	I can subtract from numbers that have zeros, such as 5,000 − 2,368.
Objective	Students subtract from numbers with zeros by regrouping across zeros.
Standard	4.NBT.4 — Fluently add and subtract multi-digit whole numbers using the standard algorithm.
Pacing	Unit 1, week 4, lesson 3 of 4
Time	About 50 minutes
Materials	Play money ($1,000, $100, $10, $1), grid paper, workbook Lesson 1.15
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Ask: What is 1,000 − 1? 10,000 − 1?
Launch	5	Read Let's Explore. Act it out: start with five $1,000 bills and pay $2,368.
Teach	15	Model the trading with money: $1,000 → ten $100 → one $100 into ten $10 → one $10 into ten $1. Then show the written algorithm. Say: Every zero we pass through becomes a 9. Show the shortcut (subtract 1 from both).
Guided Practice	10	Try It 1–3, once with each method if time allows.
Independent	12	Workbook Lesson 1.15.
Exit Ticket	3	Check.

COMMON MISCONCEPTIONS

- Writing 0 − 8 = 8.
- Changing zeros to 10 instead of 9 when regrouping across them.

SUPPORT

- Use play money for the first two workbook problems.

EXTEND

- Which is faster for 10,000 − 6,428, regrouping or the shortcut? Why?

STUDENT LESSON 1.15 · WITH ANSWERS

Lesson 1.15: Subtracting Across Zeros

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.15 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.15, Part A.
- [ ] Complete the rest of Workbook Lesson 1.15 (Parts B–E).

LET'S EXPLORE

A store had 5,000 school folders. It sold 2,368. How many folders are left?

Zeros can make subtraction look tricky. When the place you need to regroup from is 0, keep moving left until you find a digit you can regroup.

WORKED EXAMPLE · REGROUP ACROSS ZEROS

Find 5,000 − 2,368.

```
   4 9 9 10
  5,000
- 2,368
-------
  2,632
```

1. Ones: 0 − 8. The tens and hundreds are 0, so go to the thousands.
2. Think of 5,000 as 500 tens. Regroup 1 ten: 500 tens becomes 499 tens and 10 ones.
3. Now the top number reads 4 thousands, 9 hundreds, 9 tens, 10 ones.
4. Subtract each place: 10 − 8 = 2, 9 − 6 = 3, 9 − 3 = 6, 4 − 2 = 2.

Answer: 2,632 folders

A shortcut: subtract 1 from both numbers

5,000 − 2,368 has the same answer as 4,999 − 2,367, because both numbers moved down by 1. With 4,999 on top, you never need to regroup!

WORKED EXAMPLE · USE THE SHORTCUT

Find 3,000 − 1,452.

1. Subtract 1 from both: 2,999 − 1,451.

2 No regrouping: 9 − 1 = 8, 9 − 5 = 4, 9 − 4 = 5, 2 − 1 = 1.

Answer: 1,548

★ KEY IDEA

When you meet zeros, regroup from the first nonzero digit to the left. Every zero you pass becomes a 9.

✎ TRY IT

Answers are shown in color.

1. 600 − 274

Answer: 326

2. 4,000 − 1,635

Answer: 2,365

3. 20,050 − 8,317

Answer: 11,733

Workbook Answer Key · Lesson 1.15

#	Question	Answer
1	600 − 274	326
2	4,000 − 1,635	2,365
3	20,050 − 8,317	11,733
4	1,000 − 1 = ___	999
5	Check: 2,632 + 2,368 = ___	5,000
6	700 − 386 (figure)	314
7	3,000 − 1,247 (figure)	1,753
8	8,004 − 2,569 (figure)	5,435
9	60,000 − 24,318 (figure)	35,682
10	40,200 − 17,635 (figure)	22,565
11	100,000 − 45,682 (figure)	54,318
12	A farmer has 1,000 feet of fencing. She uses 387 feet. How much fencing is left?	613 feet
13	Explain how the shortcut “subtract 1 from both numbers” makes 7,000 − 2,846 easier. Then solve.	6,999 − 2,845 needs no regrouping; answer 4,154.

#	Question	Answer
14	9,000 – 3,725 (figure)	5,275

LESSON 1.16 · TEACHER PLAN
Week 4 · Lesson 4 of 4
Unit 1: Place Value, Rounding, and Whole Numbers

Lesson 1.16: Solving Multi-Step Word Problems

Student goal	I can solve two-step word problems with addition and subtraction and check that my answer is reasonable.
Objective	Students solve two-step addition and subtraction word problems and check with estimation.
Standard	4.NBT.4, 4.OA.3 — Solve multistep word problems with whole numbers; assess reasonableness of answers using estimation (4.OA.3).
Pacing	Unit 1, week 4, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 1.16, colored pencils for bar models
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	One addition and one subtraction from the warm-up.
Launch	5	Read Let's Explore. Ask: What happens first? What happens next?
Teach	15	Introduce the four-step plan. Work Example 1 with a bar model. Say: "Many problems hide a first question. Find it before you answer the big question." Work Example 2.
Guided Practice	10	Try It 1–3. Student draws a bar model for each before calculating.
Independent	12	Workbook Lesson 1.16. Assign the Unit Review pages as practice for the test.
Exit Ticket	3	Check. Tomorrow: Unit 1 Test.

COMMON MISCONCEPTIONS

- Using only two of the three numbers.
- Choosing the operation from key words alone ("more" does not always mean add).

SUPPORT	EXTEND
• Underline the question; circle the numbers; box what each number means.	• Solve a three-step problem from the Think Deeper section and write one for a sibling.

STUDENT LESSON 1.16 · WITH ANSWERS

Lesson 1.16: Solving Multi-Step Word Problems

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.16 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.16, Part A.
- [ ] Complete the rest of Workbook Lesson 1.16 (Parts B–E).

LET'S EXPLORE

A food bank had 4,250 cans. On Monday it received 1,875 more cans. On Tuesday it gave away 2,690 cans. How many cans are left?

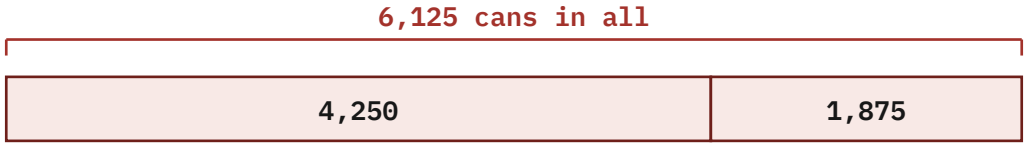
Some problems need more than one step. A plan helps.

Four-step plan

- Understand: What do you know? What do you need to find?
- Plan: Draw a bar model. Decide which operations to use and in what order.
- Solve: Do each step carefully.
- Check: Estimate. Is the answer reasonable?

WORKED EXAMPLE · TWO STEPS

Solve the food bank problem.



1 Step 1, add the new cans: 4,250 + 1,875 = 6,125.

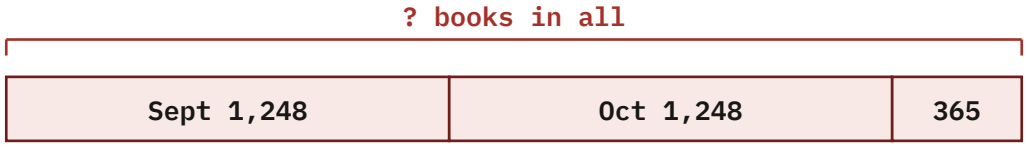
2 Step 2, subtract the cans given away: 6,125 − 2,690 = 3,435.

3 Check: 4,000 + 2,000 − 3,000 = 3,000. 3,435 is close.

Answer: 3,435 cans

WORKED EXAMPLE · COMPARE, THEN COMBINE

A bookstore sold 1,248 books in September. It sold 365 more books in October than in September. How many books did it sell in the two months?



1 Find October: 1,248 + 365 = 1,613.

2 Combine both months: 1,248 + 1,613 = 2,861.

3 Check: 1,200 + 1,600 = 2,800. Reasonable.

Answer: 2,861 books

★ KEY IDEA

Read the whole problem first. Find the hidden question (the first step), solve it, then use that answer for the next step.

✎ TRY IT

Answers are shown in color.

1. A library has 3,500 books. It buys 1,280 more and then donates 945. How many books does it have now?

Answer: 3,835

2. Ella's family drove 1,320 miles. Jon's family drove 486 fewer miles. How many miles did both families drive in all?

Answer: 2,154

3. A theater has 2,400 seats. 1,165 tickets were sold online and 738 at the door. How many seats are empty?

Answer: 497

Workbook Answer Key · Lesson 1.16

#	Question	Answer
1	A library has 3,500 books. It buys 1,280 more and then donates 945. How many books does it have now?	3,835
2	Ella's family drove 1,320 miles. Jon's family drove 486 fewer miles. How many miles did both families drive in all?	2,154
3	A theater has 2,400 seats. 1,165 tickets were sold online and 738 at the door. How many seats are empty?	497
4	6,215 − 2,480 (figure)	3,735
5	3,708 + 1,595 (figure)	5,303

#	Question	Answer
6	A school collected 2,315 pounds of paper in fall and 1,890 pounds in spring. The goal was 5,000 pounds. How many more pounds do they need?	795 pounds
7	Biloxi's minor-league ballpark had 4,860 fans on Friday and 5,215 on Saturday. Sunday had 2,080 fewer fans than Saturday. How many fans came in all three days?	13,210 fans Sunday = 5,215 − 2,080 = 3,135.
8	Mr. Davis had $9,000. He paid $3,475 for a used car and $1,260 for repairs. How much money does he have left?	$4,265
9	Write your own two-step word problem using the numbers 2,500, 1,340, and 875. Solve it.	Answers vary. Example: 2,500 + 1,340 − 875 = 2,965.
10	A tank holds 10,000 gallons. It has 6,450 gallons. 1,775 more gallons are added. How many more gallons until it is full?	1,775 gallons

UNIT REVIEW · ANSWER KEY
Study before the unit test

About 40 minutes

Unit 1 Review — Answer Key

Use the review as homework after Lesson 1.16 or as a study session. Any section with more than one mistake shows what to reteach before the test.

Answers

#	Question	Answer
1	What number is 10 times as much as 7,000?	70,000
2	Write the value of the 4 in 348,129.	40,000
3	Write 506,030 in expanded form.	500,000 + 6,000 + 30
4	Write 72,408 in word form.	seventy-two thousand four hundred eight
5	8,600 = ___ hundreds	86
6	Write >, <, or =: 409,511 ___ 409,151	>
7	Order from least to greatest: 58,302; 58,032; 53,820	53,820; 58,032; 58,302
8	Round 6,749 to the nearest hundred.	6,700
9	Round 425,600 to the nearest ten thousand.	430,000
10	Estimate 5,780 + 2,315.	6,000 + 2,000 = 8,000
11	Is 7,050 − 3,980 = 4,070 reasonable?	No. 7,000 − 4,000 = 3,000. Exact: 3,070.
12	37,865 + 24,579 (figure)	62,444
13	80,000 − 36,452 (figure)	43,548
14	A Mississippi county had 18,245 voters in one election and 21,610 in the next. How many more voters were there in the second election?	3,365 more voters

UNIT TEST · ANSWER KEY
At the end of the unit

About 45 minutes

Unit 1 Test — Answer Key

Allow about 45 minutes. Students may not use a calculator. Read word problems aloud if reading is a barrier; the test measures math.

Answers

#	Question	Answer
1	What number is 10 times as much as 300?	3,000
2	In 66,000, how many times greater is the first 6 than the second 6?	10 times
3	What is the value of the 9 in 592,407?	90,000
4	Write 40,000 + 3,000 + 80 + 7 in standard form.	43,087
5	Write 305,920 in expanded form.	300,000 + 5,000 + 900 + 20
6	Write “two hundred eleven thousand, four” in standard form.	211,004
7	5,300 = ___ hundreds	53
8	Write >, <, or =: 712,368 ___ 712,638	<
9	Order from greatest to least: 29,450; 92,450; 29,540	92,450; 29,540; 29,450
10	Round 4,382 to the nearest hundred.	4,400
11	Round 76,519 to the nearest thousand.	77,000
12	Round 849,000 to the nearest hundred thousand.	800,000
13	Round 35,000 to the nearest ten thousand.	40,000
14	Estimate 6,820 − 2,195 by rounding to the nearest thousand.	7,000 − 2,000 = 5,000
15	Is 3,980 + 2,090 = 8,070 reasonable? Explain.	No. 4,000 + 2,000 = 6,000. Exact: 6,070.
16	5,687 + 3,849 (figure)	9,536
17	62,408 + 19,795 (figure)	82,203
18	8,214 − 3,657 (figure)	4,557
19	50,000 − 27,384 (figure)	22,616

#	Question	Answer
20	A charity walk raised $12,450 the first year and $3,875 more the second year. How much did it raise in the two years together?	$28,775 Second year = $16,325. 1 pt plan/model, 1 pt computation, 1 pt answer with label.
21	A warehouse had 9,000 boxes. It shipped 2,736 boxes on Monday and 3,108 boxes on Tuesday. How many boxes are left? Show an estimate to check.	3,156 boxes Estimate: 9,000 − 3,000 − 3,000 = 3,000.

Scoring

Score	What it means / next step
25–29 (85%+)	Mastered. Move on to Unit 2.
21–24 (70–84%)	Proficient. Review missed items with the student and correct them together.
Below 21	Reteach the weakest part (A, B, C, or D) using its lessons, then give a short retest.

UNIT 2 · TEACHER OVERVIEW

Multiplication

Comparing with multiplication, place-value patterns, area models, partial products, and the standard algorithm.

Multiplication is a fast way to combine equal groups, and it can also compare amounts. In this unit you will multiply numbers up to four digits by one-digit numbers and two-digit numbers by two-digit numbers, using models that show why the methods work.

Big ideas

- Use multiplication to compare amounts ("times as many").
- Multiply by 10, 100, 1,000 and multiples of ten using place value.
- Multiply with area models, partial products, and the standard algorithm.
- Multiply two two-digit numbers.
- Solve multi-step word problems and check with estimates.

Standards

Code	Standard
4.OA.1	Interpret a multiplication equation as a comparison.
4.OA.2	Multiply or divide to solve word problems involving multiplicative comparison.
4.OA.3	Solve multistep word problems; assess reasonableness using estimation.
4.NBT.5	Multiply up to four digits by one digit and two two-digit numbers using place value strategies, equations, arrays, and area models.

What students should already know

- Multiplication facts to 10 × 10 (fluent recall).
- Place value to 1,000,000 (Unit 1).
- Adding multi-digit numbers (Unit 1).

Materials for the unit

- Grid paper

- Base-ten blocks
- Paper strips for bar models
- Multiplication fact cards for daily warm-ups

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 4	2.1	Multiplication as Comparison
Week 1	Lesson 2 of 4	2.2	Multiplying by 10, 100, and 1,000
Week 1	Lesson 3 of 4	2.3	Multiplying with Area Models
Week 1	Lesson 4 of 4	2.4	Partial Products
Week 1	End of week	Check	Weekly Check 4: Multiplying by One Digit
Week 2	Lesson 1 of 4	2.5	Multiplying Three-Digit Numbers
Week 2	Lesson 2 of 4	2.6	The Standard Algorithm for Multiplication
Week 2	Lesson 3 of 4	2.7	Multiplying Four-Digit Numbers
Week 2	Lesson 4 of 4	2.8	Estimating Products
Week 2	End of week	Check	Weekly Check 5: Multiplying Two-Digit Numbers
Week 3	Lesson 1 of 4	2.9	Multiplying Multiples of Ten
Week 3	Lesson 2 of 4	2.10	Two-Digit by Two-Digit: Area Models
Week 3	Lesson 3 of 4	2.11	Two-Digit by Two-Digit: Partial Products
Week 3	Lesson 4 of 4	2.12	Choosing and Checking a Strategy
Week 3	End of week	Check	Weekly Check 6: Multiplication Problem Solving
Week 4	Lesson 1 of 4	2.13	Multiplication Word Problems
Week 4	Lesson 2 of 4	2.14	Multi-Step Problems with Multiplication
Week 4	Lesson 3 of 4	2.15	Comparison Word Problems
Week 4	Lesson 4 of 4	2.16	Multiplication Problem Solving
Week 4	Before the test	Review	Unit 2 Review
Week 4	End of unit	Test	Unit 2 Test

TEACHING THIS UNIT AT HOME

If multiplication facts are not automatic yet, spend 5 minutes a day on them. Facts make every lesson in this unit easier.

UNIT VOCABULARY

product	The answer to a multiplication problem.
factor	A number being multiplied.
area model	A rectangle split into parts to show multiplication.
partial products	Smaller products added to find the whole product.
estimate	A close answer found with rounded numbers.

LESSON 2.1 · TEACHER PLAN
Week 1 · Lesson 1 of 4
Unit 2: Multiplication

Lesson 2.1: Multiplication as Comparison

Student goal	I can read and write multiplication equations that compare two amounts, like “24 is 3 times as many as 8.”
Objective	Students interpret multiplication as comparison and solve “times as many” problems with bar models.
Standard	4.OA.1, 4.OA.2 — Interpret a multiplication equation as a comparison; multiply to solve word problems involving multiplicative comparison.
Pacing	Unit 2, week 1, lesson 1 of 4
Time	About 50 minutes
Materials	Paper strips (cut equal lengths), workbook Lesson 2.1
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Quick facts: 6 × 7, 8 × 4, 9 × 3.
Launch	5	Read Let's Explore. Ask: What does ‘3 times as many’ mean?
Teach	15	Lay one paper strip for Ellie, then three equal strips for her sister. Say: “Times as many means copies of the smaller amount.” Work both examples. Read the Watch Out together.
Guided Practice	10	Try It 1–3. For each word problem, draw the bars first.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Confusing “times as many” with “more than.”
- Putting the comparison backwards (6 is 3 times as many as 18).

SUPPORT	EXTEND
• Act out with paper strips or cubes.	• Write your own “times as many” story about your family. • 18 is how many times as many as 3?

STUDENT LESSON 2.1 · WITH ANSWERS

Lesson 2.1: Multiplication as Comparison

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.1 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.1, Part A.
- [ ] Complete the rest of Workbook Lesson 2.1 (Parts B–F).

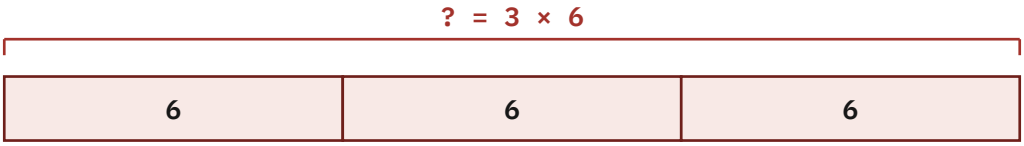
LET'S EXPLORE

Ellie has 6 stickers. Her big sister has 3 times as many stickers. How many stickers does her sister have?

You know multiplication as equal groups: 3 groups of 6. Multiplication can also compare two amounts.



Ellie: 6 stickers



Sister: 3 times as many

The sister's bar is 3 times as long as Ellie's bar. 3 × 6 = 18, so her sister has 18 stickers. We say: 18 is 3 times as many as 6.

WORKED EXAMPLE · WRITE A COMPARISON EQUATION

Write an equation for: 35 is 5 times as many as 7.

1 "Is" means equals.

2 "5 times as many as 7" means 5 × 7.

Answer: 35 = 5 × 7

WORKED EXAMPLE · FIND THE UNKNOWN AMOUNT

A tree is 4 times as tall as a bush. The bush is 2 feet tall. How tall is the tree?

1 The tree is the bigger amount. It is 4 times the bush.

2 4 × 2 = 8.

Answer: 8 feet

! WATCH OUT

"4 times as many" is not the same as "4 more." 4 more than 2 is 6. 4 times as many as 2 is 8.

★ KEY IDEA

In a comparison, one amount is a number of times as many as another. Use a bar model: the larger bar is made of copies of the smaller bar.

✎ TRY IT

Answers are shown in color.

1. Write an equation: 42 is 6 times as many as 7.

Answer: 42 = 6 × 7

2. Sam ran 3 miles. Ana ran 5 times as far. How far did Ana run?

Answer: 15 miles

3. Which is greater: 2 more than 9, or 2 times as many as 9?

Answer: 2 times as many (18 > 11)

Workbook Answer Key · Lesson 2.1

#	Question	Answer
1	Write an equation: 42 is 6 times as many as 7.	42 = 6 × 7
2	Sam ran 3 miles. Ana ran 5 times as far. How far did Ana run?	15 miles
3	Which is greater: 2 more than 9, or 2 times as many as 9?	2 times as many (18 > 11)
4	6 × 7 = ___	42
5	8 × 4 = ___	32
6	24 is 4 times as many as 6.	24 = 4 × 6
7	56 is 8 times as many as 7.	56 = 8 × 7
8	30 is 3 times as many as 10.	30 = 3 × 10
9	45 is 9 times as many as 5.	45 = 9 × 5
10	63 = 7 × 9	63 is 7 times as many as 9 (or 9 times as many as 7).

#	Question	Answer
11	20 = 5 × 4	20 is 5 times as many as 4 (or 4 times as many as 5).
12	A cat weighs 9 pounds. A dog weighs 4 times as much. How much does the dog weigh?	36 pounds
13	Lillie read 7 books this month. Ellie read 3 times as many. How many books did Ellie read?	21 books
14	A red ribbon is 8 inches long. A blue ribbon is 6 times as long. How long is the blue ribbon?	48 inches
15	Ben has 5 more cards than Zoe. Zoe has 5 cards. How many does Ben have? Is this a times-as-many problem?	10 cards; no, it is an additive comparison.
16	A puppy is 3 months old. Its mother is 8 times as old. How old is the mother? Write an equation.	24 months; 8 × 3 = 24

LESSON 2.2 · TEACHER PLAN
Week 1 · Lesson 2 of 4
Unit 2: Multiplication

Lesson 2.2: Multiplying by 10, 100, and 1,000

Student goal	I can use place value patterns to multiply by 10, 100, and 1,000 and by multiples of 10.
Objective	Students multiply one-digit numbers by multiples of 10, 100, and 1,000 using place value.
Standard	4.NBT.1, 4.NBT.5 — Use place value understanding and properties of operations to perform multi-digit arithmetic; recognize the ten-times relationship.
Pacing	Unit 2, week 1, lesson 2 of 4
Time	About 50 minutes
Materials	Place-value chart, base-ten blocks, workbook Lesson 2.2
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Review: what is 10 times as much as 34?
Launch	5	Read Let's Explore. Student solves 7 boxes, then 70 boxes.
Teach	15	Build 3 × 4 with rods (3 groups of 4 tens = 12 tens = 120). Say: "Multiply the fact, then name the unit." Fill in the table together and look for the pattern.
Guided Practice	10	Try It 1–3. Then a speed round: you say a problem, student answers.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Counting zeros only in the factors and missing the extra zero from facts like 5 × 8 = 40.

SUPPORT

- Say the unit out loud: "5 times 8 hundreds is 40 hundreds."

EXTEND

- Find 30 × 40. How does it relate to 3 × 4?

STUDENT LESSON 2.2 · WITH ANSWERS

Lesson 2.2: Multiplying by 10, 100, and 1,000

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.2 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.2, Part A.
- [ ] Complete the rest of Workbook Lesson 2.2 (Parts B–F).

LET'S EXPLORE

A box holds 100 crayons. How many crayons are in 7 boxes? In 70 boxes?

In Unit 1 you learned that each place is worth 10 times the place to its right. That makes multiplying by 10, 100, and 1,000 fast.

Fact	× 10	× 100	× 1,000
7 × 1 = 7	7 × 10 = 70	7 × 100 = 700	7 × 1,000 = 7,000
4 × 6 = 24	4 × 60 = 240	4 × 600 = 2,400	4 × 6,000 = 24,000

When you multiply by 10, every digit moves one place to the left. Multiplying by 100 moves digits two places. Multiplying by 1,000 moves them three places.

WORKED EXAMPLE · USE A BASIC FACT

Find 5 × 800.

1 800 is 8 hundreds.

2 5 × 8 hundreds = 40 hundreds.

3 40 hundreds = 4,000.

Answer: 4,000

! WATCH OUT

Count the zeros carefully when the basic fact already ends in zero: 5 × 800 = 4,000 has three zeros, because 5 × 8 = 40.

★ KEY IDEA

To multiply by a multiple of 10, 100, or 1,000: multiply the basic fact, then think about place value (how many tens, hundreds, or thousands).

✎ TRY IT

Answers are shown in color.

1. 9 × 100

Answer: 900

2. 6 × 70

Answer: 420

3. 8 × 5,000

Answer: 40,000

Workbook Answer Key · Lesson 2.2

#	Question	Answer
1	9 × 100	900
2	6 × 70	420
3	8 × 5,000	40,000
4	10 × 34	340
5	How many hundreds are in 2,500?	25
6	3 × 10	30
7	3 × 100	300
8	3 × 1,000	3,000
9	7 × 40	280
10	6 × 300	1,800
11	9 × 2,000	18,000
12	5 × 60	300
13	4 × 500	2,000
14	8 × 7,000	56,000
15	2 × 50,000	100,000
16	6 × ___ = 3,600	600

#	Question	Answer
17	___ × 90 = 720	8
18	5 × ___ = 2,000	400
19	7 × ___ = 280	40
20	A store sells pencils in packs of 100. How many pencils are in 9 packs?	900 pencils
21	Each ticket to the museum costs $20. How much do 8 tickets cost?	$160
22	Explain why 4 × 500 = 2,000 has three zeros.	4 × 5 = 20 already has a zero, and ×100 adds two more places: 20 hundreds = 2,000.

LESSON 2.3 · TEACHER PLAN
Week 1 · Lesson 3 of 4
Unit 2: Multiplication

Lesson 2.3: Multiplying with Area Models

Student goal	I can use an area model to multiply a two-digit number by a one-digit number.
Objective	Students multiply 2-digit by 1-digit numbers with area models and the distributive property.
Standard	4.NBT.5 — Multiply a whole number of up to four digits by a one-digit whole number, using strategies based on place value; illustrate with rectangular arrays and area models.
Pacing	Unit 2, week 1, lesson 3 of 4
Time	About 50 minutes
Materials	Grid paper, base-ten blocks, workbook Lesson 2.3
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Facts: 7 × 30 and 7 × 4. Ask: What is 7 × 34? (add them)
Launch	5	Read Let's Explore. Ask: Would you want to count 92 plants one by one?
Teach	15	Build 4 × 23 with blocks: 4 rows of 2 rods and 3 units. Push the rods together, then the units. Draw the area model next to the blocks. Say: The rectangle is a sketch of the blocks. Work the example (6 × 47).
Guided Practice	10	Try It 1–3 with area models.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Multiplying only the tens (6 × 40) and forgetting the ones.
- Adding instead of multiplying inside the boxes.

SUPPORT	EXTEND
• Use grid paper so each box is neat.	• Use an area model for 3 × 125 (split into 100 + 20 + 5).

STUDENT LESSON 2.3 · WITH ANSWERS

Lesson 2.3: Multiplying with Area Models

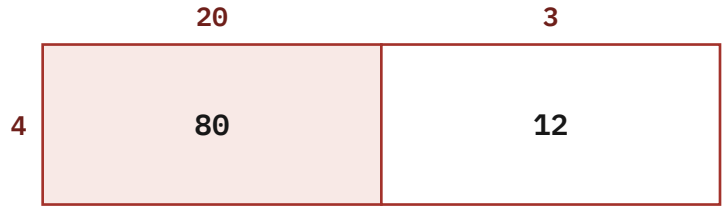
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.3 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.3, Part A.
- [ ] Complete the rest of Workbook Lesson 2.3 (Parts B–F).

LET'S EXPLORE

A garden has 4 rows with 23 tomato plants in each row. How many plants are there?

An array of 4 rows of 23 would take a long time to draw. Instead, split 23 into 20 + 3 and draw a rectangle.



4 × 23 split into 4 × 20 and 4 × 3.

The big part is 4 × 20 = 80. The small part is 4 × 3 = 12. Add the parts: 80 + 12 = 92 plants. Breaking a number apart like this uses the distributive property.

WORKED EXAMPLE · USE AN AREA MODEL

Find 6 × 47.



1. Split 47 into 40 + 7.
2. 6 × 40 = 240.
3. 6 × 7 = 42.
4. Add: 240 + 42 = 282.

Answer: 282

★ KEY IDEA

Break the larger factor into tens and ones. Multiply each part. Add the parts.

✎ TRY IT

Answers are shown in color.

1. 3 × 25

Answer: 75

2. 5 × 64

Answer: 320

3. 8 × 39

Answer: 312

Workbook Answer Key · Lesson 2.3

#	Question	Answer
1	3 × 25	75
2	5 × 64	320
3	8 × 39	312
4	7 × 30	210
5	7 × 4	28
6	3 × 42 (figure)	126 (120 + 6)
7	5 × 36 (figure)	180 (150 + 30)
8	7 × 28 (figure)	196 (140 + 56)
9	4 × 93 (figure)	372 (360 + 12)
10	6 × 54	324
11	9 × 31	279
12	2 × 87	174
13	8 × 46	368
14	A bookshelf has 5 shelves. Each shelf holds 38 books. How many books can it hold?	190 books
15	Use an area model to find 7 × 65.	455 (420 + 35)

LESSON 2.4 · TEACHER PLAN
Week 1 · Lesson 4 of 4
Unit 2: Multiplication

Lesson 2.4: Partial Products

Student goal	I can multiply using partial products and record my work in a vertical list.
Objective	Students use partial products to multiply 2- and 3-digit numbers by 1-digit numbers.
Standard	4.NBT.5 — Multiply a whole number of up to four digits by a one-digit whole number using strategies based on place value.
Pacing	Unit 2, week 1, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.4, grid paper
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Area model warm-up (3 × 24).
Launch	5	Read Let's Explore. Ask: Can we write the boxes as a list?
Teach	15	Draw 6 × 47 as an area model, then write each box as a line of partial products. Say: "Every box becomes a line. Label each line." Work Example 2 (three-digit).
Guided Practice	10	Try It 1–3. Student labels each partial product.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Multiplying the digit instead of its value (5 × 6 instead of 5 × 60).

SUPPORT

- Keep the area model beside the list for the first few problems.

EXTEND

- Find 6 × 1,234 with partial products.

STUDENT LESSON 2.4 · WITH ANSWERS

Lesson 2.4: Partial Products

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.4 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.4, Part A.
- [ ] Complete the rest of Workbook Lesson 2.4 (Parts B–E).

LET'S EXPLORE

You found 6 × 47 with an area model. Is there a way to write the same steps without drawing a rectangle?

Partial products is the area model written as a list. Each box of the area model becomes one line.

WORKED EXAMPLE · TWO-DIGIT BY ONE-DIGIT

Find 6 × 47 with partial products.

$$\begin{array}{rr} & 47 \\ \times & 6 \\ \hline 6 \times 7 & 42 \\ 6 \times 40 & 240 \\ \hline & 282 \end{array}$$

1. Multiply the ones: 6 × 7 = 42.
2. Multiply the tens: 6 × 40 = 240.
3. Add the partial products: 42 + 240 = 282.

Answer: 282

WORKED EXAMPLE · THREE-DIGIT BY ONE-DIGIT

Find 3 × 254 with partial products.

```
      254
    ×   3
3 × 4  12
3 × 50 150
3 × 200 600
      762
```

1 3 × 4 = 12

2 3 × 50 = 150

3 3 × 200 = 600

4 Add: 12 + 150 + 600 = 762

Answer: 762

★ KEY IDEA

Each partial product matches one part of the area model. Write the factors you multiplied next to each line so you can check your work.

✎ TRY IT

Answers are shown in color.

1. 4 × 73

Answer: 292

2. 7 × 126

Answer: 882

3. 5 × 308

Answer: 1,540

Workbook Answer Key · Lesson 2.4

#	Question	Answer
1	4 × 73	292
2	7 × 126	882
3	5 × 308	1,540
4	9 × 60	540
5	Area model: 3 × 24	72

#	Question	Answer
6	36 × 4 (figure)	144
7	58 × 7 (figure)	406
8	92 × 6 (figure)	552
9	143 × 2 (figure)	286
10	215 × 5 (figure)	1,075
11	407 × 8 (figure)	3,256
12	Jada found 5 × 64 by writing 5 × 4 = 20 and 5 × 6 = 30, then 20 + 30 = 50. What did she do wrong?	She multiplied 5 × 6 instead of 5 × 60. Correct: 20 + 300 = 320.
13	Use partial products to find 4 × 386. (figure)	1,544

WEEKLY CHECK · ANSWER KEY
After Lesson 2.4

About 20 minutes

Weekly Check 4: Multiplying by One Digit — Answer Key

Answers

#	Question	Answer
1	Write an equation: 54 is 6 times as many as 9.	54 = 6 × 9
2	7 × 300	2,100
3	Draw an area model for 4 × 57.	228 (200 + 28)
4	Use partial products: 3 × 246 (figure)	738
5	495 × 6 (figure)	2,970
6	2,307 × 4 (figure)	9,228
7	Estimate 8 × 612.	8 × 600 = 4,800
8	A dog is 5 times as heavy as a cat. The cat weighs 11 pounds. How much does the dog weigh?	55 pounds

LESSON 2.5 · TEACHER PLAN
Week 2 · Lesson 1 of 4
Unit 2: Multiplication

Lesson 2.5: Multiplying Three-Digit Numbers

Student goal	I can multiply a three-digit number by a one-digit number with an area model or partial products.
Objective	Students multiply 3-digit by 1-digit numbers using area models and partial products.
Standard	4.NBT.5 — Multiply a whole number of up to four digits by a one-digit whole number.
Pacing	Unit 2, week 2, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.5, grid paper
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	8 × 400 and 8 × 30.
Launch	5	Read Let's Explore (just estimate: about 200 × 50).
Teach	15	Draw the three-part area model for 5 × 324. Say: "Hundreds, tens, ones: three boxes, three partial products." Work Example 2 and discuss the zero.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Writing 7 × 609 as 7 × 69 (dropping the zero's place).
- Misaligning partial products.

SUPPORT

- Use grid paper to line up partial products.

EXTEND

- Find 9 × 999 in your head. (9 × 1,000 − 9)

STUDENT LESSON 2.5 · WITH ANSWERS

Lesson 2.5: Multiplying Three-Digit Numbers

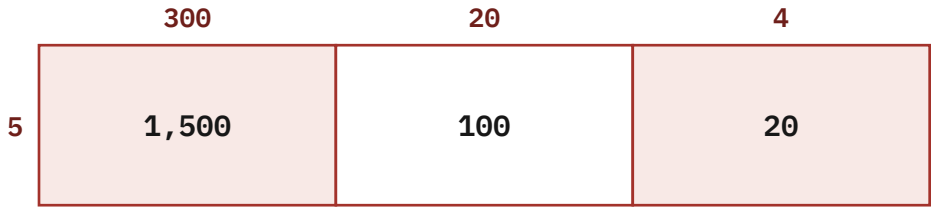
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.5 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.5, Part A.
- [ ] Complete the rest of Workbook Lesson 2.5 (Parts B–E).

LET'S EXPLORE

A school bus can carry 48 students. A Mississippi school district has 215 buses. How could we find how many students they can carry?

Three-digit numbers split into three parts: hundreds, tens, and ones. The area model has three boxes.



5 × 324 = 1,500 + 100 + 20 = 1,620

WORKED EXAMPLE · AREA MODEL AND PARTIAL PRODUCTS

Find 5 × 324.

1 5 × 300 = 1,500

2 5 × 20 = 100

3 5 × 4 = 20

4 Add: 1,500 + 100 + 20 = 1,620

Answer: 1,620

WORKED EXAMPLE · A NUMBER WITH A ZERO

Find 7 × 609.

1 609 = 600 + 9. There are no tens, so there are only two parts.

2 7 × 600 = 4,200

3 7 × 9 = 63

4 4,200 + 63 = 4,263

Answer: 4,263

★ KEY IDEA

A zero in a factor means that part of the product is zero. You can skip it, but keep the other places lined up.

✎ TRY IT

Answers are shown in color.

1. 3 × 412

Answer: 1,236

2. 6 × 275

Answer: 1,650

3. 4 × 708

Answer: 2,832

Workbook Answer Key · Lesson 2.5

#	Question	Answer
1	3 × 412	1,236
2	6 × 275	1,650
3	4 × 708	2,832
4	8 × 400	3,200
5	8 × 30	240
6	132 × 3 (figure)	396
7	241 × 4 (figure)	964
8	526 × 6 (figure)	3,156
9	380 × 7 (figure)	2,660
10	904 × 5 (figure)	4,520
11	657 × 9 (figure)	5,913
12	A theater has 8 sections with 125 seats each. How many seats are there?	1,000 seats
13	Each bag of rice weighs 5 pounds. A food pantry has 248 bags. How many pounds of rice is that?	1,240 pounds

#	Question	Answer
14	6 × 437 (figure)	2,622

LESSON 2.6 · TEACHER PLAN
Week 2 · Lesson 2 of 4
Unit 2: Multiplication

Lesson 2.6: The Standard Algorithm for Multiplication

Student goal	I can multiply a multi-digit number by a one-digit number using the standard algorithm.
Objective	Students use the standard algorithm to multiply by one-digit numbers.
Standard	4.NBT.5 — Multiply a whole number of up to four digits by a one-digit whole number.
Pacing	Unit 2, week 2, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.6, grid paper
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Partial products for 3 × 52.
Launch	5	Read Let's Explore.
Teach	15	Write 6 × 47 with partial products, then right beside it with the standard algorithm. Point out where the 4 tens went. Say: “Multiply, then add what you carried.” Work Example 2.
Guided Practice	10	Try It 1–3. Student checks one with partial products.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Adding the carried digit before multiplying.
- Forgetting the carried digit.

SUPPORT	EXTEND
• Keep partial products as a check. • Write carried digits small and in a different color.	• Multiply 4-digit numbers: 7 × 3,486.

STUDENT LESSON 2.6 · WITH ANSWERS

Lesson 2.6: The Standard Algorithm for Multiplication

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.6 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.6, Part A.
- [ ] Complete the rest of Workbook Lesson 2.6 (Parts B–E).

LET'S EXPLORE

Partial products work every time, but they take a lot of lines. Can we write 6 × 47 on just one line?

The standard algorithm does the same multiplying as partial products, but it regroups as it goes so the answer fits on one line.

WORKED EXAMPLE · STANDARD ALGORITHM

Find 6 × 47.

$$\begin{array}{r} 47 \\ \times\ \ 6 \\ \hline 282 \end{array}$$

1 Ones: 6 × 7 = 42. Write 2 in the ones place. Regroup 4 tens: write a small 4 above the tens.

2 Tens: 6 × 4 tens = 24 tens. Add the regrouped 4 tens: 28 tens.

3 Write 28 in front of the 2.

Answer: 282

WORKED EXAMPLE · THREE DIGITS

Find 4 × 386.

$$\begin{array}{r} 386 \\ \times\ \ 4 \\ \hline 1,544 \end{array}$$

1 Ones: 4 × 6 = 24. Write 4, regroup 2 tens.

2 Tens: 4 × 8 = 32, plus 2 = 34 tens. Write 4, regroup 3 hundreds.

3 Hundreds: 4 × 3 = 12, plus 3 = 15 hundreds. Write 15.

Answer: 1,544

! WATCH OUT

Multiply first, then add the regrouped number. Do not add the regrouped digit before multiplying.

★ KEY IDEA

Standard algorithm: multiply the ones, regroup; multiply the tens, add the regrouped tens, regroup; keep going left.

✎ TRY IT

Answers are shown in color.

1. 5 × 68

Answer: 340

2. 3 × 457

Answer: 1,371

3. 8 × 209

Answer: 1,672

Workbook Answer Key · Lesson 2.6

#	Question	Answer
1	5 × 68	340
2	3 × 457	1,371
3	8 × 209	1,672
4	Partial products: 3 × 52	156
5	7 × 8 + 5	61
6	29 × 3 (figure)	87
7	64 × 7 (figure)	448
8	85 × 6 (figure)	510
9	318 × 4 (figure)	1,272
10	472 × 5 (figure)	2,360

#	Question	Answer
11	836 × 9 (figure)	7,524
12	Solve 7 × 245 both ways. Do you get the same answer?	1,715 both ways
13	6 × 584 (figure)	3,504

LESSON 2.7 · TEACHER PLAN
Week 2 · Lesson 3 of 4
Unit 2: Multiplication

Lesson 2.7: Multiplying Four-Digit Numbers

Student goal	I can multiply a four-digit number by a one-digit number.
Objective	Students multiply 4-digit by 1-digit numbers and check with estimates.
Standard	4.NBT.5 — Multiply a whole number of up to four digits by a one-digit whole number.
Pacing	Unit 2, week 2, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.7
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	4 × 3,000 and 5 × 216.
Launch	5	Read Let's Explore; estimate first (6 × 1,000 = 6,000; 6 × 1,300 = 7,800).
Teach	15	Work Example 1 slowly. Say: One more place, one more step. Work Example 2. Model estimating first.
Guided Practice	10	Try It 1–3 with an estimate for each.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Losing track of the regrouped digit over a zero.

SUPPORT

- Partial products are always allowed.

EXTEND

- Find 9 × 9,999.

STUDENT LESSON 2.7 · WITH ANSWERS

Lesson 2.7: Multiplying Four-Digit Numbers

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.7 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.7, Part A.
- [ ] Complete the rest of Workbook Lesson 2.7 (Parts B–E).

LET'S EXPLORE

A farm ships 1,250 crates of sweet potatoes each week. How many crates does it ship in 6 weeks?

The same steps work for four-digit numbers. There is one more place, so one more step.

WORKED EXAMPLE · FOUR DIGITS

Find 6 × 1,250.

$$\begin{array}{r} 1{,}250 \\ \times \quad 6 \\ \hline 7{,}500 \end{array}$$

1. Ones: 6 × 0 = 0.
2. Tens: 6 × 5 = 30 tens. Write 0, regroup 3 hundreds.
3. Hundreds: 6 × 2 = 12, plus 3 = 15 hundreds. Write 5, regroup 1 thousand.
4. Thousands: 6 × 1 = 6, plus 1 = 7 thousands.

Answer: 7,500 crates

WORKED EXAMPLE · CHECK WITH AN ESTIMATE

Find 4 × 2,397 and check.

1. Standard algorithm gives 9,588.
2. Estimate: 4 × 2,000 = 8,000 and 4 × 2,400 = 9,600. 9,588 is close.

Answer: 9,588

★ KEY IDEA

Estimate before you multiply. If your answer is far from the estimate, check each step.

✎ TRY IT

Answers are shown in color.

1. 3 × 2,143

Answer: 6,429

2. 5 × 1,608

Answer: 8,040

3. 7 × 3,425

Answer: 23,975

Workbook Answer Key · Lesson 2.7

#	Question	Answer
1	3 × 2,143	6,429
2	5 × 1,608	8,040
3	7 × 3,425	23,975
4	4 × 3,000	12,000
5	5 × 216	1,080
6	1,324 × 2 (figure)	2,648
7	2,516 × 3 (figure)	7,548
8	3,047 × 4 (figure)	12,188
9	1,689 × 5 (figure)	8,445
10	4,372 × 6 (figure)	26,232
11	2,958 × 8 (figure)	23,664
12	A plane flies 2,475 miles each trip. How far does it fly in 4 trips?	9,900 miles
13	A school orders 3 boxes of paper with 5,000 sheets each and 2 boxes with 2,500 sheets each. How many sheets in all?	20,000 sheets
14	7 × 1,836 (figure)	12,852

LESSON 2.8 · TEACHER PLAN
Week 2 · Lesson 4 of 4
Unit 2: Multiplication

Lesson 2.8: Estimating Products

Student goal	I can round to estimate products and decide if an answer is reasonable.
Objective	Students estimate products and judge reasonableness.
Standard	4.NBT.3, 4.OA.3 — Round to any place; assess reasonableness using estimation.
Pacing	Unit 2, week 2, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.8, store flyer (optional)
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Round 467; 8 × 600.
Launch	5	Read Let's Explore.
Teach	15	Work both examples. Say: Round, then use a basic fact. Discuss overestimate vs. underestimate.
Guided Practice	10	Try It 1–3. Then Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Rounding both factors when one is already a single digit.

SUPPORT

- Round only the bigger number.

EXTEND

- Estimate 38 × 52 two ways.

STUDENT LESSON 2.8 · WITH ANSWERS

Lesson 2.8: Estimating Products

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.8 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.8, Part A.
- [ ] Complete the rest of Workbook Lesson 2.8 (Parts B–F).

LET'S EXPLORE

A ticket costs $38. About how much will 7 tickets cost? Is $200 enough?

To estimate a product, round the larger factor to its greatest place, then multiply in your head.

WORKED EXAMPLE · ESTIMATE

Estimate 7 × 38.

1 Round 38 to 40.

2 7 × 40 = 280.

Answer: about $280, so $200 is not enough

WORKED EXAMPLE · OVER OR UNDER?

Estimate 4 × 2,684. Is the exact answer more or less than the estimate?

1 Round 2,684 to 3,000.

2 4 × 3,000 = 12,000.

3 We rounded up, so the exact answer is less than 12,000. (Exact: 10,736.)

Answer: about 12,000

★ KEY IDEA

Rounding up gives an overestimate. Rounding down gives an underestimate.

✎ TRY IT

Answers are shown in color.

1. Estimate 6 × 52

Answer: 6 × 50 = 300

2. Estimate 3 × 789

Answer: 3 × 800 = 2,400

3. Is 5 × 412 = 20,060 reasonable?

Answer: No. 5 × 400 = 2,000. Exact 2,060.

Workbook Answer Key · Lesson 2.8

#	Question	Answer
1	Estimate 6 × 52	6 × 50 = 300
2	Estimate 3 × 789	3 × 800 = 2,400
3	Is 5 × 412 = 20,060 reasonable?	No. 5 × 400 = 2,000. Exact 2,060.
4	Round 467 to the nearest hundred.	500
5	8 × 600	4,800
6	4 × 63	about 240
7	9 × 28	about 270
8	5 × 347	about 1,500
9	6 × 812	about 4,800
10	3 × 4,250	about 12,000
11	7 × 1,960	about 14,000
12	8 × 49 = 392 A. Yes B. No	Yes (8 × 50 = 400)
13	6 × 315 = 1,290 A. Yes B. No	No (6 × 300 = 1,800; exact 1,890)
14	4 × 2,105 = 8,420 A. Yes B. No	Yes (4 × 2,000 = 8,000)
15	9 × 711 = 63,999 A. Yes B. No	No (9 × 700 = 6,300; exact 6,399)
16	A camp needs 6 tents that cost $189 each. The camp has $1,000. Use an estimate to decide if that is enough. Then find the exact cost.	Estimate 6 × 200 = 1,200 (overestimate) — too close to tell; exact 6 × 189 = $1,134, so not enough.
17	Estimate 5 × 693. Is the exact answer more or less than your estimate?	5 × 700 = 3,500; exact is less (3,465).

WEEKLY CHECK · ANSWER KEY
After Lesson 2.8
About 20 minutes

Weekly Check 5: Multiplying Two-Digit Numbers — Answer Key

Answers

#	Question	Answer
1	60 × 70	4,200
2	80 × 50	4,000
3	Use an area model: 23 × 15	345
4	Use partial products: 42 × 36 (figure)	1,512
5	Estimate 58 × 31.	60 × 30 = 1,800
6	25 × 28	700
7	Is 47 × 22 = 1,834 reasonable? Explain.	No. 50 × 20 = 1,000. Exact: 1,034.

LESSON 2.9 · TEACHER PLAN
Week 3 · Lesson 1 of 4
Unit 2: Multiplication

Lesson 2.9: Multiplying Multiples of Ten

Student goal	I can multiply two multiples of ten, like 30 × 40, using basic facts and place value.
Objective	Students multiply pairs of multiples of ten.
Standard	4.NBT.5 — Multiply two two-digit numbers using strategies based on place value.
Pacing	Unit 2, week 3, lesson 1 of 4
Time	About 50 minutes
Materials	Base-ten flats, workbook Lesson 2.9
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	6 × 9 and 6 × 90.
Launch	5	Read Let's Explore.
Teach	15	Show that a ten × a ten is a hundred flat (10 rows of 10). Say: "Tens times tens makes hundreds." Fill in the table; work both examples.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Writing 30 × 40 = 120.

SUPPORT

- Use a 10 × 10 grid to show tens × tens = hundreds.

EXTEND

- Find 300 × 40.

STUDENT LESSON 2.9 · WITH ANSWERS

Lesson 2.9: Multiplying Multiples of Ten

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.9 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.9, Part A.
- [ ] Complete the rest of Workbook Lesson 2.9 (Parts B–E).

LET'S EXPLORE

A classroom has 30 rows of seats in an auditorium with 40 seats in each row. How many seats is that?

30 × 40 means 3 tens times 4 tens. 3 × 4 = 12, and tens times tens makes hundreds. So 30 × 40 = 12 hundreds = 1,200.

Problem	Basic fact	Think	Product
30 × 40	3 × 4 = 12	tens × tens = hundreds	1,200
20 × 60	2 × 6 = 12	12 hundreds	1,200
50 × 80	5 × 8 = 40	40 hundreds	4,000

WORKED EXAMPLE · TENS TIMES TENS

Find 70 × 90.

1 7 × 9 = 63.

2 tens × tens = hundreds: 63 hundreds.

Answer: 6,300

WORKED EXAMPLE · TENS TIMES ONES

Find 40 × 6.

1 4 × 6 = 24.

2 tens × ones = tens: 24 tens.

Answer: 240

★ KEY IDEA

Multiply the basic fact. Then count: tens × tens = hundreds. A quick check: the product has as many zeros as the two factors together, plus any zero from the fact.

✎ TRY IT

Answers are shown in color.

1. 20 × 30

Answer: 600

2. 60 × 50

Answer: 3,000

3. 80 × 70

Answer: 5,600

Workbook Answer Key · Lesson 2.9

#	Question	Answer
1	20 × 30	600
2	60 × 50	3,000
3	80 × 70	5,600
4	6 × 9	54
5	6 × 90	540
6	10 × 40	400
7	20 × 20	400
8	30 × 50	1,500
9	40 × 60	2,400
10	70 × 30	2,100
11	90 × 90	8,100
12	50 × 40	2,000
13	60 × 80	4,800
14	___ × 40 = 2,800	70
15	50 × ___ = 3,500	70

#	Question	Answer
16	Explain how you know 40 × 50 = 2,000.	4 × 5 = 20; tens × tens = hundreds; 20 hundreds = 2,000.

LESSON 2.10 · TEACHER PLAN
Week 3 · Lesson 2 of 4
Unit 2: Multiplication

Lesson 2.10: Two-Digit by Two-Digit: Area Models

Student goal	I can use an area model to multiply two two-digit numbers.
Objective	Students multiply 2-digit by 2-digit numbers with area models.
Standard	4.NBT.5 — Multiply two two-digit numbers using strategies based on place value, illustrated with area models.
Pacing	Unit 2, week 3, lesson 2 of 4
Time	About 50 minutes
Materials	Grid paper, workbook Lesson 2.10
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	20 × 40 and 7 × 60.
Launch	5	Read Let's Explore.
Teach	15	Draw 23 × 14 on grid paper (23 by 14 squares) and split it into four parts. Say: "Split both numbers, four boxes, add them all." Work Example 2.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Making only two boxes (splitting just one factor) and multiplying wrong parts.

SUPPORT

- Use grid paper to draw the real rectangle once.

EXTEND

- Find 99 × 99 using 100 × 99 − 99.

STUDENT LESSON 2.10 · WITH ANSWERS

Lesson 2.10: Two-Digit by Two-Digit: Area Models

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.10 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.10, Part A.
- [ ] Complete the rest of Workbook Lesson 2.10 (Parts B–E).

LET'S EXPLORE

An orchard has 23 rows of pecan trees with 14 trees in each row. How many trees are there?

When both factors have two digits, split both into tens and ones. The area model has four boxes.

	20	3
10	200	30
4	80	12

23 × 14: split 23 into 20 + 3 and 14 into 10 + 4.

WORKED EXAMPLE · FOUR BOXES

Find 23 × 14.

1 10 × 20 = 200

2 10 × 3 = 30

3 4 × 20 = 80

4 4 × 3 = 12

5 Add all four: 200 + 30 + 80 + 12 = 322

Answer: 322 trees

WORKED EXAMPLE · ANOTHER ONE

Find 36 × 25.

	30	6
20	600	120
5	150	30

1 20 × 30 = 600 and 20 × 6 = 120

2 5 × 30 = 150 and 5 × 6 = 30

3 600 + 120 + 150 + 30 = 900

Answer: 900

★ KEY IDEA

Two-digit × two-digit makes four partial products. Add all four.

✎ TRY IT

Answers are shown in color.

1. 12 × 13

Answer: 156

2. 21 × 34

Answer: 714

3. 45 × 16

Answer: 720

Workbook Answer Key · Lesson 2.10

#	Question	Answer
1	12 × 13	156
2	21 × 34	714
3	45 × 16	720
4	20 × 40	800
5	7 × 60	420
6	13 × 22 (figure)	286
7	31 × 15 (figure)	465

#	Question	Answer
8	24 × 26 (figure)	624
9	42 × 17 (figure)	714
10	18 × 21	378
11	35 × 14	490
12	Use an area model to find 27 × 32.	864

LESSON 2.11 · TEACHER PLAN
Week 3 · Lesson 3 of 4
Unit 2: Multiplication

Lesson 2.11: Two-Digit by Two-Digit: Partial Products

Student goal	I can multiply two two-digit numbers using partial products.
Objective	Students multiply 2-digit by 2-digit numbers using partial products.
Standard	4.NBT.5 — Multiply two two-digit numbers using strategies based on place value.
Pacing	Unit 2, week 3, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.11
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	30 × 50; area model 12 × 14.
Launch	5	Read Let's Explore.
Teach	15	Draw the area model for 34 × 26, then write each box as a labeled line. Say: "Four boxes become four lines."
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Forgetting one of the four products.

SUPPORT

- Check off each box of the area model as you write its line.

EXTEND

- Try the standard algorithm for 2-digit × 2-digit (preview of Grade 5).

STUDENT LESSON 2.11 · WITH ANSWERS

Lesson 2.11: Two-Digit by Two-Digit: Partial Products

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.11 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.11, Part A.
- [ ] Complete the rest of Workbook Lesson 2.11 (Parts B–E).

LET'S EXPLORE

Can the four boxes of the area model be written as a list, like we did with one-digit numbers?

Yes! Each box becomes one partial product. Label each line so you know which digits you multiplied.

WORKED EXAMPLE · PARTIAL PRODUCTS

Find 34 × 26.

	34
	× 26
6 × 4	24
6 × 30	180
20 × 4	80
20 × 30	600
	884

1. 6 × 4 = 24
2. 6 × 30 = 180
3. 20 × 4 = 80
4. 20 × 30 = 600
5. Add: 24 + 180 + 80 + 600 = 884

Answer: 884

★ KEY IDEA

Multiply each part of one factor by each part of the other: ones × ones, ones × tens, tens × ones, tens × tens. Then add.

! WATCH OUT

Line up the partial products by place value before adding.

✎ TRY IT

Answers are shown in color.

1. 24 × 13

Answer: 312

2. 52 × 41

Answer: 2,132

3. 67 × 28

Answer: 1,876

Workbook Answer Key · Lesson 2.11

#	Question	Answer
1	24 × 13	312
2	52 × 41	2,132
3	67 × 28	1,876
4	30 × 50	1,500
5	Area model: 12 × 14	168
6	23 × 12 (figure)	276
7	41 × 32 (figure)	1,312
8	56 × 24 (figure)	1,344
9	73 × 45 (figure)	3,285
10	68 × 39 (figure)	2,652
11	94 × 57 (figure)	5,358
12	A bakery makes 48 trays of cookies. Each tray holds 24 cookies. How many cookies is that?	1,152 cookies
13	46 × 35 (figure)	1,610

LESSON 2.12 · TEACHER PLAN
Week 3 · Lesson 4 of 4
Unit 2: Multiplication

Lesson 2.12: Choosing and Checking a Strategy

Student goal	I can choose a multiplication strategy and check my answer with an estimate.
Objective	Students choose efficient multiplication strategies and check with estimates.
Standard	4.NBT.5, 4.OA.3 — Multiply two two-digit numbers using strategies based on place value and properties of operations.
Pacing	Unit 2, week 3, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.12
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	25 × 4 and 40 × 30.
Launch	5	Read Let's Explore. Ask the student to guess the three ways.
Teach	15	Show 25 × 16 three ways. Say: "The best strategy depends on the numbers." Work both examples.
Guided Practice	10	Try It 1–3. Then Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Believing there is only one right way.

SUPPORT

- Default to partial products when unsure.

EXTEND

- Find 99 × 25 in your head.

STUDENT LESSON 2.12 · WITH ANSWERS

Lesson 2.12: Choosing and Checking a Strategy

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.12 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.12, Part A.
- [ ] Complete the rest of Workbook Lesson 2.12 (Parts B–E).

LET'S EXPLORE

Three students solved 25 × 16 in different ways and all got 400. How could that be?

There is more than one good way to multiply. Choose the one that fits the numbers.

- Area model: great for seeing the parts.
- Partial products: organized and easy to check.
- Mental math with friendly numbers: 25 × 16 = 25 × 4 × 4 = 100 × 4 = 400.
- Standard algorithm: quick for one-digit factors.

WORKED EXAMPLE · FRIENDLY NUMBERS

Find 25 × 12.

1 25 × 4 = 100, and 12 = 4 × 3.

2 So 25 × 12 = 100 × 3.

Answer: 300

WORKED EXAMPLE · ESTIMATE TO CHECK

Is 38 × 21 = 798 reasonable?

1 Estimate: 40 × 20 = 800.

2 798 is close to 800.

Answer: Yes

★ KEY IDEA

Pick a strategy that fits the numbers, then estimate to check.

✎ TRY IT

Answers are shown in color.

1. 50 × 18

Answer: 900

2. 19 × 30 (hint: 20 × 30 − 30)

Answer: 570

3. Is 42 × 31 = 1,302 reasonable?

Answer: Yes. 40 × 30 = 1,200.

Workbook Answer Key · Lesson 2.12

#	Question	Answer
1	50 × 18	900
2	19 × 30 (hint: 20 × 30 − 30)	570
3	Is 42 × 31 = 1,302 reasonable?	Yes. 40 × 30 = 1,200.
4	25 × 4	100
5	40 × 30	1,200
6	25 × 24	600
7	15 × 22	330
8	49 × 20	980
9	33 × 33	1,089
10	64 × 15	960
11	81 × 19	1,539
12	53 × 48 = 2,544 A. Yes B. No	Yes (50 × 50 = 2,500)
13	29 × 61 = 2,769 A. Yes B. No	No (30 × 60 = 1,800; exact 1,769)
14	Which strategy would you use for 25 × 36? Solve it and explain.	900. Example: 25 × 4 = 100, 36 = 4 × 9, so 100 × 9 = 900.

WEEKLY CHECK · ANSWER KEY
After Lesson 2.12
About 20 minutes

Weekly Check 6: Multiplication Problem Solving — Answer Key

Answers

#	Question	Answer
1	Mom buys 4 shirts at $13 and a hat for $17. Total?	$69
2	A bakery made 32 trays of 18 rolls. How many rolls?	576 rolls
3	A whale is 3 times as long as a 25-foot boat. How long is the whale?	75 feet
4	A tree is 12 feet taller than a 25-foot house. How tall is the tree?	37 feet
5	A garden is 23 feet by 14 feet. What is its area?	322 square feet
6	A store had 600 cans. It sold 24 cases of 12 cans. How many are left?	312 cans

LESSON 2.13 · TEACHER PLAN
Week 4 · Lesson 1 of 4
Unit 2: Multiplication

Lesson 2.13: Multiplication Word Problems

Student goal	I can solve word problems that use multiplication, including equal groups, arrays, and area.
Objective	Students solve multiplication word problems.
Standard	4.OA.3, 4.NBT.5 — Solve multistep word problems with whole numbers; multiply to solve problems.
Pacing	Unit 2, week 4, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.13
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	12 × 5; 30 × 12.
Launch	5	Read Let's Explore.
Teach	15	Work both examples. Ask the student to name the clue in each. Say: "Equal groups, arrays, and area all mean multiply."
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Adding the numbers in a word problem without thinking.

SUPPORT

- Draw a quick picture or array.

EXTEND

- Write a word problem for a sibling.

STUDENT LESSON 2.13 · WITH ANSWERS

Lesson 2.13: Multiplication Word Problems

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.13 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.13, Part A.
- [ ] Complete the rest of Workbook Lesson 2.13 (Parts B–D).

LET'S EXPLORE

A pack of trading cards has 12 cards. How many cards are in 15 packs?

Look for equal groups, rows and columns, or length and width. These are clues to multiply.

WORKED EXAMPLE · EQUAL GROUPS

15 packs with 12 cards each. How many cards?

1 Equal groups: 15 groups of 12.

2 15 × 12 = 180.

Answer: 180 cards

WORKED EXAMPLE · AREA

A rug is 9 feet long and 12 feet wide. What is its area?

1 Area of a rectangle = length × width.

2 9 × 12 = 108.

Answer: 108 square feet

★ KEY IDEA

Ask: Are there equal groups? Rows and columns? Length and width? If yes, multiply.

✎ TRY IT

Answers are shown in color.

1. A team bought 14 jerseys at $23 each. What was the cost?

Answer: $322

2. A parking lot has 18 rows of 25 spaces. How many spaces?

Answer: 450 spaces

3. A garden is 16 feet by 7 feet. What is its area?

Answer: 112 square feet

Workbook Answer Key · Lesson 2.13

#	Question	Answer
1	A team bought 14 jerseys at $23 each. What was the cost?	$322
2	A parking lot has 18 rows of 25 spaces. How many spaces?	450 spaces
3	A garden is 16 feet by 7 feet. What is its area?	112 square feet
4	12 × 5	60
5	30 × 12	360
6	A school bus holds 48 students. How many students can 6 buses hold?	288 students
7	A box of crayons has 24 crayons. How many crayons are in 35 boxes?	840 crayons
8	A chicken farm collects 365 eggs a day. How many eggs in 7 days?	2,555 eggs
9	A field is 32 yards long and 18 yards wide. What is its area?	576 square yards
10	A movie theater has 22 rows of 19 seats. How many seats?	418 seats
11	Each of 4 classes collected 1,215 cans. How many cans in all?	4,860 cans
12	Write and solve a word problem for 13 × 24.	Answers vary; product 312.

LESSON 2.14 · TEACHER PLAN
Week 4 · Lesson 2 of 4
Unit 2: Multiplication

Lesson 2.14: Multi-Step Problems with Multiplication

Student goal	I can solve multi-step word problems that use multiplication with addition or subtraction.
Objective	Students solve two-step problems involving multiplication.
Standard	4.OA.3 — Solve multistep word problems posed with whole numbers using the four operations; represent with equations with a letter for the unknown.
Pacing	Unit 2, week 4, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.14
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	16 × 5; 400 − 286.
Launch	5	Read Let's Explore.
Teach	15	Model the four-step plan (Understand, Plan, Solve, Check). Write one equation per step.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Doing only the first step.

SUPPORT

- Underline each question hidden in the problem.

EXTEND

- Write a three-step problem.

STUDENT LESSON 2.14 · WITH ANSWERS

Lesson 2.14: Multi-Step Problems with Multiplication

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.14 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.14, Part A.
- [ ] Complete the rest of Workbook Lesson 2.14 (Parts B–D).

LET'S EXPLORE

Mom buys 3 shirts for $14 each and a pair of shoes for $39. How much does she spend?

Many real problems mix operations. Find the hidden first step, solve it, then finish.

WORKED EXAMPLE · MULTIPLY, THEN ADD

3 shirts at $14 and shoes for $39. Total?

1 Shirts: 3 × 14 = 42.

2 Add the shoes: 42 + 39 = 81.

Answer: $81

WORKED EXAMPLE · MULTIPLY, THEN SUBTRACT

A store had 500 bottles of water. It sold 18 cases of 24 bottles. How many bottles are left?

1 Sold: 18 × 24 = 432.

2 Left: 500 − 432 = 68.

Answer: 68 bottles

★ KEY IDEA

Write an equation for each step. Use a letter for the unknown: 3 × 14 + 39 = t.

✎ TRY IT

Answers are shown in color.

1. 4 tickets at $12 and a $9 parking fee. Total?

Answer: $57

2. A farmer had 1,000 plants. He planted 25 rows of 36. How many plants are left?

Answer: 100 plants

3. Ana earns $15 a week for 12 weeks and spends $95. How much is left?

Answer: $85

Workbook Answer Key · Lesson 2.14

#	Question	Answer
1	4 tickets at $12 and a $9 parking fee. Total?	$57
2	A farmer had 1,000 plants. He planted 25 rows of 36. How many plants are left?	100 plants
3	Ana earns $15 a week for 12 weeks and spends $95. How much is left?	$85
4	16 × 5	80
5	400 − 286	114
6	Grandma bought 5 bags of apples with 12 apples each. She used 23 apples for pies. How many apples are left?	37 apples
7	A class sold 36 boxes of popcorn at $5 each and 48 candy bars at $2 each. How much money did they raise?	$276
8	A library has 14 shelves with 45 books each. It buys 120 more books. How many books does it have?	750 books
9	Dad drives 38 miles to work and 38 miles home, 5 days a week. How many miles is that each week?	380 miles
10	Tickets cost $18 for adults and $9 for kids. What do 2 adults and 3 kids pay?	$63

LESSON 2.15 · TEACHER PLAN
Week 4 · Lesson 3 of 4
Unit 2: Multiplication

Lesson 2.15: Comparison Word Problems

Student goal	I can tell the difference between “times as many” and “more than” problems and solve both.
Objective	Students distinguish and solve multiplicative and additive comparison problems.
Standard	4.OA.2 — Multiply or divide to solve word problems involving multiplicative comparison, distinguishing from additive comparison.
Pacing	Unit 2, week 4, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.15
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	6 × 25 vs 6 more than 25.
Launch	5	Read Let's Explore.
Teach	15	Work both examples with bar models. Say: “Times means copies; more means extra.”
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Treating every comparison as addition.

SUPPORT

- Draw bars for every problem.

EXTEND

- Create a sorting game: times-as-many cards vs more-than cards.

STUDENT LESSON 2.15 · WITH ANSWERS

Lesson 2.15: Comparison Word Problems

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.15 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.15, Part A.
- [ ] Complete the rest of Workbook Lesson 2.15 (Parts B–D).

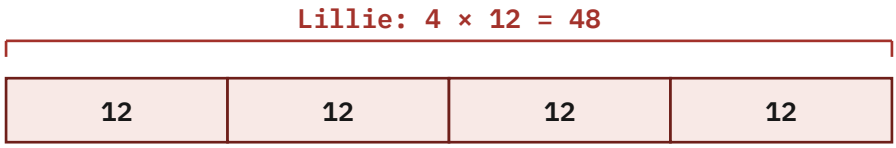
LET'S EXPLORE

Lillie has 4 times as many shells as Ellie. Ellie has 12 shells. How many shells do they have together?

Some comparison problems use multiplication (times as many) and some use addition or subtraction (more than, fewer than). Read carefully!

WORKED EXAMPLE · TIMES AS MANY, THEN ADD

Ellie has 12 shells. Lillie has 4 times as many. How many in all?



1 Lillie: 4 × 12 = 48.

2 Together: 12 + 48 = 60.

Answer: 60 shells

WORKED EXAMPLE · MORE THAN

A blue whale is 98 feet long. A school bus is 58 feet shorter. How long is the bus?

1 "Shorter" means subtract: 98 − 58 = 40.

Answer: 40 feet

! WATCH OUT

The word "times" means multiply. The words "more" and "fewer" by themselves mean add or subtract.

✎ TRY IT

Answers are shown in color.

1. A pencil costs 30¢. A pen costs 3 times as much. How much does the pen cost?

Answer: 90¢

2. Tom has 25 marbles. Kim has 8 more than Tom. How many does Kim have?

Answer: 33 marbles

3. A giraffe is 18 feet tall. That is 3 times as tall as a man. How tall is the man?

Answer: 6 feet

Workbook Answer Key · Lesson 2.15

#	Question	Answer
1	A pencil costs 30¢. A pen costs 3 times as much. How much does the pen cost?	90¢
2	Tom has 25 marbles. Kim has 8 more than Tom. How many does Kim have?	33 marbles
3	A giraffe is 18 feet tall. That is 3 times as tall as a man. How tall is the man?	6 feet
4	6 × 25	150
5	6 more than 25	31
6	A horse eats 8 times as much hay as a goat. A goat eats 3 pounds. How much does the horse eat?	24 pounds
7	Max is 9 years old. His dad is 27 years older. How old is his dad?	36 years old
8	A fourth grader reads 45 pages. A seventh grader reads 3 times as many. How many pages?	135 pages
9	A red car costs $3,500 more than a blue car. The blue car costs $12,000. Cost of the red car?	$15,500
10	Ellie collected 15 leaves. Lillie collected 6 times as many. How many did they collect together?	105 leaves
11	A tower is 5 times as tall as a 40-foot house. How tall is the tower?	200 feet
12	Write one “times as many” problem and one “more than” problem using 7 and 21.	Example: 21 is 3 times as many as 7; 21 is 14 more than 7.

LESSON 2.16 · TEACHER PLAN
Week 4 · Lesson 4 of 4
Unit 2: Multiplication

Lesson 2.16: Multiplication Problem Solving

Student goal	I can solve a variety of multiplication problems and explain my thinking.
Objective	Students apply multiplication strategies in mixed practice and problem solving.
Standard	4.NBT.5, 4.OA.3 — Multiply multi-digit numbers; solve multistep word problems.
Pacing	Unit 2, week 4, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.16; Unit 2 Review pages
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	8 × 8 and 20 × 7.
Launch	5	Read Let's Explore. Plan the reunion together.
Teach	15	Work the two examples as a plan. Review each strategy from the unit with one quick problem.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook Lesson 2.16. Assign the Unit Review for homework.
Exit Ticket	3	Check. Tomorrow: Unit 2 Test.

COMMON MISCONCEPTIONS

- Rushing through multi-step problems.

SUPPORT

- Allow area models on any problem.

EXTEND

- Plan a real family event with numbers.

STUDENT LESSON 2.16 · WITH ANSWERS

Lesson 2.16: Multiplication Problem Solving

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.16 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.16, Part A.
- [ ] Complete the rest of Workbook Lesson 2.16 (Parts B–E).

LET'S EXPLORE

You are planning a family reunion for 64 people. Each table seats 8. Each person needs 2 cups. Cups come in packs of 20. What do you need to buy?

Today you will use everything from this unit: comparisons, place value patterns, area models, partial products, the standard algorithm, and estimation.

WORKED EXAMPLE · PLAN A PARTY

How many cups are needed for 64 people if each person gets 2 cups?

1 64 × 2 = 128 cups.

Answer: 128 cups

WORKED EXAMPLE · PACKS

Cups come in packs of 20. Is 6 packs enough?

1 6 × 20 = 120.

2 120 is less than 128.

Answer: No. 7 packs (140 cups) are needed.

★ KEY IDEA

Read carefully, choose a strategy, show your work, and check with an estimate.

✎ TRY IT

Answers are shown in color.

1. How many tables for 64 people at 8 per table?

Answer: 8 tables

2. Plates come in packs of 25. How many plates in 3 packs?

Answer: 75 plates

3. Each person eats 3 rolls. How many rolls for 64 people?

Answer: 192 rolls

Workbook Answer Key · Lesson 2.16

#	Question	Answer
1	How many tables for 64 people at 8 per table?	8 tables
2	Plates come in packs of 25. How many plates in 3 packs?	75 plates
3	Each person eats 3 rolls. How many rolls for 64 people?	192 rolls
4	8 × 8	64
5	20 × 7	140
6	57 × 8 (figure)	456
7	1,406 × 7 (figure)	9,842
8	38 × 24 (figure)	912
9	65 × 43 (figure)	2,795
10	Write an equation: 72 is 9 times as many as 8.	72 = 9 × 8
11	Estimate 6 × 4,890.	6 × 5,000 = 30,000
12	A fruit stand sells 36 baskets of peaches a day for 14 days. Each basket has 25 peaches. How many baskets does it sell? How many peaches?	504 baskets; 12,600 peaches
13	What was the most useful strategy in this unit for you? Show it with 28 × 15.	Answers vary; 28 × 15 = 420.

UNIT REVIEW · ANSWER KEY
Study before the unit test

About 40 minutes

Unit 2 Review — Answer Key

Use as homework or a study session. Reteach any section with more than one mistake.

Answers

#	Question	Answer
1	9 × 700	6,300
2	40 × 80	3,200
3	Write an equation: 36 is 4 times as many as 9.	36 = 4 × 9
4	A rope is 7 feet long. Another is 6 times as long. How long is it?	42 feet
5	76 × 4 (figure)	304
6	528 × 6 (figure)	3,168
7	3,209 × 7 (figure)	22,463
8	43 × 27 (figure)	1,161
9	Estimate 7 × 892.	7 × 900 = 6,300
10	A store sells 24 cartons of milk each day. How many cartons in 31 days?	744 cartons
11	Ticket prices: $16 adults, $8 kids. What do 3 adults and 4 kids pay?	$80

UNIT TEST · ANSWER KEY
At the end of the unit

About 45 minutes

Unit 2 Test — Answer Key

Allow about 45 minutes. No calculator. Any correct strategy earns full credit if work is shown.

Answers

#	Question	Answer
1	6 × 4,000	24,000
2	70 × 30	2,100
3	Write an equation: 48 is 6 times as many as 8.	48 = 6 × 8
4	A cat is 4 years old. A horse is 5 times as old. How old is the horse?	20 years
5	Is this “times as many” or “more than”? Ben has 6 more pens than Ana.	more than (additive)
6	Use an area model: 5 × 67	335
7	Use partial products: 34 × 21 (figure)	714
8	8 × 457 (figure)	3,656
9	6 × 2,385 (figure)	14,310
10	56 × 42 (figure)	2,352
11	Estimate 9 × 512. Is the exact answer more or less?	4,500 (9 × 500); exact is more (4,608)
12	A school orders 28 boxes of markers. Each box has 12 markers. The art room already has 75 markers. How many markers will it have?	411 markers
13	A tree farm plants 35 rows of 24 pine trees. It sells 290 trees. How many trees are left?	550 trees

Scoring

Score	What it means / next step
20–23 (85%+)	Mastered. Move on to Unit 3.
16–19	Proficient. Correct missed items together.
Below 16	Reteach weakest part and retest.

UNIT 3 · TEACHER OVERVIEW

Division

Sharing and grouping, dividing with place value, partial quotients, long division, and remainders.

Division splits a total into equal groups. In this unit you will use multiplication facts and place value to divide numbers up to four digits by one-digit numbers, and decide what to do with leftovers.

Big ideas

- Explain division as sharing and as grouping.
- Divide multiples of 10, 100, and 1,000.
- Divide with area models, partial quotients, and long division.
- Find and check remainders.
- Decide what a remainder means in a word problem.

Standards

Code	Standard
4.NBT.6	Find whole-number quotients and remainders with up to four-digit dividends and one-digit divisors, using strategies based on place value.
4.OA.2	Multiply or divide to solve word problems involving multiplicative comparison.
4.OA.3	Solve multistep word problems, including problems in which remainders must be interpreted.

What students should already know

- Multiplication facts (fluent).
- Multiplying by one digit (Unit 2).
- Subtraction with regrouping (Unit 1).

Materials for the unit

- Counters (beans or buttons work)

- Paper plates or cups for sharing
- Grid paper
- Divide-Multiply-Subtract-Bring down card

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 4	3.1	What Division Means
Week 1	Lesson 2 of 4	3.2	Dividing Multiples of 10, 100, and 1,000
Week 1	Lesson 3 of 4	3.3	Dividing with Area Models
Week 1	Lesson 4 of 4	3.4	Partial Quotients
Week 1	End of week	Check	Weekly Check 7: Dividing
Week 2	Lesson 1 of 4	3.5	Understanding Remainders
Week 2	Lesson 2 of 4	3.6	Dividing Larger Numbers
Week 2	Lesson 3 of 4	3.7	What Do I Do With the Remainder?
Week 2	Lesson 4 of 4	3.8	Division Problem Solving
Week 2	Before the test	Review	Unit 3 Review
Week 2	End of unit	Test	Unit 3 Test

TEACHING THIS UNIT AT HOME

Unit 3 spans winter break (Dec 21 – Jan 1). Over the break, practice division facts with cards for a few minutes on most days.

UNIT VOCABULARY

dividend	The number being divided.
divisor	The number you divide by.
quotient	The answer to a division problem.
remainder	The amount left over.

LESSON 3.1 · TEACHER PLAN
Week 1 · Lesson 1 of 4
Unit 3: Division

Lesson 3.1: What Division Means

Student goal	I can explain division as sharing equally or making equal groups, and use multiplication facts to divide.
Objective	Students understand division as sharing and grouping and use fact families.
Standard	4.OA.2, 4.NBT.6 — Find whole-number quotients using strategies based on the relationship between multiplication and division.
Pacing	Unit 3, week 1, lesson 1 of 4
Time	About 50 minutes
Materials	24 counters or cookies, cups or paper plates, workbook Lesson 3.1
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	6 × 9 and 7 × ? = 49.
Launch	5	Read Let's Explore; share 24 counters onto 4 plates.
Teach	15	Act out both meanings: sharing (4 plates) and grouping (bags of 4). Say: "Division undoes multiplication." Introduce dividend, divisor, quotient.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking division always means sharing.
- Mixing up the dividend and divisor.

SUPPORT

- Use counters for every problem.

EXTEND

- Why can't you divide 24 cookies among 0 children?

STUDENT LESSON 3.1 · WITH ANSWERS

Lesson 3.1: What Division Means

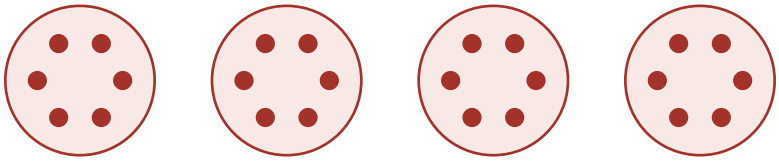
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.1 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.1, Part A.
- [ ] Complete the rest of Workbook Lesson 3.1 (Parts B–F).

LET'S EXPLORE

Grandma baked 24 cookies for 4 grandchildren. How many cookies does each child get if she shares them fairly?

Division answers two kinds of questions. Sharing: 24 cookies shared by 4 children, how many each? Grouping: 24 cookies put in bags of 4, how many bags? Both are 24 ÷ 4 = 6.



24 shared into 4 equal groups: 6 in each group.

Division is the opposite of multiplication. If you know 4 × 6 = 24, you know 24 ÷ 4 = 6 and 24 ÷ 6 = 4.

Dividend	÷	Divisor	=	Quotient
24	÷	4	=	6

WORKED EXAMPLE · USE A FACT FAMILY

Find 56 ÷ 7.

1 Think: 7 × ? = 56.

2 7 × 8 = 56.

Answer: 8

★ KEY IDEA

To divide, think of the related multiplication fact: 56 ÷ 7 = ? is the same as 7 × ? = 56.

✎ TRY IT

Answers are shown in color.

1. 36 ÷ 9

Answer: 4

2. Write the fact family for 5, 8, and 40.

Answer: 5 × 8 = 40, 8 × 5 = 40, 40 ÷ 5 = 8, 40 ÷ 8 = 5

3. 42 apples go in bags of 6. How many bags?

Answer: 7 bags

Workbook Answer Key · Lesson 3.1

#	Question	Answer
1	36 ÷ 9	4
2	Write the fact family for 5, 8, and 40.	5 × 8 = 40, 8 × 5 = 40, 40 ÷ 5 = 8, 40 ÷ 8 = 5
3	42 apples go in bags of 6. How many bags?	7 bags
4	6 × 9	54
5	7 × ___ = 49	7
6	18 ÷ 3	6
7	45 ÷ 5	9
8	63 ÷ 9	7
9	32 ÷ 4	8
10	72 ÷ 8	9
11	49 ÷ 7	7
12	54 ÷ 6	9
13	81 ÷ 9	9
14	Write the fact family for 6, 7, 42.	6 × 7 = 42, 7 × 6 = 42, 42 ÷ 6 = 7, 42 ÷ 7 = 6
15	Write the fact family for 3, 9, 27.	3 × 9 = 27, 9 × 3 = 27, 27 ÷ 3 = 9, 27 ÷ 9 = 3
16	35 crayons are shared by 5 friends. How many crayons does each friend get?	7 crayons
17	48 chairs are put in rows of 8. How many rows?	6 rows
18	Write a sharing story and a grouping story for 30 ÷ 6.	Answers vary; quotient 5.

LESSON 3.2 · TEACHER PLAN
Week 1 · Lesson 2 of 4
Unit 3: Division

Lesson 3.2: Dividing Multiples of 10, 100, and 1,000

Student goal	I can use basic facts and place value to divide numbers like 2,400 ÷ 6.
Objective	Students divide multiples of 10, 100, and 1,000 by one-digit numbers.
Standard	4.NBT.6 — Find quotients using strategies based on place value.
Pacing	Unit 3, week 1, lesson 2 of 4
Time	About 50 minutes
Materials	Base-ten blocks, workbook Lesson 3.2
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	48 ÷ 8; hundreds in 3,600.
Launch	5	Read Let's Explore.
Teach	15	Show 24 flats shared into 6 groups (4 flats each). Say: "Name the unit, divide the fact, name the unit again." Work the example.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Dropping or adding zeros.

SUPPORT

- Check every answer by multiplying.

EXTEND

- Find 36,000 ÷ 40 (preview).

STUDENT LESSON 3.2 · WITH ANSWERS

Lesson 3.2: Dividing Multiples of 10, 100, and 1,000

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.2 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.2, Part A.
- [ ] Complete the rest of Workbook Lesson 3.2 (Parts B–E).

LET'S EXPLORE

A school has 2,400 pencils to share equally among 6 classrooms. How many pencils does each room get?

Use a basic fact and think about the unit. 2,400 is 24 hundreds. 24 hundreds ÷ 6 = 4 hundreds = 400.

Basic fact	Tens	Hundreds	Thousands
24 ÷ 6 = 4	240 ÷ 6 = 40	2,400 ÷ 6 = 400	24,000 ÷ 6 = 4,000

WORKED EXAMPLE · THINK IN UNITS

Find 3,500 ÷ 7.

1 3,500 = 35 hundreds.

2 35 ÷ 7 = 5, so 35 hundreds ÷ 7 = 5 hundreds.

Answer: 500

! WATCH OUT

Check by multiplying: 7 × 500 = 3,500. If you get a different number, recount the zeros.

★ KEY IDEA

Name the dividend in tens, hundreds, or thousands, use the basic fact, then rename the quotient.

✎ TRY IT

Answers are shown in color.

1. 320 ÷ 4

Answer: 80

2. 4,500 ÷ 9

Answer: 500

3. 6,000 ÷ 3

Answer: 2,000

Workbook Answer Key · Lesson 3.2

#	Question	Answer
1	320 ÷ 4	80
2	4,500 ÷ 9	500
3	6,000 ÷ 3	2,000
4	48 ÷ 8	6
5	How many hundreds are in 3,600?	36
6	80 ÷ 4	20
7	150 ÷ 5	30
8	210 ÷ 7	30
9	640 ÷ 8	80
10	1,200 ÷ 3	400
11	2,800 ÷ 4	700
12	5,400 ÷ 9	600
13	3,000 ÷ 5	600
14	48,000 ÷ 6	8,000
15	35,000 ÷ 7	5,000
16	A farm packs 1,800 eggs equally into 6 crates. How many eggs are in each crate?	300 eggs
17	Explain how 42 ÷ 7 helps you find 4,200 ÷ 7.	4,200 is 42 hundreds; 42 ÷ 7 = 6, so 6 hundreds = 600.

LESSON 3.3 · TEACHER PLAN
Week 1 · Lesson 3 of 4
Unit 3: Division

Lesson 3.3: Dividing with Area Models

Student goal	I can use an area model to divide a two- or three-digit number by a one-digit number.
Objective	Students divide using area models and the distributive property.
Standard	4.NBT.6 — Find whole-number quotients up to four-digit dividends and one-digit divisors using place value strategies and area models.
Pacing	Unit 3, week 1, lesson 3 of 4
Time	About 50 minutes
Materials	Grid paper, workbook Lesson 3.3
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	90 ÷ 3 and 6 × 40.
Launch	5	Read Let's Explore. Draw the rectangle with the unknown side.
Teach	15	Work Example 1. Say: Find a big friendly piece of the dividend first. Work Example 2 (three digits).
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Splitting the dividend into parts that are not divisible by the divisor (e.g., 84 = 70 + 14 for ÷ 4).

SUPPORT

- Use multiples tables (4, 8, 12, … 40, 80) as a reference.

EXTEND

- Find 1,248 ÷ 4 with an area model.

STUDENT LESSON 3.3 · WITH ANSWERS

Lesson 3.3: Dividing with Area Models

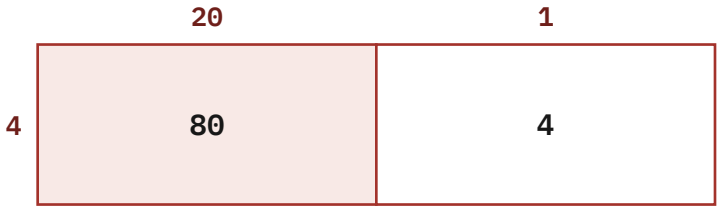
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.3 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.3, Part A.
- [ ] Complete the rest of Workbook Lesson 3.3 (Parts B–E).

LET'S EXPLORE

A rectangle has an area of 84 square feet. One side is 4 feet long. How long is the other side?

Division can be pictured as a rectangle where you know the area (the dividend) and one side (the divisor). The missing side is the quotient.



84 ÷ 4: 80 ÷ 4 = 20 and 4 ÷ 4 = 1, so the side is 20 + 1 = 21.

WORKED EXAMPLE · SPLIT THE DIVIDEND INTO FRIENDLY PARTS

Find 84 ÷ 4.

1 Split 84 into 80 + 4, numbers that are easy to divide by 4.

2 80 ÷ 4 = 20.

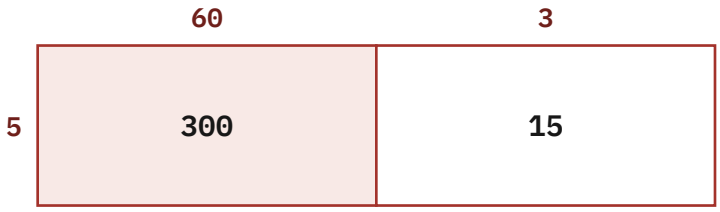
3 4 ÷ 4 = 1.

4 Add: 20 + 1 = 21.

Answer: 21

WORKED EXAMPLE · THREE DIGITS

Find 315 ÷ 5.



1 Split 315 into 300 + 15.

2 300 ÷ 5 = 60.

3 15 ÷ 5 = 3.

4 60 + 3 = 63.

Answer: 63

★ KEY IDEA

Break the dividend into parts that are easy to divide (multiples of the divisor). Divide each part, then add the answers.

✎ TRY IT

Answers are shown in color.

1. 96 ÷ 3

Answer: 32

2. 78 ÷ 6

Answer: 13

3. 426 ÷ 6

Answer: 71

Workbook Answer Key · Lesson 3.3

#	Question	Answer
1	96 ÷ 3	32
2	78 ÷ 6	13
3	426 ÷ 6	71
4	90 ÷ 3	30
5	6 × 40	240
6	68 ÷ 2	34
7	93 ÷ 3	31

#	Question	Answer
8	85 ÷ 5	17
9	72 ÷ 4	18
10	256 ÷ 8	32
11	434 ÷ 7	62
12	A garden has an area of 96 square yards. It is 8 yards wide. How long is it?	12 yards
13	Use an area model to find 252 ÷ 6.	42 (240 ÷ 6 = 40, 12 ÷ 6 = 2)

LESSON 3.4 · TEACHER PLAN
Week 1 · Lesson 4 of 4
Unit 3: Division

Lesson 3.4: Partial Quotients

Student goal	I can divide using partial quotients and check my answer with multiplication.
Objective	Students divide 3-digit numbers by 1-digit numbers using partial quotients.
Standard	4.NBT.6 — Find whole-number quotients using strategies based on place value and the relationship between multiplication and division.
Pacing	Unit 3, week 1, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 3.4, a multiples chart (×10, ×20, ×5)
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	10 × 7 and 20 × 8.
Launch	5	Read Let's Explore.
Teach	15	Work the example in the table. Say: Take away big easy groups first. Show a second way (20 groups first) to get the same answer.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Subtracting a multiple bigger than what is left.

SUPPORT

- Write a 'menu' of easy multiples first (10×, 20×, 5×, 2×).

EXTEND

- Divide 1,512 ÷ 6 with partial quotients.

STUDENT LESSON 3.4 · WITH ANSWERS

Lesson 3.4: Partial Quotients

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.4 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.4, Part A.
- [ ] Complete the rest of Workbook Lesson 3.4 (Parts B–D).

LET'S EXPLORE

There are 156 students going on a field trip. Each van holds 6 students. How many vans are needed?

With partial quotients, you take away big, easy groups first, then smaller groups, until nothing (or a little) is left.

WORKED EXAMPLE · PARTIAL QUOTIENTS

Find 156 ÷ 6.

1. Take away 10 groups of 6: 156 − 60 = 96. (10)
2. Take away 10 more groups: 96 − 60 = 36. (10)
3. Take away 6 groups: 36 − 36 = 0. (6)
4. Add the groups: 10 + 10 + 6 = 26.
5. Check: 26 × 6 = 156. ✓

Answer: 26 vans

Left to divide	Subtract	Groups of 6
156	− 60 (10 × 6)	10
96	− 60 (10 × 6)	10
36	− 36 (6 × 6)	6
0		Total: 26

★ KEY IDEA

Any easy multiple works: 10 groups, 20 groups, 5 groups. Different students may take different steps and still get the same quotient.

✎ TRY IT

Answers are shown in color.

1. 144 ÷ 4

Answer: 36

2. 215 ÷ 5

Answer: 43

3. 738 ÷ 9

Answer: 82

Workbook Answer Key · Lesson 3.4

#	Question	Answer
1	144 ÷ 4	36
2	215 ÷ 5	43
3	738 ÷ 9	82
4	10 × 7	70
5	20 × 8	160
6	84 ÷ 3	28
7	132 ÷ 4	33
8	195 ÷ 5	39
9	378 ÷ 7	54
10	504 ÷ 8	63
11	846 ÷ 9	94
12	Divide 288 ÷ 6 with partial quotients. Check by multiplying.	48; 48 × 6 = 288

WEEKLY CHECK · ANSWER KEY
After Lesson 3.4

About 15 minutes

Weekly Check 7: Dividing — Answer Key

Answers

#	Question	Answer
1	63 ÷ 9	7
2	2,800 ÷ 7	400
3	Use an area model: 96 ÷ 4	24
4	Partial quotients: 175 ÷ 5	35
5	29 ÷ 6	4 R5
6	Is 42 ÷ 5 = 7 R7 correct? Explain.	No; remainder must be < 5. 42 ÷ 5 = 8 R2.

LESSON 3.5 · TEACHER PLAN
Week 2 · Lesson 1 of 4
Unit 3: Division

Lesson 3.5: Understanding Remainders

Student goal	I can find a quotient with a remainder and explain what the remainder means.
Objective	Students find quotients with remainders and check them.
Standard	4.NBT.6 — Find whole-number quotients and remainders.
Pacing	Unit 3, week 2, lesson 1 of 4
Time	About 50 minutes
Materials	Counters, workbook Lesson 3.5
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	7 × 6; 45 − 42.
Launch	5	Read Let's Explore. Act it out with 26 counters.
Teach	15	Work Example 1. Say: Make as many full groups as you can; what's left is the remainder. Read the Watch Out. Work Example 2.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- A remainder that is bigger than the divisor.

SUPPORT

- Use counters.

EXTEND

- Find a number that leaves remainder 1 when divided by 2, 3, and 4. (13, 25, ...)

STUDENT LESSON 3.5 · WITH ANSWERS

Lesson 3.5: Understanding Remainders

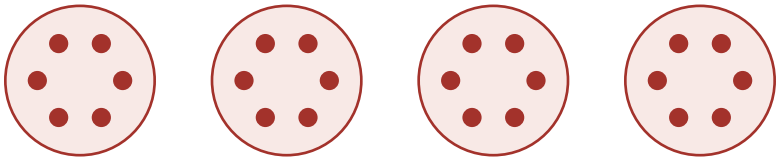
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.5 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.5, Part A.
- [ ] Complete the rest of Workbook Lesson 3.5 (Parts B–E).

LET'S EXPLORE

Ellie has 26 stickers to share equally among 4 friends. How many does each friend get? Are any left?

Sometimes the groups come out even and sometimes there are some left over. The leftover amount is the remainder.



26 ÷ 4: each friend gets 6 (24 stickers), and 2 are left.

WORKED EXAMPLE · WRITE A REMAINDER

Find 26 ÷ 4.

1 4 × 6 = 24, the closest product that is not too big.

2 26 − 24 = 2 left over.

3 Write the answer as 6 R2.

Answer: 6 R2

! WATCH OUT

The remainder must always be less than the divisor. If you have 5 left when dividing by 4, you can make another group!

WORKED EXAMPLE · CHECK A REMAINDER

Check 47 ÷ 5 = 9 R2.

1 Multiply: 9 × 5 = 45.

2 Add the remainder: 45 + 2 = 47. ✓

Answer: Correct

★ KEY IDEA

Quotient × divisor + remainder = dividend.

✎ TRY IT

Answers are shown in color.

1. 17 ÷ 3

Answer: 5 R2

2. 38 ÷ 6

Answer: 6 R2

3. 50 ÷ 7

Answer: 7 R1

Workbook Answer Key · Lesson 3.5

#	Question	Answer
1	17 ÷ 3	5 R2
2	38 ÷ 6	6 R2
3	50 ÷ 7	7 R1
4	7 × 6	42
5	45 − 42	3
6	13 ÷ 2	6 R1
7	23 ÷ 5	4 R3
8	31 ÷ 4	7 R3
9	44 ÷ 6	7 R2
10	58 ÷ 8	7 R2
11	67 ÷ 9	7 R4
12	40 ÷ 7	5 R5
13	29 ÷ 3	9 R2
14	Is 53 ÷ 6 = 8 R5 correct? Show how you checked.	Yes: 8 × 6 = 48, 48 + 5 = 53

#	Question	Answer
15	Is 35 ÷ 4 = 7 R7 correct? Explain.	No. The remainder 7 is bigger than 4. 35 ÷ 4 = 8 R3.
16	Find 61 ÷ 8 and check your answer.	7 R5; 7 × 8 + 5 = 61

LESSON 3.6 · TEACHER PLAN
Week 2 · Lesson 2 of 4
Unit 3: Division

Lesson 3.6: Dividing Larger Numbers

Student goal	I can divide up to four-digit numbers by one-digit numbers using place value, step by step.
Objective	Students divide up to 4-digit numbers by 1-digit numbers with long division.
Standard	4.NBT.6 — Find whole-number quotients with up to four-digit dividends and one-digit divisors.
Pacing	Unit 3, week 2, lesson 2 of 4
Time	About 50 minutes
Materials	Grid paper, workbook Lesson 3.6, “Divide, Multiply, Subtract, Bring down” card
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	36 ÷ 4; 2,400 ÷ 6.
Launch	5	Read Let's Explore; estimate first (1,200 ÷ 4 = 300).
Teach	15	Work the example on grid paper, one digit per box. Say: “Divide, multiply, subtract, bring down.” Point out the zero in the quotient.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Forgetting zeros in the quotient.
- Misaligning digits.

SUPPORT	EXTEND
• Grid paper; estimate first so a missing zero is noticed. • Partial quotients are always allowed.	• Divide 9,999 ÷ 9 and 9,990 ÷ 9.

STUDENT LESSON 3.6 · WITH ANSWERS

Lesson 3.6: Dividing Larger Numbers

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.6 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.6, Part A.
- [ ] Complete the rest of Workbook Lesson 3.6 (Parts B–D).

LET'S EXPLORE

A charity has $1,236 to split equally among 4 families. How much does each family get?

Long division divides one place at a time, starting at the greatest place. Repeat four steps: divide, multiply, subtract, bring down.

WORKED EXAMPLE · LONG DIVISION

Find 1,236 ÷ 4.

$$4\overline{)1236}\quad \text{quotient: } 309$$

1. 1 thousand ÷ 4: not enough. Use 12 hundreds instead. 12 ÷ 4 = 3. Write 3 over the hundreds. 3 × 4 = 12, 12 − 12 = 0.
2. Bring down the 3 tens. 3 ÷ 4 = 0. Write 0 over the tens. 0 × 4 = 0, 3 − 0 = 3.
3. Bring down the 6 ones: 36. 36 ÷ 4 = 9. Write 9 over the ones. 9 × 4 = 36, 36 − 36 = 0.
4. Check: 309 × 4 = 1,236. ✓

Answer: $309

! WATCH OUT

When a place cannot be divided (like 3 ÷ 4), write a 0 in the quotient. Skipping it would give 39, which is far too small.

★ KEY IDEA

Divide, multiply, subtract, bring down. Repeat until every digit has been brought down.

✎ TRY IT

Answers are shown in color.

1. 846 ÷ 2

Answer: 423

2. 1,575 ÷ 5

Answer: 315

3. 2,418 ÷ 6

Answer: 403

Workbook Answer Key · Lesson 3.6

#	Question	Answer
1	846 ÷ 2	423
2	1,575 ÷ 5	315
3	2,418 ÷ 6	403
4	36 ÷ 4	9
5	Is 2,400 ÷ 6 = 40 or 400?	400
6	368 ÷ 2 (figure)	184
7	645 ÷ 3 (figure)	215
8	928 ÷ 4 (figure)	232
9	1,715 ÷ 5 (figure)	343
10	3,612 ÷ 6 (figure)	602
11	2,856 ÷ 7 (figure)	408
12	4,096 ÷ 8 (figure)	512
13	7,254 ÷ 9 (figure)	806
14	2,128 ÷ 7 (figure)	304

LESSON 3.7 · TEACHER PLAN
Week 2 · Lesson 3 of 4
Unit 3: Division

Lesson 3.7: What Do I Do With the Remainder?

Student goal	I can decide what a remainder means in a word problem.
Objective	Students interpret remainders in context.
Standard	4.OA.3 — Solve multistep word problems with whole-number answers, including problems in which remainders must be interpreted.
Pacing	Unit 3, week 2, lesson 3 of 4
Time	About 50 minutes
Materials	Counters, workbook Lesson 3.7
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	38 ÷ 5; 62 ÷ 7.
Launch	5	Read Let's Explore. Ask: Can we have 6 R3 cars?
Teach	15	Discuss the three remainder cases using the table. Work the example (round up).
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Writing remainders in answers to story problems without thinking about them.

SUPPORT

- Act out each story with counters.

EXTEND

- Write one story for each kind of remainder.

STUDENT LESSON 3.7 · WITH ANSWERS

Lesson 3.7: What Do I Do With the Remainder?

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.7 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.7, Part A.
- [ ] Complete the rest of Workbook Lesson 3.7 (Parts B–D).

LET'S EXPLORE

27 people are going to the zoo. Each car holds 4 people. How many cars do they need?

27 ÷ 4 = 6 R3. But the answer is not "6 R3 cars"! The remainder means something different in different stories.

What to do with the remainder	Example (27 ÷ 4 = 6 R3)	Answer
Round up: everyone needs a spot	Cars for 27 people, 4 per car	7 cars
Drop it: only full groups count	Bags of 4 apples from 27 apples	6 full bags
The remainder is the answer	Apples left after making bags of 4	3 apples

WORKED EXAMPLE · ROUND UP

A teacher has 75 students. Each table seats 8. How many tables?

1 75 ÷ 8 = 9 R3.

2 3 students still need seats, so add a table.

Answer: 10 tables

★ KEY IDEA

Always ask: what does the remainder mean in this story?

✎ TRY IT

Answers are shown in color.

1. 50 photos go in an album, 6 per page. How many pages are needed?

Answer: 9 pages

2. 50 crayons go in boxes of 6. How many full boxes?

Answer: 8 full boxes

3. 50 crayons in boxes of 6. How many crayons are left over?

Answer: 2 crayons

Workbook Answer Key · Lesson 3.7

#	Question	Answer
1	50 photos go in an album, 6 per page. How many pages are needed?	9 pages
2	50 crayons go in boxes of 6. How many full boxes?	8 full boxes
3	50 crayons in boxes of 6. How many crayons are left over?	2 crayons
4	38 ÷ 5	7 R3
5	62 ÷ 7	8 R6
6	A boat holds 6 people. 39 people want to ride. How many trips?	7 trips (39 ÷ 6 = 6 R3; round up)
7	Ribbons are cut into 8-inch pieces from 70 inches. How many full pieces?	8 pieces (70 ÷ 8 = 8 R6; drop)
8	A baker fills boxes of 6 muffins from 100 muffins. How many muffins are left out of boxes?	4 muffins (100 ÷ 6 = 16 R4; the remainder is the answer)
9	4 friends share $30 equally. How much does each get, and how much is left?	$7 each, $2 left (30 ÷ 4 = 7 R2)
10	Write a story for 45 ÷ 7 where the answer is 7.	Example: 45 people, 7 per van; 7 vans needed.

LESSON 3.8 · TEACHER PLAN
Week 2 · Lesson 4 of 4
Unit 3: Division

Lesson 3.8: Division Problem Solving

Student goal	I can solve word problems with division and check my answers with multiplication.
Objective	Students solve one- and two-step division problems.
Standard	4.OA.3, 4.NBT.6 — Solve multistep word problems with whole numbers, including interpreting remainders.
Pacing	Unit 3, week 2, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 3.8
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	720 ÷ 8; 91 ÷ 7.
Launch	5	Read Let's Explore.
Teach	15	Work both examples. Say: Check every division with multiplication.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook Lesson 3.8 and the Unit 3 Review.
Exit Ticket	3	Check. Tomorrow: Unit 3 Test.

COMMON MISCONCEPTIONS

- Choosing division when a problem needs multiplication.

SUPPORT

- Draw a bar model.

EXTEND

- Plan equal shares for a real family trip budget.

STUDENT LESSON 3.8 · WITH ANSWERS

Lesson 3.8: Division Problem Solving

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.8 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.8, Part A.
- [ ] Complete the rest of Workbook Lesson 3.8 (Parts B–D).

LET'S EXPLORE

A family drives 1,248 miles in 4 days, the same distance each day. How far do they drive each day?

Look for clues for division: shared equally, each, per, how many groups.

WORKED EXAMPLE · EQUAL SHARING

1,248 miles in 4 equal days.

1 1,248 ÷ 4 = 312.

2 Check: 312 × 4 = 1,248.

Answer: 312 miles each day

WORKED EXAMPLE · TWO STEPS

A class earns $96 at a bake sale and $120 at a car wash. They split the money equally among 6 field-trip groups. How much per group?

1 Add: 96 + 120 = 216.

2 Divide: 216 ÷ 6 = 36.

Answer: $36

★ KEY IDEA

Multiply to check every division answer.

✎ TRY IT

Answers are shown in color.

1. A rope 96 feet long is cut into 8 equal pieces. How long is each piece?

Answer: 12 feet

2. There are 365 days in a year. About how many weeks is that? (7 days in a week)

Answer: 52 weeks and 1 day (52 R1)

3. Mom buys 3 packs of 12 water bottles and splits them among 4 coolers. How many per cooler?

Answer: 9 bottles

Workbook Answer Key · Lesson 3.8

#	Question	Answer
1	A rope 96 feet long is cut into 8 equal pieces. How long is each piece?	12 feet
2	There are 365 days in a year. About how many weeks is that? (7 days in a week)	52 weeks and 1 day (52 R1)
3	Mom buys 3 packs of 12 water bottles and splits them among 4 coolers. How many per cooler?	9 bottles
4	720 ÷ 8	90
5	91 ÷ 7	13
6	A school collected 1,530 cans and packed them equally into 5 boxes. How many cans in each box?	306 cans
7	Dad has 250 tomato plants. He plants them in rows of 9. How many full rows? How many plants are left?	27 full rows, 7 left
8	A pack of 8 juice boxes costs $4. How many packs are needed for 60 kids?	8 packs (60 ÷ 8 = 7 R4; round up)
9	Lillie read 156 pages in 6 days and Ellie read 112 pages in 4 days. Who read more pages each day?	Ellie: 28 per day vs Lillie 26 per day
10	2,184 ÷ 6. Check with multiplication.	364; 364 × 6 = 2,184

UNIT REVIEW · ANSWER KEY
Study before the unit test

About 35 minutes

Unit 3 Review — Answer Key

Homework or a study session before the test.

Answers

#	Question	Answer
1	54 ÷ 6	9
2	3,200 ÷ 8	400
3	Write the fact family for 4, 9, 36.	4 × 9 = 36, 9 × 4 = 36, 36 ÷ 4 = 9, 36 ÷ 9 = 4
4	84 ÷ 6 (figure)	14
5	347 ÷ 5 (figure)	69 R2
6	1,638 ÷ 7 (figure)	234
7	4,520 ÷ 8 (figure)	565
8	58 students ride in vans that hold 8. How many vans?	8 vans
9	58 cookies go in bags of 8. How many full bags?	7 bags

UNIT TEST · ANSWER KEY
At the end of the unit

About 40 minutes

Unit 3 Test — Answer Key

Allow about 40 minutes. Any correct method earns full credit if work is shown.

Answers

#	Question	Answer
1	72 ÷ 8	9
2	4,500 ÷ 5	900
3	360 ÷ 6	60
4	Write a division fact related to 7 × 8 = 56.	56 ÷ 7 = 8 or 56 ÷ 8 = 7
5	96 ÷ 4 (figure)	24
6	275 ÷ 6 (figure)	45 R5
7	1,206 ÷ 3 (figure)	402
8	3,549 ÷ 7 (figure)	507
9	5,000 ÷ 8 (figure)	625
10	Check 83 ÷ 9 = 9 R2 by multiplying. Is it correct?	Yes: 9 × 9 = 81, 81 + 2 = 83
11	125 people are going to a game. Each van holds 9 people. How many vans are needed? Explain the remainder.	14 vans (125 ÷ 9 = 13 R8; round up)
12	A store has 3 boxes of 144 markers. It puts them into packs of 6. How many packs can it make?	72 packs (432 ÷ 6)

Scoring

Score	What it means / next step
19–22 (85%+)	Mastered.
16–18	Proficient; correct missed items.
Below 16	Reteach and retest.

UNIT 4 · TEACHER OVERVIEW

Factors, Multiples, and Patterns

Factor pairs, prime and composite numbers, multiples, divisibility, and patterns.

Factors and multiples show how numbers are built from other numbers. Patterns show how numbers grow and change by rules. These ideas prepare you for fractions in the next units.

Big ideas

- Find all factor pairs of a number to 100.
- Tell whether a number is prime or composite.
- Find multiples and common multiples.
- Use divisibility rules.
- Extend number and shape patterns and find rules.

Standards

Code	Standard
4.OA.4	Find all factor pairs for a whole number 1–100; recognize multiples; determine prime or composite.
4.OA.5	Generate a number or shape pattern that follows a given rule; identify apparent features of the pattern.

What students should already know

- Multiplication and division facts.
- Dividing with remainders (Unit 3).

Materials for the unit

- Square tiles (or grid paper squares)
- Hundred chart
- Crayons
- Pattern blocks or paper shapes

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 4	4.1	Factors and Factor Pairs
Week 1	Lesson 2 of 4	4.2	Prime and Composite Numbers
Week 1	Lesson 3 of 4	4.3	Multiples
Week 1	Lesson 4 of 4	4.4	Divisibility Rules
Week 1	End of week	Check	Weekly Check 8: Factors and Multiples
Week 2	Lesson 1 of 4	4.5	Number Patterns
Week 2	Lesson 2 of 4	4.6	Shape Patterns
Week 2	Lesson 3 of 4	4.7	Input-Output Tables
Week 2	Lesson 4 of 4	4.8	Factors, Multiples, and Patterns in Problems
Week 2	Before the test	Review	Unit 4 Review
Week 2	End of unit	Test	Unit 4 Test

TEACHING THIS UNIT AT HOME

Hunt for patterns at home: tiles, fabric, calendars, house numbers. Ask: what's the rule? What comes next?

UNIT VOCABULARY

factor	A number that divides another with no remainder.
multiple	The product of a number and a whole number.
prime number	A number with exactly two factors.
composite number	A number with more than two factors.
pattern	A predictable arrangement of numbers or shapes.

LESSON 4.1 · TEACHER PLAN
Week 1 · Lesson 1 of 4
Unit 4: Factors, Multiples, and Patterns

Lesson 4.1: Factors and Factor Pairs

Student goal	I can find all the factor pairs of a whole number from 1 to 100.
Objective	Students find factor pairs for whole numbers 1–100.
Standard	4.OA.4 — Find all factor pairs for a whole number in the range 1–100; recognize that a whole number is a multiple of each of its factors.
Pacing	Unit 4, week 1, lesson 1 of 4
Time	About 50 minutes
Materials	Square tiles or grid paper, workbook Lesson 4.1
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	4 × 6; 45 ÷ 5.
Launch	5	Read Let's Explore. Build arrays with 12 tiles.
Teach	15	Record each rectangle as a factor pair. Say: "Test 1, 2, 3 in order and stop when the pairs meet." Work the 36 example.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Missing factor pairs by testing out of order.
- Listing a pair twice.

SUPPORT

- Use tiles to build every rectangle.

EXTEND

- Which number under 50 has the most factors?

STUDENT LESSON 4.1 · WITH ANSWERS

Lesson 4.1: Factors and Factor Pairs

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.1 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.1, Part A.
- [ ] Complete the rest of Workbook Lesson 4.1 (Parts B–F).

LET'S EXPLORE

Ellie wants to arrange 12 chairs in equal rows with no chairs left over. How many different ways can she do it?

Every rectangle you can make with 12 squares shows a factor pair of 12.

Rows	In each row	Factor pair
1	12	1 × 12
2	6	2 × 6
3	4	3 × 4



3 rows of 4 chairs

The factors of 12 are 1, 2, 3, 4, 6, and 12. (4 rows of 3 is the same pair as 3 rows of 4, just turned.)

WORKED EXAMPLE · FIND ALL FACTOR PAIRS

Find all the factor pairs of 36.

1 Start with 1: 1 × 36.

2 Try 2: 2 × 18. Try 3: 3 × 12. Try 4: 4 × 9. Try 5: no. Try 6: 6 × 6.

3 When the factors meet or start repeating, you are done.

Answer: 1 × 36, 2 × 18, 3 × 12, 4 × 9, 6 × 6

★ KEY IDEA

Test 1, 2, 3, … in order. Stop when the pairs meet in the middle. List the factors from least to greatest.

✎ TRY IT

Answers are shown in color.

1. List the factor pairs of 18.

Answer: 1 × 18, 2 × 9, 3 × 6

2. List all the factors of 20.

Answer: 1, 2, 4, 5, 10, 20

3. Is 7 a factor of 42?

Answer: Yes (6 × 7 = 42)

Workbook Answer Key · Lesson 4.1

#	Question	Answer
1	List the factor pairs of 18.	1 × 18, 2 × 9, 3 × 6
2	List all the factors of 20.	1, 2, 4, 5, 10, 20
3	Is 7 a factor of 42?	Yes (6 × 7 = 42)
4	4 × 6	24
5	45 ÷ 5	9
6	8	1 × 8, 2 × 4
7	15	1 × 15, 3 × 5
8	16	1 × 16, 2 × 8, 4 × 4
9	24	1 × 24, 2 × 12, 3 × 8, 4 × 6
10	30	1 × 30, 2 × 15, 3 × 10, 5 × 6
11	48	1 × 48, 2 × 24, 3 × 16, 4 × 12, 6 × 8
12	Is 6 a factor of 54? A. Yes B. No	Yes
13	Is 8 a factor of 60? A. Yes B. No	No
14	Is 9 a factor of 81? A. Yes B. No	Yes

#	Question	Answer
15	Is 4 a factor of 98? A. Yes B. No	No
16	A gardener has 28 plants. What are all the ways she can plant them in equal rows?	1 × 28, 2 × 14, 4 × 7 (and turned around)
17	List all the factors of 40.	1, 2, 4, 5, 8, 10, 20, 40

LESSON 4.2 · TEACHER PLAN
Week 1 · Lesson 2 of 4
Unit 4: Factors, Multiples, and Patterns

Lesson 4.2: Prime and Composite Numbers

Student goal	I can tell whether a number is prime or composite.
Objective	Students classify numbers as prime or composite.
Standard	4.OA.4 — Determine whether a given whole number in the range 1–100 is prime or composite.
Pacing	Unit 4, week 1, lesson 2 of 4
Time	About 50 minutes
Materials	Square tiles, hundred chart, workbook Lesson 4.2
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Factors of 10 and 11.
Launch	5	Read Let's Explore with 7 tiles.
Teach	15	Build rectangles for 7, 9, 13, 15. Say: "Only one rectangle means prime."
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking 1 is prime.
- Thinking all odd numbers are prime.

SUPPORT

- Use tiles.

EXTEND

- Sieve of Eratosthenes on a hundred chart: cross out multiples of 2, 3, 5, 7.

STUDENT LESSON 4.2 · WITH ANSWERS

Lesson 4.2: Prime and Composite Numbers

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.2 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.2, Part A.
- [ ] Complete the rest of Workbook Lesson 4.2 (Parts B–E).

LET'S EXPLORE

Try to make a rectangle with 7 tiles that is more than one row. Can you?

Some numbers can only make one rectangle: a single row. Those numbers are prime. Numbers that can make more than one rectangle are composite.

Number	Factors	Prime or composite?
7	1, 7	prime
9	1, 3, 9	composite
13	1, 13	prime
15	1, 3, 5, 15	composite

WORKED EXAMPLE · PRIME OR COMPOSITE?

Is 21 prime or composite?

1 $21 = 3 \times 7$.

2 21 has more than two factors: 1, 3, 7, 21.

Answer: composite

! WATCH OUT

1 is neither prime nor composite. It has only one factor. 2 is the only even prime number.

★ KEY IDEA

Exactly two factors → prime. More than two → composite.

✎ TRY IT

Answers are shown in color.

1. Is 17 prime or composite?

Answer: prime

2. Is 27 prime or composite?

Answer: composite

3. List the prime numbers between 1 and 20.

Answer: 2, 3, 5, 7, 11, 13, 17, 19

Workbook Answer Key · Lesson 4.2

#	Question	Answer
1	Is 17 prime or composite?	prime
2	Is 27 prime or composite?	composite
3	List the prime numbers between 1 and 20.	2, 3, 5, 7, 11, 13, 17, 19
4	Factors of 10	1, 2, 5, 10
5	Factors of 11	1, 11
6	5	prime
7	12	composite
8	19	prime
9	25	composite
10	29	prime
11	33	composite
12	41	prime
13	49	composite
14	51	composite
15	53	prime
16	Mia says all odd numbers are prime. Give an example that shows she is wrong.	Examples: 9, 15, 21, 25
17	Is 39 prime or composite? Explain.	Composite: 3 × 13 = 39.

LESSON 4.3 · TEACHER PLAN
Week 1 · Lesson 3 of 4
Unit 4: Factors, Multiples, and Patterns

Lesson 4.3: Multiples

Student goal	I can find multiples of a number and tell whether a number is a multiple of another number.
Objective	Students find multiples and decide whether one number is a multiple of another.
Standard	4.OA.4 — Recognize that a whole number is a multiple of each of its factors; determine whether a whole number is a multiple of a given one-digit number.
Pacing	Unit 4, week 1, lesson 3 of 4
Time	About 50 minutes
Materials	Hundred chart, crayons, workbook Lesson 4.3
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Skip-count by 9s.
Launch	5	Read Let's Explore.
Teach	15	Color the multiples of 6 on a hundred chart; then circle the multiples of 4 to find common multiples. Say: "Factor and multiple are two sides of one fact."
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Confusing factors and multiples.

SUPPORT

- Use the phrase: factors are small, multiples are many.

EXTEND

- Find the first three common multiples of 4 and 6.

STUDENT LESSON 4.3 · WITH ANSWERS

Lesson 4.3: Multiples

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.3 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.3, Part A.
- [ ] Complete the rest of Workbook Lesson 4.3 (Parts B–F).

LET'S EXPLORE

A bus comes every 15 minutes, starting at 8:00. Will there be a bus at 9:00?

Skip counting gives multiples. The multiples of 15 are 15, 30, 45, 60, ... A bus comes at 8:15, 8:30, 8:45, and 9:00.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Multiples of 6 on a hundred chart.

WORKED EXAMPLE · IS IT A MULTIPLE?

Is 72 a multiple of 8?

1 Ask: 8 × ? = 72.

2 8 × 9 = 72.

Answer: Yes

★ KEY IDEA

A number is a multiple of each of its factors. 72 is a multiple of 8, and 8 is a factor of 72.

✎ TRY IT

Answers are shown in color.

1. List the first 6 multiples of 7.

Answer: 7, 14, 21, 28, 35, 42

2. Is 54 a multiple of 6?

Answer: Yes

3. Is 46 a multiple of 4?

Answer: No

Workbook Answer Key · Lesson 4.3

#	Question	Answer
1	List the first 6 multiples of 7.	7, 14, 21, 28, 35, 42
2	Is 54 a multiple of 6?	Yes
3	Is 46 a multiple of 4?	No
4	Count by 9s to 45.	9, 18, 27, 36, 45
5	Is 5 a factor of 35?	Yes
6	3	3, 6, 9, 12, 15
7	6	6, 12, 18, 24, 30
8	8	8, 16, 24, 32, 40
9	11	11, 22, 33, 44, 55
10	12	12, 24, 36, 48, 60
11	25	25, 50, 75, 100, 125
12	Is 36 a multiple of 9? A. Yes B. No	Yes
13	Is 50 a multiple of 6? A. Yes B. No	No
14	Is 84 a multiple of 7? A. Yes B. No	Yes

#	Question	Answer
15	Is 90 a multiple of 4? A. Yes B. No	No
16	Ellie has art every 3 days and piano every 4 days. Both are on day 12. On what day are both on again?	Day 24 Common multiples of 3 and 4: 12, 24, 36.
17	Name a number that is a multiple of both 5 and 6.	30 (or 60, 90, ...)

LESSON 4.4 · TEACHER PLAN
Week 1 · Lesson 4 of 4
Unit 4: Factors, Multiples, and Patterns

Lesson 4.4: Divisibility Rules

Student goal	I can use divisibility rules for 2, 3, 5, 9, and 10 to find factors quickly.
Objective	Students use divisibility rules to identify factors.
Standard	4.OA.4 — Determine whether a whole number is a multiple of a one-digit number.
Pacing	Unit 4, week 1, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 4.4, rule card
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Digit sums.
Launch	5	Read Let's Explore.
Teach	15	Teach each rule with one example. Make a rule card. Work the 135 example.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Applying the digit-sum rule to 5 or 10.

SUPPORT

- Keep the rule card visible.

EXTEND

- Why does the 9 rule work? (Each place value is one more than a multiple of 9.)

STUDENT LESSON 4.4 · WITH ANSWERS

Lesson 4.4: Divisibility Rules

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.4 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.4, Part A.
- [ ] Complete the rest of Workbook Lesson 4.4 (Parts B–E).

LET'S EXPLORE

Is 348 divisible by 3? You could divide... or you could use a shortcut.

Divisibility rules let you test for factors without dividing.

A number is divisible by...	if...	Example
2	its ones digit is even (0, 2, 4, 6, 8)	348 ✓
3	the sum of its digits is divisible by 3	3 + 4 + 8 = 15 ✓
5	its ones digit is 0 or 5	345 ✓
9	the sum of its digits is divisible by 9	3 + 4 + 2 = 9, so 342 ✓
10	its ones digit is 0	340 ✓

WORKED EXAMPLE · USE THE RULES

Which of 2, 3, 5, 9, 10 divide 135?

1 Ones digit 5: divisible by 5, not by 2 or 10.

2 Digit sum 1 + 3 + 5 = 9: divisible by 3 and by 9.

Answer: 3, 5, and 9

★ KEY IDEA

Even ones digit → 2. Ends in 0 or 5 → 5. Ends in 0 → 10. Digit sum divisible by 3 or 9 → 3 or 9.

✎ TRY IT

Answers are shown in color.

1. Is 726 divisible by 3?

Answer: Yes (7 + 2 + 6 = 15)

2. Is 415 divisible by 2?

Answer: No

3. Is 918 divisible by 9?

Answer: Yes (9 + 1 + 8 = 18)

Workbook Answer Key · Lesson 4.4

#	Question	Answer
1	Is 726 divisible by 3?	Yes (7 + 2 + 6 = 15)
2	Is 415 divisible by 2?	No
3	Is 918 divisible by 9?	Yes (9 + 1 + 8 = 18)
4	4 + 7 + 2	13
5	Is 36 even?	Yes
6	60	2, 3, 5, 10
7	81	3, 9
8	115	5
9	222	2, 3
10	405	3, 5, 9
11	1000	2, 5, 10
12	A teacher has 234 stickers. Can she share them equally among 9 students with none left?	Yes; 2 + 3 + 4 = 9, so 234 ÷ 9 = 26
13	Write a 3-digit number that is divisible by 2, 3, and 5.	Any multiple of 30, e.g. 120, 330

WEEKLY CHECK · ANSWER KEY
After Lesson 4.4

About 15 minutes

Weekly Check 8: Factors and Multiples — Answer Key

Answers

#	Question	Answer
1	List the factor pairs of 32.	1 × 32, 2 × 16, 4 × 8
2	Is 23 prime or composite?	prime
3	Is 57 prime or composite?	composite (3 × 19)
4	List the first five multiples of 9.	9, 18, 27, 36, 45
5	Is 96 a multiple of 8?	Yes
6	Which of 2, 3, 5, 9, 10 divide 270?	2, 3, 5, 9, 10

LESSON 4.5 · TEACHER PLAN
Week 2 · Lesson 1 of 4
Unit 4: Factors, Multiples, and Patterns

Lesson 4.5: Number Patterns

Student goal	I can make and extend a number pattern that follows a rule.
Objective	Students generate and extend number patterns and identify features.
Standard	4.OA.5 — Generate a number or shape pattern that follows a given rule; identify apparent features not explicit in the rule.
Pacing	Unit 4, week 2, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 4.5
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Skip-count from 3 by 4s.
Launch	5	Read Let's Explore. Fill the table.
Teach	15	Work both examples. Say: “Find the change; then look for hidden features.”
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Finding the change only between the first two terms.

SUPPORT

- Write the change between each pair of terms.

EXTEND

- Patterns that alternate: add 2, add 3, add 2, ...

STUDENT LESSON 4.5 · WITH ANSWERS

Lesson 4.5: Number Patterns

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.5 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.5, Part A.
- [ ] Complete the rest of Workbook Lesson 4.5 (Parts B–F).

LET'S EXPLORE

A savings jar starts with $5. Each week $3 is added. How much will be in the jar after 6 weeks?

A rule tells how to get each term from the one before. Rule: start at 5, add 3.

Week	0	1	2	3	4	5	6
Dollars	5	8	11	14	17	20	23

WORKED EXAMPLE · FIND THE RULE

What is the rule? 4, 9, 14, 19, ...

1 9 − 4 = 5, 14 − 9 = 5, 19 − 14 = 5.

Answer: Start at 4, add 5. Next terms: 24, 29

WORKED EXAMPLE · NOTICE FEATURES

Rule: start at 4, add 5. Which terms are odd?

1 4 (even), 9 (odd), 14 (even), 19 (odd).

2 Even + odd = odd, odd + odd = even.

Answer: Every other term is odd.

★ KEY IDEA

Find the change between terms to find the rule. Then look for features the rule doesn't say out loud, like odd and even.

✎ TRY IT

Answers are shown in color.

1. Rule: start at 7, add 4. Write the first 5 terms.

Answer: 7, 11, 15, 19, 23

2. Find the rule: 50, 44, 38, 32

Answer: Start at 50, subtract 6

3. Next two terms: 3, 6, 12, 24, ___, ___

Answer: 48, 96 (multiply by 2)

Workbook Answer Key · Lesson 4.5

#	Question	Answer
1	Rule: start at 7, add 4. Write the first 5 terms.	7, 11, 15, 19, 23
2	Find the rule: 50, 44, 38, 32	Start at 50, subtract 6
3	Next two terms: 3, 6, 12, 24, ___, ___	48, 96 (multiply by 2)
4	Count by 4s from 3: 3, 7, ___, ___	11, 15
5	100 − 8	92
6	Start at 2, add 6	2, 8, 14, 20, 26, 32
7	Start at 90, subtract 7	90, 83, 76, 69, 62, 55
8	Start at 1, multiply by 3	1, 3, 9, 27, 81, 243
9	5, 13, 21, 29	add 8; 37
10	100, 88, 76, 64	subtract 12; 52
11	2, 4, 8, 16	multiply by 2; 32
12	11, 22, 33, 44	add 11; 55
13	Rule: start at 1, add 2. Will 50 ever be a term? Explain.	No. All terms are odd (1, 3, 5, ...).
14	Make your own pattern with a rule. Write 5 terms.	Answers vary.

LESSON 4.6 · TEACHER PLAN
Week 2 · Lesson 2 of 4
Unit 4: Factors, Multiples, and Patterns

Lesson 4.6: Shape Patterns

Student goal	I can extend and describe a repeating or growing shape pattern.
Objective	Students extend repeating and growing shape patterns.
Standard	4.OA.5 — Generate a shape pattern that follows a given rule; identify apparent features.
Pacing	Unit 4, week 2, lesson 2 of 4
Time	About 50 minutes
Materials	Pattern blocks or colored paper shapes, workbook Lesson 4.6
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	17 ÷ 3; next in 6, 9, 12.
Launch	5	Read Let's Explore. Build the border with blocks.
Teach	15	Work both examples. Say: Find the core, then divide.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Counting the core wrong.

SUPPORT

- Circle each core with a pencil.

EXTEND

- Find the 100th shape in a 7-shape core.

STUDENT LESSON 4.6 · WITH ANSWERS

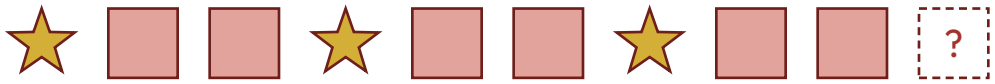
Lesson 4.6: Shape Patterns

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.6 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.6, Part A.
- [ ] Complete the rest of Workbook Lesson 4.6 (Parts B–E).

LET'S EXPLORE

A quilt border goes star, square, square, star, square, square, … What shape is 20th?



The core of this repeating pattern is star, square, square, 3 shapes long. Every 3rd shape is a square that ends the core.

WORKED EXAMPLE · USE DIVISION

What is the 20th shape?

1 The core has 3 shapes.

2 20 ÷ 3 = 6 R2: 6 full cores, then 2 more shapes.

3 The 2nd shape in the core is a square.

Answer: square

WORKED EXAMPLE · GROWING PATTERN

Figure 1 has 1 square, Figure 2 has 3, Figure 3 has 5. How many in Figure 10?

1 The rule is add 2.

2 Figure 10 = 1 + 9 × 2 = 19.

Answer: 19 squares

★ KEY IDEA

For repeating patterns, find the core and use division. For growing patterns, find the rule.

✎ TRY IT

Answers are shown in color.

1. Core: circle, triangle. What is the 15th shape?

Answer: circle (15 is odd)

2. Core: A, B, C, D. What is the 22nd letter?

Answer: B (22 ÷ 4 = 5 R2)

3. Growing: 4, 8, 12, … squares. How many in Figure 7?

Answer: 28

Workbook Answer Key · Lesson 4.6

#	Question	Answer
1	Core: circle, triangle. What is the 15th shape?	circle (15 is odd)
2	Core: A, B, C, D. What is the 22nd letter?	B (22 ÷ 4 = 5 R2)
3	Growing: 4, 8, 12, … squares. How many in Figure 7?	28
4	17 ÷ 3	5 R2
5	Next: 6, 9, 12, ___	15
6	(figure)	triangle
7	(figure)	square
8	Core: red, blue, blue, yellow. What color is the 30th bead?	blue (30 ÷ 4 = 7 R2)
9	Core: □ □ ○. How many circles are in the first 24 shapes?	8 circles
10	A tower adds 3 blocks each day. It has 5 blocks on day 1. How many blocks on day 8?	26 blocks
11	Toothpick squares in a row: 4, 7, 10, … toothpicks. How many for 6 squares?	19 toothpicks
12	Make a repeating pattern with a core of 3 shapes. What is the 12th shape?	Answers vary; the 12th shape is the last shape of the core.

LESSON 4.7 · TEACHER PLAN
Week 2 · Lesson 3 of 4
Unit 4: Factors, Multiples, and Patterns

Lesson 4.7: Input-Output Tables

Student goal	I can find and use a rule for an input-output table.
Objective	Students find and apply rules for input-output tables.
Standard	4.OA.5 — Generate a number pattern that follows a given rule.
Pacing	Unit 4, week 2, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 4.7
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	8 × 7; 45 + 9.
Launch	5	Read Let's Explore. Play 'guess my rule'.
Teach	15	Work the example. Say: Test the rule on every pair.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Checking only the first pair.

SUPPORT

- Write the operation between each input and output.

EXTEND

- Two-step rules: × 2 then + 1.

STUDENT LESSON 4.7 · WITH ANSWERS

Lesson 4.7: Input-Output Tables

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.7 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.7, Part A.
- [ ] Complete the rest of Workbook Lesson 4.7 (Parts B–D).

LET'S EXPLORE

A machine turns 3 into 12, 5 into 20, and 8 into 32. What does it turn 10 into?

Input	3	5	8	10
Output	12	20	32	?

Each output is 4 times the input. The rule is multiply by 4, so 10 becomes 40.

WORKED EXAMPLE · FIND THE RULE

Inputs 2, 4, 7 → outputs 9, 11, 14.

1 Each output is 7 more than its input.

2 Check: 2 + 7 = 9, 4 + 7 = 11, 7 + 7 = 14.

Answer: Add 7

★ KEY IDEA

Test your rule on every pair in the table, not just one.

✎ TRY IT

Answers are shown in color.

1. Rule: × 6. Input 7 → ?

Answer: 42

2. Inputs 10, 20, 30 → outputs 5, 15, 25. Rule?

Answer: subtract 5

3. Rule: ÷ 3. Output 9 → input?

Answer: 27

Workbook Answer Key · Lesson 4.7

#	Question	Answer
1	Rule: × 6. Input 7 → ?	42
2	Inputs 10, 20, 30 → outputs 5, 15, 25. Rule?	subtract 5
3	Rule: ÷ 3. Output 9 → input?	27
4	8 × 7	56
5	45 + 9	54
6	In: 1, 2, 3, 4 → Out: 5, 10, 15, ___	× 5; 20
7	In: 6, 9, 12, 15 → Out: 2, 3, 4, ___	÷ 3; 5
8	In: 12, 20, 33 → Out: 21, 29, ___	+ 9; 42
9	In: 50, 40, 30 → Out: 38, 28, ___	− 12; 18
10	Make a table with the rule × 8. Use 4 inputs.	Answers vary, e.g. 1→8, 2→16, 5→40, 9→72.

LESSON 4.8 · TEACHER PLAN
Week 2 · Lesson 4 of 4
Unit 4: Factors, Multiples, and Patterns

Lesson 4.8: Factors, Multiples, and Patterns in Problems

Student goal	I can use factors, multiples, and patterns to solve problems.
Objective	Students apply factors, multiples, and patterns to solve problems.
Standard	4.OA.4, 4.OA.5 — Find factor pairs; recognize multiples; generate patterns.
Pacing	Unit 4, week 2, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 4.8, Unit 4 Review
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Factors of 24; multiples of 6.
Launch	5	Read Let's Explore.
Teach	15	Work both examples. Discuss: factors or multiples?
Guided Practice	10	Try It 1–3.
Independent	12	Workbook Lesson 4.8 and the Unit 4 Review.
Exit Ticket	3	Check. Tomorrow: Unit 4 Test.

COMMON MISCONCEPTIONS

- Using factors when the question needs multiples.

SUPPORT

- List both and look for matches.

EXTEND

- Find the least common multiple of 4, 6, and 9.

STUDENT LESSON 4.8 · WITH ANSWERS

Lesson 4.8: Factors, Multiples, and Patterns in Problems

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.8 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.8, Part A.
- [ ] Complete the rest of Workbook Lesson 4.8 (Parts B–D).

LET'S EXPLORE

Hot dogs come in packs of 10 and buns in packs of 8. What is the fewest of each you can buy to have the same number of hot dogs and buns?

Common multiples answer "at the same time" or "the same number" questions. Factors answer "equal rows" and "equal groups" questions.

WORKED EXAMPLE · COMMON MULTIPLE

Packs of 10 hot dogs and 8 buns.

1 Multiples of 10: 10, 20, 30, 40.

2 Multiples of 8: 8, 16, 24, 32, 40.

3 The first match is 40.

Answer: 4 packs of hot dogs and 5 packs of buns

WORKED EXAMPLE · FACTORS

48 students need equal teams of more than 5 and fewer than 10 students. What team sizes work?

1 Factors of 48: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48.

2 Between 5 and 10: 6 and 8.

Answer: teams of 6 or 8

★ KEY IDEA

Same number or same time → common multiples. Equal rows or equal teams → factors.

✎ TRY IT

Answers are shown in color.

1. Juice comes in packs of 6 and cups in packs of 9. Fewest of each to match?

Answer: 3 packs of juice, 2 packs of cups (18)

2. Can 52 chairs be put in equal rows of 4? Of 5?

Answer: Rows of 4: yes (13); rows of 5: no

3. Is 91 prime?

Answer: No (7 × 13)

Workbook Answer Key · Lesson 4.8

#	Question	Answer
1	Juice comes in packs of 6 and cups in packs of 9. Fewest of each to match?	3 packs of juice, 2 packs of cups (18)
2	Can 52 chairs be put in equal rows of 4? Of 5?	Rows of 4: yes (13); rows of 5: no
3	Is 91 prime?	No (7 × 13)
4	Factors of 24	1, 2, 3, 4, 6, 8, 12, 24
5	Multiples of 6 to 36	6, 12, 18, 24, 30, 36
6	Two lighthouses flash every 6 and every 10 seconds. They flash together now. In how many seconds will they flash together again?	30 seconds
7	Grandpa has 36 tomato plants. List every way he can plant them in equal rows.	1 × 36, 2 × 18, 3 × 12, 4 × 9, 6 × 6
8	A pattern starts at 3 and adds 4. Is 50 in the pattern? Explain.	No. Terms are 3, 7, 11, … always odd.
9	Name all prime numbers between 40 and 60.	41, 43, 47, 53, 59
10	Write a problem that uses common multiples. Solve it.	Answers vary.

UNIT REVIEW · ANSWER KEY
Study before the unit test

About 30 minutes

Unit 4 Review — Answer Key

Homework or a study session before the test.

Answers

#	Question	Answer
1	Factors of 42	1, 2, 3, 6, 7, 14, 21, 42
2	Prime or composite: 31	prime
3	First 5 multiples of 12	12, 24, 36, 48, 60
4	Is 7 a factor of 84?	Yes
5	Rule: start at 6, add 7. First five terms.	6, 13, 20, 27, 34
6	Core: ○ △ □. What is the 14th shape?	△ (14 ÷ 3 = 4 R2)
7	In: 2, 5, 9 → Out: 16, 40, 72. Rule?	× 8
8	Is 100 in the pattern 1, 4, 7, 10, …? Explain.	Yes: 100 = 1 + 33 × 3.

UNIT TEST · ANSWER KEY
At the end of the unit
About 35 minutes

Unit 4 Test — Answer Key

Answers

#	Question	Answer
1	List all factors of 28.	1, 2, 4, 7, 14, 28
2	List the factor pairs of 45.	1 × 45, 3 × 15, 5 × 9
3	Prime or composite: 47	prime
4	Prime or composite: 63	composite
5	List the first six multiples of 7.	7, 14, 21, 28, 35, 42
6	Is 72 a multiple of 9?	Yes
7	Is 4 a factor of 54?	No
8	Which of 2, 3, 5, 9, 10 divide 135?	3, 5, 9
9	Rule: start at 8, add 6. Write five terms.	8, 14, 20, 26, 32
10	Find the rule: 81, 72, 63, 54	subtract 9
11	Core: red, red, blue. What is the 17th color?	red (17 ÷ 3 = 5 R2)
12	In: 3, 6, 9 → Out: 1, 2, 3. Rule?	÷ 3
13	Rule: start at 2, add 4. Will the terms be odd or even? Why?	Even: even + even = even.
14	Rolls come in packs of 12 and patties in packs of 8. What is the fewest of each pack to have the same number of rolls and patties?	2 packs of rolls, 3 packs of patties (24)
15	A class of 30 students needs equal teams with at least 3 students each. List all the possible team sizes.	3, 5, 6, 10, 15, 30

Scoring

Score	What it means / next step
17–19 (85%+)	Mastered.
14–16	Proficient; correct missed items.
Below 14	Reteach and retest.

UNIT 5 · TEACHER OVERVIEW

Fraction Equivalence and Comparison

Fractions on number lines, equivalent fractions, simplest form, and comparing and ordering fractions.

Fractions are numbers that name parts of a whole. In this unit you will find different names for the same fraction, simplify fractions, and compare fractions with different numerators and denominators.

Big ideas

- Explain fractions with models and number lines, including fractions greater than 1.
- Find fractions of a set.
- Make equivalent fractions by multiplying or dividing.
- Compare fractions with benchmarks, common denominators, and common numerators.
- Explain why comparisons need the same whole.

Standards

Code	Standard
4.NF.1	Explain why a/b is equivalent to (n×a)/(n×b) using visual models; generate equivalent fractions.
4.NF.2	Compare two fractions with different numerators and denominators using common denominators, numerators, or benchmarks; same whole; record with >, =, <.

What students should already know

- Factors and multiples (Unit 4).
- Grade 3 fractions: unit fractions, fractions on a number line.

Materials for the unit

- Paper strips for fraction strips (make a set: halves to twelfths)
- Paper circles
- Ruler marked in eighths

- Counters

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 4	5.1	Fractions and Unit Fractions
Week 1	Lesson 2 of 4	5.2	Fractions on a Number Line
Week 1	Lesson 3 of 4	5.3	Fractions Equal to and Greater Than 1
Week 1	Lesson 4 of 4	5.4	Fractions of a Set
Week 1	End of week	Check	Weekly Check 9: Understanding Fractions
Week 2	Lesson 1 of 4	5.5	Equivalent Fractions with Models
Week 2	Lesson 2 of 4	5.6	Equivalent Fractions by Multiplying
Week 2	Lesson 3 of 4	5.7	Simplest Form
Week 2	Lesson 4 of 4	5.8	Equivalent Fractions on a Number Line
Week 2	End of week	Check	Weekly Check 10: Equivalent Fractions
Week 3	Lesson 1 of 4	5.9	Comparing: Same Denominator or Same Numerator
Week 3	Lesson 2 of 4	5.10	Comparing with Benchmarks
Week 3	Lesson 3 of 4	5.11	Comparing with Common Denominators
Week 3	Lesson 4 of 4	5.12	Comparing with Common Numerators
Week 3	End of week	Check	Weekly Check 11: Comparing Fractions
Week 4	Lesson 1 of 4	5.13	Ordering Fractions
Week 4	Lesson 2 of 4	5.14	Comparing Fractions of the Same Whole
Week 4	Lesson 3 of 4	5.15	Fraction Comparison Word Problems
Week 4	Lesson 4 of 4	5.16	Explaining Fraction Reasoning
Week 4	Before the test	Review	Unit 5 Review
Week 4	End of unit	Test	Unit 5 Test

TEACHING THIS UNIT AT HOME

Make a set of paper fraction strips together (halves through twelfths). Cook together and talk about measuring cups: which is more, one-third cup or one-fourth cup?

UNIT VOCABULARY

fraction	A number naming part of a whole.
numerator	Top number: parts counted.
denominator	Bottom number: equal parts in the whole.
equivalent fractions	Fractions naming the same amount.
simplest form	No common factor but 1.
benchmark fraction	Friendly fraction like 0, 1/2, 1.

LESSON 5.1 · TEACHER PLAN
Week 1 · Lesson 1 of 4

Unit 5: Fraction Equivalence and Comparison

Lesson 5.1: Fractions and Unit Fractions

Student goal	I can name a fraction as a number of equal parts of a whole, like $^{3}/_{4}$ is 3 of 4 equal parts.
Objective	Students name fractions as counts of unit fractions.
Standard	4.NF.1 (review of 3.NF.1) — Review of fraction meaning; build toward 4.NF.1 equivalence.
Pacing	Unit 5, week 1, lesson 1 of 4
Time	About 50 minutes
Materials	Paper strips, crayons, workbook Lesson 5.1
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	12 ÷ 4.
Launch	5	Read Let's Explore. Fold a strip into 8 equal parts and shade 3.
Teach	15	Say: "The bottom number is how many equal parts; the top is how many we count." Work the example. Discuss the Watch Out with an unequal drawing.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Counting all the parts as the numerator.
- Accepting unequal parts.

SUPPORT

- Fold paper strips.

EXTEND

- Is the same fraction always the same amount of pizza? (preview of Lesson 5.14)

STUDENT LESSON 5.1 · WITH ANSWERS

Lesson 5.1: Fractions and Unit Fractions

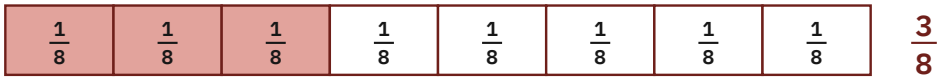
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.1 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.1, Part A.
- [ ] Complete the rest of Workbook Lesson 5.1 (Parts B–E).

LET'S EXPLORE

A pan of cornbread is cut into 8 equal pieces. You eat 3. What fraction of the pan did you eat?

A fraction names equal parts of a whole. The denominator tells how many equal parts the whole is cut into. The numerator tells how many parts we are counting.

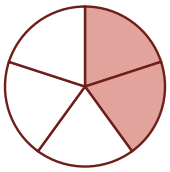


3 of 8 equal parts: $^{3}/_{8}$

Each piece is $^{1}/_{8}$ of the pan. $^{1}/_{8}$ is a unit fraction. Three pieces are 3 copies of $^{1}/_{8}$, so you ate $^{3}/_{8}$.

WORKED EXAMPLE · NAME A FRACTION

What fraction of the circle is shaded?



1 The circle has 5 equal parts, so the denominator is 5.

2 2 parts are shaded, so the numerator is 2.

Answer: $^{2}/_{5}$

! WATCH OUT

The parts must be equal. A shape cut into 4 unequal pieces does not show fourths.

★ KEY IDEA

A fraction $\frac{a}{b}$ means a copies of the unit fraction $\frac{1}{b}$.

✎ TRY IT

Answers are shown in color.

1. What fraction of a pizza is 1 slice if the pizza has 6 equal slices?

Answer: $\frac{1}{6}$

2. How many $\frac{1}{10}$s make $\frac{7}{10}$?

Answer: 7

3. A bar is cut into 4 equal parts with 3 shaded. Name the fraction.

Answer: $\frac{3}{4}$

Workbook Answer Key · Lesson 5.1

#	Question	Answer
1	What fraction of a pizza is 1 slice if the pizza has 6 equal slices?	$\frac{1}{6}$
2	How many $\frac{1}{10}$s make $\frac{7}{10}$?	7
3	A bar is cut into 4 equal parts with 3 shaded. Name the fraction.	$\frac{3}{4}$
4	12 ÷ 4	3
5	Which is a fraction: 3 or one-half?	one-half
6	(figure)	$\frac{1}{3}$
7	(figure)	$\frac{5}{6}$
8	(figure)	$\frac{2}{8}$
9	(figure)	$\frac{7}{10}$
10	4 of 5 equal parts	$\frac{4}{5}$
11	1 of 12 equal parts	$\frac{1}{12}$
12	6 copies of $\frac{1}{8}$	$\frac{6}{8}$
13	3 copies of $\frac{1}{4}$	$\frac{3}{4}$
14	Draw a bar and shade $\frac{2}{3}$. Label the unit fractions.	Bar in 3 equal parts, 2 shaded, each labeled $\frac{1}{3}$

LESSON 5.2 · TEACHER PLAN
Week 1 · Lesson 2 of 4
Unit 5: Fraction Equivalence and Comparison

Lesson 5.2: Fractions on a Number Line

Student goal	I can place fractions on a number line.
Objective	Students locate fractions on a number line.
Standard	4.NF.1 — Understand a fraction as a number on the number line.
Pacing	Unit 5, week 1, lesson 2 of 4
Time	About 50 minutes
Materials	Ruler, workbook Lesson 5.2
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	How many $^{1}/_{5}$s in $^{4}/_{5}$?
Launch	5	Read Let's Explore.
Teach	15	Draw 0 to 1 and split into 4 equal parts with a ruler. Say: "Count spaces, not marks." Work the example.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Counting tick marks (including 0) instead of spaces.

SUPPORT

- Use a paper strip laid along the line.

EXTEND

- Place $^{5}/_{4}$ on a number line from 0 to 2.

STUDENT LESSON 5.2 · WITH ANSWERS

Lesson 5.2: Fractions on a Number Line

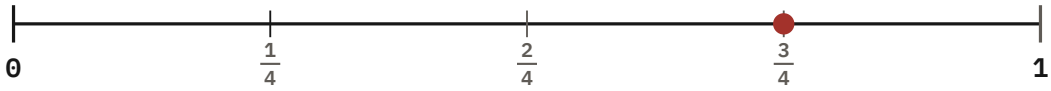
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.2 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.2, Part A.
- [ ] Complete the rest of Workbook Lesson 5.2 (Parts B–E).

LET'S EXPLORE

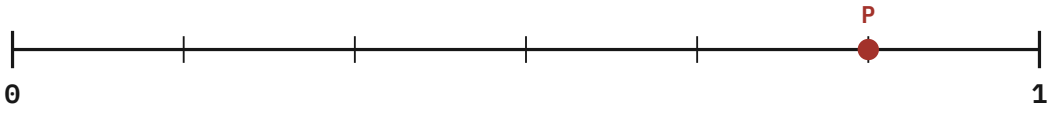
Where would $^{3}/_{4}$ be on a number line from 0 to 1?

On a number line, the whole is the distance from 0 to 1. Split it into equal parts, and count parts from 0. To show $^{3}/_{4}$, split 0 to 1 into 4 equal parts and count 3.



WORKED EXAMPLE · NAME THE POINT

What fraction is at point P?



1 0 to 1 is split into 6 equal parts.

2 P is 5 parts from 0.

Answer: $^{5}/_{6}$

★ KEY IDEA

Count the spaces between marks, not the marks, to find the denominator.

✎ TRY IT

Answers are shown in color.

1. A line from 0 to 1 is split into 3 equal parts. What fraction is at the first mark after 0?
Answer: $^{1}/_{3}$
2. On a line split into 8 parts, how many parts from 0 is $^{5}/_{8}$?
Answer: 5
3. What fraction is halfway between 0 and 1?
Answer: $^{1}/_{2}$

Workbook Answer Key · Lesson 5.2

#	Question	Answer
1	A line from 0 to 1 is split into 3 equal parts. What fraction is at the first mark after 0?	$^{1}/_{3}$
2	On a line split into 8 parts, how many parts from 0 is $^{5}/_{8}$?	5
3	What fraction is halfway between 0 and 1?	$^{1}/_{2}$
4	How many $^{1}/_{5}$s in $^{4}/_{5}$?	4
5	Count by 2s to 12.	2, 4, 6, 8, 10, 12
6	Point A (figure)	$^{2}/_{3}$
7	Point B (figure)	$^{1}/_{4}$
8	Point C (figure)	$^{7}/_{8}$
9	Point D (figure)	$^{3}/_{5}$
10	Draw a number line from 0 to 1. Mark $^{1}/_{6}$, $^{3}/_{6}$, and $^{5}/_{6}$.	Line with 6 equal parts; points at 1st, 3rd, 5th marks
11	Explain how to find $^{2}/_{5}$ on a number line.	Split 0–1 into 5 equal parts; count 2 parts from 0.

LESSON 5.3 · TEACHER PLAN
Week 1 · Lesson 3 of 4
Unit 5: Fraction Equivalence and Comparison

Lesson 5.3: Fractions Equal to and Greater Than 1

Student goal	I can show fractions equal to 1 and greater than 1, like $^{4}/_{4}$ and $^{5}/_{4}$.
Objective	Students represent fractions equal to and greater than 1.
Standard	4.NF.1 — Understand fractions greater than 1 and mixed numbers (foundation for 4.NF.3).
Pacing	Unit 5, week 1, lesson 3 of 4
Time	About 50 minutes
Materials	Fraction strips, workbook Lesson 5.3
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	13 ÷ 4.
Launch	5	Read Let's Explore with paper 'pies'.
Teach	15	Build 5/4 with strips: one whole and one fourth more. Say: "How many wholes fit? What is left?" Work the example.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking a fraction can never be greater than 1.

SUPPORT

- Use strips and count wholes.

EXTEND

- Which is greater, 11/4 or 3? Why?

STUDENT LESSON 5.3 · WITH ANSWERS

Lesson 5.3: Fractions Equal to and Greater Than 1

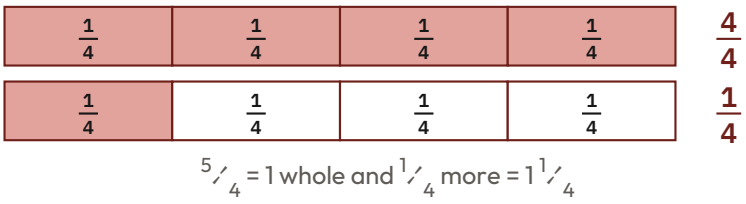
STUDENT'S ASSIGNMENT

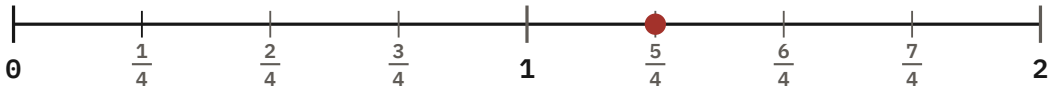
- [ ] Read Lesson 5.3 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.3, Part A.
- [ ] Complete the rest of Workbook Lesson 5.3 (Parts B–E).

LET'S EXPLORE

Mom cut 2 pies into fourths. The family ate 5 pieces. Is that more or less than one whole pie?

When the numerator equals the denominator, the fraction equals 1 whole: $^4/_4 = 1$. When the numerator is bigger, the fraction is greater than 1.





WORKED EXAMPLE · WRITE AS A MIXED NUMBER

Write $^7/_3$ as a mixed number.

1 3 thirds make 1 whole. 7 thirds = 3 + 3 + 1.

2 That is 2 wholes and 1 third.

Answer: $2\ ^1/_3$

★ KEY IDEA

Same numerator and denominator = 1 whole. To change an improper fraction to a mixed number, see how many wholes fit.

✎ TRY IT

Answers are shown in color.

1. $^{6}/_{6}$ = ___

Answer: 1

2. Write $^{9}/_{4}$ as a mixed number.

Answer: $2\,^{1}/_{4}$

3. Write $1\,^{2}/_{5}$ as a fraction.

Answer: $^{7}/_{5}$

Workbook Answer Key · Lesson 5.3

#	Question	Answer
1	$^{6}/_{6}$ = ___	1
2	Write $^{9}/_{4}$ as a mixed number.	$2\,^{1}/_{4}$
3	Write $1\,^{2}/_{5}$ as a fraction.	$^{7}/_{5}$
4	13 ÷ 4	3 R1
5	Name the fraction: 3 of 3 equal parts	$^{3}/_{3}$
6	$^{5}/_{3}$	$1\,^{2}/_{3}$
7	$^{7}/_{2}$	$3\,^{1}/_{2}$
8	$^{11}/_{4}$	$2\,^{3}/_{4}$
9	$^{13}/_{5}$	$2\,^{3}/_{5}$
10	$1\,^{1}/_{2}$	$^{3}/_{2}$
11	$2\,^{3}/_{4}$	$^{11}/_{4}$
12	$3\,^{1}/_{3}$	$^{10}/_{3}$
13	$1\,^{5}/_{8}$	$^{13}/_{8}$
14	Draw a model of $^{8}/_{3}$. Write it as a mixed number.	$2\,^{2}/_{3}$

LESSON 5.4 · TEACHER PLAN
Week 1 · Lesson 4 of 4
Unit 5: Fraction Equivalence and Comparison

Lesson 5.4: Fractions of a Set

Student goal	I can find a fraction of a group of objects.
Objective	Students find fractions of sets.
Standard	4.NF.4 (preview), 3.NF.1 — Understand fractions of a set (supports 4.NF.4).
Pacing	Unit 5, week 1, lesson 4 of 4
Time	About 50 minutes
Materials	24 counters, workbook Lesson 5.4
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	24 ÷ 6; 3 × 7.
Launch	5	Read Let's Explore with 12 counters.
Teach	15	Divide counters into 3 equal groups. Say: "Divide by the bottom, multiply by the top." Work the example.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Dividing by the numerator.

SUPPORT

- Use counters.

EXTEND

- If $\frac{3}{4}$ of a number is 18, what is the number? (24)

STUDENT LESSON 5.4 · WITH ANSWERS

Lesson 5.4: Fractions of a Set

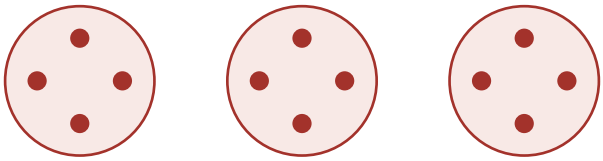
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.4 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.4, Part A.
- [ ] Complete the rest of Workbook Lesson 5.4 (Parts B–E).

LET'S EXPLORE

There are 12 eggs in a carton. $^1/_3$ of them are brown. How many brown eggs are there?

A fraction can name part of a set. To find $^1/_3$ of 12, split the 12 into 3 equal groups. One group is $^1/_3$.



12 eggs in 3 equal groups: $^1/_3$ of 12 = 4.

WORKED EXAMPLE · MORE THAN ONE PART

Find $^2/_3$ of 12.

1 $^1/_3$ of 12 = 12 ÷ 3 = 4.

2 $^2/_3$ is 2 of those groups: 2 × 4 = 8.

Answer: 8

★ KEY IDEA

To find a fraction of a set: divide by the denominator to find one part, then multiply by the numerator.

✎ TRY IT

Answers are shown in color.

1. $^1/_4$ of 20

Answer: 5

2. $^3/_5$ of 15

Answer: 9

3. Lillie has 18 books. $^{2}/_{6}$ are mysteries. How many mysteries?

Answer: 6

Workbook Answer Key · Lesson 5.4

#	Question	Answer
1	$^{1}/_{4}$ of 20	5
2	$^{3}/_{5}$ of 15	9
3	Lillie has 18 books. $^{2}/_{6}$ are mysteries. How many mysteries?	6
4	24 ÷ 6	4
5	3 × 7	21
6	$^{1}/_{2}$ of 16	8
7	$^{1}/_{5}$ of 30	6
8	$^{2}/_{3}$ of 21	14
9	$^{3}/_{4}$ of 24	18
10	$^{5}/_{6}$ of 36	30
11	$^{4}/_{9}$ of 27	12
12	A class of 28 students went on a trip. $^{3}/_{4}$ of them rode the bus. How many rode the bus?	21 students
13	Ellie picked 40 strawberries and ate $^{1}/_{8}$ of them. How many are left?	35 strawberries
14	Draw a picture to find $^{2}/_{5}$ of 10.	4

WEEKLY CHECK · ANSWER KEY
After Lesson 5.4

About 15 minutes

Weekly Check 9: Understanding Fractions — Answer Key

Answers

#	Question	Answer
1	Name the shaded fraction. (figure)	$\frac{5}{8}$
2	Name point P. (figure)	$\frac{3}{4}$
3	Write $\frac{11}{3}$ as a mixed number.	$3\frac{2}{3}$
4	Write $2\frac{1}{4}$ as a fraction.	$\frac{9}{4}$
5	$\frac{2}{3}$ of 18	12
6	$\frac{3}{3}$ = ___	1

LESSON 5.5 · TEACHER PLAN
Week 2 · Lesson 1 of 4
Unit 5: Fraction Equivalence and Comparison

Lesson 5.5: Equivalent Fractions with Models

Student goal	I can use models to show that two fractions are equivalent.
Objective	Students show equivalent fractions with area and bar models.
Standard	4.NF.1 — Explain why a fraction a/b is equivalent to (n×a)/(n×b) using visual fraction models.
Pacing	Unit 5, week 2, lesson 1 of 4
Time	About 50 minutes
Materials	Fraction strips (make from paper), workbook Lesson 5.5
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Shade half of 6 parts.
Launch	5	Read Let's Explore.
Teach	15	Lay 1/2, 2/4, 4/8 strips under each other. Say: "More pieces, smaller pieces, same amount." Work the example by cutting a strip.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking bigger numbers mean a bigger fraction.

SUPPORT

- Fraction strips for every problem.

EXTEND

- Find three fractions equivalent to 2/3.

STUDENT LESSON 5.5 · WITH ANSWERS

Lesson 5.5: Equivalent Fractions with Models

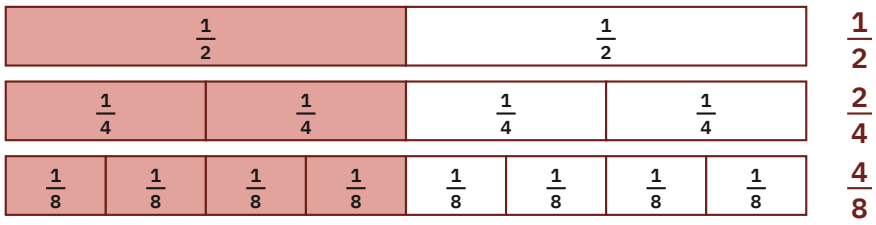
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.5 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.5, Part A.
- [ ] Complete the rest of Workbook Lesson 5.5 (Parts B–E).

LET'S EXPLORE

Ellie ate $^{1}/_{2}$ of a sandwich. Lillie ate $^{2}/_{4}$ of the same size sandwich. Who ate more?

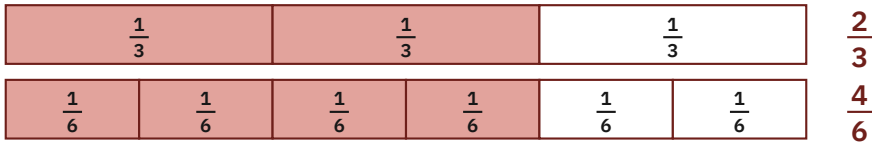
Look at the bars. $^{1}/_{2}$ and $^{2}/_{4}$ cover the same length. They are equivalent fractions: different names for the same amount.



When you cut each part into 2 smaller parts, you get twice as many parts, and each part is half as big. The amount does not change.

WORKED EXAMPLE · FIND AN EQUIVALENT FRACTION

Use a model to find a fraction equal to $^{2}/_{3}$ with sixths.



1 Draw thirds. Cut each third in half.

2 2 shaded thirds become 4 shaded sixths.

Answer: $^{2}/_{3} = {}^{4}/_{6}$

★ KEY IDEA

Equivalent fractions cover the same amount of the same whole.

✎ TRY IT

Answers are shown in color.

1. $\frac{1}{3} = \frac{?}{6}$

Answer: $\frac{2}{6}$

2. $\frac{3}{4} = \frac{?}{8}$

Answer: $\frac{6}{8}$

3. Are $\frac{2}{5}$ and $\frac{4}{10}$ equivalent?

Answer: Yes

Workbook Answer Key · Lesson 5.5

#	Question	Answer
1	$\frac{1}{3} = \frac{?}{6}$	$\frac{2}{6}$
2	$\frac{3}{4} = \frac{?}{8}$	$\frac{6}{8}$
3	Are $\frac{2}{5}$ and $\frac{4}{10}$ equivalent?	Yes
4	Shade half of a bar of 6 parts. How many parts?	3
5	2 × 4	8
6	(figure)	$\frac{1}{2} = \frac{3}{6}$
7	(figure)	$\frac{2}{4} = \frac{4}{8}$
8	(figure)	$\frac{1}{5} = \frac{2}{10}$
9	(figure)	$\frac{2}{3} = \frac{8}{12}$
10	$\frac{1}{4} = \frac{?}{8}$	2
11	$\frac{3}{5} = \frac{?}{10}$	6
12	$\frac{1}{3} = \frac{?}{12}$	4
13	$\frac{5}{6} = \frac{?}{12}$	10
14	Draw models to show $\frac{3}{4} = \frac{6}{8}$.	Two equal bars: 4 parts with 3 shaded; 8 parts with 6 shaded

LESSON 5.6 · TEACHER PLAN
Week 2 · Lesson 2 of 4
Unit 5: Fraction Equivalence and Comparison

Lesson 5.6: Equivalent Fractions by Multiplying

Student goal	I can multiply the numerator and denominator by the same number to make an equivalent fraction.
Objective	Students generate equivalent fractions by multiplying.
Standard	4.NF.1 — Explain why a/b = (n×a)/(n×b).
Pacing	Unit 5, week 2, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.6
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	6 × 4.
Launch	5	Read Let's Explore.
Teach	15	Connect the model from 5.5 to the multiplication table. Say: "What you do to the bottom, you do to the top."
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Adding the same number to top and bottom.

SUPPORT

- Show with strips why adding fails (1/2 vs 2/3).

EXTEND

- Why is multiplying by 4/4 the same as multiplying by 1?

STUDENT LESSON 5.6 · WITH ANSWERS

Lesson 5.6: Equivalent Fractions by Multiplying

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.6 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.6, Part A.
- [ ] Complete the rest of Workbook Lesson 5.6 (Parts B–F).

LET'S EXPLORE

Is there a way to find a fraction equal to $^{3}/_{5}$ with a denominator of 20 without drawing?

In the models, cutting every part into 4 pieces multiplies the number of parts by 4 and the number of shaded parts by 4.

	Numerator	Denominator
$^{3}/_{5}$	3	5
× 4	× 4	× 4
$^{12}/_{20}$	12	20

WORKED EXAMPLE · MULTIPLY

Find a fraction equal to $^{2}/_{3}$ with denominator 12.

1 $3 \times 4 = 12$, so multiply by 4.

2 Multiply the numerator too: $2 \times 4 = 8$.

Answer: $^{2}/_{3} = {}^{8}/_{12}$

! WATCH OUT

Multiply the numerator and the denominator by the same number. Multiplying both by 4 is like multiplying by $^{4}/_{4}$, which equals 1.

★ KEY IDEA

$$\frac{a}{b} = \frac{n \times a}{n \times b}$$

✎ TRY IT

Answers are shown in color.

1. $\frac{1}{4} = \frac{?}{12}$

Answer: 3

2. $\frac{5}{6} = \frac{?}{18}$

Answer: 15

3. Write two fractions equivalent to $\frac{2}{5}$.

Answer: e.g. $\frac{4}{10}$, $\frac{6}{15}$

Workbook Answer Key · Lesson 5.6

#	Question	Answer
1	$\frac{1}{4} = \frac{?}{12}$	3
2	$\frac{5}{6} = \frac{?}{18}$	15
3	Write two fractions equivalent to $\frac{2}{5}$.	e.g. $\frac{4}{10}$, $\frac{6}{15}$
4	6×4	24
5	$\frac{1}{2} = \frac{?}{4}$	2
6	$\frac{1}{3} = \frac{?}{6}$	2
7	$\frac{3}{4} = \frac{?}{12}$	9
8	$\frac{2}{5} = \frac{?}{20}$	8
9	$\frac{5}{8} = \frac{?}{16}$	10
10	$\frac{1}{6} = \frac{?}{30}$	5
11	$\frac{4}{7} = \frac{?}{21}$	12
12	$\frac{1}{2} = \frac{6}{?}$	12
13	$\frac{3}{10} = \frac{30}{?}$	100
14	$\frac{3}{4}$	e.g. $\frac{6}{8}$, $\frac{9}{12}$, $\frac{12}{16}$
15	Is $\frac{3}{8} = \frac{9}{24}$? Explain.	Yes. $3 \times 3 = 9$ and $8 \times 3 = 24$.

LESSON 5.7 · TEACHER PLAN
Week 2 · Lesson 3 of 4
Unit 5: Fraction Equivalence and Comparison

Lesson 5.7: Simplest Form

Student goal	I can divide the numerator and denominator by a common factor to write a fraction in simplest form.
Objective	Students write fractions in simplest form.
Standard	4.NF.1 — Recognize and generate equivalent fractions.
Pacing	Unit 5, week 2, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.7
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Factors of 12 and 16.
Launch	5	Read Let's Explore.
Teach	15	Work both examples. Say: Divide top and bottom by the same factor.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Dividing only the numerator.
- Stopping before simplest form.

SUPPORT

- List factors of both numbers.

EXTEND

- Simplify 36/48 in one step.

STUDENT LESSON 5.7 · WITH ANSWERS

Lesson 5.7: Simplest Form

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.7 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.7, Part A.
- [ ] Complete the rest of Workbook Lesson 5.7 (Parts B–E).

LET'S EXPLORE

A recipe calls for $^{6}/_{8}$ of a cup of milk. Is there a simpler name for that amount?

You can also go the other way: divide the numerator and denominator by a common factor. 6 and 8 are both divisible by 2.

WORKED EXAMPLE · SIMPLIFY

Write $^{6}/_{8}$ in simplest form.

1 $6 \div 2 = 3$ and $8 \div 2 = 4$.

2 3 and 4 have no common factor but 1.

Answer: $^{3}/_{4}$

WORKED EXAMPLE · USE THE GREATEST COMMON FACTOR

Write $^{12}/_{18}$ in simplest form.

1 Factors of 12: 1, 2, 3, 4, 6, 12. Factors of 18: 1, 2, 3, 6, 9, 18.

2 The greatest common factor is 6.

3 $12 \div 6 = 2$, $18 \div 6 = 3$.

Answer: $^{2}/_{3}$

★ KEY IDEA

Divide the numerator and denominator by a common factor. Keep going until 1 is the only common factor.

✎ TRY IT

Answers are shown in color.

1. Simplify $^{4}/_{10}$

Answer: $\frac{2}{5}$

2. Simplify $\frac{9}{12}$

Answer: $\frac{3}{4}$

3. Simplify $\frac{8}{16}$

Answer: $\frac{1}{2}$

Workbook Answer Key · Lesson 5.7

#	Question	Answer
1	Simplify $\frac{4}{10}$	$\frac{2}{5}$
2	Simplify $\frac{9}{12}$	$\frac{3}{4}$
3	Simplify $\frac{8}{16}$	$\frac{1}{2}$
4	Factors of 12	1, 2, 3, 4, 6, 12
5	Factors of 16	1, 2, 4, 8, 16
6	$\frac{2}{4}$	$\frac{1}{2}$
7	$\frac{3}{9}$	$\frac{1}{3}$
8	$\frac{5}{10}$	$\frac{1}{2}$
9	$\frac{6}{9}$	$\frac{2}{3}$
10	$\frac{8}{12}$	$\frac{2}{3}$
11	$\frac{10}{15}$	$\frac{2}{3}$
12	$\frac{14}{21}$	$\frac{2}{3}$
13	$\frac{15}{20}$	$\frac{3}{4}$
14	Is $\frac{5}{7}$ in simplest form? A. Yes B. No	Yes
15	Is $\frac{4}{6}$ in simplest form? A. Yes B. No	No ($\frac{2}{3}$)
16	Write $\frac{18}{24}$ in simplest form. Show your steps.	$\frac{3}{4}$

LESSON 5.8 · TEACHER PLAN
Week 2 · Lesson 4 of 4
Unit 5: Fraction Equivalence and Comparison

Lesson 5.8: Equivalent Fractions on a Number Line

Student goal	I can show equivalent fractions at the same point on a number line.
Objective	Students show equivalent fractions on number lines.
Standard	4.NF.1 — Recognize equivalent fractions as the same point on a number line.
Pacing	Unit 5, week 2, lesson 4 of 4
Time	About 50 minutes
Materials	Ruler, workbook Lesson 5.8
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	2/5 = ?/10.
Launch	5	Read Let's Explore with a ruler.
Teach	15	Draw three stacked number lines (halves, fourths, eighths) the same length. Say: “Same point, same number.”
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Drawing number lines of different lengths.

SUPPORT

- Use a ruler marked in eighths.

EXTEND

- Find all fractions with denominators up to 12 equal to 1/2.

STUDENT LESSON 5.8 · WITH ANSWERS

Lesson 5.8: Equivalent Fractions on a Number Line

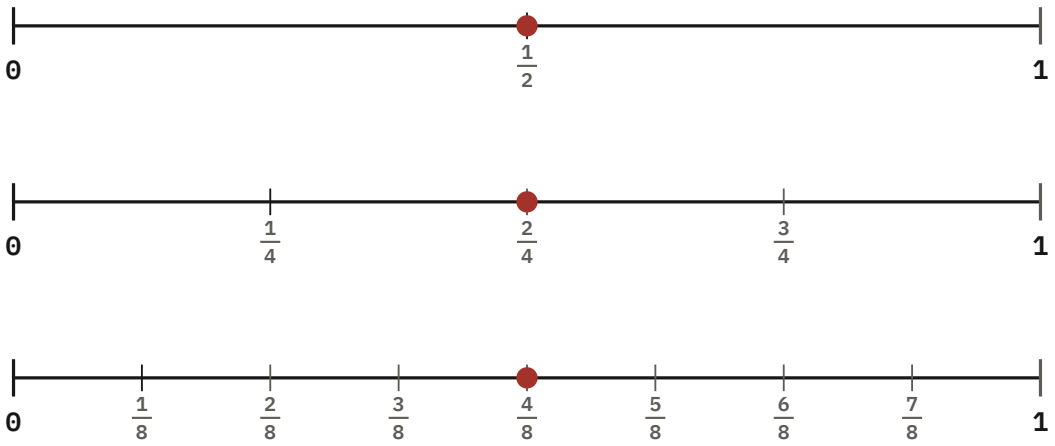
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.8 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.8, Part A.
- [ ] Complete the rest of Workbook Lesson 5.8 (Parts B–E).

LET'S EXPLORE

On a ruler, the $^{1}/_{2}$-inch mark is the same mark as $^{2}/_{4}$ and $^{4}/_{8}$. Why?

Equivalent fractions land on the same point of a number line, because they are the same distance from 0.



WORKED EXAMPLE · FIND THE SAME POINT

Which eighths fraction is at the same point as $^{3}/_{4}$?

1 Each fourth is 2 eighths.

2 3 fourths = 6 eighths.

Answer: $^{6}/_{8}$

★ KEY IDEA

Equivalent fractions are the same number, so they are at the same point on a number line.

✎ TRY IT

Answers are shown in color.

1. Which sixths fraction is at the same point as $^{1}/_{3}$?

Answer: $^{2}/_{6}$

2. Which tenths fraction is at the same point as $^{1}/_{2}$?

Answer: $^{5}/_{10}$

3. Is $^{2}/_{3}$ at the same point as $^{4}/_{6}$?

Answer: Yes

Workbook Answer Key · Lesson 5.8

#	Question	Answer
1	Which sixths fraction is at the same point as $^{1}/_{3}$?	$^{2}/_{6}$
2	Which tenths fraction is at the same point as $^{1}/_{2}$?	$^{5}/_{10}$
3	Is $^{2}/_{3}$ at the same point as $^{4}/_{6}$?	Yes
4	$^{2}/_{5} = {}^{?}/_{10}$	4
5	Simplify $^{6}/_{10}$	$^{3}/_{5}$
6	(figure)	$^{1}/_{4} = {}^{2}/_{8}$
7	(figure)	$^{2}/_{3} = {}^{4}/_{6}$
8	Draw a number line from 0 to 1 in sixths. Circle the fractions equal to $^{1}/_{2}$ and $^{1}/_{3}$.	$^{3}/_{6}$ and $^{2}/_{6}$
9	Explain why $^{3}/_{5}$ and $^{6}/_{10}$ are at the same point.	Each fifth is 2 tenths, so 3 fifths = 6 tenths: same distance from 0.

WEEKLY CHECK · ANSWER KEY
After Lesson 5.8

About 15 minutes

Weekly Check 10: Equivalent Fractions — Answer Key

Answers

#	Question	Answer
1	$\frac{2}{3} = \frac{?}{9}$	6
2	$\frac{3}{5} = \frac{?}{20}$	12
3	Simplify $\frac{8}{10}$	$\frac{4}{5}$
4	Simplify $\frac{9}{15}$	$\frac{3}{5}$
5	Write two fractions equivalent to $\frac{1}{4}$.	e.g. $\frac{2}{8}$, $\frac{3}{12}$
6	Are $\frac{3}{4}$ and $\frac{9}{12}$ equivalent? Explain.	Yes; × 3

LESSON 5.9 · TEACHER PLAN
Week 3 · Lesson 1 of 4
Unit 5: Fraction Equivalence and Comparison

Lesson 5.9: Comparing: Same Denominator or Same Numerator

Student goal	I can compare fractions that have the same denominator or the same numerator.
Objective	Students compare fractions with like numerators or denominators.
Standard	4.NF.2 — Compare two fractions with different numerators and denominators; recognize comparisons are valid only for the same whole.
Pacing	Unit 5, week 3, lesson 1 of 4
Time	About 50 minutes
Materials	Fraction strips, workbook Lesson 5.9
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Which piece is bigger?
Launch	5	Read Let's Explore.
Teach	15	Compare with strips. Say: Same size pieces? Count them. Same number of pieces? Compare their size.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking $\frac{1}{8} > \frac{1}{4}$ because 8 > 4.

SUPPORT

- Strips for every comparison.

EXTEND

- Order 1/2, 1/3, 1/4, 1/5, 1/6.

STUDENT LESSON 5.9 · WITH ANSWERS

Lesson 5.9: Comparing: Same Denominator or Same Numerator

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.9 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.9, Part A.
- [ ] Complete the rest of Workbook Lesson 5.9 (Parts B–D).

LET'S EXPLORE

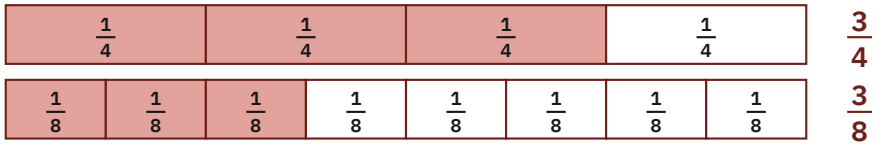
Which is more, $^{3}/_{8}$ of a pizza or $^{5}/_{8}$ of the same pizza? Which is more, $^{3}/_{4}$ or $^{3}/_{8}$?

Same denominator

When the denominators are the same, the pieces are the same size. More pieces means more: $^{5}/_{8} > {}^{3}/_{8}$.

Same numerator

When the numerators are the same, you have the same number of pieces. Compare the size of the pieces: fourths are bigger than eighths, so $^{3}/_{4} > {}^{3}/_{8}$.



! WATCH OUT

A bigger denominator means smaller pieces. $^{1}/_{8}$ is less than $^{1}/_{4}$, even though $8 > 4$.

★ KEY IDEA

Same denominator: compare numerators. Same numerator: the smaller denominator is the bigger fraction.

✎ TRY IT

Answers are shown in color.

1. $^{2}/_{6}$ ◯ $^{5}/_{6}$

Answer: <

2. $\frac{4}{5}$ ◯ $\frac{4}{10}$

Answer: >

3. $\frac{1}{3}$ ◯ $\frac{1}{12}$

Answer: >

Workbook Answer Key · Lesson 5.9

#	Question	Answer
1	$\frac{2}{6}$ ◯ $\frac{5}{6}$	<
2	$\frac{4}{5}$ ◯ $\frac{4}{10}$	>
3	$\frac{1}{3}$ ◯ $\frac{1}{12}$	>
4	Which is bigger, 1 of 3 equal pieces or 1 of 6 equal pieces of the same cake?	1 of 3
5	Simplify $\frac{4}{8}$	$\frac{1}{2}$
6	$\frac{3}{5}$ ◯ $\frac{4}{5}$	<
7	$\frac{7}{10}$ ◯ $\frac{3}{10}$	>
8	$\frac{2}{3}$ ◯ $\frac{2}{5}$	>
9	$\frac{5}{12}$ ◯ $\frac{5}{6}$	<
10	$\frac{1}{2}$ ◯ $\frac{1}{4}$	>
11	$\frac{6}{8}$ ◯ $\frac{6}{8}$	=
12	$\frac{9}{12}$ ◯ $\frac{7}{12}$	>
13	$\frac{3}{10}$ ◯ $\frac{3}{4}$	<
14	Explain why $\frac{2}{3} > \frac{2}{5}$.	Both have 2 pieces, but thirds are bigger than fifths.

LESSON 5.10 · TEACHER PLAN
Week 3 · Lesson 2 of 4
Unit 5: Fraction Equivalence and Comparison

Lesson 5.10: Comparing with Benchmarks

Student goal	I can compare fractions by thinking about 0, $^1/_2$, and 1.
Objective	Students compare fractions using benchmarks.
Standard	4.NF.2 — Compare two fractions by comparing to a benchmark fraction such as 1/2.
Pacing	Unit 5, week 3, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.10
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Half of 10, half of 7.
Launch	5	Read Let's Explore.
Teach	15	Sort fractions into less than half, half, more than half. Work the 'close to 1' example.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Not knowing how to find half of an odd denominator.

SUPPORT

- Use strips folded in half.

EXTEND

- Compare 13/25 and 12/23.

STUDENT LESSON 5.10 · WITH ANSWERS

Lesson 5.10: Comparing with Benchmarks

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.10 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.10, Part A.
- [ ] Complete the rest of Workbook Lesson 5.10 (Parts B–E).

LET'S EXPLORE

Which is greater, $\frac{3}{8}$ or $\frac{4}{6}$? They have different numerators and denominators!

Compare each fraction to $\frac{1}{2}$. $\frac{3}{8}$ is less than $\frac{4}{8}$, which is half. $\frac{4}{6}$ is more than $\frac{3}{6}$, which is half. So $\frac{3}{8} < \frac{4}{6}$.

Less than half	Exactly half	More than half
numerator less than half of denominator	numerator is half of denominator	numerator more than half of denominator
$\frac{2}{5}, \frac{3}{8}, \frac{1}{4}$	$\frac{3}{6}, \frac{5}{10}$	$\frac{3}{5}, \frac{5}{8}, \frac{3}{4}$

WORKED EXAMPLE · CLOSE TO 1

Compare $\frac{7}{8}$ and $\frac{9}{10}$.

1 Both are one piece away from 1.

2 $\frac{7}{8}$ is missing $\frac{1}{8}$; $\frac{9}{10}$ is missing $\frac{1}{10}$.

3 $\frac{1}{10}$ is the smaller gap, so $\frac{9}{10}$ is closer to 1.

Answer: $\frac{7}{8} < \frac{9}{10}$

★ KEY IDEA

Ask: Is each fraction less than, equal to, or more than $\frac{1}{2}$? Near 1, compare the missing pieces.

✎ TRY IT

Answers are shown in color.

1. $\frac{2}{6}$ ◯ $\frac{5}{8}$

Answer: <

2. $\frac{4}{7}$ ◯ $\frac{1}{3}$

Answer: >

3. $\frac{5}{6}$ ◯ $\frac{7}{8}$

Answer: <

Workbook Answer Key · Lesson 5.10

#	Question	Answer
1	$\frac{2}{6}$ ◯ $\frac{5}{8}$	<
2	$\frac{4}{7}$ ◯ $\frac{1}{3}$	>
3	$\frac{5}{6}$ ◯ $\frac{7}{8}$	<
4	Half of 10	5
5	Half of 7	3.5 (3 and a half)
6	$\frac{3}{7}$	less
7	$\frac{5}{9}$	greater
8	$\frac{6}{12}$	equal
9	$\frac{2}{5}$	less
10	$\frac{7}{12}$	greater
11	$\frac{4}{10}$	less
12	$\frac{3}{8}$ ◯ $\frac{2}{3}$	<
13	$\frac{5}{6}$ ◯ $\frac{2}{5}$	>
14	$\frac{1}{4}$ ◯ $\frac{4}{7}$	<
15	$\frac{11}{12}$ ◯ $\frac{9}{10}$	>
16	Compare $\frac{4}{9}$ and $\frac{6}{10}$. Explain using $\frac{1}{2}$.	$\frac{4}{9} < \frac{6}{10}$: $\frac{4}{9}$ is less than half and $\frac{6}{10}$ is more than half.

LESSON 5.11 · TEACHER PLAN
Week 3 · Lesson 3 of 4
Unit 5: Fraction Equivalence and Comparison

Lesson 5.11: Comparing with Common Denominators

Student goal	I can compare fractions by renaming them with a common denominator.
Objective	Students compare fractions using common denominators.
Standard	4.NF.2 — Compare two fractions by creating common denominators.
Pacing	Unit 5, week 3, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.11
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Common multiple of 4 and 6.
Launch	5	Read Let's Explore.
Teach	15	Work both examples. Say: Make the pieces the same size, then count.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Changing only the denominator.

SUPPORT

- Use the multiplication table to find common multiples.

EXTEND

- Compare 5/7 and 7/9.

STUDENT LESSON 5.11 · WITH ANSWERS

Lesson 5.11: Comparing with Common Denominators

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.11 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.11, Part A.
- [ ] Complete the rest of Workbook Lesson 5.11 (Parts B–D).

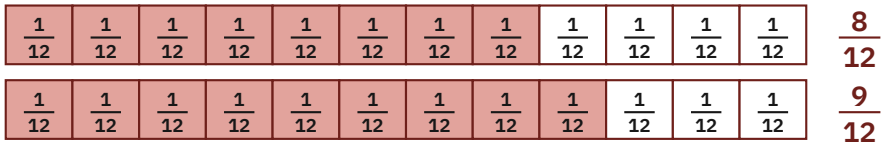
LET'S EXPLORE

Compare $^{2}/_{3}$ and $^{3}/_{4}$. Neither benchmarks nor matching numbers help much. What can we do?

Rename both fractions so they have the same denominator. Then the pieces are the same size and you can compare numerators.

WORKED EXAMPLE · COMMON DENOMINATOR

Compare $^{2}/_{3}$ and $^{3}/_{4}$.



1 Multiples of 3: 3, 6, 9, 12. Multiples of 4: 4, 8, 12. Use 12.

2 $^{2}/_{3} = {}^{8}/_{12}$ (× 4).

3 $^{3}/_{4} = {}^{9}/_{12}$ (× 3).

4 $8 < 9$.

Answer: $^{2}/_{3} < {}^{3}/_{4}$

WORKED EXAMPLE · ONE DENOMINATOR IS A MULTIPLE OF THE OTHER

Compare $^{3}/_{5}$ and $^{7}/_{10}$.

1 10 is a multiple of 5. $^{3}/_{5} = {}^{6}/_{10}$.

2 $6 < 7$.

Answer: $^{3}/_{5} < {}^{7}/_{10}$

★ KEY IDEA

Find a common multiple of the denominators. Rename both fractions, then compare numerators.

✎ TRY IT

Answers are shown in color.

1. $\frac{1}{2}$ ◯ $\frac{3}{5}$

Answer: <

2. $\frac{5}{6}$ ◯ $\frac{3}{4}$

Answer: >

3. $\frac{2}{4}$ ◯ $\frac{5}{10}$

Answer: =

Workbook Answer Key · Lesson 5.11

#	Question	Answer
1	$\frac{1}{2}$ ◯ $\frac{3}{5}$	<
2	$\frac{5}{6}$ ◯ $\frac{3}{4}$	>
3	$\frac{2}{4}$ ◯ $\frac{5}{10}$	=
4	Common multiple of 4 and 6	12
5	$\frac{1}{3} = \frac{?}{12}$	4
6	$\frac{1}{3}$ ◯ $\frac{2}{6}$	=
7	$\frac{3}{4}$ ◯ $\frac{5}{8}$	>
8	$\frac{2}{5}$ ◯ $\frac{3}{10}$	>
9	$\frac{4}{6}$ ◯ $\frac{2}{3}$	=
10	$\frac{3}{8}$ ◯ $\frac{1}{3}$	>
11	$\frac{5}{12}$ ◯ $\frac{1}{2}$	<
12	$\frac{7}{10}$ ◯ $\frac{4}{5}$	<
13	$\frac{2}{3}$ ◯ $\frac{5}{9}$	>
14	Compare $\frac{5}{8}$ and $\frac{2}{3}$. Show the common denominator.	$\frac{15}{24} < \frac{16}{24}$, so $\frac{5}{8} < \frac{2}{3}$

LESSON 5.12 · TEACHER PLAN
Week 3 · Lesson 4 of 4
Unit 5: Fraction Equivalence and Comparison

Lesson 5.12: Comparing with Common Numerators

Student goal	I can compare fractions by renaming them with a common numerator.
Objective	Students compare fractions using common numerators.
Standard	4.NF.2 — Compare two fractions by creating common numerators.
Pacing	Unit 5, week 3, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.12
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	3/4 = 6/?.
Launch	5	Read Let's Explore.
Teach	15	Work the example. List all four strategies so far: same denominator, same numerator, benchmark, common denominator/numerator.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Forgetting that larger denominators mean smaller pieces.

SUPPORT

- Strips.

EXTEND

- Make a strategy poster.

STUDENT LESSON 5.12 · WITH ANSWERS

Lesson 5.12: Comparing with Common Numerators

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.12 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.12, Part A.
- [ ] Complete the rest of Workbook Lesson 5.12 (Parts B–E).

LET'S EXPLORE

Compare $\frac{2}{5}$ and $\frac{4}{9}$. Is there a quick way?

Sometimes it is easier to make the numerators match. $\frac{2}{5} = \frac{4}{10}$. Now compare $\frac{4}{10}$ and $\frac{4}{9}$: same number of pieces, and ninths are bigger than tenths.

WORKED EXAMPLE · COMMON NUMERATOR

Compare $\frac{2}{5}$ and $\frac{4}{9}$.

1 $\frac{2}{5} = \frac{4}{10}$.

2 4 tenths vs 4 ninths: ninths are bigger pieces.

Answer: $\frac{2}{5} < \frac{4}{9}$

★ KEY IDEA

If the numerators are easy to match, rename to a common numerator. Then the smaller denominator is the greater fraction.

✎ TRY IT

Answers are shown in color.

1. $\frac{1}{3}$ ◯ $\frac{2}{7}$

Answer: >

2. $\frac{3}{4}$ ◯ $\frac{6}{9}$

Answer: >

3. $\frac{2}{3}$ ◯ $\frac{4}{5}$

Answer: <

Workbook Answer Key · Lesson 5.12

#	Question	Answer
1	$\frac{1}{3}$ ◯ $\frac{2}{7}$	>
2	$\frac{3}{4}$ ◯ $\frac{6}{9}$	>
3	$\frac{2}{3}$ ◯ $\frac{4}{5}$	<
4	$\frac{3}{4} = \frac{6}{?}$	8
5	$\frac{4}{7}$ ◯ $\frac{4}{9}$	>
6	$\frac{1}{4}$ ◯ $\frac{2}{9}$	>
7	$\frac{2}{3}$ ◯ $\frac{4}{7}$	>
8	$\frac{3}{5}$ ◯ $\frac{6}{11}$	>
9	$\frac{1}{2}$ ◯ $\frac{3}{5}$	<
10	$\frac{2}{7}$ ◯ $\frac{4}{15}$	>
11	$\frac{3}{8}$ ◯ $\frac{6}{15}$	<
12	Compare $\frac{5}{6}$ and $\frac{5}{8}$. Which strategy is fastest? Why?	Same numerator: $\frac{5}{6} > \frac{5}{8}$ because sixths are bigger.
13	Compare $\frac{3}{7}$ and $\frac{6}{13}$.	$\frac{3}{7} = \frac{6}{14} < \frac{6}{13}$

WEEKLY CHECK · ANSWER KEY
After Lesson 5.12

About 15 minutes

Weekly Check 11: Comparing Fractions — Answer Key

Answers

#	Question	Answer
1	$\frac{3}{7}$ ◯ $\frac{5}{7}$	<
2	$\frac{2}{5}$ ◯ $\frac{2}{9}$	>
3	$\frac{3}{8}$ ◯ $\frac{4}{6}$	<
4	$\frac{5}{6}$ ◯ $\frac{7}{9}$	>
5	$\frac{2}{3}$ ◯ $\frac{8}{12}$	=
6	$\frac{4}{10}$ ◯ $\frac{1}{2}$	<

LESSON 5.13 · TEACHER PLAN
Week 4 · Lesson 1 of 4
Unit 5: Fraction Equivalence and Comparison

Lesson 5.13: Ordering Fractions

Student goal	I can order three or more fractions from least to greatest.
Objective	Students order three or more fractions.
Standard	4.NF.2 — Compare fractions; record with symbols.
Pacing	Unit 5, week 4, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.13
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	2/3 vs 5/8.
Launch	5	Read Let's Explore.
Teach	15	Work both examples. Discuss which strategy fits.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Ordering by numerator or denominator alone.

SUPPORT

- Rename all to a common denominator.

EXTEND

- Order five fractions.

STUDENT LESSON 5.13 · WITH ANSWERS

Lesson 5.13: Ordering Fractions

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.13 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.13, Part A.
- [ ] Complete the rest of Workbook Lesson 5.13 (Parts B–E).

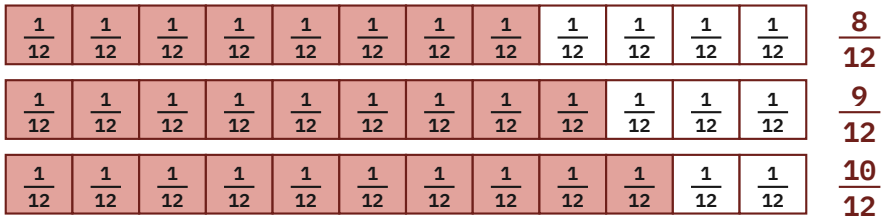
LET'S EXPLORE

Three friends ran $^{3}/_{4}$ mile, $^{2}/_{3}$ mile, and $^{5}/_{6}$ mile. Who ran farthest? Who ran the least?

Ordering fractions is comparing again and again. A common denominator lets you compare all of them at once.

WORKED EXAMPLE · ORDER

Order $^{3}/_{4}$, $^{2}/_{3}$, $^{5}/_{6}$ from least to greatest.



1 Common denominator 12.

2 $^{3}/_{4} = {}^{9}/_{12}$, $^{2}/_{3} = {}^{8}/_{12}$, $^{5}/_{6} = {}^{10}/_{12}$.

3 $8 < 9 < 10$.

Answer: $^{2}/_{3}$, $^{3}/_{4}$, $^{5}/_{6}$

WORKED EXAMPLE · USE BENCHMARKS

Order $^{1}/_{5}$, $^{7}/_{8}$, $^{1}/_{2}$.

1 $^{1}/_{5}$ is near 0, $^{1}/_{2}$ is half, $^{7}/_{8}$ is near 1.

Answer: $^{1}/_{5}$, $^{1}/_{2}$, $^{7}/_{8}$

★ KEY IDEA

Use the fastest strategy: benchmarks if the fractions are far apart, a common denominator if they are close.

✎ TRY IT

Answers are shown in color.

1. Order least to greatest: $\frac{1}{2}, \frac{1}{3}, \frac{1}{6}$

Answer: $\frac{1}{6}, \frac{1}{3}, \frac{1}{2}$

2. Order greatest to least: $\frac{3}{5}, \frac{7}{10}, \frac{1}{2}$

Answer: $\frac{7}{10}, \frac{3}{5}, \frac{1}{2}$

3. Which is greatest: $\frac{5}{8}, \frac{3}{4}, \frac{2}{3}$?

Answer: $\frac{3}{4}$

Workbook Answer Key · Lesson 5.13

#	Question	Answer
1	Order least to greatest: $\frac{1}{2}, \frac{1}{3}, \frac{1}{6}$	$\frac{1}{6}, \frac{1}{3}, \frac{1}{2}$
2	Order greatest to least: $\frac{3}{5}, \frac{7}{10}, \frac{1}{2}$	$\frac{7}{10}, \frac{3}{5}, \frac{1}{2}$
3	Which is greatest: $\frac{5}{8}, \frac{3}{4}, \frac{2}{3}$?	$\frac{3}{4}$
4	$\frac{2}{3} \bigcirc \frac{5}{8}$	>
5	$\frac{1}{2} = \frac{?}{12}$	6
6	$\frac{1}{4}, \frac{3}{8}, \frac{1}{2}$	$\frac{1}{4}, \frac{3}{8}, \frac{1}{2}$
7	$\frac{2}{3}, \frac{1}{6}, \frac{5}{12}$	$\frac{1}{6}, \frac{5}{12}, \frac{2}{3}$
8	$\frac{4}{5}, \frac{7}{10}, \frac{1}{2}$	$\frac{1}{2}, \frac{7}{10}, \frac{4}{5}$
9	$\frac{3}{4}, \frac{5}{6}, \frac{7}{12}$	$\frac{7}{12}, \frac{3}{4}, \frac{5}{6}$
10	Lillie used $\frac{2}{3}$ cup of flour, Ellie used $\frac{3}{4}$ cup, and Mom used $\frac{5}{8}$ cup. Who used the most? The least?	Most: Ellie; least: Mom
11	Order $\frac{3}{10}, \frac{1}{2}, \frac{2}{5}$ from greatest to least.	$\frac{1}{2}, \frac{2}{5}, \frac{3}{10}$

LESSON 5.14 · TEACHER PLAN
Week 4 · Lesson 2 of 4
Unit 5: Fraction Equivalence and Comparison

Lesson 5.14: Comparing Fractions of the Same Whole

Student goal	I can explain why fractions can only be compared when they refer to the same whole.
Objective	Students recognize that fraction comparisons require the same whole.
Standard	4.NF.2 — Recognize that comparisons are valid only when the two fractions refer to the same whole.
Pacing	Unit 5, week 4, lesson 2 of 4
Time	About 50 minutes
Materials	Two paper circles of different sizes, workbook Lesson 5.14
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	1/3 of 18 and 1/2 of 18.
Launch	5	Read Let's Explore with a big and small paper circle.
Teach	15	Cut half of the small circle and a third of the big circle; compare. Say: “Same whole first, then compare.”
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Always trusting the symbol comparison regardless of the whole.

SUPPORT

- Use real objects of different sizes.

EXTEND

- Create a tricky fraction riddle.

STUDENT LESSON 5.14 · WITH ANSWERS

Lesson 5.14: Comparing Fractions of the Same Whole

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.14 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.14, Part A.
- [ ] Complete the rest of Workbook Lesson 5.14 (Parts B–E).

LET'S EXPLORE

Ellie ate $\frac{1}{2}$ of a small pizza. Lillie ate $\frac{1}{3}$ of a large pizza. Ellie says she ate more because $\frac{1}{2} > \frac{1}{3}$. Is she right?

Not necessarily! $\frac{1}{2} > \frac{1}{3}$ is only true when both fractions are of the same whole. Half of a small pizza could be less than a third of a huge pizza.

★ KEY IDEA

Comparisons are only valid when the two fractions refer to the same whole.

WORKED EXAMPLE · DIFFERENT WHOLES

Is $\frac{1}{2}$ of 10 more or less than $\frac{1}{3}$ of 30?

1 $\frac{1}{2}$ of 10 = 5.

2 $\frac{1}{3}$ of 30 = 10.

Answer: Less: 5 < 10

✎ TRY IT

Answers are shown in color.

1. $\frac{1}{4}$ of 40 or $\frac{1}{2}$ of 12. Which is more?

Answer: $\frac{1}{4}$ of 40 (10 > 6)

2. Can you say $\frac{2}{3} > \frac{1}{2}$ for any two pizzas?

Answer: Only if the pizzas are the same size.

3. $\frac{3}{4}$ of 8 vs $\frac{3}{4}$ of 20

Answer: $\frac{3}{4}$ of 20 is more (15 > 6)

Workbook Answer Key · Lesson 5.14

#	Question	Answer
1	$\frac{1}{4}$ of 40 or $\frac{1}{2}$ of 12. Which is more?	$\frac{1}{4}$ of 40 (10 > 6)
2	Can you say $\frac{2}{3} > \frac{1}{2}$ for any two pizzas?	Only if the pizzas are the same size.
3	$\frac{3}{4}$ of 8 vs $\frac{3}{4}$ of 20	$\frac{3}{4}$ of 20 is more (15 > 6)
4	$\frac{1}{3}$ of 18	6
5	$\frac{1}{2}$ of 18	9
6	$\frac{1}{2}$ of 8 or $\frac{1}{4}$ of 24	$\frac{1}{4}$ of 24 (6 > 4)
7	$\frac{2}{3}$ of 9 or $\frac{1}{2}$ of 14	$\frac{1}{2}$ of 14 (7 > 6)
8	$\frac{3}{5}$ of 10 or $\frac{1}{2}$ of 10	$\frac{3}{5}$ of 10 (6 > 5)
9	$\frac{1}{6}$ of 60 or $\frac{5}{6}$ of 6	$\frac{1}{6}$ of 60 (10 > 5)
10	Draw two different-size rectangles where $\frac{1}{4}$ of one is bigger than $\frac{1}{2}$ of the other.	Drawings vary: a large rectangle's fourth vs a small rectangle's half.
11	Explain why the size of the whole matters when comparing fractions.	The same fraction of a bigger whole is a bigger amount.

LESSON 5.15 · TEACHER PLAN
Week 4 · Lesson 3 of 4
Unit 5: Fraction Equivalence and Comparison

Lesson 5.15: Fraction Comparison Word Problems

Student goal	I can solve word problems by comparing fractions and explain my answer.
Objective	Students solve comparison word problems with fractions.
Standard	4.NF.2 — Compare two fractions; justify conclusions.
Pacing	Unit 5, week 4, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.15
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Two quick comparisons.
Launch	5	Read Let's Explore.
Teach	15	Work both examples. Require a sentence answer.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Answering with just a symbol.

SUPPORT

- Underline the question.

EXTEND

- Make a fraction comparison game with cards.

STUDENT LESSON 5.15 · WITH ANSWERS

Lesson 5.15: Fraction Comparison Word Problems

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.15 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.15, Part A.
- [ ] Complete the rest of Workbook Lesson 5.15 (Parts B–D).

LET'S EXPLORE

Ellie practiced piano $^{3}/_{4}$ hour. Lillie practiced $^{5}/_{6}$ hour. Who practiced longer?

Decide which strategy to use, compare, then answer in a sentence.

WORKED EXAMPLE · COMPARE TIMES

$^{3}/_{4}$ hour vs $^{5}/_{6}$ hour.

1 Common denominator 12: $^{9}/_{12}$ vs $^{10}/_{12}$.

Answer: Lillie practiced longer.

WORKED EXAMPLE · MISSING PIECE

A recipe needs $^{2}/_{3}$ cup of sugar. You have $^{3}/_{4}$ cup. Do you have enough?

1 $^{2}/_{3} = {}^{8}/_{12}$, $^{3}/_{4} = {}^{9}/_{12}$.

2 $9 > 8$.

Answer: Yes, you have enough.

★ KEY IDEA

Answer word problems with a complete sentence and make sure the wholes are the same.

✎ TRY IT

Answers are shown in color.

1. A blue ribbon is $^{5}/_{8}$ yard. A red ribbon is $^{1}/_{2}$ yard. Which is longer?

Answer: the blue ribbon

2. Ellie ate $^{2}/_{6}$ of her sandwich. Lillie ate $^{1}/_{3}$ of hers (same size). Who ate more?

Answer: They ate the same amount.

3. A trail is $^{7}/_{10}$ mile. Another is $^{3}/_{5}$ mile. Which is shorter?

Answer: the $^{3}/_{5}$-mile trail

Workbook Answer Key · Lesson 5.15

#	Question	Answer
1	A blue ribbon is $^{5}/_{8}$ yard. A red ribbon is $^{1}/_{2}$ yard. Which is longer?	the blue ribbon
2	Ellie ate $^{2}/_{6}$ of her sandwich. Lillie ate $^{1}/_{3}$ of hers (same size). Who ate more?	They ate the same amount.
3	A trail is $^{7}/_{10}$ mile. Another is $^{3}/_{5}$ mile. Which is shorter?	the $^{3}/_{5}$-mile trail
4	$^{2}/_{5}$ ◯ $^{1}/_{2}$	<
5	$^{4}/_{6}$ ◯ $^{2}/_{3}$	=
6	Dad walked $^{4}/_{5}$ mile. Mom walked $^{7}/_{10}$ mile. Who walked farther?	Dad walked farther.
7	A plant grew $^{3}/_{8}$ inch in week 1 and $^{1}/_{4}$ inch in week 2. Which week did it grow more?	Week 1.
8	Lillie finished $^{5}/_{12}$ of her project. Ellie finished $^{1}/_{2}$ of hers. Who has more left to do?	Lillie has more left.
9	A jug has $^{2}/_{3}$ gallon of tea. A recipe needs $^{5}/_{6}$ gallon. Is there enough?	No. $^{2}/_{3} = {}^{4}/_{6} < {}^{5}/_{6}$.
10	Write a word problem comparing $^{3}/_{4}$ and $^{2}/_{3}$. Solve it.	Answers vary.

LESSON 5.16 · TEACHER PLAN
Week 4 · Lesson 4 of 4
Unit 5: Fraction Equivalence and Comparison

Lesson 5.16: Explaining Fraction Reasoning

Student goal	I can explain and justify my thinking about equivalent fractions and comparisons.
Objective	Students justify equivalence and comparisons with models, numbers, and words.
Standard	4.NF.1, 4.NF.2 — Justify conclusions about equivalence and comparison.
Pacing	Unit 5, week 4, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.16, Unit 5 Review
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Simplify 10/12.
Launch	5	Read Let's Explore.
Teach	15	Work the example. Model a three-part explanation.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook Lesson 5.16 and the Unit 5 Review.
Exit Ticket	3	Check. Tomorrow: Unit 5 Test.

COMMON MISCONCEPTIONS

- Giving answers without reasons.

SUPPORT

- Sentence frames: 'I know ... because ...'

EXTEND

- Teach a sibling one strategy.

STUDENT LESSON 5.16 · WITH ANSWERS

Lesson 5.16: Explaining Fraction Reasoning

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.16 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.16, Part A.
- [ ] Complete the rest of Workbook Lesson 5.16 (Parts B–D).

LET'S EXPLORE

Sam says $\frac{3}{6} > \frac{1}{2}$ because 3 > 1. How would you convince Sam he is wrong?

Mathematicians explain their thinking with models, numbers, and words. A strong explanation uses at least two of these.

WORKED EXAMPLE · JUSTIFY

Show that $\frac{3}{6} = \frac{1}{2}$.

1 Model: 3 of 6 equal parts covers the same length as 1 of 2 equal parts.

2 Numbers: $\frac{1}{2} = \frac{1 \times 3}{2 \times 3} = \frac{3}{6}$.

3 Words: The pieces are 3 times smaller, but there are 3 times as many.

Answer: They are equal.

★ KEY IDEA

Use a model, numbers, and words to justify your answer.

✎ TRY IT

Answers are shown in color.

1. Explain why $\frac{2}{8} = \frac{1}{4}$.

Answer: 2 eighths cover the same length as 1 fourth; 1×2/4×2.

2. Explain why $\frac{5}{6} < \frac{7}{8}$.

Answer: Each is one piece from 1; an eighth is a smaller gap.

3. Is $\frac{4}{10} > \frac{1}{2}$? Explain.

Answer: No; $\frac{5}{10} = \frac{1}{2}$ and 4 < 5.

Workbook Answer Key · Lesson 5.16

#	Question	Answer
1	Explain why $^{2}/_{8} = {}^{1}/_{4}$.	2 eighths cover the same length as 1 fourth; 1×2/4×2.
2	Explain why $^{5}/_{6} < {}^{7}/_{8}$.	Each is one piece from 1; an eighth is a smaller gap.
3	Is $^{4}/_{10} > {}^{1}/_{2}$? Explain.	No; $^{5}/_{10} = {}^{1}/_{2}$ and 4 < 5.
4	Simplify $^{10}/_{12}$	$^{5}/_{6}$
5	$^{3}/_{9}$ ◯ $^{1}/_{3}$	=
6	Is $^{4}/_{6}$ equivalent to $^{2}/_{3}$?	Yes; $^{2}/_{3} = {}^{4}/_{6}$ (× 2); model shows same length.
7	Which is greater: $^{5}/_{8}$ or $^{3}/_{5}$?	$^{5}/_{8}$ ($^{25}/_{40}$ vs $^{24}/_{40}$)
8	Maria says $^{1}/_{4}$ of a pizza is always less than $^{1}/_{3}$ of a pizza. When is she right? When could she be wrong?	Right if the pizzas are the same size; wrong if the $^{1}/_{4}$ pizza is much larger.
9	Which comparison strategy do you like best? Show it with an example.	Answers vary.

UNIT REVIEW · ANSWER KEY
Study before the unit test

About 35 minutes

Unit 5 Review — Answer Key

Homework or a study session before the test.

Answers

#	Question	Answer
1	Write $^{13}/_{4}$ as a mixed number.	$3\,^{1}/_{4}$
2	$^{3}/_{5}$ of 25	15
3	Name point Q. (figure)	$^{5}/_{6}$
4	$^{3}/_{4} = {}^{?}/_{16}$	12
5	Simplify $^{12}/_{16}$	$^{3}/_{4}$
6	$^{5}/_{9}$ ◯ $^{2}/_{5}$	>
7	$^{3}/_{4}$ ◯ $^{7}/_{10}$	>
8	Order least to greatest: $^{3}/_{5}$, $^{1}/_{2}$, $^{7}/_{10}$	$^{1}/_{2}$, $^{3}/_{5}$, $^{7}/_{10}$
9	Is $^{1}/_{2}$ of a small cake always more than $^{1}/_{4}$ of a big cake? Explain.	No; the wholes are different sizes.

UNIT TEST · ANSWER KEY
At the end of the unit
About 40 minutes

Unit 5 Test — Answer Key

Answers

#	Question	Answer
1	Name the shaded fraction. (figure)	$\frac{3}{10}$
2	Name point R. (figure)	$\frac{2}{5}$
3	Write $\frac{17}{5}$ as a mixed number.	$3\frac{2}{5}$
4	$\frac{3}{4}$ of 32	24
5	$\frac{2}{5} = \frac{?}{15}$	6
6	$\frac{5}{6} = \frac{20}{?}$	24
7	Simplify $\frac{15}{25}$	$\frac{3}{5}$
8	Simplify $\frac{14}{20}$	$\frac{7}{10}$
9	$\frac{4}{9} \bigcirc \frac{7}{9}$	<
10	$\frac{3}{4} \bigcirc \frac{3}{7}$	>
11	$\frac{5}{8} \bigcirc \frac{2}{6}$	>
12	$\frac{2}{3} \bigcirc \frac{7}{10}$	<
13	$\frac{9}{12} \bigcirc \frac{3}{4}$	=
14	$\frac{5}{6} \bigcirc \frac{4}{5}$	>
15	Order from least to greatest and explain your strategy: $\frac{2}{3}$, $\frac{5}{12}$, $\frac{3}{4}$	$\frac{5}{12}$, $\frac{2}{3}$, $\frac{3}{4}$ ($\frac{5}{12}$, $\frac{8}{12}$, $\frac{9}{12}$)
16	Ellie says $\frac{4}{8} > \frac{1}{2}$ because 4 > 1. Use a model and numbers to show whether she is right.	She is wrong: $\frac{4}{8} = \frac{1}{2}$; model shows equal lengths.

Scoring

Score	What it means / next step
17–20 (85%+)	Mastered.
14–16	Proficient; correct missed items.
Below 14	Reteach and retest.

Fraction Operations

Adding and subtracting fractions and mixed numbers, and multiplying fractions by whole numbers.

Now that you can compare fractions, you will compute with them. You will add and subtract fractions and mixed numbers with like denominators, regroup wholes, multiply fractions by whole numbers, and solve real problems with measurements and recipes.

Big ideas

- Decompose fractions in more than one way.
- Add and subtract fractions with like denominators.
- Add and subtract mixed numbers, regrouping when needed.
- Multiply a fraction by a whole number.
- Solve multi-step word problems with fractions and measurements.

Standards

Code	Standard
4.NF.3	Understand a fraction a/b with a > 1 as a sum of fractions 1/b; add and subtract fractions and mixed numbers with like denominators; solve word problems.
4.NF.4	Apply and extend previous understandings of multiplication to multiply a fraction by a whole number.
4.MD.2	Solve word problems involving distances and measurements with fractions.

What students should already know

- Equivalent fractions and comparing (Unit 5).
- Multiplication (Unit 2).

Materials for the unit

- Fraction strips
- Inch rulers (marked to eighths)

- Measuring cups
- Paper ‘pizzas’

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 4	6.1	Decomposing Fractions
Week 1	Lesson 2 of 4	6.2	Adding Fractions with Like Denominators
Week 1	Lesson 3 of 4	6.3	Subtracting Fractions with Like Denominators
Week 1	Lesson 4 of 4	6.4	Fraction Word Problems
Week 1	End of week	Check	Weekly Check 12: Adding and Subtracting Fractions
Week 2	Lesson 1 of 4	6.5	Mixed Numbers and Improper Fractions
Week 2	Lesson 2 of 4	6.6	Adding Mixed Numbers
Week 2	Lesson 3 of 4	6.7	Subtracting Mixed Numbers
Week 2	Lesson 4 of 4	6.8	Subtracting Mixed Numbers with Regrouping
Week 2	End of week	Check	Weekly Check 13: Mixed Numbers
Week 3	Lesson 1 of 4	6.9	Fractions as Multiples of Unit Fractions
Week 3	Lesson 2 of 4	6.10	Multiplying a Fraction by a Whole Number
Week 3	Lesson 3 of 4	6.11	Models for Fraction Multiplication
Week 3	Lesson 4 of 4	6.12	Fraction Multiplication Word Problems
Week 3	End of week	Check	Weekly Check 14: Multiplying Fractions
Week 4	Lesson 1 of 4	6.13	Fractions and Measurement
Week 4	Lesson 2 of 4	6.14	Multi-Step Fraction Problems
Week 4	Lesson 3 of 4	6.15	Estimating with Fractions
Week 4	Lesson 4 of 4	6.16	Fraction Operations Problem Solving
Week 4	Before the test	Review	Unit 6 Review
Week 4	End of unit	Test	Unit 6 Test

TEACHING THIS UNIT AT HOME

Spring break falls between Weeks 3 and 4 of this unit (Mar 15–19). Cooking together is the best fraction practice there is.

UNIT VOCABULARY

decompose	Break a number into parts.
mixed number	A whole number and a fraction.
improper fraction	A fraction ≥ 1.
like denominators	Same denominators.

LESSON 6.1 · TEACHER PLAN
Week 1 · Lesson 1 of 4

Unit 6: Fraction Operations

Lesson 6.1: Decomposing Fractions

Student goal	I can break a fraction into a sum of smaller fractions with the same denominator, like $\frac{3}{8} = \frac{1}{8} + \frac{2}{8}$.
Objective	Students decompose fractions into sums with like denominators.
Standard	4.NF.3b — Decompose a fraction into a sum of fractions with the same denominator in more than one way; justify with a model.
Pacing	Unit 6, week 1, lesson 1 of 4
Time	About 50 minutes
Materials	Fraction strips, workbook Lesson 6.1
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	5 = 2 + ?
Launch	5	Read Let's Explore with 8 paper slices.
Teach	15	Split 5 eighths into groups several ways. Say: "The size of the pieces stays the same; only how we group them changes."
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Changing the denominator when decomposing (e.g., 5/8 = 2/4 + 3/4).

SUPPORT

- Use strips.

EXTEND

- How many ways can you decompose 6/8 into two parts?

STUDENT LESSON 6.1 · WITH ANSWERS

Lesson 6.1: Decomposing Fractions

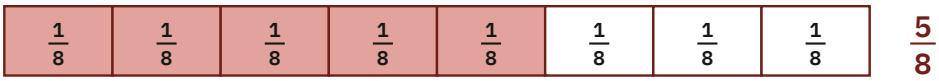
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.1 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.1, Part A.
- [ ] Complete the rest of Workbook Lesson 6.1 (Parts B–F).

LET'S EXPLORE

A pizza is cut into 8 slices. Ellie and Lillie together eat $^{5}/_{8}$ of it. How might they have shared those slices?

$^{5}/_{8}$ is 5 eighths. You can split 5 eighths into parts in many ways, just like you can split 5 into 2 + 3 or 1 + 4.



Ways to decompose five-eighths
$^{5}/_{8} = {}^{1}/_{8} + {}^{1}/_{8} + {}^{1}/_{8} + {}^{1}/_{8} + {}^{1}/_{8}$
$^{5}/_{8} = {}^{2}/_{8} + {}^{3}/_{8}$
$^{5}/_{8} = {}^{1}/_{8} + {}^{4}/_{8}$

WORKED EXAMPLE · DECOMPOSE A MIXED NUMBER

Decompose $1\,{}^{1}/_{4}$ into fourths.

1 1 whole = $^{4}/_{4}$.

2 $1\,{}^{1}/_{4} = {}^{4}/_{4} + {}^{1}/_{4} = {}^{5}/_{4}$.

Answer: $^{1}/_{4} + {}^{1}/_{4} + {}^{1}/_{4} + {}^{1}/_{4} + {}^{1}/_{4}$

★ KEY IDEA

When you decompose a fraction, the denominator stays the same. The numerators add up to the original numerator.

✎ TRY IT

Answers are shown in color.

1. Decompose $\frac{4}{6}$ two ways.

Answer: e.g. $\frac{1}{6} + \frac{3}{6}; \frac{2}{6} + \frac{2}{6}$

2. $\frac{7}{10} = \frac{3}{10} + \frac{?}{10}$

Answer: 4

3. Write $\frac{3}{5}$ as a sum of unit fractions.

Answer: $\frac{1}{5} + \frac{1}{5} + \frac{1}{5}$

Workbook Answer Key · Lesson 6.1

#	Question	Answer
1	Decompose $\frac{4}{6}$ two ways.	e.g. $\frac{1}{6} + \frac{3}{6}; \frac{2}{6} + \frac{2}{6}$
2	$\frac{7}{10} = \frac{3}{10} + \frac{?}{10}$	4
3	Write $\frac{3}{5}$ as a sum of unit fractions.	$\frac{1}{5} + \frac{1}{5} + \frac{1}{5}$
4	5 = 2 + ___	3
5	Name 3 of 8 equal parts.	$\frac{3}{8}$
6	$\frac{5}{6} = \frac{2}{6} + \frac{?}{6}$	3
7	$\frac{7}{8} = \frac{3}{8} + \frac{?}{8}$	4
8	$\frac{9}{10} = \frac{4}{10} + \frac{?}{10}$	5
9	$\frac{4}{5} = \frac{1}{5} + \frac{?}{5}$	3
10	$\frac{11}{12} = \frac{6}{12} + \frac{?}{12}$	5
11	$\frac{11}{12} = \frac{5}{12} + \frac{?}{12}$	6
12	$\frac{4}{5}$	Answers vary, numerators must add to 4
13	$\frac{6}{8}$	Answers vary, numerators must add to 6
14	$1\frac{2}{3}$ into thirds	$\frac{3}{3} + \frac{2}{3} (= \frac{5}{3})$
15	Show $\frac{7}{12}$ as a sum of two fractions.	Answers vary, e.g. $\frac{3}{12} + \frac{4}{12}$

LESSON 6.2 · TEACHER PLAN
Week 1 · Lesson 2 of 4
Unit 6: Fraction Operations

Lesson 6.2: Adding Fractions with Like Denominators

Student goal	I can add fractions that have the same denominator.
Objective	Students add fractions with like denominators.
Standard	4.NF.3a, 4.NF.3d — Understand addition of fractions as joining parts referring to the same whole.
Pacing	Unit 6, week 1, lesson 2 of 4
Time	About 50 minutes
Materials	Fraction strips, workbook Lesson 6.2
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Decompose 3/5.
Launch	5	Read Let's Explore.
Teach	15	Join strips: 2 fifths and 1 fifth. Say: "Add the count, keep the size." Work both examples; read Watch Out.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Adding denominators.

SUPPORT

- Say it in words: "3 eighths plus 4 eighths."

EXTEND

- Add three fractions: 1/8 + 3/8 + 5/8.

STUDENT LESSON 6.2 · WITH ANSWERS

Lesson 6.2: Adding Fractions with Like Denominators

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.2 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.2, Part A.
- [ ] Complete the rest of Workbook Lesson 6.2 (Parts B–E).

LET'S EXPLORE

Ellie walked $\frac{2}{5}$ mile to the park and $\frac{1}{5}$ mile to the store. How far did she walk?

Fifths plus fifths makes fifths. Add the numerators: 2 fifths + 1 fifth = 3 fifths. The denominator tells the size of the pieces, so it does not change.



WORKED EXAMPLE · ADD

$\frac{2}{5} + \frac{1}{5}$

1 Add the numerators: 2 + 1 = 3.

2 Keep the denominator: fifths.

Answer: $\frac{3}{5}$

WORKED EXAMPLE · SUM GREATER THAN 1

$\frac{5}{6} + \frac{4}{6}$

1 5 + 4 = 9 sixths.

2 $\frac{9}{6} = \frac{6}{6} + \frac{3}{6} = 1\frac{3}{6}$.

Answer: $\frac{9}{6} = 1\frac{3}{6}$

! WATCH OUT

Do not add the denominators! $\frac{1}{4} + \frac{1}{4}$ is $\frac{2}{4}$, not $\frac{2}{8}$. Two quarters of a dollar are 2 fourths of a dollar, not 2 eighths.

★ KEY IDEA

Add the numerators. Keep the denominator.

✎ TRY IT

Answers are shown in color.

1. $\frac{3}{8} + \frac{4}{8}$

Answer: $\frac{7}{8}$

2. $\frac{1}{3} + \frac{2}{3}$

Answer: 1

3. $\frac{7}{10} + \frac{5}{10}$

Answer: $1\frac{2}{10}$

Workbook Answer Key · Lesson 6.2

#	Question	Answer
1	$\frac{3}{8} + \frac{4}{8}$	$\frac{7}{8}$
2	$\frac{1}{3} + \frac{2}{3}$	1
3	$\frac{7}{10} + \frac{5}{10}$	$1\frac{2}{10}$
4	$\frac{3}{5} = \frac{1}{5} + \frac{?}{5}$	2
5	4 + 3	7
6	$\frac{1}{4} + \frac{2}{4}$	$\frac{3}{4}$
7	$\frac{2}{6} + \frac{3}{6}$	$\frac{5}{6}$
8	$\frac{4}{8} + \frac{3}{8}$	$\frac{7}{8}$
9	$\frac{5}{9} + \frac{2}{9}$	$\frac{7}{9}$
10	$\frac{3}{5} + \frac{4}{5}$	$1\frac{2}{5}$
11	$\frac{7}{12} + \frac{8}{12}$	$1\frac{3}{12}$
12	$\frac{6}{10} + \frac{4}{10}$	1
13	$\frac{2}{3} + \frac{2}{3}$	$1\frac{1}{3}$
14	Lillie poured $\frac{3}{8}$ cup of milk and then $\frac{4}{8}$ cup more. How much milk did she pour?	$\frac{7}{8}$ cup
15	A board is painted $\frac{2}{6}$ red and $\frac{3}{6}$ blue. What fraction is painted?	$\frac{5}{6}$

#	Question	Answer
16	Explain why $^{2}/_{7} + {}^{3}/_{7}$ is not $^{5}/_{14}$.	The pieces stay sevenths; only the count changes: 5 sevenths.

LESSON 6.3 · TEACHER PLAN
Week 1 · Lesson 3 of 4
Unit 6: Fraction Operations

Lesson 6.3: Subtracting Fractions with Like Denominators

Student goal	I can subtract fractions that have the same denominator.
Objective	Students subtract fractions with like denominators.
Standard	4.NF.3a, 4.NF.3d — Understand subtraction of fractions as separating parts of the same whole.
Pacing	Unit 6, week 1, lesson 3 of 4
Time	About 50 minutes
Materials	Fraction strips, workbook Lesson 6.3
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	2/7 + 3/7.
Launch	5	Read Let's Explore.
Teach	15	Remove 3 eighth-strips from 7. Work both examples. Say: To take from 1, rename 1.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Subtracting denominators to get 0.

SUPPORT

- Strips.

EXTEND

- What fraction plus 3/10 makes 1?

STUDENT LESSON 6.3 · WITH ANSWERS

Lesson 6.3: Subtracting Fractions with Like Denominators

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.3 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.3, Part A.
- [ ] Complete the rest of Workbook Lesson 6.3 (Parts B–E).

LET'S EXPLORE

There was $\frac{7}{8}$ of a pie left. The family ate $\frac{3}{8}$ of the pie. How much pie is left now?

Subtracting works the same way: subtract the numerators and keep the denominator.



$\frac{7}{8}$

WORKED EXAMPLE · SUBTRACT

$\frac{7}{8} - \frac{3}{8}$

1 $7 - 3 = 4$ eighths.

Answer: $\frac{4}{8} = \frac{1}{2}$

WORKED EXAMPLE · SUBTRACT FROM 1

$1 - \frac{2}{5}$

1 $1 = \frac{5}{5}$.

2 $5 - 2 = 3$ fifths.

Answer: $\frac{3}{5}$

★ KEY IDEA

Subtract the numerators. Keep the denominator. To subtract from 1, rename 1 as a fraction with the same denominator.

✎ TRY IT

Answers are shown in color.

1. $\frac{5}{6} - \frac{2}{6}$

Answer: $\frac{3}{6}$

2. $\frac{9}{10} - \frac{4}{10}$

Answer: $\frac{5}{10}$

3. $1 - \frac{3}{4}$

Answer: $\frac{1}{4}$

Workbook Answer Key · Lesson 6.3

#	Question	Answer
1	$\frac{5}{6} - \frac{2}{6}$	$\frac{3}{6}$
2	$\frac{9}{10} - \frac{4}{10}$	$\frac{5}{10}$
3	$1 - \frac{3}{4}$	$\frac{1}{4}$
4	$\frac{2}{7} + \frac{3}{7}$	$\frac{5}{7}$
5	8 – 5	3
6	$\frac{3}{4} - \frac{1}{4}$	$\frac{2}{4}$
7	$\frac{5}{6} - \frac{3}{6}$	$\frac{2}{6}$
8	$\frac{7}{8} - \frac{2}{8}$	$\frac{5}{8}$
9	$\frac{4}{9} - \frac{4}{9}$	0
10	$\frac{9}{10} - \frac{5}{10}$	$\frac{4}{10}$
11	$\frac{11}{12} - \frac{6}{12}$	$\frac{5}{12}$
12	$1 - \frac{1}{3}$	$\frac{2}{3}$
13	$1 - \frac{5}{8}$	$\frac{3}{8}$
14	A ribbon is $\frac{9}{10}$ meter long. Ellie cuts off $\frac{4}{10}$ meter. How much is left?	$\frac{5}{10}$ meter
15	Lillie has 1 whole sandwich. She eats $\frac{3}{8}$ of it. How much is left?	$\frac{5}{8}$ of the sandwich
16	$\frac{11}{12} - \frac{4}{12}$	$\frac{7}{12}$

LESSON 6.4 · TEACHER PLAN
Week 1 · Lesson 4 of 4
Unit 6: Fraction Operations

Lesson 6.4: Fraction Word Problems

Student goal	I can solve word problems by adding and subtracting fractions with like denominators.
Objective	Students solve fraction word problems.
Standard	4.NF.3d — Solve word problems involving addition and subtraction of fractions referring to the same whole and having like denominators.
Pacing	Unit 6, week 1, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 6.4
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Two computations.
Launch	5	Read Let's Explore.
Teach	15	Work both examples with bar models.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Choosing the wrong operation.

SUPPORT

- Draw the model first.

EXTEND

- Write a two-step fraction story.

STUDENT LESSON 6.4 · WITH ANSWERS

Lesson 6.4: Fraction Word Problems

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.4 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.4, Part A.
- [ ] Complete the rest of Workbook Lesson 6.4 (Parts B–D).

LET'S EXPLORE

A recipe uses $^{3}/_{4}$ cup of flour for the cookies and $^{2}/_{4}$ cup for the frosting. How much flour is that? Is it more than 1 cup?

Use a bar model or number line, decide whether to add or subtract, then answer in a sentence.

WORKED EXAMPLE · ADD

Flour for cookies and frosting.

1 $^{3}/_{4} + ^{2}/_{4} = ^{5}/_{4} = 1\,^{1}/_{4}$.

Answer: $1\,^{1}/_{4}$ cups, more than 1 cup

WORKED EXAMPLE · COMPARE BY SUBTRACTING

Lillie ran $^{7}/_{10}$ mile. Ellie ran $^{4}/_{10}$ mile. How much farther did Lillie run?

1 $^{7}/_{10} - ^{4}/_{10} = ^{3}/_{10}$.

Answer: $^{3}/_{10}$ mile farther

★ KEY IDEA

Joining → add. Taking away or "how much more" → subtract. The pieces must be the same size.

✎ TRY IT

Answers are shown in color.

1. $^{2}/_{6}$ of a garden is beans and $^{3}/_{6}$ is corn. What fraction is beans and corn?

Answer: $^{5}/_{6}$

2. A jug had $^{7}/_{8}$ gallon of juice. $^{5}/_{8}$ gallon was poured out. How much is left?

Answer: $^{2}/_{8}$ gallon

3. Dad painted $^{3}/_{5}$ of a fence. How much more to finish?

Answer: $\frac{2}{5}$

Workbook Answer Key · Lesson 6.4

#	Question	Answer
1	$\frac{2}{6}$ of a garden is beans and $\frac{3}{6}$ is corn. What fraction is beans and corn?	$\frac{5}{6}$
2	A jug had $\frac{7}{8}$ gallon of juice. $\frac{5}{8}$ gallon was poured out. How much is left?	$\frac{2}{8}$ gallon
3	Dad painted $\frac{3}{5}$ of a fence. How much more to finish?	$\frac{2}{5}$
4	$\frac{4}{6} + \frac{3}{6}$	$1\frac{1}{6}$
5	$\frac{9}{10} - \frac{2}{10}$	$\frac{7}{10}$
6	Ellie read $\frac{2}{8}$ of her book on Monday and $\frac{3}{8}$ on Tuesday. What fraction did she read?	$\frac{5}{8}$
7	A cake was $\frac{5}{6}$ left. Lillie ate $\frac{1}{6}$. What fraction is left?	$\frac{4}{6}$
8	Mom bought $\frac{3}{4}$ pound of grapes and $\frac{3}{4}$ pound of cherries. How many pounds of fruit?	$1\frac{2}{4}$ pounds
9	A path is $\frac{9}{10}$ mile. You walked $\frac{6}{10}$ mile. How far is left?	$\frac{3}{10}$ mile
10	Write and solve a story for $\frac{5}{12} + \frac{4}{12}$.	$\frac{9}{12}$

WEEKLY CHECK · ANSWER KEY
After Lesson 6.4

About 15 minutes

Weekly Check 12: Adding and Subtracting Fractions — Answer Key

Answers

#	Question	Answer
1	$\frac{3}{9} + \frac{4}{9}$	$\frac{7}{9}$
2	$\frac{7}{10} - \frac{3}{10}$	$\frac{4}{10}$
3	$1 - \frac{3}{8}$	$\frac{5}{8}$
4	$\frac{5}{6} + \frac{4}{6}$	$1\frac{3}{6}$
5	Decompose $\frac{4}{5}$ as a sum of two fractions.	Answers vary
6	Mom had $\frac{7}{8}$ yard of fabric and used $\frac{4}{8}$. How much is left?	$\frac{3}{8}$ yard

LESSON 6.5 · TEACHER PLAN
Week 2 · Lesson 1 of 4
Unit 6: Fraction Operations

Lesson 6.5: Mixed Numbers and Improper Fractions

Student goal	I can rename mixed numbers as fractions and fractions as mixed numbers.
Objective	Students convert between mixed numbers and fractions.
Standard	4.NF.3b — Decompose and compose fractions greater than 1.
Pacing	Unit 6, week 2, lesson 1 of 4
Time	About 50 minutes
Materials	Fraction strips, workbook Lesson 6.5
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	19 ÷ 4.
Launch	5	Read Let's Explore.
Teach	15	Build 2 3/4 with strips and count fourths. Work both examples.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Multiplying the numerator instead of the whole.

SUPPORT

- Draw wholes.

EXTEND

- Write 4 7/8 as eighths in your head.

STUDENT LESSON 6.5 · WITH ANSWERS

Lesson 6.5: Mixed Numbers and Improper Fractions

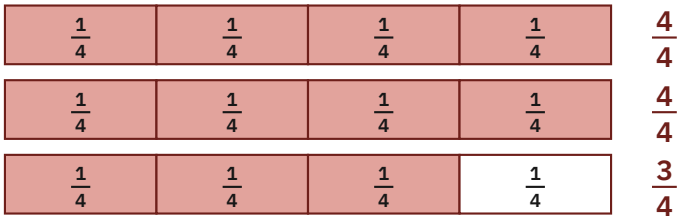
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.5 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.5, Part A.
- [ ] Complete the rest of Workbook Lesson 6.5 (Parts B–E).

LET'S EXPLORE

A carpenter needs $2\,{}^{3}\!/_{4}$ feet of board. How many fourths of a foot is that?

Each whole is ${}^{4}\!/_{4}$. Two wholes are ${}^{8}\!/_{4}$. Add the ${}^{3}\!/_{4}$: $2\,{}^{3}\!/_{4} = {}^{11}\!/_{4}$.



WORKED EXAMPLE · MIXED NUMBER TO FRACTION

Write $3\,{}^{2}\!/_{5}$ as a fraction.

1 3 wholes = 3 × 5 = 15 fifths.

2 15 + 2 = 17 fifths.

Answer: ${}^{17}\!/_{5}$

WORKED EXAMPLE · FRACTION TO MIXED NUMBER

Write ${}^{23}\!/_{6}$ as a mixed number.

1 How many 6s are in 23? 23 ÷ 6 = 3 R5.

2 3 wholes and 5 sixths.

Answer: $3\,{}^{5}\!/_{6}$

★ KEY IDEA

Mixed → fraction: multiply the whole by the denominator, add the numerator. Fraction → mixed: divide the numerator by the denominator.

✎ TRY IT

Answers are shown in color.

1. $1\frac{3}{8} = ?$

Answer: $\frac{11}{8}$

2. $\frac{14}{3} = ?$

Answer: $4\frac{2}{3}$

3. $4\frac{1}{2} = ?$

Answer: $\frac{9}{2}$

Workbook Answer Key · Lesson 6.5

#	Question	Answer
1	$1\frac{3}{8} = ?$	$\frac{11}{8}$
2	$\frac{14}{3} = ?$	$4\frac{2}{3}$
3	$4\frac{1}{2} = ?$	$\frac{9}{2}$
4	$19 \div 4$	4 R3
5	$3 \times 6 + 2$	20
6	$1\frac{2}{3}$	$\frac{5}{3}$
7	$2\frac{1}{4}$	$\frac{9}{4}$
8	$3\frac{3}{5}$	$\frac{18}{5}$
9	$2\frac{5}{8}$	$\frac{21}{8}$
10	$4\frac{1}{6}$	$\frac{25}{6}$
11	$5\frac{2}{3}$	$\frac{17}{3}$
12	$\frac{7}{4}$	$1\frac{3}{4}$
13	$\frac{11}{3}$	$3\frac{2}{3}$
14	$\frac{13}{5}$	$2\frac{3}{5}$
15	$\frac{17}{6}$	$2\frac{5}{6}$
16	$\frac{25}{8}$	$3\frac{1}{8}$
17	$\frac{31}{10}$	$3\frac{1}{10}$
18	Which is greater, $2\frac{1}{3}$ or $\frac{8}{3}$? Explain.	$\frac{8}{3} = 2\frac{2}{3} > 2\frac{1}{3}$

LESSON 6.6 · TEACHER PLAN
Week 2 · Lesson 2 of 4
Unit 6: Fraction Operations

Lesson 6.6: Adding Mixed Numbers

Student goal	I can add mixed numbers with like denominators.
Objective	Students add mixed numbers with like denominators.
Standard	4.NF.3c — Add and subtract mixed numbers with like denominators.
Pacing	Unit 6, week 2, lesson 2 of 4
Time	About 50 minutes
Materials	Fraction strips, workbook Lesson 6.6
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	9/8 as a mixed number.
Launch	5	Read Let's Explore.
Teach	15	Work both examples. Say: Wholes with wholes, fractions with fractions, then regroup.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Leaving answers like 3 7/5.

SUPPORT

- Strips.

EXTEND

- Add three mixed numbers.

STUDENT LESSON 6.6 · WITH ANSWERS

Lesson 6.6: Adding Mixed Numbers

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.6 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.6, Part A.
- [ ] Complete the rest of Workbook Lesson 6.6 (Parts B–E).

LET'S EXPLORE

Ellie practiced piano $1\frac{1}{4}$ hours on Saturday and $2\frac{2}{4}$ hours on Sunday. How long did she practice?

Add the whole numbers. Add the fractions. If the fractions add to 1 or more, regroup.

WORKED EXAMPLE · ADD

$1\frac{1}{4} + 2\frac{2}{4}$

1 Wholes: $1 + 2 = 3$.

2 Fractions: $\frac{1}{4} + \frac{2}{4} = \frac{3}{4}$.

Answer: $3\frac{3}{4}$ hours

WORKED EXAMPLE · REGROUP

$2\frac{3}{5} + 1\frac{4}{5}$

1 Wholes: $2 + 1 = 3$.

2 Fractions: $\frac{3}{5} + \frac{4}{5} = \frac{7}{5} = 1\frac{2}{5}$.

3 $3 + 1\frac{2}{5} = 4\frac{2}{5}$.

Answer: $4\frac{2}{5}$

★ KEY IDEA

Add wholes, add fractions, then regroup fractions that make a whole.

✎ TRY IT

Answers are shown in color.

1. $3\frac{1}{6} + 1\frac{4}{6}$

Answer: $4\frac{5}{6}$

2. $2\frac{5}{8} + 1\frac{6}{8}$

Answer: $4\frac{3}{8}$

3. $4\frac{1}{2} + 3\frac{1}{2}$

Answer: 8

Workbook Answer Key · Lesson 6.6

#	Question	Answer
1	$3\frac{1}{6} + 1\frac{4}{6}$	$4\frac{5}{6}$
2	$2\frac{5}{8} + 1\frac{6}{8}$	$4\frac{3}{8}$
3	$4\frac{1}{2} + 3\frac{1}{2}$	8
4	$\frac{9}{8}$ as a mixed number	$1\frac{1}{8}$
5	$\frac{3}{5} + \frac{2}{5}$	1
6	$1\frac{1}{3} + 2\frac{1}{3}$	$3\frac{2}{3}$
7	$2\frac{2}{6} + 1\frac{3}{6}$	$3\frac{5}{6}$
8	$3\frac{3}{8} + 2\frac{4}{8}$	$5\frac{7}{8}$
9	$1\frac{4}{5} + 1\frac{3}{5}$	$3\frac{2}{5}$
10	$4\frac{7}{10} + 2\frac{5}{10}$	$7\frac{2}{10}$
11	$5\frac{2}{4} + 3\frac{2}{4}$	9
12	A bean plant was $4\frac{3}{8}$ inches tall. It grew $1\frac{6}{8}$ inches. How tall is it now?	$6\frac{1}{8}$ inches
13	$3\frac{5}{6} + 2\frac{4}{6}$	$6\frac{3}{6}$ $(= 6\frac{1}{2})$

LESSON 6.7 · TEACHER PLAN
Week 2 · Lesson 3 of 4
Unit 6: Fraction Operations

Lesson 6.7: Subtracting Mixed Numbers

Student goal	I can subtract mixed numbers with like denominators when no regrouping is needed.
Objective	Students subtract mixed numbers without regrouping.
Standard	4.NF.3c — Add and subtract mixed numbers with like denominators.
Pacing	Unit 6, week 2, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 6.7
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	7/8 − 4/8.
Launch	5	Read Let's Explore.
Teach	15	Work both examples.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Subtracting fractions in the wrong order.

SUPPORT

- Strips.

EXTEND

- Check by adding.

STUDENT LESSON 6.7 · WITH ANSWERS

Lesson 6.7: Subtracting Mixed Numbers

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.7 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.7, Part A.
- [ ] Complete the rest of Workbook Lesson 6.7 (Parts B–E).

LET'S EXPLORE

A rope is $5\frac{7}{8}$ feet long. You cut off $2\frac{3}{8}$ feet. How much is left?

Subtract the fractions. Subtract the wholes.

WORKED EXAMPLE · SUBTRACT

$5\frac{7}{8} - 2\frac{3}{8}$

1 Fractions: $\frac{7}{8} - \frac{3}{8} = \frac{4}{8}$.

2 Wholes: $5 - 2 = 3$.

Answer: $3\frac{4}{8} = 3\frac{1}{2}$ feet

WORKED EXAMPLE · USE FRACTIONS

$\frac{15}{4} - \frac{6}{4}$

1 $\frac{15}{4} - \frac{6}{4} = \frac{9}{4}$.

2 $\frac{9}{4} = 2\frac{1}{4}$.

Answer: $2\frac{1}{4}$

★ KEY IDEA

Subtract fractions, then wholes. You can also change both to fractions first.

✎ TRY IT

Answers are shown in color.

1. $4\frac{5}{6} - 1\frac{2}{6}$

Answer: $3\frac{3}{6}$

2. $7\frac{9}{10} - 3\frac{4}{10}$

Answer: $4\frac{5}{10}$

3. $3\frac{2}{3} - 2$

Answer: $1\frac{2}{3}$

Workbook Answer Key · Lesson 6.7

#	Question	Answer
1	$4\frac{5}{6} - 1\frac{2}{6}$	$3\frac{3}{6}$
2	$7\frac{9}{10} - 3\frac{4}{10}$	$4\frac{5}{10}$
3	$3\frac{2}{3} - 2$	$1\frac{2}{3}$
4	$\frac{7}{8} - \frac{4}{8}$	$\frac{3}{8}$
5	$9 - 4$	5
6	$3\frac{3}{4} - 1\frac{1}{4}$	$2\frac{2}{4}$
7	$5\frac{4}{5} - 2\frac{1}{5}$	$3\frac{3}{5}$
8	$6\frac{5}{6} - 3\frac{2}{6}$	$3\frac{3}{6}$
9	$8\frac{7}{8} - 4\frac{3}{8}$	$4\frac{4}{8}$
10	$4\frac{9}{10} - 2\frac{4}{10}$	$2\frac{5}{10}$
11	$9\frac{2}{3} - 5\frac{1}{3}$	$4\frac{1}{3}$
12	Lillie's backpack weighs $8\frac{5}{6}$ pounds. She takes out a $2\frac{3}{6}$-pound book. How much does it weigh now?	$6\frac{2}{6}$ pounds
13	$6\frac{7}{12} - 2\frac{5}{12}$	$4\frac{2}{12}$

LESSON 6.8 · TEACHER PLAN
Week 2 · Lesson 4 of 4
Unit 6: Fraction Operations

Lesson 6.8: Subtracting Mixed Numbers with Regrouping

Student goal	I can regroup a whole to subtract mixed numbers, like $3\frac{1}{4} - 1\frac{3}{4}$.
Objective	Students subtract mixed numbers with regrouping.
Standard	4.NF.3c — Add and subtract mixed numbers with like denominators by replacing each with an equivalent fraction or using properties.
Pacing	Unit 6, week 2, lesson 4 of 4
Time	About 50 minutes
Materials	Fraction strips, workbook Lesson 6.8
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Regroup 2 1/3.
Launch	5	Read Let's Explore with paper pizzas.
Teach	15	Cut one whole into fourths to show regrouping. Work both examples. Alternative: change both to fractions (13/4 – 7/4).
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Subtracting the smaller fraction from the larger (3/4 – 1/4) instead of regrouping.

SUPPORT

- Change to improper fractions instead.

EXTEND

- Explain two ways to solve 5 1/6 – 2 5/6.

STUDENT LESSON 6.8 · WITH ANSWERS

Lesson 6.8: Subtracting Mixed Numbers with Regrouping

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.8 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.8, Part A.
- [ ] Complete the rest of Workbook Lesson 6.8 (Parts B–D).

LET'S EXPLORE

There are $3\frac{1}{4}$ pizzas. The family eats $1\frac{3}{4}$ pizzas. How much is left? Can you take $\frac{3}{4}$ from $\frac{1}{4}$?

You can't take $\frac{3}{4}$ from $\frac{1}{4}$. Regroup one whole as $\frac{4}{4}$: $3\frac{1}{4} = 2\frac{5}{4}$.

WORKED EXAMPLE · REGROUP

$3\frac{1}{4} - 1\frac{3}{4}$

1 Regroup: $3\frac{1}{4} = 2\frac{5}{4}$.

2 Fractions: $\frac{5}{4} - \frac{3}{4} = \frac{2}{4}$.

3 Wholes: $2 - 1 = 1$.

Answer: $1\frac{2}{4} = 1\frac{1}{2}$ pizzas

WORKED EXAMPLE · SUBTRACT FROM A WHOLE NUMBER

$4 - 1\frac{2}{3}$

1 Regroup: $4 = 3\frac{3}{3}$.

2 $\frac{3}{3} - \frac{2}{3} = \frac{1}{3}$; $3 - 1 = 2$.

Answer: $2\frac{1}{3}$

★ KEY IDEA

When the first fraction is too small, trade 1 whole for $\frac{d}{d}$ and add it to the fraction.

✎ TRY IT

Answers are shown in color.

1. $4\frac{1}{5} - 2\frac{3}{5}$
Answer: $1\frac{3}{5}$
2. $5\frac{2}{6} - 1\frac{5}{6}$
Answer: $3\frac{3}{6}$
3. $3 - 1\frac{3}{8}$
Answer: $1\frac{5}{8}$

Workbook Answer Key · Lesson 6.8

#	Question	Answer
1	$4\frac{1}{5} - 2\frac{3}{5}$	$1\frac{3}{5}$
2	$5\frac{2}{6} - 1\frac{5}{6}$	$3\frac{3}{6}$
3	$3 - 1\frac{3}{8}$	$1\frac{5}{8}$
4	$2\frac{1}{3} = 1\frac{?}{3}$	4
5	$1 = \frac{?}{6}$	6
6	$4\frac{1}{3} - 1\frac{2}{3}$	$2\frac{2}{3}$
7	$5\frac{1}{4} - 2\frac{3}{4}$	$2\frac{2}{4}$
8	$6\frac{2}{5} - 3\frac{4}{5}$	$2\frac{3}{5}$
9	$3\frac{3}{8} - 1\frac{7}{8}$	$1\frac{4}{8}$
10	$7\frac{1}{6} - 4\frac{5}{6}$	$2\frac{2}{6}$
11	$8\frac{3}{10} - 2\frac{9}{10}$	$5\frac{4}{10}$
12	$5 - 2\frac{1}{4}$	$2\frac{3}{4}$
13	$2 - \frac{5}{8}$	$1\frac{3}{8}$
14	$6\frac{2}{10} - 3\frac{7}{10}$	$2\frac{5}{10}$

WEEKLY CHECK · ANSWER KEY
After Lesson 6.8

About 15 minutes

Weekly Check 13: Mixed Numbers — Answer Key

Answers

#	Question	Answer
1	Write $3\frac{2}{5}$ as a fraction.	$\frac{17}{5}$
2	Write $\frac{22}{4}$ as a mixed number.	$5\frac{2}{4}$
3	$2\frac{3}{8} + 1\frac{7}{8}$	$4\frac{2}{8}$
4	$5\frac{4}{6} - 2\frac{1}{6}$	$3\frac{3}{6}$
5	$4\frac{1}{3} - 1\frac{2}{3}$	$2\frac{2}{3}$
6	$3 - 1\frac{3}{4}$	$1\frac{1}{4}$

LESSON 6.9 · TEACHER PLAN
Week 3 · Lesson 1 of 4
Unit 6: Fraction Operations

Lesson 6.9: Fractions as Multiples of Unit Fractions

Student goal	I can write a fraction as a whole number times a unit fraction, like $^{5}/_{4} = 5 \times {}^{1}/_{4}$.
Objective	Students express fractions as multiples of unit fractions.
Standard	4.NF.4a — Understand a fraction a/b as a multiple of 1/b.
Pacing	Unit 6, week 3, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 6.9, measuring cups
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Repeated addition.
Launch	5	Read Let's Explore with a 1/3 cup scoop.
Teach	15	Jump by thirds on a number line. Say: "Fractions are copies of unit fractions."
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Multiplying the denominator by the whole number.

SUPPORT

- Number line jumps.

EXTEND

- Write 12/4 as a multiple of 1/4 and as a whole number.

STUDENT LESSON 6.9 · WITH ANSWERS

Lesson 6.9: Fractions as Multiples of Unit Fractions

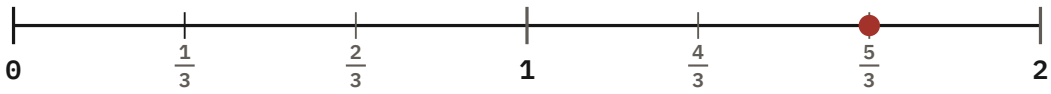
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.9 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.9, Part A.
- [ ] Complete the rest of Workbook Lesson 6.9 (Parts B–E).

LET'S EXPLORE

How many $^{1}/_{3}$-cup scoops make $^{5}/_{3}$ cups?

$^{5}/_{3}$ is 5 thirds: $^{1}/_{3} + {}^{1}/_{3} + {}^{1}/_{3} + {}^{1}/_{3} + {}^{1}/_{3}$. Repeated addition is multiplication, so $^{5}/_{3} = 5 \times {}^{1}/_{3}$.



WORKED EXAMPLE · WRITE AS A MULTIPLE

Write $^{7}/_{8}$ as a multiple of a unit fraction.

1 7 eighths = 7 copies of 1 eighth.

Answer: $^{7}/_{8} = 7 \times {}^{1}/_{8}$

★ KEY IDEA

$^{a}/_{b} = a \times {}^{1}/_{b}$

✎ TRY IT

Answers are shown in color.

1. $^{4}/_{5} =$ ___ $\times {}^{1}/_{5}$

Answer: 4

2. $6 \times {}^{1}/_{10} = ?$

Answer: $^{6}/_{10}$

3. $^{9}/_{4} = 9 \times ?$

Answer: $^{1}/_{4}$

Workbook Answer Key · Lesson 6.9

#	Question	Answer
1	$\frac{4}{5} = ___ \times \frac{1}{5}$	4
2	$6 \times \frac{1}{10} = ?$	$\frac{6}{10}$
3	$\frac{9}{4} = 9 \times ?$	$\frac{1}{4}$
4	$3 + 3 + 3 + 3 = ___ \times 3$	4
5	$\frac{1}{6} + \frac{1}{6} + \frac{1}{6}$	$\frac{3}{6}$
6	$\frac{3}{4}$	$3 \times \frac{1}{4}$
7	$\frac{5}{6}$	$5 \times \frac{1}{6}$
8	$\frac{2}{3}$	$2 \times \frac{1}{3}$
9	$\frac{7}{10}$	$7 \times \frac{1}{10}$
10	$\frac{9}{5}$	$9 \times \frac{1}{5}$
11	$\frac{11}{8}$	$11 \times \frac{1}{8}$
12	$4 \times \frac{1}{7}$	$\frac{4}{7}$
13	$3 \times \frac{1}{12}$	$\frac{3}{12}$
14	$6 \times \frac{1}{5}$	$\frac{6}{5}$
15	$10 \times \frac{1}{3}$	$\frac{10}{3}$
16	Draw a number line to show $\frac{4}{3} = 4 \times \frac{1}{3}$.	Four jumps of $\frac{1}{3}$ from 0 to $\frac{4}{3}$

LESSON 6.10 · TEACHER PLAN
Week 3 · Lesson 2 of 4
Unit 6: Fraction Operations

Lesson 6.10: Multiplying a Fraction by a Whole Number

Student goal	I can multiply a whole number by a fraction, like $3 \times \frac{2}{5}$.
Objective	Students multiply a fraction by a whole number.
Standard	4.NF.4b — Understand a multiple of a/b as a multiple of 1/b; multiply a fraction by a whole number.
Pacing	Unit 6, week 3, lesson 2 of 4
Time	About 50 minutes
Materials	Fraction strips, workbook Lesson 6.10
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	7/8 = 7 × ?
Launch	5	Read Let's Explore.
Teach	15	Lay out 4 groups of 2/3 with strips and count thirds. Say: "Multiply the count, keep the size."
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Multiplying the denominator too.

SUPPORT

- Repeated addition first.

EXTEND

- What is 12 × 3/4? (9)

STUDENT LESSON 6.10 · WITH ANSWERS

Lesson 6.10: Multiplying a Fraction by a Whole Number

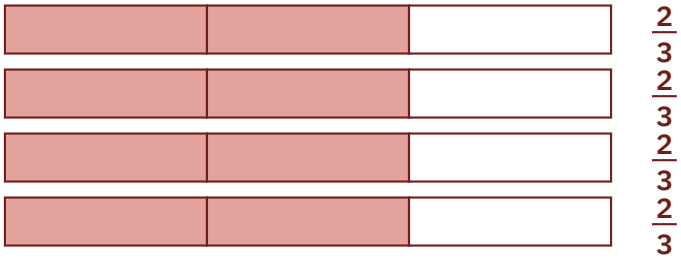
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.10 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.10, Part A.
- [ ] Complete the rest of Workbook Lesson 6.10 (Parts B–D).

LET'S EXPLORE

Each batch of muffins uses $2/3$ cup of oil. How much oil is needed for 4 batches?

$4 \times 2/3$ means 4 groups of $2/3$. That is 4 × 2 thirds = 8 thirds.



WORKED EXAMPLE · MULTIPLY

$4 \times 2/3$

1 $4 \times 2/3 = 4 \times 2 \times 1/3 = 8 \times 1/3$.

2 $8/3 = 2\ 2/3$.

Answer: $2\ 2/3$ cups

! WATCH OUT

Multiply the whole number by the numerator only. The size of the pieces (the denominator) does not change.

★ KEY IDEA

$n \times a/b = n \times a/b$

✎ TRY IT

Answers are shown in color.

1. $3 \times {}^{2}/_{5}$

Answer: ${}^{6}/_{5} = 1\,{}^{1}/_{5}$

2. $5 \times {}^{3}/_{4}$

Answer: ${}^{15}/_{4} = 3\,{}^{3}/_{4}$

3. $6 \times {}^{1}/_{2}$

Answer: 3

Workbook Answer Key · Lesson 6.10

#	Question	Answer
1	$3 \times {}^{2}/_{5}$	${}^{6}/_{5} = 1\,{}^{1}/_{5}$
2	$5 \times {}^{3}/_{4}$	${}^{15}/_{4} = 3\,{}^{3}/_{4}$
3	$6 \times {}^{1}/_{2}$	3
4	${}^{7}/_{8} = 7 \times$ ___	${}^{1}/_{8}$
5	4×3	12
6	$2 \times {}^{3}/_{8}$	${}^{6}/_{8}$
7	$3 \times {}^{2}/_{7}$	${}^{6}/_{7}$
8	$4 \times {}^{3}/_{5}$	$2\,{}^{2}/_{5}$
9	$6 \times {}^{2}/_{3}$	4
10	$5 \times {}^{5}/_{6}$	$4\,{}^{1}/_{6}$
11	$8 \times {}^{3}/_{4}$	6
12	$7 \times {}^{2}/_{10}$	$1\,{}^{4}/_{10}$
13	$9 \times {}^{1}/_{3}$	3
14	Draw a model for $3 \times {}^{3}/_{4}$. Find the product.	${}^{9}/_{4} = 2\,{}^{1}/_{4}$

LESSON 6.11 · TEACHER PLAN
Week 3 · Lesson 3 of 4
Unit 6: Fraction Operations

Lesson 6.11: Models for Fraction Multiplication

Student goal	I can use number lines and area models to multiply fractions by whole numbers.
Objective	Students model fraction multiplication.
Standard	4.NF.4b — Multiply a fraction by a whole number using visual models.
Pacing	Unit 6, week 3, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 6.11
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	2 × 3/4.
Launch	5	Read Let's Explore.
Teach	15	Draw frog hops on a number line. Work the example two ways.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Counting tick marks instead of jumps.

SUPPORT

- Number line with labels.

EXTEND

- Make your own hop story.

STUDENT LESSON 6.11 · WITH ANSWERS

Lesson 6.11: Models for Fraction Multiplication

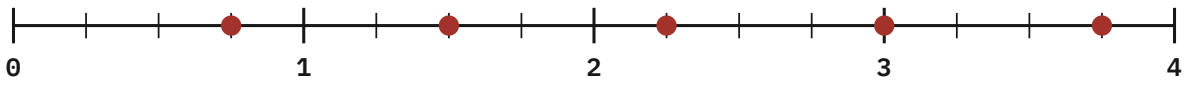
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.11 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.11, Part A.
- [ ] Complete the rest of Workbook Lesson 6.11 (Parts B–E).

LET'S EXPLORE

A frog jumps $3/4$ meter each hop. Where is it after 5 hops?

Each hop is $3/4$. Five hops: $5 \times 3/4 = 15/4 = 3\,3/4$ meters.



WORKED EXAMPLE · TWO WAYS

$6 \times 2/5$

1 Repeated addition: $2/5 + 2/5 + 2/5 + 2/5 + 2/5 + 2/5 = 12/5$.

2 Multiply: $6 \times 2 = 12$, so $12/5 = 2\,2/5$.

Answer: $2\,2/5$

★ KEY IDEA

A number line shows equal jumps; the product is where you land.

✎ TRY IT

Answers are shown in color.

1. $4 \times 3/8$

Answer: $12/8 = 1\,4/8$

2. $3 \times 5/6$

Answer: $15/6 = 2\,3/6$

3. A snail moves $2/10$ meter per minute. How far in 7 minutes?

Answer: $14/10$ meters

Workbook Answer Key · Lesson 6.11

#	Question	Answer
1	$4 \times {}^{3}/_{8}$	${}^{12}/_{8} = 1\,{}^{4}/_{8}$
2	$3 \times {}^{5}/_{6}$	${}^{15}/_{6} = 2\,{}^{3}/_{6}$
3	A snail moves ${}^{2}/_{10}$ meter per minute. How far in 7 minutes?	${}^{14}/_{10}$ meters
4	$2 \times {}^{3}/_{4}$	${}^{6}/_{4}$
5	${}^{10}/_{4}$ as a mixed number	$2\,{}^{2}/_{4}$
6	$3 \times {}^{2}/_{6}$	1
7	$4 \times {}^{2}/_{3}$	$2\,{}^{2}/_{3}$
8	$5 \times {}^{3}/_{10}$	$1\,{}^{5}/_{10}$
9	$2 \times {}^{7}/_{8}$	$1\,{}^{6}/_{8}$
10	Each lap is ${}^{3}/_{8}$ mile. Ellie runs 6 laps. How far does she run?	${}^{18}/_{8} = 2\,{}^{2}/_{8}$ miles
11	Show $4 \times {}^{2}/_{5}$ on a number line.	${}^{8}/_{5} = 1\,{}^{3}/_{5}$

LESSON 6.12 · TEACHER PLAN
Week 3 · Lesson 4 of 4
Unit 6: Fraction Operations

Lesson 6.12: Fraction Multiplication Word Problems

Student goal	I can solve word problems that multiply a fraction by a whole number.
Objective	Students solve word problems multiplying fractions by whole numbers.
Standard	4.NF.4c — Solve word problems involving multiplication of a fraction by a whole number.
Pacing	Unit 6, week 3, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 6.12
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	3 × 3/5.
Launch	5	Read Let's Explore.
Teach	15	Work both examples. Ask: Between which whole numbers is the answer?
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Adding instead of multiplying.

SUPPORT

- Draw equal groups.

EXTEND

- Double a real recipe.

STUDENT LESSON 6.12 · WITH ANSWERS

Lesson 6.12: Fraction Multiplication Word Problems

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.12 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.12, Part A.
- [ ] Complete the rest of Workbook Lesson 6.12 (Parts B–D).

LET'S EXPLORE

Each student needs $\frac{3}{8}$ yard of ribbon for a project. There are 5 students. How much ribbon is needed? Between which two whole numbers is the answer?

WORKED EXAMPLE · RIBBON

$5 \times \frac{3}{8}$

1 $5 \times 3 = 15$, so $\frac{15}{8}$.

2 $\frac{15}{8} = 1\frac{7}{8}$: between 1 and 2 yards.

Answer: $1\frac{7}{8}$ yards

WORKED EXAMPLE · RECIPE

A recipe uses $\frac{2}{3}$ cup of sugar. Mom triples the recipe. How much sugar?

1 $3 \times \frac{2}{3} = \frac{6}{3} = 2$.

Answer: 2 cups

★ KEY IDEA

Equal groups of a fraction → multiply. Then think: between which whole numbers does the answer lie?

✎ TRY IT

Answers are shown in color.

1. A plant grows $\frac{3}{4}$ inch a week. How much in 6 weeks?

Answer: $\frac{18}{4} = 4\frac{2}{4}$ inches

2. 8 friends each eat $\frac{1}{4}$ pizza. How much pizza?

Answer: 2 pizzas

3. A bottle holds $\frac{5}{6}$ liter. How much do 4 bottles hold?
Answer: $\frac{20}{6} = 3\frac{2}{6}$ liters

Workbook Answer Key · Lesson 6.12

#	Question	Answer
1	A plant grows $\frac{3}{4}$ inch a week. How much in 6 weeks?	$\frac{18}{4} = 4\frac{2}{4}$ inches
2	8 friends each eat $\frac{1}{4}$ pizza. How much pizza?	2 pizzas
3	A bottle holds $\frac{5}{6}$ liter. How much do 4 bottles hold?	$\frac{20}{6} = 3\frac{2}{6}$ liters
4	$3 \times \frac{3}{5}$	$\frac{9}{5}$
5	$\frac{9}{5}$ as a mixed number	$1\frac{4}{5}$
6	Ellie walks $\frac{5}{8}$ mile each day. How far does she walk in 4 days?	$\frac{20}{8} = 2\frac{4}{8}$ miles
7	A dog eats $\frac{2}{3}$ can of food a day. How many cans in a week?	$\frac{14}{3} = 4\frac{2}{3}$ cans
8	Lillie fills 9 cups with $\frac{3}{4}$ cup of juice each. How much juice? Between which two whole numbers?	$\frac{27}{4} = 6\frac{3}{4}$ cups; between 6 and 7
9	A board is cut into 6 pieces, each $\frac{5}{12}$ foot. How long was the board (no waste)?	$\frac{30}{12} = 2\frac{6}{12}$ feet
10	Write a word problem for $7 \times \frac{2}{5}$. Solve it.	$\frac{14}{5} = 2\frac{4}{5}$

WEEKLY CHECK · ANSWER KEY
After Lesson 6.12

About 15 minutes

Weekly Check 14: Multiplying Fractions — Answer Key

Answers

#	Question	Answer
1	$\frac{7}{9} = ___ \times \frac{1}{9}$	7
2	$3 \times \frac{2}{7}$	$\frac{6}{7}$
3	$5 \times \frac{3}{4}$	$3\frac{3}{4}$
4	$8 \times \frac{1}{2}$	4
5	A cat eats $\frac{1}{3}$ can of food a day. How much in 5 days?	$1\frac{2}{3}$ cans
6	Between which two whole numbers is $7 \times \frac{3}{5}$?	4 and 5 ($4\frac{1}{5}$)

LESSON 6.13 · TEACHER PLAN
Week 4 · Lesson 1 of 4
Unit 6: Fraction Operations

Lesson 6.13: Fractions and Measurement

Student goal	I can measure lengths to the nearest half, fourth, and eighth inch and add or subtract the measurements.
Objective	Students measure to fractional inches and compute with measurements.
Standard	4.MD.2, 4.NF.3d — Use the four operations to solve word problems involving distances and measurement with fractions.
Pacing	Unit 6, week 4, lesson 1 of 4
Time	About 50 minutes
Materials	Inch rulers, small objects, workbook Lesson 6.13
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Equivalent fractions on a ruler.
Launch	5	Read Let's Explore.
Teach	15	Find halves, fourths, eighths on a real ruler. Measure three objects together. Work the example.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Starting measurement at 1 instead of 0.

SUPPORT

- Color-code ruler marks.

EXTEND

- Measure 5 objects and order them.

STUDENT LESSON 6.13 · WITH ANSWERS

Lesson 6.13: Fractions and Measurement

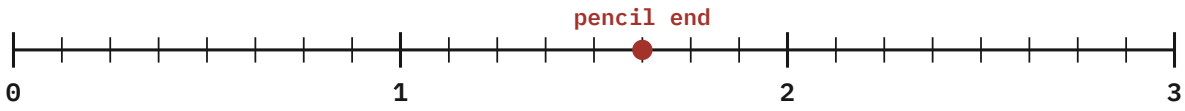
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.13 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.13, Part A.
- [ ] Complete the rest of Workbook Lesson 6.13 (Parts B–E).

LET'S EXPLORE

How long is your pencil? Can you measure it more exactly than “about 6 inches”?

An inch ruler splits each inch into halves, fourths, and eighths. The longest marks between inches are halves, then fourths, then eighths.



The pencil ends at 13 eighths of an inch past 0: $13/8 = 1\,5/8$ inches.

WORKED EXAMPLE · ADD MEASUREMENTS

Two sticks are $2\,3/8$ in. and $1\,7/8$ in. Laid end to end, how long are they?

1 $2\,3/8 + 1\,7/8 = 3\,10/8 = 4\,2/8$.

Answer: $4\,2/8 = 4\,1/4$ inches

★ KEY IDEA

Read a ruler like a number line: count the wholes, then count the small parts.

✎ TRY IT

Answers are shown in color.

1. How many eighths are in $1/2$ inch?

Answer: 4

2. $3\,1/4$ in. − $1\,3/4$ in.

Answer: $1\,2/4$ in.

3. Which is longer: $5/8$ in. or $3/4$ in.?

Answer: $\frac{3}{4}$ in.

Workbook Answer Key · Lesson 6.13

#	Question	Answer
1	How many eighths are in $\frac{1}{2}$ inch?	4
2	$3\frac{1}{4}$ in. − $1\frac{3}{4}$ in.	$1\frac{2}{4}$ in.
3	Which is longer: $\frac{5}{8}$ in. or $\frac{3}{4}$ in.?	$\frac{3}{4}$ in.
4	$\frac{1}{4} = \frac{?}{8}$	2
5	$\frac{6}{8} = \frac{?}{4}$	3
6	Use a ruler to measure your pencil to the nearest eighth inch.	Answers vary
7	Measure your thumb to the nearest fourth inch.	Answers vary
8	Measure a crayon to the nearest half inch.	Answers vary
9	Measure the width of this workbook to the nearest inch.	about $8\frac{1}{2}$ in. (8.5 in.)
10	$1\frac{3}{4}$ in. + $2\frac{2}{4}$ in.	$4\frac{1}{4}$ in.
11	$5\frac{5}{8}$ in. − $2\frac{7}{8}$ in.	$2\frac{6}{8}$ in.
12	A ribbon is $4\frac{1}{8}$ in. long. Ellie cuts off $1\frac{5}{8}$ in. How much is left?	$2\frac{4}{8}$ in. = $2\frac{1}{2}$ in.

LESSON 6.14 · TEACHER PLAN
Week 4 · Lesson 2 of 4
Unit 6: Fraction Operations

Lesson 6.14: Multi-Step Fraction Problems

Student goal	I can solve multi-step word problems with fractions.
Objective	Students solve multi-step fraction problems.
Standard	4.NF.3d, 4.NF.4c — Solve word problems with fraction addition, subtraction, and multiplication by whole numbers.
Pacing	Unit 6, week 4, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 6.14
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	4 × 3/5.
Launch	5	Read Let's Explore.
Teach	15	Work both examples. Write an equation per step.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Stopping after the first step.

SUPPORT

- Underline each question.

EXTEND

- Plan a recipe for 12 from a recipe for 4.

STUDENT LESSON 6.14 · WITH ANSWERS

Lesson 6.14: Multi-Step Fraction Problems

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.14 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.14, Part A.
- [ ] Complete the rest of Workbook Lesson 6.14 (Parts B–D).

LET'S EXPLORE

Lillie needs 3 pieces of ribbon, each $\frac{3}{4}$ foot long. She has a ribbon that is 3 feet long. How much will be left?

WORKED EXAMPLE · MULTIPLY, THEN SUBTRACT

Ribbon problem.

1 Needed: $3 \times \frac{3}{4} = \frac{9}{4} = 2\frac{1}{4}$ feet.

2 Left: $3 - 2\frac{1}{4} = \frac{3}{4}$ foot.

Answer: $\frac{3}{4}$ foot

WORKED EXAMPLE · ADD, THEN COMPARE

Ellie ran $\frac{5}{8}$ mile and walked $\frac{6}{8}$ mile. Lillie ran $1\frac{2}{8}$ miles. Who went farther?

1 Ellie: $\frac{5}{8} + \frac{6}{8} = \frac{11}{8} = 1\frac{3}{8}$.

2 $1\frac{3}{8} > 1\frac{2}{8}$.

Answer: Ellie went farther.

★ KEY IDEA

Find the hidden first step. Write each step as an equation.

✎ TRY IT

Answers are shown in color.

1. A pitcher holds 2 quarts. Mom pours 4 cups of $\frac{1}{4}$ quart each. How much is left in the pitcher?

Answer: 1 quart

2. Dad hiked $2\frac{3}{10}$ miles in the morning and $1\frac{9}{10}$ in the afternoon. How far in all?

Answer: $4\frac{2}{10}$ miles

3. 6 kids each eat $^{2}/_{8}$ of a pie. Is 1 pie enough?
Answer: No; they need $^{12}/_{8} = 1\,^{4}/_{8}$ pies.

Workbook Answer Key · Lesson 6.14

#	Question	Answer
1	A pitcher holds 2 quarts. Mom pours 4 cups of $^{1}/_{4}$ quart each. How much is left in the pitcher?	1 quart
2	Dad hiked $2\,^{3}/_{10}$ miles in the morning and $1\,^{9}/_{10}$ in the afternoon. How far in all?	$4\,^{2}/_{10}$ miles
3	6 kids each eat $^{2}/_{8}$ of a pie. Is 1 pie enough?	No; they need $^{12}/_{8} = 1\,^{4}/_{8}$ pies.
4	$4 \times {}^{3}/_{5}$	$^{12}/_{5}$
5	$3 - 1\,^{2}/_{5}$	$1\,^{3}/_{5}$
6	A baker uses $^{3}/_{4}$ cup of butter in each of 3 cakes. She had 4 cups. How much is left?	$1\,^{3}/_{4}$ cups
7	Ellie read $1\,^{1}/_{4}$ hours on Saturday and $2\,^{2}/_{4}$ on Sunday. Lillie read 4 hours. Who read more, and by how much?	Lillie, by $^{1}/_{4}$ hour
8	A garden row is 10 feet. Each pepper plant needs $1\,^{1}/_{2}$ feet. Can 6 plants fit? Explain.	Yes; $6 \times 1\,^{1}/_{2} = 9$ feet ≤ 10.
9	Write a two-step fraction problem. Solve it.	Answers vary.

LESSON 6.15 · TEACHER PLAN
Week 4 · Lesson 3 of 4
Unit 6: Fraction Operations

Lesson 6.15: Estimating with Fractions

Student goal	I can estimate sums, differences, and products with fractions using benchmarks.
Objective	Students estimate with fractions and mixed numbers.
Standard	4.NF.3d, 4.NF.4c — Use benchmark fractions and estimation to assess reasonableness.
Pacing	Unit 6, week 4, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 6.15
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Closer to 0, 1/2, or 1?
Launch	5	Read Let's Explore.
Teach	15	Work the examples. Say: Round, compute, then check the exact answer.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Rounding the numerator only.

SUPPORT

- Number line from 0 to 1.

EXTEND

- Estimate the total length of 3 objects.

STUDENT LESSON 6.15 · WITH ANSWERS

Lesson 6.15: Estimating with Fractions

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.15 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.15, Part A.
- [ ] Complete the rest of Workbook Lesson 6.15 (Parts B–E).

LET'S EXPLORE

About how much is $2\,{}^{7}\!/_{8} + 3\,{}^{1}\!/_{10}$? Do you need an exact answer to know it's about 6?

Round each fraction or mixed number to the nearest whole or half. $2\,{}^{7}\!/_{8}$ is about 3. $3\,{}^{1}\!/_{10}$ is about 3. So the sum is about 6.

WORKED EXAMPLE · ESTIMATE A PRODUCT

About how much is $4 \times {}^{7}\!/_{8}$?

1 ${}^{7}\!/_{8}$ is close to 1.

2 $4 \times 1 = 4$, a little less.

Answer: a little less than 4

★ KEY IDEA

Round fractions to 0, ${}^{1}\!/_{2}$, or 1 (and mixed numbers to the nearest whole or half) to estimate.

✎ TRY IT

Answers are shown in color.

1. Estimate $5\,{}^{1}\!/_{6} - 1\,{}^{5}\!/_{6}$

Answer: about $5 - 2 = 3$

2. Estimate $6 \times {}^{4}\!/_{9}$

Answer: about $6 \times {}^{1}\!/_{2} = 3$

3. Is $1\,{}^{3}\!/_{8} + 2\,{}^{1}\!/_{8}$ more or less than 4?

Answer: less (exact $3\,{}^{4}\!/_{8}$)

Workbook Answer Key · Lesson 6.15

#	Question	Answer
1	Estimate $5\frac{1}{6} - 1\frac{5}{6}$	about 5 − 2 = 3
2	Estimate $6 \times \frac{4}{9}$	about $6 \times \frac{1}{2} = 3$
3	Is $1\frac{3}{8} + 2\frac{1}{8}$ more or less than 4?	less (exact $3\frac{4}{8}$)
4	Is $\frac{5}{6}$ closer to $\frac{1}{2}$ or 1?	1
5	Is $\frac{1}{8}$ closer to 0 or $\frac{1}{2}$?	0
6	$3\frac{4}{5} + 2\frac{1}{5}$	about 4 + 2 = 6
7	$6\frac{1}{10} - 2\frac{9}{10}$	about 6 − 3 = 3
8	$8 \times \frac{5}{12}$	about $8 \times \frac{1}{2} = 4$
9	$\frac{11}{12} + \frac{1}{12} + \frac{5}{6}$	about 1 + 0 + 1 = 2 (exact $1\frac{5}{6}$)
10	Sam says $2\frac{1}{4} + 1\frac{3}{4} = 3\frac{4}{8}$. Use an estimate to decide if he is right.	No; about 2 + 2 = 4. Exact: 4.
11	Estimate $5 \times 1\frac{7}{8}$.	about 5 × 2 = 10

LESSON 6.16 · TEACHER PLAN
Week 4 · Lesson 4 of 4
Unit 6: Fraction Operations

Lesson 6.16: Fraction Operations Problem Solving

Student goal	I can choose the right operation and solve fraction problems from start to finish.
Objective	Students apply fraction operations in problem solving.
Standard	4.NF.3, 4.NF.4 — 4.NF.3 and 4.NF.4 review.
Pacing	Unit 6, week 4, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 6.16, Unit 6 Review
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Two warm-ups.
Launch	5	Read Let's Explore.
Teach	15	Work the example.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook Lesson 6.16 and the Unit 6 Review.
Exit Ticket	3	Check. Tomorrow: Unit 6 Test.

COMMON MISCONCEPTIONS

- Mixing up operations.

SUPPORT

- Bar models.

EXTEND

- Make a real fruit salad!

STUDENT LESSON 6.16 · WITH ANSWERS

Lesson 6.16: Fraction Operations Problem Solving

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.16 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.16, Part A.
- [ ] Complete the rest of Workbook Lesson 6.16 (Parts B–E).

LET'S EXPLORE

Plan a fruit salad: it needs fractions of cups of different fruits for a group. How much fruit do you need?

This lesson pulls together adding, subtracting, and multiplying fractions.

WORKED EXAMPLE · FRUIT SALAD

Each serving: $\frac{3}{4}$ cup grapes and $\frac{2}{4}$ cup berries. Make 4 servings. How much fruit?

1 One serving: $\frac{3}{4} + \frac{2}{4} = \frac{5}{4}$.

2 Four servings: $4 \times \frac{5}{4} = \frac{20}{4} = 5$.

Answer: 5 cups

★ KEY IDEA

Read, plan, solve, check.

✎ TRY IT

Answers are shown in color.

1. $4 \times \frac{3}{4} + 4 \times \frac{2}{4}$

Answer: 5

2. $3\frac{1}{3} - 1\frac{2}{3}$

Answer: $1\frac{2}{3}$

3. $\frac{5}{6} + \frac{5}{6} + \frac{5}{6}$

Answer: $2\frac{3}{6}$

Workbook Answer Key · Lesson 6.16

#	Question	Answer
1	$4 \times \frac{3}{4} + 4 \times \frac{2}{4}$	5
2	$3\frac{1}{3} - 1\frac{2}{3}$	$1\frac{2}{3}$
3	$\frac{5}{6} + \frac{5}{6} + \frac{5}{6}$	$2\frac{3}{6}$
4	$1\frac{3}{5} + 2\frac{4}{5}$	$4\frac{2}{5}$
5	$3 \times \frac{4}{5}$	$2\frac{2}{5}$
6	$\frac{5}{8} + \frac{6}{8}$	$1\frac{3}{8}$
7	$\frac{11}{12} - \frac{7}{12}$	$\frac{4}{12}$
8	$4\frac{2}{6} - 1\frac{5}{6}$	$2\frac{3}{6}$
9	$6 \times \frac{3}{10}$	$1\frac{8}{10}$
10	Write $\frac{19}{4}$ as a mixed number.	$4\frac{3}{4}$
11	Decompose $\frac{5}{6}$ two ways.	Answers vary
12	A class makes 6 posters. Each poster uses $\frac{2}{3}$ yard of paper. The roll has 5 yards. How much paper is left?	1 yard
13	Which fraction operation was hardest for you? Make up and solve one problem of that type.	Answers vary.

UNIT REVIEW · ANSWER KEY
Study before the unit test

About 35 minutes

Unit 6 Review — Answer Key

Homework or a study session before the test.

Answers

#	Question	Answer
1	$\frac{4}{7} + \frac{5}{7}$	$1\frac{2}{7}$
2	$\frac{9}{10} - \frac{6}{10}$	$\frac{3}{10}$
3	$3\frac{5}{8} + 2\frac{6}{8}$	$6\frac{3}{8}$
4	$6\frac{1}{4} - 2\frac{3}{4}$	$3\frac{2}{4}$
5	$4 \times \frac{5}{6}$	$3\frac{2}{6}$
6	$\frac{9}{10} = 9 \times$ ___	$\frac{1}{10}$
7	Lillie runs $\frac{3}{4}$ mile each day for 5 days. Ellie runs $3\frac{1}{4}$ miles in all. Who runs farther, and by how much?	Lillie, by $\frac{2}{4}$ mile ($3\frac{3}{4}$ vs $3\frac{1}{4}$)
8	A board is $6\frac{2}{8}$ feet. Dad cuts off $2\frac{5}{8}$ feet. How much is left?	$3\frac{5}{8}$ feet

UNIT TEST · ANSWER KEY
At the end of the unit
About 45 minutes

Unit 6 Test — Answer Key

Answers

#	Question	Answer
1	$\frac{2}{8} + \frac{5}{8}$	$\frac{7}{8}$
2	$\frac{11}{12} - \frac{4}{12}$	$\frac{7}{12}$
3	$1 - \frac{2}{9}$	$\frac{7}{9}$
4	$\frac{4}{5} + \frac{4}{5}$	$1\frac{3}{5}$
5	Decompose $\frac{5}{8}$ in two different ways.	Answers vary
6	Write $4\frac{3}{4}$ as a fraction.	$\frac{19}{4}$
7	Write $\frac{17}{3}$ as a mixed number.	$5\frac{2}{3}$
8	$2\frac{4}{6} + 3\frac{5}{6}$	$6\frac{3}{6}$
9	$7\frac{2}{5} - 3\frac{4}{5}$	$3\frac{3}{5}$
10	$4 - 2\frac{1}{3}$	$1\frac{2}{3}$
11	$\frac{8}{3} = 8 \times$ ___	$\frac{1}{3}$
12	$6 \times \frac{2}{5}$	$2\frac{2}{5}$
13	$9 \times \frac{2}{3}$	6
14	Between which two whole numbers is $5 \times \frac{5}{6}$?	4 and 5 ($4\frac{1}{6}$)
15	Ellie needs 4 pieces of string, each $\frac{5}{8}$ yard. She has 3 yards. How much will be left?	$\frac{4}{8}$ yard (= $\frac{1}{2}$)
16	Lillie hiked $2\frac{3}{4}$ miles Saturday and $1\frac{3}{4}$ miles Sunday. Her goal was 5 miles. Did she reach it? Explain.	No; $4\frac{2}{4}$ miles, $\frac{2}{4}$ mile short.

Scoring

Score	What it means / next step
17–20 (85%+)	Mastered.
14–16	Proficient; correct missed items.
Below 14	Reteach and retest.

UNIT 7 · TEACHER OVERVIEW

Decimals

Tenths and hundredths, equivalent decimals, comparing decimals, and money.

Decimals are another way to write fractions with denominators of 10 and 100. You will connect decimals to fractions, place value, number lines, and money.

Big ideas

- Write tenths and hundredths as fractions and decimals.
- Show that tenths and hundredths can be equivalent.
- Add tenths and hundredths.
- Compare and order decimals.
- Add and subtract money amounts and make change.

Standards

Code	Standard
4.NF.5	Express a fraction with denominator 10 as an equivalent fraction with denominator 100; add fractions with denominators 10 and 100.
4.NF.6	Use decimal notation for fractions with denominators 10 or 100.
4.NF.7	Compare two decimals to hundredths; record with >, =, <.
4.MD.2	Solve word problems involving money, including problems with decimals.

What students should already know

- Fractions and equivalence (Units 5–6).
- Place value (Unit 1).

Materials for the unit

- Hundred grids
- Real or play money
- Place-value chart with a decimal point

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 4	7.1	Tenths
Week 1	Lesson 2 of 4	7.2	Decimals Greater Than 1
Week 1	Lesson 3 of 4	7.3	Tenths on a Number Line
Week 1	Lesson 4 of 4	7.4	Decimal Place Value
Week 1	End of week	Check	Weekly Check 15: Tenths
Week 2	Lesson 1 of 4	7.5	Hundredths
Week 2	Lesson 2 of 4	7.6	Equivalent Tenths and Hundredths
Week 2	Lesson 3 of 4	7.7	Adding Tenths and Hundredths
Week 2	Lesson 4 of 4	7.8	Decimals in Different Forms
Week 2	End of week	Check	Weekly Check 16: Hundredths
Week 3	Lesson 1 of 4	7.9	Comparing Decimals with Models
Week 3	Lesson 2 of 4	7.10	Comparing Decimals with Place Value
Week 3	Lesson 3 of 4	7.11	Ordering Decimals
Week 3	Lesson 4 of 4	7.12	Decimals and Fractions on a Number Line
Week 3	End of week	Check	Weekly Check 17: Comparing Decimals
Week 4	Lesson 1 of 4	7.13	Money and Decimals
Week 4	Lesson 2 of 4	7.14	Adding Money
Week 4	Lesson 3 of 4	7.15	Subtracting Money and Making Change
Week 4	Lesson 4 of 4	7.16	Decimal and Money Problem Solving
Week 4	Before the test	Review	Unit 7 Review
Week 4	End of unit	Test	Unit 7 Test

TEACHING THIS UNIT AT HOME

Let your child handle money this unit: count change, read receipts, and compare prices per item at the store.

UNIT VOCABULARY

decimal	A number written with a decimal point.
tenth	1 of 10 equal parts.
hundredth	1 of 100 equal parts.
decimal point	Separates wholes from parts.

LESSON 7.1 · TEACHER PLAN
Week 1 · Lesson 1 of 4
Unit 7: Decimals

Lesson 7.1: Tenths

Student goal	I can write tenths as fractions and as decimals, like $^{3}/_{10}$ = 0.3.
Objective	Students write tenths as fractions and decimals.
Standard	4.NF.6 — Use decimal notation for fractions with denominators 10 or 100.
Pacing	Unit 7, week 1, lesson 1 of 4
Time	About 50 minutes
Materials	Paper strips cut into 10 parts, workbook Lesson 7.1
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Simplify 5/10.
Launch	5	Read Let's Explore.
Teach	15	Shade tenths on a strip. Write fraction and decimal side by side. Say: "The decimal point separates wholes from parts."
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Writing 3/10 as 3.10 or 0.03.

SUPPORT

- Always say 'tenths' aloud.

EXTEND

- Write 10/10 as a decimal (1.0).

STUDENT LESSON 7.1 · WITH ANSWERS

Lesson 7.1: Tenths

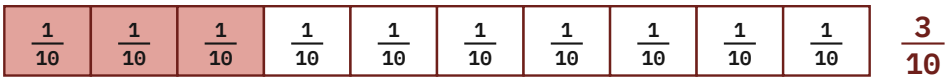
STUDENT'S ASSIGNMENT

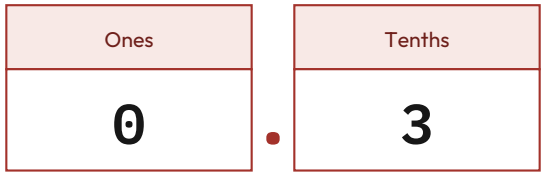
- [ ] Read Lesson 7.1 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.1, Part A.
- [ ] Complete the rest of Workbook Lesson 7.1 (Parts B–F).

LET'S EXPLORE

A candy bar is split into 10 equal pieces. You eat 3 pieces. How can you write the amount you ate without a fraction bar?

When a whole is split into 10 equal parts, each part is one tenth. Three tenths can be written as the fraction $^{3}/_{10}$ or the decimal 0.3.





0.3: 0 ones and 3 tenths

The decimal point separates whole numbers from parts of a whole. The first place after the decimal point is the tenths place. We read 0.3 as "three tenths."

WORKED EXAMPLE · WRITE AS A DECIMAL

Write $^{7}/_{10}$ as a decimal.

1 7 tenths.

2 Write 7 in the tenths place.

Answer: 0.7

★ KEY IDEA

$^{n}/_{10}$ = 0.n. The tenths place is the first place to the right of the decimal point.

✎ TRY IT

Answers are shown in color.

1. Write $^{9}/_{10}$ as a decimal.

Answer: 0.9

2. Write 0.4 as a fraction.

Answer: $^{4}/_{10}$

3. How do you read 0.6?

Answer: six tenths

Workbook Answer Key · Lesson 7.1

#	Question	Answer
1	Write $^{9}/_{10}$ as a decimal.	0.9
2	Write 0.4 as a fraction.	$^{4}/_{10}$
3	How do you read 0.6?	six tenths
4	Simplify $^{5}/_{10}$	$^{1}/_{2}$
5	How many tenths make 1 whole?	10
6	(figure)	$^{2}/_{10} = 0.2$
7	(figure)	$^{5}/_{10} = 0.5$
8	(figure)	$^{8}/_{10} = 0.8$
9	(figure)	$^{1}/_{10} = 0.1$
10	$^{6}/_{10}$	0.6
11	$^{3}/_{10}$	0.3
12	$^{9}/_{10}$	0.9
13	$^{4}/_{10}$	0.4
14	0.1	$^{1}/_{10}$
15	0.7	$^{7}/_{10}$
16	0.2	$^{2}/_{10}$
17	0.5	$^{5}/_{10}$
18	Write “eight tenths” as a fraction and as a decimal.	$^{8}/_{10} = 0.8$

LESSON 7.2 · TEACHER PLAN
Week 1 · Lesson 2 of 4
Unit 7: Decimals

Lesson 7.2: Decimals Greater Than 1

Student goal	I can write mixed numbers with tenths as decimals, like $2\ ^{4}/_{10} = 2.4$.
Objective	Students write numbers greater than 1 with tenths as decimals.
Standard	4.NF.6 — Use decimal notation for fractions with denominators 10.
Pacing	Unit 7, week 1, lesson 2 of 4
Time	About 50 minutes
Materials	Fraction strips, workbook Lesson 7.2
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	0.6 as a fraction.
Launch	5	Read Let's Explore.
Teach	15	Model 2 4/10 with strips; write 2.4. Say: "'and' means the decimal point."
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Saying 'and' in whole numbers.
- Writing 2.4 as 24.

SUPPORT

- Use the chart.

EXTEND

- Write 105/10 as a decimal.

STUDENT LESSON 7.2 · WITH ANSWERS

Lesson 7.2: Decimals Greater Than 1

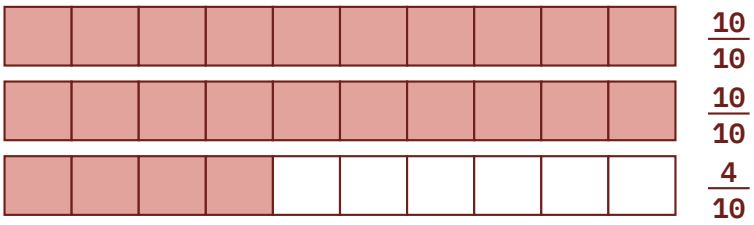
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.2 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.2, Part A.
- [ ] Complete the rest of Workbook Lesson 7.2 (Parts B–F).

LET'S EXPLORE

A recipe needs 2 and 4 tenths cups of flour. How can you write that as a decimal?

The whole number goes to the left of the decimal point. The tenths go to the right. $2\,{}^{4}\!/_{10} = 2.4$, read "two and four tenths." The word and stands for the decimal point.



Tens	Ones		Tenths
	2	.	4

WORKED EXAMPLE · WRITE AS A DECIMAL

Write $13\,{}^{6}\!/_{10}$ as a decimal.

1 13 wholes, 6 tenths.

Answer: 13.6

★ KEY IDEA

Say "and" only for the decimal point: 13.6 is "thirteen and six tenths."

✎ TRY IT

Answers are shown in color.

1. $5\,^{9}/_{10}$ as a decimal

Answer: 5.9

2. 4.3 as a mixed number

Answer: $4\,^{3}/_{10}$

3. Read 20.7 aloud. Write the words.

Answer: twenty and seven tenths

Workbook Answer Key · Lesson 7.2

#	Question	Answer
1	$5\,^{9}/_{10}$ as a decimal	5.9
2	4.3 as a mixed number	$4\,^{3}/_{10}$
3	Read 20.7 aloud. Write the words.	twenty and seven tenths
4	Write 0.6 as a fraction.	$^{6}/_{10}$
5	$^{17}/_{10}$ as a mixed number	$1\,^{7}/_{10}$
6	$1\,^{2}/_{10}$	1.2
7	$3\,^{5}/_{10}$	3.5
8	$6\,^{1}/_{10}$	6.1
9	$12\,^{8}/_{10}$	12.8
10	2.7	$2\,^{7}/_{10}$
11	9.3	$9\,^{3}/_{10}$
12	15.5	$15\,^{5}/_{10}$
13	4.9	$4\,^{9}/_{10}$
14	7.4	seven and four tenths
15	31.2	thirty-one and two tenths
16	Write $^{26}/_{10}$ as a mixed number and a decimal.	$2\,^{6}/_{10} = 2.6$

LESSON 7.3 · TEACHER PLAN
Week 1 · Lesson 3 of 4
Unit 7: Decimals

Lesson 7.3: Tenths on a Number Line

Student goal	I can place and name decimals in tenths on a number line.
Objective	Students locate tenths on number lines.
Standard	4.NF.6 — Locate decimals on a number line.
Pacing	Unit 7, week 1, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 7.3
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Count by tenths.
Launch	5	Read Let's Explore.
Teach	15	Draw 0 to 2 in tenths. Say: Count jumps of one tenth.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Counting marks instead of jumps.

SUPPORT

- Label every mark.

EXTEND

- Place 0.25 between 0.2 and 0.3.

STUDENT LESSON 7.3 · WITH ANSWERS

Lesson 7.3: Tenths on a Number Line

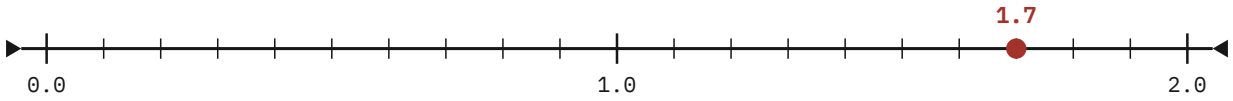
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.3 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.3, Part A.
- [ ] Complete the rest of Workbook Lesson 7.3 (Parts B–D).

LET'S EXPLORE

A walking trail has markers every tenth of a mile. Lillie stops at the 7th marker after the 1-mile sign. Where is she?

A number line from 0 to 1 split into 10 equal parts shows tenths. 1.7 is 7 tenths past 1.



WORKED EXAMPLE · NAME THE POINT

What decimal is at point A?



1 Each jump is 0.1.

2 A is 4 jumps past 3.

Answer: 3.4

★ KEY IDEA

Count jumps of 0.1 from the whole number.

✎ TRY IT

Answers are shown in color.

1. What decimal is halfway between 0 and 1?

Answer: 0.5

2. What decimal is 3 tenths more than 2?

Answer: 2.3

3. Which whole number is 6.8 closer to?

Answer: 7

Workbook Answer Key · Lesson 7.3

#	Question	Answer
1	What decimal is halfway between 0 and 1?	0.5
2	What decimal is 3 tenths more than 2?	2.3
3	Which whole number is 6.8 closer to?	7
4	Count by tenths from 0.6 to 1.2.	0.6, 0.7, 0.8, 0.9, 1.0, 1.1, 1.2
5	Point A (figure)	0.3
6	Point B (figure)	1.8
7	Point C (figure)	4.5
8	Point D (figure)	7.1
9	Draw a number line from 5 to 6 in tenths. Mark 5.4 and 5.9.	Ten equal parts; points at the 4th and 9th marks

LESSON 7.4 · TEACHER PLAN
Week 1 · Lesson 4 of 4
Unit 7: Decimals

Lesson 7.4: Decimal Place Value

Student goal	I can tell the place and value of digits in a decimal.
Objective	Students identify place and value in decimals.
Standard	4.NF.6 — Understand decimal place value to tenths.
Pacing	Unit 7, week 1, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 7.4
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Value of 4 in 345.
Launch	5	Read Let's Explore.
Teach	15	Extend the place-value chart to tenths.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Calling the tenths place ‘oneths’.

SUPPORT

- Use the chart.

EXTEND

- What is 10 tenths + 10 tenths?

STUDENT LESSON 7.4 · WITH ANSWERS

Lesson 7.4: Decimal Place Value

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.4 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.4, Part A.
- [ ] Complete the rest of Workbook Lesson 7.4 (Parts B–E).

LET'S EXPLORE

In 45.7, the 4, the 5, and the 7 all have different values. What is each worth?

Place value continues to the right of the decimal point. Each place is still 10 times the place to its right: 1 one = 10 tenths.

Tens	Ones		Tenths
4	5	.	7

WORKED EXAMPLE · VALUE OF A DIGIT

What is the value of each digit in 45.7?

1 4 is in the tens place: 40.

2 5 is in the ones place: 5.

3 7 is in the tenths place: 0.7 ($^{7}/_{10}$).

Answer: 40 + 5 + 0.7

★ KEY IDEA

Ones, then the decimal point, then tenths, then hundredths. 10 tenths = 1 one.

✎ TRY IT

Answers are shown in color.

1. Value of the 3 in 6.3

Answer: 0.3 ($^{3}/_{10}$)

2. Value of the 6 in 6.3

Answer: 6

3. Write 20 + 8 + 0.1 as a decimal.

Answer: 28.1

Workbook Answer Key · Lesson 7.4

#	Question	Answer
1	Value of the 3 in 6.3	0.3 ($^{3}/_{10}$)
2	Value of the 6 in 6.3	6
3	Write 20 + 8 + 0.1 as a decimal.	28.1
4	Value of the 4 in 345	40
5	How many tenths in 1?	10
6	3.8	0.8
7	25.1	20
8	17.4	7
9	40.6	0.6
10	30 + 2 + 0.9	32.9
11	5 + 0.4	5.4
12	60 + 0.3	60.3
13	8 ones and 2 tenths	8.2
14	Write 73.5 in expanded form.	70 + 3 + 0.5

WEEKLY CHECK · ANSWER KEY
After Lesson 7.4

About 15 minutes

Weekly Check 15: Tenths — Answer Key

Answers

#	Question	Answer
1	$^{7}/_{10}$ as a decimal	0.7
2	3.9 as a mixed number	$3\,^{9}/_{10}$
3	Write 15.6 in words.	fifteen and six tenths
4	Value of the 4 in 2.4	0.4
5	Name point A. (figure)	6.3
6	Write 40 + 2 + 0.8 as a decimal.	42.8

LESSON 7.5 · TEACHER PLAN
Week 2 · Lesson 1 of 4
Unit 7: Decimals

Lesson 7.5: Hundredths

Student goal	I can write hundredths as fractions and decimals, like $^{37}/_{100} = 0.37$.
Objective	Students write hundredths as fractions and decimals.
Standard	4.NF.6 — Use decimal notation for fractions with denominator 100.
Pacing	Unit 7, week 2, lesson 1 of 4
Time	About 50 minutes
Materials	Hundred grids, coins, workbook Lesson 7.5
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Cents in a dollar.
Launch	5	Read Let's Explore with coins.
Teach	15	Shade 37 squares on a hundred grid. Work the example and Watch Out using pennies and dimes.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Writing 5/100 as 0.5.

SUPPORT

- Use money: dimes are tenths, pennies are hundredths.

EXTEND

- Write 125 hundredths as a decimal.

STUDENT LESSON 7.5 · WITH ANSWERS

Lesson 7.5: Hundredths

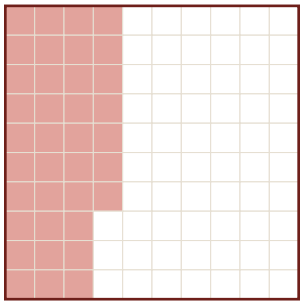
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.5 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.5, Part A.
- [ ] Complete the rest of Workbook Lesson 7.5 (Parts B–E).

LET'S EXPLORE

A dollar is 100 cents. If you have 37 cents, what part of a dollar do you have?

Split a whole into 100 equal parts. Each part is one hundredth. 37 hundredths = $^{37}/_{100}$ = 0.37.



37 of 100 squares shaded: 0.37

Ones		Tenths	Hundredths
0	.	3	7

WORKED EXAMPLE · HUNDREDTHS WITH A ZERO

Write $^{5}/_{100}$ as a decimal.

1 5 hundredths: 0 tenths and 5 hundredths.

2 Write a 0 in the tenths place.

Answer: 0.05

! WATCH OUT

0.5 and 0.05 are different! 0.5 is 5 tenths (50 cents). 0.05 is 5 hundredths (5 cents).

★ KEY IDEA

Hundredths use two places after the decimal point.

✎ TRY IT

Answers are shown in color.

1. $^{48}/_{100}$ as a decimal

Answer: 0.48

2. 0.09 as a fraction

Answer: $^{9}/_{100}$

3. $^{3}/_{100}$ as a decimal

Answer: 0.03

Workbook Answer Key · Lesson 7.5

#	Question	Answer
1	$^{48}/_{100}$ as a decimal	0.48
2	0.09 as a fraction	$^{9}/_{100}$
3	$^{3}/_{100}$ as a decimal	0.03
4	How many cents in a dollar?	100
5	$^{7}/_{10}$ as a decimal	0.7
6	(figure)	$^{25}/_{100} = 0.25$
7	(figure)	$^{62}/_{100} = 0.62$
8	(figure)	$^{8}/_{100} = 0.08$
9	(figure)	$^{90}/_{100} = 0.90$
10	$^{13}/_{100}$	0.13
11	$^{70}/_{100}$	0.70
12	$^{4}/_{100}$	0.04
13	$^{99}/_{100}$	0.99
14	Explain the difference between 0.4 and 0.04.	0.4 is 4 tenths (40 hundredths); 0.04 is 4 hundredths.

LESSON 7.6 · TEACHER PLAN
Week 2 · Lesson 2 of 4
Unit 7: Decimals

Lesson 7.6: Equivalent Tenths and Hundredths

Student goal	I can show that tenths and hundredths can be equivalent, like $^{3}/_{10} = {}^{30}/_{100}$ and 0.3 = 0.30.
Objective	Students show tenths/hundredths equivalence.
Standard	4.NF.5, 4.NF.6 — Express a fraction with denominator 10 as an equivalent fraction with denominator 100.
Pacing	Unit 7, week 2, lesson 2 of 4
Time	About 50 minutes
Materials	Dimes and pennies, hundred grids, workbook Lesson 7.6
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	1/2 = ?/10.
Launch	5	Read Let's Explore with real coins.
Teach	15	Shade 3 columns and count squares (30). Say: "Each tenth is 10 hundredths."
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking 0.30 is greater than 0.3.

SUPPORT

- Coins.

EXTEND

- Write 1.5 as hundredths (150/100).

STUDENT LESSON 7.6 · WITH ANSWERS

Lesson 7.6: Equivalent Tenths and Hundredths

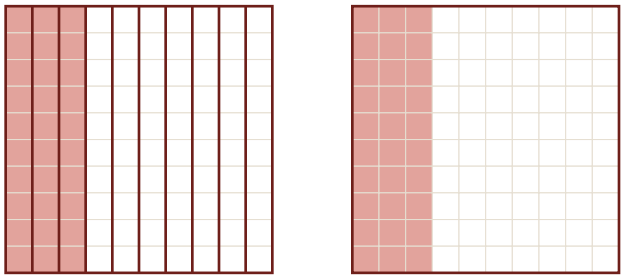
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.6 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.6, Part A.
- [ ] Complete the rest of Workbook Lesson 7.6 (Parts B–E).

LET'S EXPLORE

Is 3 dimes the same amount of money as 30 pennies?

Yes! 3 dimes = 30 cents = 30 pennies. In the same way, $^{3}/_{10} = {}^{30}/_{100}$: each tenth is 10 hundredths.



0.3 (3 columns) = 0.30 (30 squares)

WORKED EXAMPLE · RENAME TENTHS AS HUNDREDTHS

$^{6}/_{10} = {}^{?}/_{100}$

1 Multiply top and bottom by 10.

Answer: $^{60}/_{100}$; 0.6 = 0.60

★ KEY IDEA

A zero at the end of a decimal does not change its value: 0.3 = 0.30.

✎ TRY IT

Answers are shown in color.

1. $^{9}/_{10} = {}^{?}/_{100}$

Answer: 90

2. $^{40}/_{100} = {}^{?}/_{10}$

Answer: 4

3. 0.5 = 0.___

Answer: 50

Workbook Answer Key · Lesson 7.6

#	Question	Answer
1	$^{9}/_{10} = {}^{?}/_{100}$	90
2	$^{40}/_{100} = {}^{?}/_{10}$	4
3	0.5 = 0.___	50
4	$^{1}/_{2} = {}^{?}/_{10}$	5
5	How many pennies equal 1 dime?	10
6	$^{2}/_{10} = {}^{?}/_{100}$	20
7	$^{7}/_{10} = {}^{?}/_{100}$	70
8	$^{1}/_{10} = {}^{?}/_{100}$	10
9	$^{50}/_{100} = {}^{?}/_{10}$	5
10	$^{80}/_{100} = {}^{?}/_{10}$	8
11	$^{30}/_{100} = {}^{?}/_{10}$	3
12	0.6 and 0.60 A. equal B. not equal	equal
13	0.6 and 0.06 A. equal B. not equal	not equal
14	2.4 and 2.40 A. equal B. not equal	equal
15	0.70 and 0.07 A. equal B. not equal	not equal
16	Use money to explain why $^{5}/_{10} = {}^{50}/_{100}$.	5 dimes = 50 cents = 50 pennies.

LESSON 7.7 · TEACHER PLAN
Week 2 · Lesson 3 of 4
Unit 7: Decimals

Lesson 7.7: Adding Tenths and Hundredths

Student goal	I can add a fraction in tenths to a fraction in hundredths, like $^{3}/_{10} + {}^{4}/_{100}$.
Objective	Students add fractions with denominators 10 and 100.
Standard	4.NF.5 — Add two fractions with respective denominators 10 and 100.
Pacing	Unit 7, week 2, lesson 3 of 4
Time	About 50 minutes
Materials	Coins, workbook Lesson 7.7
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	4/10 = ?/100.
Launch	5	Read Let's Explore with coins.
Teach	15	Work both examples. Say: Make the pieces the same size first.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Adding 3/10 + 4/100 = 7/110.

SUPPORT

- Use money.

EXTEND

- Add three: 3/10 + 4/100 + 1/10.

STUDENT LESSON 7.7 · WITH ANSWERS

Lesson 7.7: Adding Tenths and Hundredths

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.7 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.7, Part A.
- [ ] Complete the rest of Workbook Lesson 7.7 (Parts B–E).

LET'S EXPLORE

You have 3 dimes and 4 pennies. What part of a dollar is that?

To add tenths and hundredths, first make the denominators the same. Change tenths to hundredths.

WORKED EXAMPLE · ADD

$\frac{3}{10} + \frac{4}{100}$

1 $\frac{3}{10} = \frac{30}{100}$.

2 $\frac{30}{100} + \frac{4}{100} = \frac{34}{100}$.

Answer: $\frac{34}{100} = 0.34$

WORKED EXAMPLE · ADD WITH REGROUPING

$\frac{7}{10} + \frac{45}{100}$

1 $\frac{7}{10} = \frac{70}{100}$.

2 $\frac{70}{100} + \frac{45}{100} = \frac{115}{100} = 1\frac{15}{100}$.

Answer: 1.15

★ KEY IDEA

Rename tenths as hundredths, then add the numerators.

✎ TRY IT

Answers are shown in color.

1. $\frac{2}{10} + \frac{5}{100}$

Answer: $\frac{25}{100}$

2. $\frac{6}{10} + \frac{30}{100}$

Answer: $^{90}/_{100}$

3. $^{8}/_{10} + {}^{35}/_{100}$

Answer: $^{115}/_{100} = 1\,{}^{15}/_{100}$

Workbook Answer Key · Lesson 7.7

#	Question	Answer
1	$^{2}/_{10} + {}^{5}/_{100}$	$^{25}/_{100}$
2	$^{6}/_{10} + {}^{30}/_{100}$	$^{90}/_{100}$
3	$^{8}/_{10} + {}^{35}/_{100}$	$^{115}/_{100} = 1\,{}^{15}/_{100}$
4	$^{4}/_{10} = {}^{?}/_{100}$	40
5	$^{23}/_{100} + {}^{41}/_{100}$	$^{64}/_{100}$
6	$^{1}/_{10} + {}^{6}/_{100}$	$^{16}/_{100}$
7	$^{4}/_{10} + {}^{22}/_{100}$	$^{62}/_{100}$
8	$^{5}/_{10} + {}^{50}/_{100}$	1
9	$^{7}/_{10} + {}^{3}/_{100}$	$^{73}/_{100}$
10	$^{9}/_{10} + {}^{15}/_{100}$	$1\,{}^{5}/_{100}$
11	$^{6}/_{10} + {}^{40}/_{100}$	1
12	Ellie walked $^{5}/_{10}$ mile. Then she walked $^{25}/_{100}$ mile. How far did she walk? Write a fraction and a decimal.	$^{75}/_{100} = 0.75$ mile
13	$^{2}/_{10} + {}^{68}/_{100}$	$^{88}/_{100} = 0.88$

LESSON 7.8 · TEACHER PLAN
Week 2 · Lesson 4 of 4
Unit 7: Decimals

Lesson 7.8: Decimals in Different Forms

Student goal	I can write decimals in standard, word, fraction, and expanded form.
Objective	Students write decimals in multiple forms.
Standard	4.NF.6 — Read and write decimals to hundredths.
Pacing	Unit 7, week 2, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 7.8
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	45/100 as a decimal.
Launch	5	Read Let's Explore.
Teach	15	Fill the table together. Work the example.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Reading 6.08 as ‘six and eight tenths’.

SUPPORT

- Look at the last place to name the decimal.

EXTEND

- Find a real decimal in a sports page.

STUDENT LESSON 7.8 · WITH ANSWERS

Lesson 7.8: Decimals in Different Forms

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.8 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.8, Part A.
- [ ] Complete the rest of Workbook Lesson 7.8 (Parts B–F).

LET'S EXPLORE

The fastest 100-meter dash time at a school meet was 13.45 seconds. How do you say that number?

Form	13.45
Standard	13.45
Word	thirteen and forty-five hundredths
Fraction / mixed number	$13\,\frac{45}{100}$
Expanded	10 + 3 + 0.4 + 0.05

WORKED EXAMPLE · WRITE IN WORDS

Write 6.08 in words.

1 Whole number: six.

2 “and” for the point.

3 08 hundredths = eight hundredths.

Answer: six and eight hundredths

★ KEY IDEA

Read the decimal part as a whole number, then say the last place: 0.45 is “forty-five hundredths.”

✎ TRY IT

Answers are shown in color.

1. Write 2.7 in words.

Answer: two and seven tenths

2. Write “nine and twelve hundredths” as a decimal.

Answer: 9.12

3. Write 5.36 in expanded form.

Answer: 5 + 0.3 + 0.06

Workbook Answer Key · Lesson 7.8

#	Question	Answer
1	Write 2.7 in words.	two and seven tenths
2	Write “nine and twelve hundredths” as a decimal.	9.12
3	Write 5.36 in expanded form.	5 + 0.3 + 0.06
4	$^{45}/_{100}$ as a decimal	0.45
5	0.3 = 0.___	30
6	four and six tenths	4.6
7	twenty and three hundredths	20.03
8	$7\,^{81}/_{100}$	7.81
9	1 + 0.5 + 0.02	1.52
10	0.29	twenty-nine hundredths
11	14.6	fourteen and six tenths
12	38.04	30 + 8 + 0.04
13	6.75	6 + 0.7 + 0.05
14	Write 12.09 in words and as a mixed number.	twelve and nine hundredths; $12\,^{9}/_{100}$

WEEKLY CHECK · ANSWER KEY
After Lesson 7.8

About 15 minutes

Weekly Check 16: Hundredths — Answer Key

Answers

#	Question	Answer
1	$\frac{29}{100}$ as a decimal	0.29
2	$\frac{6}{100}$ as a decimal	0.06
3	$\frac{4}{10} = \frac{?}{100}$	40
4	$\frac{3}{10} + \frac{52}{100}$	$\frac{82}{100}$
5	Write 7.08 in words.	seven and eight hundredths
6	Write 3.45 in expanded form.	3 + 0.4 + 0.05

LESSON 7.9 · TEACHER PLAN
Week 3 · Lesson 1 of 4
Unit 7: Decimals

Lesson 7.9: Comparing Decimals with Models

Student goal	I can compare two decimals using hundred grids and number lines.
Objective	Students compare decimals with models.
Standard	4.NF.7 — Compare two decimals to hundredths by reasoning about their size; same whole.
Pacing	Unit 7, week 3, lesson 1 of 4
Time	About 50 minutes
Materials	Hundred grids, workbook Lesson 7.9
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	0.4 = 0.40.
Launch	5	Read Let's Explore.
Teach	15	Shade 0.4 and 0.35 on grids.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking more digits means greater.

SUPPORT

- Shade both.

EXTEND

- Find a decimal between 0.4 and 0.5.

STUDENT LESSON 7.9 · WITH ANSWERS

Lesson 7.9: Comparing Decimals with Models

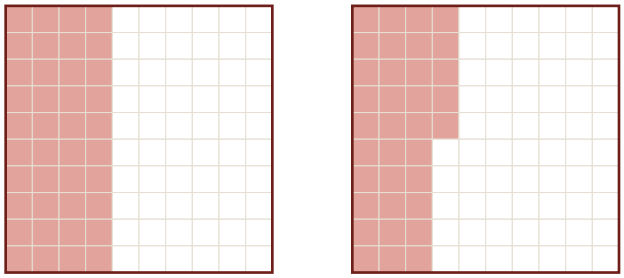
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.9 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.9, Part A.
- [ ] Complete the rest of Workbook Lesson 7.9 (Parts B–D).

LET'S EXPLORE

Is 0.4 or 0.35 greater? 35 is bigger than 4... but is 0.35 bigger than 0.4?

Models show the truth: 0.4 shades 40 squares, and 0.35 shades only 35 squares. So 0.4 > 0.35.



0.4 (left) and 0.35 (right)

! WATCH OUT

Longer decimals are not always greater. Compare the same places.

★ KEY IDEA

Decimals must refer to the same whole to be compared. Shade both on the same-size grid.

✎ TRY IT

Answers are shown in color.

1. 0.6 ◯ 0.59

Answer: >

2. 0.25 ◯ 0.3

Answer: <

3. 0.7 ◯ 0.70

Answer: =

Workbook Answer Key · Lesson 7.9

#	Question	Answer
1	0.6 ◯ 0.59	>
2	0.25 ◯ 0.3	<
3	0.7 ◯ 0.70	=
4	0.4 = 0.___	40
5	$^{35}/_{100}$ ◯ $^{40}/_{100}$	<
6	0.5 ◯ 0.45 (figure)	>
7	0.2 ◯ 0.12 (figure)	>
8	0.8 ◯ 0.80 (figure)	=
9	0.33 ◯ 0.3 (figure)	>
10	Lillie says 0.15 > 0.5 because 15 > 5. Is she right? Explain.	No: 0.5 = 0.50 and 50 hundredths > 15 hundredths.

LESSON 7.10 · TEACHER PLAN
Week 3 · Lesson 2 of 4
Unit 7: Decimals

Lesson 7.10: Comparing Decimals with Place Value

Student goal	I can compare decimals by lining up place values.
Objective	Students compare decimals using place value.
Standard	4.NF.7 — Compare decimals; record with >, =, <.
Pacing	Unit 7, week 3, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 7.10
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	0.5 vs 0.45.
Launch	5	Read Let's Explore.
Teach	15	Line up decimal points in a chart. Say: “Same places, from the left.”
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Comparing 3.09 and 3.1 as 309 vs 31.

SUPPORT

- Fill zeros.

EXTEND

- Order sports times.

STUDENT LESSON 7.10 · WITH ANSWERS

Lesson 7.10: Comparing Decimals with Place Value

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.10 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.10, Part A.
- [ ] Complete the rest of Workbook Lesson 7.10 (Parts B–D).

LET'S EXPLORE

Two runners finished in 12.48 seconds and 12.5 seconds. Who was faster (less time)?

Line up the decimal points. Write a zero so both have hundredths. Compare from the left like whole numbers.

	Tens	Ones	.	Tenths	Hundredths
12.48	1	2	.	4	8
12.50	1	2	.	5	0

WORKED EXAMPLE · COMPARE

12.48 vs 12.5

1 Tens and ones are the same.

2 Tenths: 4 < 5.

Answer: 12.48 < 12.5; the 12.48 runner was faster

★ KEY IDEA

Line up the decimal points, fill in zeros, compare from the left.

✎ TRY IT

Answers are shown in color.

1. 3.09 ◯ 3.1

Answer: <

2. 5.6 ◯ 5.06

Answer: >

3. 10.4 ◯ 9.99

Answer: >

Workbook Answer Key · Lesson 7.10

#	Question	Answer
1	3.09 ◯ 3.1	<
2	5.6 ◯ 5.06	>
3	10.4 ◯ 9.99	>
4	0.5 ◯ 0.45	>
5	Write 0.7 with hundredths.	0.70
6	2.4 ◯ 2.38	>
7	7.05 ◯ 7.5	<
8	0.9 ◯ 0.90	=
9	13.2 ◯ 12.99	>
10	4.67 ◯ 4.76	<
11	8.1 ◯ 8.01	>
12	Which is greater, 6.7 or 6.68? Explain with place value.	6.7 (6.70): 7 tenths > 6 tenths.

LESSON 7.11 · TEACHER PLAN
Week 3 · Lesson 3 of 4
Unit 7: Decimals

Lesson 7.11: Ordering Decimals

Student goal	I can order three or more decimals.
Objective	Students order decimals.
Standard	4.NF.7 — Compare decimals to hundredths.
Pacing	Unit 7, week 3, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 7.11
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	3.5 vs 3.45.
Launch	5	Read Let's Explore.
Teach	15	Work the example in a place-value chart.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Ordering by number of digits.

SUPPORT

- Fill zeros.

EXTEND

- Order 5 decimals.

STUDENT LESSON 7.11 · WITH ANSWERS

Lesson 7.11: Ordering Decimals

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.11 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.11, Part A.
- [ ] Complete the rest of Workbook Lesson 7.11 (Parts B–F).

LET'S EXPLORE

Three bean plants grew 2.35 in., 2.5 in., and 2.07 in. List them from shortest to tallest growth.

WORKED EXAMPLE · ORDER

2.35, 2.5, 2.07 from least to greatest

1 Write with hundredths: 2.35, 2.50, 2.07.

2 Compare tenths: 0 < 3 < 5.

Answer: 2.07, 2.35, 2.5

★ KEY IDEA

Write all decimals with the same number of places, then order like whole numbers.

✎ TRY IT

Answers are shown in color.

1. Order least to greatest: 0.6, 0.06, 0.66

Answer: 0.06, 0.6, 0.66

2. Order greatest to least: 1.2, 1.12, 1.21

Answer: 1.21, 1.2, 1.12

3. Which is least: 4.4, 4.04, 4.40?

Answer: 4.04

Workbook Answer Key · Lesson 7.11

#	Question	Answer
1	Order least to greatest: 0.6, 0.06, 0.66	0.06, 0.6, 0.66
2	Order greatest to least: 1.2, 1.12, 1.21	1.21, 1.2, 1.12
3	Which is least: 4.4, 4.04, 4.40?	4.04
4	3.5 ◯ 3.45	>
5	Write 1.3 with hundredths.	1.30
6	0.8, 0.18, 0.81	0.18, 0.8, 0.81
7	3.4, 3.04, 3.44	3.04, 3.4, 3.44
8	12.1, 11.9, 12.01	11.9, 12.01, 12.1
9	5.5, 5.05, 5.55	5.55, 5.5, 5.05
10	0.7, 0.69, 0.77	0.77, 0.7, 0.69
11	Long jumps: Ellie 2.48 m, Lillie 2.6 m, cousin 2.53 m. Who jumped farthest? Order the jumps.	Lillie; 2.6, 2.53, 2.48
12	Order 9.09, 9.9, 9.19 from least to greatest.	9.09, 9.19, 9.9

LESSON 7.12 · TEACHER PLAN
Week 3 · Lesson 4 of 4
Unit 7: Decimals

Lesson 7.12: Decimals and Fractions on a Number Line

Student goal	I can place decimals and fractions on the same number line and compare them.
Objective	Students compare fractions and decimals.
Standard	4.NF.6, 4.NF.7 — Use decimal notation for fractions; compare.
Pacing	Unit 7, week 3, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 7.12
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	1/4 = ?/100.
Launch	5	Read Let's Explore.
Teach	15	Convert both to hundredths.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Treating 3/4 as 0.34.

SUPPORT

- Benchmark list: 1/2 = 0.5, 1/4 = 0.25, 3/4 = 0.75.

EXTEND

- Convert 3/20 to hundredths.

STUDENT LESSON 7.12 · WITH ANSWERS

Lesson 7.12: Decimals and Fractions on a Number Line

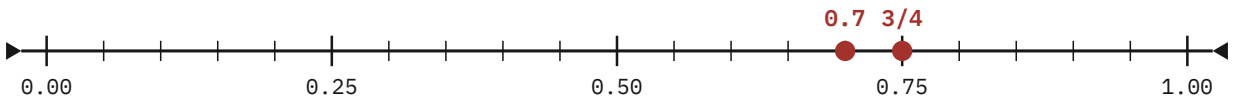
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.12 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.12, Part A.
- [ ] Complete the rest of Workbook Lesson 7.12 (Parts B–E).

LET'S EXPLORE

Which is greater, $^{3}/_{4}$ or 0.7?

Write both in hundredths. $^{3}/_{4} = {}^{75}/_{100} = 0.75$. $0.75 > 0.70$, so $^{3}/_{4} > 0.7$.



WORKED EXAMPLE · COMMON BENCHMARK DECIMALS

Write $^{1}/_{2}$, $^{1}/_{4}$, $^{1}/_{5}$ as decimals.

1 $^{1}/_{2} = {}^{50}/_{100} = 0.5$.

2 $^{1}/_{4} = {}^{25}/_{100} = 0.25$.

3 $^{1}/_{5} = {}^{2}/_{10} = 0.2$.

Answer: 0.5, 0.25, 0.2

★ KEY IDEA

Change fractions to tenths or hundredths, then compare them as decimals.

✎ TRY IT

Answers are shown in color.

1. $^{1}/_{2}$ ◯ 0.45

Answer: >

2. $^{2}/_{5}$ ◯ 0.4

Answer: =

3. 0.8 ◯ $^{3}/_{4}$

Answer: >

Workbook Answer Key · Lesson 7.12

#	Question	Answer
1	$^{1}/_{2}$ ◯ 0.45	>
2	$^{2}/_{5}$ ◯ 0.4	=
3	0.8 ◯ $^{3}/_{4}$	>
4	$^{1}/_{4} = {}^{?}/_{100}$	25
5	$^{3}/_{5} = {}^{?}/_{10}$	6
6	$^{1}/_{2}$	0.5
7	$^{3}/_{4}$	0.75
8	$^{4}/_{5}$	0.8
9	$^{3}/_{10}$	0.3
10	$^{1}/_{4}$ ◯ 0.3	<
11	0.6 ◯ $^{3}/_{5}$	=
12	$^{7}/_{10}$ ◯ 0.07	>
13	0.5 ◯ $^{2}/_{5}$	>
14	Order from least to greatest: 0.6, $^{1}/_{2}$, 0.55	$^{1}/_{2}$, 0.55, 0.6

WEEKLY CHECK · ANSWER KEY
After Lesson 7.12

About 15 minutes

Weekly Check 17: Comparing Decimals — Answer Key

Answers

#	Question	Answer
1	0.6 ◯ 0.58	>
2	4.07 ◯ 4.7	<
3	2.30 ◯ 2.3	=
4	$^1/_4$ ◯ 0.2	>
5	Order: 1.5, 1.05, 1.55	1.05, 1.5, 1.55
6	Which is greatest: 0.9, 0.89, 0.09?	0.9

LESSON 7.13 · TEACHER PLAN
Week 4 · Lesson 1 of 4
Unit 7: Decimals

Lesson 7.13: Money and Decimals

Student goal	I can write money amounts as decimals and connect dollars, dimes, and pennies to ones, tenths, and hundredths.
Objective	Students relate money to decimal place value.
Standard	4.NF.6, 4.MD.2 — Use decimal notation; solve money problems.
Pacing	Unit 7, week 4, lesson 1 of 4
Time	About 50 minutes
Materials	Play or real money, workbook Lesson 7.13
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	25 + 25 + 10.
Launch	5	Read Let's Explore with money.
Teach	15	Build amounts with money and write each one. Say: “Dimes are tenths; pennies are hundredths.”
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Writing $3.5 for $3.05.

SUPPORT

- Use real coins.

EXTEND

- Fewest coins to make $0.87.

STUDENT LESSON 7.13 · WITH ANSWERS

Lesson 7.13: Money and Decimals

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.13 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.13, Part A.
- [ ] Complete the rest of Workbook Lesson 7.13 (Parts B–E).

LET'S EXPLORE

You have 2 one-dollar bills, 3 dimes, and 4 pennies. How do you write that amount?



Dollars are ones, dimes are tenths of a dollar, and pennies are hundredths of a dollar. 2 dollars, 3 dimes, 4 pennies = $2.34.

Coin or bill	Value	Decimal
dollar	100¢	$1.00
quarter	25¢	$0.25
dime	10¢	$0.10
nickel	5¢	$0.05
penny	1¢	$0.01

WORKED EXAMPLE · COUNT MONEY

3 quarters and 2 dimes

1 Quarters: 75¢.

2 Dimes: 20¢.

3 Total: 95¢.

Answer: $0.95

★ KEY IDEA

Write money with a dollar sign and two decimal places: $3.05, not $3.5 for three dollars and five cents.

✎ TRY IT

Answers are shown in color.

1. 4 dollars, 1 dime, 7 pennies

Answer: $4.17

2. 2 quarters, 1 nickel

Answer: $0.55

3. What fraction of a dollar is 1 quarter?

Answer: $\frac{25}{100} = \frac{1}{4}$

Workbook Answer Key · Lesson 7.13

#	Question	Answer
1	4 dollars, 1 dime, 7 pennies	$4.17
2	2 quarters, 1 nickel	$0.55
3	What fraction of a dollar is 1 quarter?	$\frac{25}{100} = \frac{1}{4}$
4	25 + 25 + 10	60
5	0.6 = 0.___	60
6	(figure)	$1.50
7	(figure)	$0.43
8	(figure)	$3.07
9	(figure)	$1.00
10	eight dollars and nine cents	$8.09
11	47 cents	$0.47
12	12 dollars and 50 cents	$12.50
13	6 dimes	$0.60
14	Show $1.36 two different ways with coins and bills.	Answers vary.

LESSON 7.14 · TEACHER PLAN
Week 4 · Lesson 2 of 4
Unit 7: Decimals

Lesson 7.14: Adding Money

Student goal	I can add money amounts by lining up the decimal points.
Objective	Students add money amounts.
Standard	4.MD.2 — Solve word problems involving money with decimals.
Pacing	Unit 7, week 4, lesson 2 of 4
Time	About 50 minutes
Materials	Play money, workbook Lesson 7.14
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	275 + 125.
Launch	5	Read Let's Explore.
Teach	15	Work the example with money first, then on paper.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Misaligning when one amount is whole dollars.

SUPPORT

- Write .00 on whole dollars.

EXTEND

- Plan a $20 shopping list.

STUDENT LESSON 7.14 · WITH ANSWERS

Lesson 7.14: Adding Money

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.14 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.14, Part A.
- [ ] Complete the rest of Workbook Lesson 7.14 (Parts B–E).

LET'S EXPLORE

A notebook costs $2.75 and a pack of pens costs $3.49. What is the total?

Add money like whole numbers: line up the decimal points (dollars under dollars, cents under cents), add from the right, regroup, and bring the decimal point straight down.

WORKED EXAMPLE · ADD MONEY

$2.75 + $3.49

$$\begin{array}{r} {\scriptstyle 1\ 1\ \ } \\ 275 \\ +349 \\ \hline 624 \end{array}$$

1 Think in cents: 275 + 349 = 624 cents.

2 Write the decimal point: $6.24.

Answer: $6.24

! WATCH OUT

Line up the decimal points, not the right edges, when one amount has no cents: $5 + $1.25 → $5.00 + $1.25.

★ KEY IDEA

Line up the decimal points, add like whole numbers, then place the decimal point in the answer.

✎ TRY IT

Answers are shown in color.

1. $4.20 + $3.65

Answer: $7.85

2. $1.89 + $2.47

Answer: $4.36

3. $10 + $3.75

Answer: $13.75

Workbook Answer Key · Lesson 7.14

#	Question	Answer
1	$4.20 + $3.65	$7.85
2	$1.89 + $2.47	$4.36
3	$10 + $3.75	$13.75
4	275 + 125	400
5	$1.00 = ___ cents	100
6	$3.25 + $4.50	$7.75
7	$6.48 + $2.39	$8.87
8	$12.95 + $4.06	$17.01
9	$0.89 + $0.76	$1.65
10	$15.00 + $8.45	$23.45
11	$23.60 + $17.85	$41.45
12	Ellie buys a book for $7.95 and a bookmark for $1.29. How much does she spend?	$9.24
13	Lillie earns $4.50 on Monday, $5.25 on Tuesday, and $3.75 on Wednesday. How much does she earn?	$13.50
14	$8.67 + $5.48	$14.15

LESSON 7.15 · TEACHER PLAN
Week 4 · Lesson 3 of 4
Unit 7: Decimals

Lesson 7.15: Subtracting Money and Making Change

Student goal	I can subtract money amounts and make change.
Objective	Students subtract money and make change.
Standard	4.MD.2 — Solve money word problems.
Pacing	Unit 7, week 4, lesson 3 of 4
Time	About 50 minutes
Materials	Play money, workbook Lesson 7.15
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	1000 − 365.
Launch	5	Read Let's Explore. Play store.
Teach	15	Work both examples; practice counting up with coins.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Forgetting to write $20 as $20.00.

SUPPORT

- Count up with real money.

EXTEND

- Run a pretend store for 10 minutes.

STUDENT LESSON 7.15 · WITH ANSWERS

Lesson 7.15: Subtracting Money and Making Change

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.15 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.15, Part A.
- [ ] Complete the rest of Workbook Lesson 7.15 (Parts B–E).

LET'S EXPLORE

A game costs $13.65. You pay with a $20 bill. How much change do you get?

WORKED EXAMPLE · SUBTRACT

$20.00 − $13.65

```
  2,000
- 1,365
-------
    635
```

1 Write $20 as $20.00.

2 2000 − 1365 = 635 cents.

Answer: $6.35

WORKED EXAMPLE · COUNT UP

Make change for $13.65 from $20.

1 $13.65 + $0.35 = $14.00.

2 $14.00 + $6.00 = $20.00.

3 $0.35 + $6.00 = $6.35.

Answer: $6.35

★ KEY IDEA

Subtract to find change, or count up from the price to the amount paid.

✎ TRY IT

Answers are shown in color.

1. $9.50 − $4.25

Answer: $5.25

2. $10.00 − $6.79

Answer: $3.21

3. Change from $5 for a $2.38 snack

Answer: $2.62

Workbook Answer Key · Lesson 7.15

#	Question	Answer
1	$9.50 − $4.25	$5.25
2	$10.00 − $6.79	$3.21
3	Change from $5 for a $2.38 snack	$2.62
4	1000 − 365	635
5	$2.65 + ___ = $3.00	$0.35
6	$8.75 − $3.40	$5.35
7	$6.20 − $2.85	$3.35
8	$15.00 − $7.62	$7.38
9	$20.00 − $11.49	$8.51
10	$5.05 − $1.98	$3.07
11	$50.00 − $23.75	$26.25
12	Price $4.70, paid $5.00	$0.30
13	Price $12.15, paid $20.00	$7.85
14	Price $0.68, paid $1.00	$0.32
15	Price $36.40, paid $40.00	$3.60
16	Mom pays for $27.83 of groceries with $50. How much change?	$22.17

LESSON 7.16 · TEACHER PLAN
Week 4 · Lesson 4 of 4
Unit 7: Decimals

Lesson 7.16: Decimal and Money Problem Solving

Student goal	I can solve multi-step problems with decimals and money.
Objective	Students solve multi-step decimal and money problems.
Standard	4.MD.2, 4.NF.7 — Solve word problems involving money and decimals.
Pacing	Unit 7, week 4, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 7.16, Unit 7 Review
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Two computations.
Launch	5	Read Let's Explore.
Teach	15	Work the example: estimate, then compute.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook Lesson 7.16 and the Unit 7 Review.
Exit Ticket	3	Check. Tomorrow: Unit 7 Test.

COMMON MISCONCEPTIONS

- Skipping the estimate and missing errors.

SUPPORT

- Use play money.

EXTEND

- Plan a real grocery list with a budget.

STUDENT LESSON 7.16 · WITH ANSWERS

Lesson 7.16: Decimal and Money Problem Solving

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.16 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.16, Part A.
- [ ] Complete the rest of Workbook Lesson 7.16 (Parts B–D).

LET'S EXPLORE

You have $20 for the school book fair. Books cost $6.95, $4.50, and $7.25. Can you buy all three?

WORKED EXAMPLE · ADD, THEN COMPARE

Book fair

1 $6.95 + $4.50 + $7.25 = $18.70.

2 $18.70 < $20.00.

Answer: Yes, with $1.30 left

★ KEY IDEA

Estimate first ($7 + $5 + $7 = $19), then find the exact answer.

✎ TRY IT

Answers are shown in color.

1. 3 notebooks at $1.25 each. Total?

Answer: $3.75

2. $10 − ($3.49 + $2.75)

Answer: $3.76

3. Which costs less: 4 pens at $0.60 or a 4-pack for $2.25?

Answer: the 4-pack ($2.25 < $2.40)

Workbook Answer Key · Lesson 7.16

#	Question	Answer
1	3 notebooks at $1.25 each. Total?	$3.75
2	$10 − ($3.49 + $2.75)	$3.76
3	Which costs less: 4 pens at $0.60 or a 4-pack for $2.25?	the 4-pack ($2.25 < $2.40)
4	$4.50 + $7.25	$11.75
5	$20.00 − $18.70	$1.30
6	Ellie buys 2 smoothies at $3.75 each and a muffin for $2.40. She pays with $10. How much change?	$0.10
7	Lillie saves $2.50 each week for 6 weeks. Does she have enough for a $14.95 game? Explain.	Yes; $15.00 ≥ $14.95, with $0.05 left.
8	Gas costs $3.09 a gallon in one town and $2.98 in another. Which is cheaper, and by how much?	$2.98, by $0.11
9	A family's lunch costs $8.45, $6.90, $7.25, and $5.40. What is the total?	$28.00
10	Plan a snack order under $10 from: chips $1.25, juice $1.75, apple $0.80, sandwich $4.50. Show your total and change.	Answers vary.

UNIT REVIEW · ANSWER KEY
Study before the unit test

About 35 minutes

Unit 7 Review — Answer Key

Homework or a study session before the test.

Answers

#	Question	Answer
1	$^{8}/_{10}$ as a decimal	0.8
2	$^{45}/_{100}$ as a decimal	0.45
3	0.07 as a fraction	$^{7}/_{100}$
4	$^{6}/_{10} + ^{25}/_{100}$	$^{85}/_{100}$
5	3.4 ◯ 3.38	>
6	$^{3}/_{4}$ ◯ 0.8	<
7	Order: 2.2, 2.02, 2.12	2.02, 2.12, 2.2
8	$6.85 + $2.47	$9.32
9	$20.00 − $14.36	$5.64
10	3 tickets at $4.50 each. Pay with $20. Change?	$6.50

UNIT TEST · ANSWER KEY
At the end of the unit

About 45 minutes

Unit 7 Test — Answer Key

Answers

#	Question	Answer
1	$^{3}/_{10}$ as a decimal	0.3
2	$^{57}/_{100}$ as a decimal	0.57
3	$4\,^{9}/_{100}$ as a decimal	4.09
4	Write 12.6 in words.	twelve and six tenths
5	Value of the 8 in 3.08	0.08
6	$^{7}/_{10} = {}^{?}/_{100}$	70
7	$^{4}/_{10} + {}^{35}/_{100}$	$^{75}/_{100}$
8	0.5 ◯ 0.45	>
9	6.09 ◯ 6.9	<
10	$^{1}/_{2}$ ◯ 0.50	=
11	Order least to greatest: 0.4, 0.04, 0.44	0.04, 0.4, 0.44
12	$15.75 + $8.68	$24.43
13	$30.00 − $17.46	$12.54
14	Write the amount: 5 dollars, 3 quarters, 2 pennies	$5.77
15	Ellie buys 2 books at $5.95 each and a pen for $1.35. She pays with $20. How much change does she get?	$6.75
16	Lillie ran 3.45 km Monday and 3.5 km Tuesday. On which day did she run farther? Explain using place value.	Tuesday: 3.50 > 3.45 (5 tenths > 4 tenths).

Scoring

Score	What it means / next step
20–23 (85%+)	Mastered.
16–19	Proficient; correct missed items.
Below 16	Reteach and retest.

Geometry, Angles, and Line Plots

Lines and rays, classifying shapes, symmetry, measuring and drawing angles, and line plots with fractions.

Geometry is the math of shapes and space. You will name lines and angles, sort shapes by their attributes, find lines of symmetry, measure angles with a protractor, and show measurement data on line plots.

Big ideas

- Draw and identify points, lines, segments, rays, and angles.
- Identify parallel and perpendicular lines.
- Classify triangles and quadrilaterals; find lines of symmetry.
- Measure, draw, add, and find unknown angles.
- Make and use line plots with fractional measurements.

Standards

Code	Standard
4.G.1	Draw points, lines, line segments, rays, angles, and perpendicular and parallel lines.
4.G.2	Classify two-dimensional figures by lines and angles; recognize right triangles.
4.G.3	Recognize and draw lines of symmetry.
4.MD.4	Make a line plot of measurements in fractions of a unit; solve problems with it.
4.MD.5	Understand angles and angle measurement in degrees.
4.MD.6	Measure and sketch angles with a protractor.
4.MD.7	Recognize angle measure as additive; find unknown angles.

What students should already know

- Shapes (Grade 3).
- Fractions and fraction operations (Units 5–6).

Materials for the unit

- Protractor
- Ruler
- Index cards (right-angle testers)
- Paper shapes
- Paper plate angle maker

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 4	8.1	Points, Lines, Segments, and Rays
Week 1	Lesson 2 of 4	8.2	Parallel and Perpendicular Lines
Week 1	Lesson 3 of 4	8.3	Classifying Triangles and Quadrilaterals
Week 1	Lesson 4 of 4	8.4	Lines of Symmetry
Week 1	End of week	Check	Weekly Check 18: Lines and Shapes
Week 2	Lesson 1 of 4	8.5	Angles: Right, Acute, and Obtuse
Week 2	Lesson 2 of 4	8.6	Angles and Turns
Week 2	Lesson 3 of 4	8.7	Estimating Angle Measures
Week 2	Lesson 4 of 4	8.8	Classifying Shapes by Angles
Week 2	End of week	Check	Weekly Check 19: Angles
Week 3	Lesson 1 of 4	8.9	Measuring Angles with a Protractor
Week 3	Lesson 2 of 4	8.10	Drawing Angles
Week 3	Lesson 3 of 4	8.11	Adding Angle Measures
Week 3	Lesson 4 of 4	8.12	Finding Unknown Angles
Week 3	End of week	Check	Weekly Check 20: Measuring Angles
Week 4	Lesson 1 of 4	8.13	Line Plots with Fractions
Week 4	Lesson 2 of 4	8.14	Solving Problems with Line Plots
Week 4	Lesson 3 of 4	8.15	Geometry Review: Attributes of Shapes
Week 4	Lesson 4 of 4	8.16	Geometry and Data Problem Solving
Week 4	Before the test	Review	Unit 8 Review
Week 4	End of unit	Test	Unit 8 Test

TEACHING THIS UNIT AT HOME

Hunt for angles and symmetry at home and outside: roof lines, letters on signs, leaves, and tiles.

UNIT VOCABULARY

angle	Two rays with a shared endpoint.
degree	Unit for measuring angles.
parallel	Never meet.
perpendicular	Meet at right angles.
line of symmetry	Folds a figure into matching halves.
line plot	Number line with X marks for data.

LESSON 8.1 · TEACHER PLAN
Week 1 · Lesson 1 of 4
Unit 8: Geometry, Angles, and Line Plots

Lesson 8.1: Points, Lines, Segments, and Rays

Student goal	I can draw and identify points, lines, line segments, and rays.
Objective	Students identify and draw points, lines, segments, and rays.
Standard	4.G.1 — Draw points, lines, line segments, rays, angles, and perpendicular and parallel lines; identify these in two-dimensional figures.
Pacing	Unit 8, week 1, lesson 1 of 4
Time	About 50 minutes
Materials	Ruler, string, flashlight, workbook Lesson 8.1
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Name shapes.
Launch	5	Read Let's Explore. Shine a flashlight; hold a string.
Teach	15	Draw each figure with a ruler and label it. Say: "Count the endpoints."
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Naming a ray from the wrong end.

SUPPORT

- Use a flashlight for rays.

EXTEND

- Find 5 real-world examples of each.

STUDENT LESSON 8.1 · WITH ANSWERS

Lesson 8.1: Points, Lines, Segments, and Rays

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.1 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.1, Part A.
- [ ] Complete the rest of Workbook Lesson 8.1 (Parts B–E).

LET'S EXPLORE

A flashlight beam starts at the flashlight and shines out. A piece of string has two ends. A railroad track seems to go on forever. How are these alike and different?

Geometry has its own vocabulary for straight paths. The difference is in the endpoints.



Name	Endpoints	How to write it
point A	—	A
line segment AB	2	AB with a bar on top
ray CD	1 (C)	CD with an arrow on top (start at the endpoint)
line EF	0	EF with arrows both ways on top

WORKED EXAMPLE · NAME IT

A path starts at point G, goes through H, and keeps going forever. What is it?

1 It has one endpoint (G).

2 It goes on forever in one direction.

Answer: ray GH

★ KEY IDEA

Line: no endpoints. Ray: one endpoint. Segment: two endpoints.

✎ TRY IT

Answers are shown in color.

1. How many endpoints does a ray have?

Answer: 1

2. A pencil's edge from tip to eraser is like a ___.

Answer: line segment

3. Name the figure: two arrows, through points P and Q.

Answer: line PQ

Workbook Answer Key · Lesson 8.1

#	Question	Answer
1	How many endpoints does a ray have?	1
2	A pencil's edge from tip to eraser is like a ___.	line segment
3	Name the figure: two arrows, through points P and Q.	line PQ
4	How many sides does a triangle have?	3
5	Name a shape with 4 sides.	square, rectangle, …
6	(figure)	line segment
7	(figure)	ray
8	(figure)	line
9	(figure)	point
10	Draw line segment XY.	segment with endpoints X and Y
11	Draw ray BA.	ray starting at B through A
12	Explain the difference between a ray and a line segment.	A ray has one endpoint and goes on forever; a segment has two endpoints.

LESSON 8.2 · TEACHER PLAN
Week 1 · Lesson 2 of 4
Unit 8: Geometry, Angles, and Line Plots

Lesson 8.2: Parallel and Perpendicular Lines

Student goal	I can identify and draw parallel, perpendicular, and intersecting lines.
Objective	Students identify and draw parallel, perpendicular, and intersecting lines.
Standard	4.G.1 — Draw and identify perpendicular and parallel lines.
Pacing	Unit 8, week 1, lesson 2 of 4
Time	About 50 minutes
Materials	Ruler, index card (for right angles), workbook Lesson 8.2
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Endpoints review.
Launch	5	Read Let's Explore. Look around the room.
Teach	15	Draw each pair. Use an index card corner to check right angles.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking any crossing lines are perpendicular.

SUPPORT

- Index card corner as a right-angle checker.

EXTEND

- Find perpendicular lines in letters of the alphabet.

STUDENT LESSON 8.2 · WITH ANSWERS

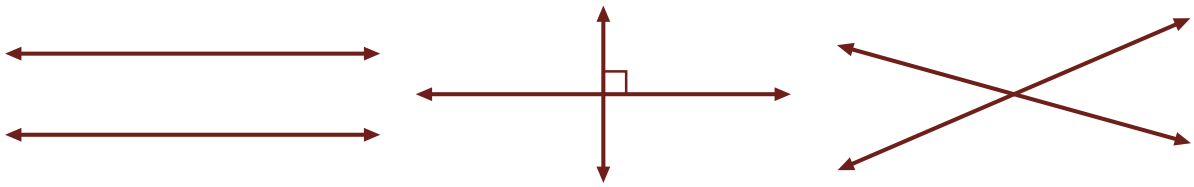
Lesson 8.2: Parallel and Perpendicular Lines

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.2 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.2, Part A.
- [ ] Complete the rest of Workbook Lesson 8.2 (Parts B–E).

LET'S EXPLORE

Look at a notebook page and a window frame. Which lines never meet? Which lines meet in square corners?



parallel perpendicular intersecting

Parallel lines are like railroad tracks: they never meet. Perpendicular lines meet at a square corner, marked with a small square. All perpendicular lines are intersecting, but not all intersecting lines are perpendicular.

WORKED EXAMPLE · FIND THEM IN A SHAPE

In a rectangle, which sides are parallel? Which are perpendicular?



1 Opposite sides never meet: parallel.

2 Sides that meet at corners make right angles: perpendicular.

Answer: Opposite sides are parallel; touching sides are perpendicular.

★ KEY IDEA

Parallel: never meet. Perpendicular: meet at right angles.

✎ TRY IT

Answers are shown in color.

1. Are the two rails of a ladder parallel or perpendicular?

Answer: parallel

2. The corner of a book shows ___ lines.

Answer: perpendicular

3. Can parallel lines intersect?

Answer: No

Workbook Answer Key · Lesson 8.2

#	Question	Answer
1	Are the two rails of a ladder parallel or perpendicular?	parallel
2	The corner of a book shows ___ lines.	perpendicular
3	Can parallel lines intersect?	No
4	How many endpoints does a line have?	0
5	Draw a ray.	ray
6	(figure)	perpendicular
7	(figure)	parallel
8	(figure)	intersecting
9	(figure)	parallel
10	Name something with parallel lines.	e.g. railroad tracks, notebook lines
11	Name something with perpendicular lines.	e.g. window corner, plus sign
12	Draw two parallel lines and a third line perpendicular to both.	Like a ladder rung across rails

LESSON 8.3 · TEACHER PLAN
Week 1 · Lesson 3 of 4
Unit 8: Geometry, Angles, and Line Plots

Lesson 8.3: Classifying Triangles and Quadrilaterals

Student goal	I can sort triangles and quadrilaterals by their sides and angles.
Objective	Students classify triangles and quadrilaterals.
Standard	4.G.2 — Classify two-dimensional figures based on parallel or perpendicular lines and angles of a specified size; recognize right triangles.
Pacing	Unit 8, week 1, lesson 3 of 4
Time	About 50 minutes
Materials	Paper shapes or pattern blocks, workbook Lesson 8.3
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Review.
Launch	5	Read Let's Explore.
Teach	15	Sort paper shapes into groups. Discuss shapes that fit more than one group. Say: “Which features does it have?”
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking a square is not a rectangle.

SUPPORT

- Use a feature checklist.

EXTEND

- Make a Venn diagram of rectangles and rhombuses.

STUDENT LESSON 8.3 · WITH ANSWERS

Lesson 8.3: Classifying Triangles and Quadrilaterals

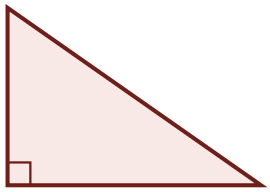
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.3 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.3, Part A.
- [ ] Complete the rest of Workbook Lesson 8.3 (Parts B–E).

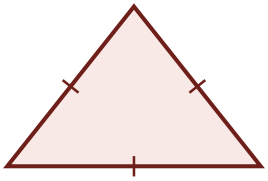
LET'S EXPLORE

A square is a rectangle. Is a rectangle always a square?

Triangles



right triangle (one right angle)

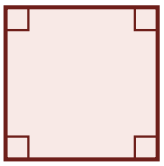


equilateral (all sides equal; all angles acute)



obtuse triangle (one obtuse angle)

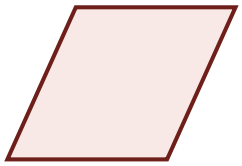
Quadrilaterals



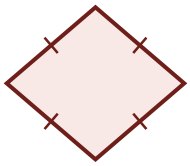
square



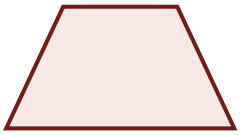
rectangle



parallelogram



rhombus



trapezoid

Shapes can belong to more than one group. A square has 4 right angles (so it is a rectangle), 4 equal sides (so it is a rhombus), and 2 pairs of parallel sides (so it is a parallelogram).

★ KEY IDEA

Classify by sides (equal? parallel? perpendicular?) and by angles (right, acute, obtuse).

✎ TRY IT

Answers are shown in color.

1. Name a quadrilateral with 4 right angles and 4 equal sides.

Answer: square

2. Does a trapezoid always have parallel sides?

Answer: Yes, at least one pair

3. Is every rectangle a square?

Answer: No

Workbook Answer Key · Lesson 8.3

#	Question	Answer
1	Name a quadrilateral with 4 right angles and 4 equal sides.	square
2	Does a trapezoid always have parallel sides?	Yes, at least one pair
3	Is every rectangle a square?	No
4	What do perpendicular lines make?	right angles
5	How many sides does a quadrilateral have?	4
6	(figure)	right triangle
7	(figure)	parallelogram
8	(figure)	trapezoid
9	(figure)	rhombus
10	Every square is a rectangle. A. True B. False	True
11	Every rhombus is a square. A. True B. False	False
12	A triangle can have two right angles. A. True B. False	False
13	A rectangle has 2 pairs of parallel sides. A. True B. False	True
14	Draw a quadrilateral that is a parallelogram but not a rectangle.	A slanted parallelogram

LESSON 8.4 · TEACHER PLAN
Week 1 · Lesson 4 of 4
Unit 8: Geometry, Angles, and Line Plots

Lesson 8.4: Lines of Symmetry

Student goal	I can find and draw lines of symmetry.
Objective	Students identify and draw lines of symmetry.
Standard	4.G.3 — Recognize a line of symmetry; identify line-symmetric figures and draw lines of symmetry.
Pacing	Unit 8, week 1, lesson 4 of 4
Time	About 50 minutes
Materials	Paper shapes to fold, mirror (optional), workbook Lesson 8.4
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Review.
Launch	5	Read Let's Explore. Fold paper shapes.
Teach	15	Fold a rectangle both ways and on the diagonal. Say: "If it matches exactly, it's symmetric."
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking the diagonal of a rectangle is a line of symmetry.

SUPPORT

- Fold paper.

EXTEND

- Find symmetry in nature: leaves, butterflies.

STUDENT LESSON 8.4 · WITH ANSWERS

Lesson 8.4: Lines of Symmetry

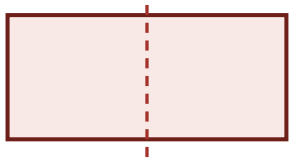
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.4 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.4, Part A.
- [ ] Complete the rest of Workbook Lesson 8.4 (Parts B–E).

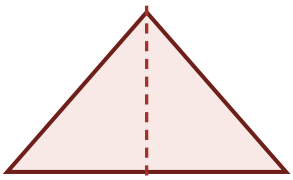
LET'S EXPLORE

Fold a paper heart in half. Do the two halves match exactly?

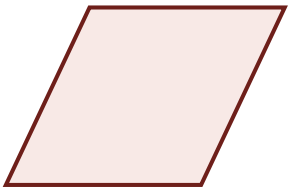
If you can fold a figure so the two halves match exactly, the fold line is a line of symmetry.



rectangle: 2 lines (one shown)



equilateral triangle: 3 lines (one shown)



parallelogram: no lines of symmetry

WORKED EXAMPLE · COUNT LINES OF SYMMETRY

How many lines of symmetry does a square have?

1 Fold top to bottom: match.

2 Fold left to right: match.

3 Fold along both diagonals: match.

Answer: 4

! WATCH OUT

A diagonal fold of a rectangle (that isn't a square) does not make matching halves.

★ KEY IDEA

Imagine folding. If the halves match exactly, it's a line of symmetry.

✎ TRY IT

Answers are shown in color.

1. How many lines of symmetry does the letter A have?

Answer: 1

2. Does the letter Z have a line of symmetry?

Answer: No

3. How many lines of symmetry does a circle have?

Answer: infinitely many

Workbook Answer Key · Lesson 8.4

#	Question	Answer
1	How many lines of symmetry does the letter A have?	1
2	Does the letter Z have a line of symmetry?	No
3	How many lines of symmetry does a circle have?	infinitely many
4	Every square is a ___.	rectangle (and rhombus)
5	Name a shape with no right angles.	e.g. equilateral triangle
6	(figure) A. Yes B. No	Yes
7	(figure) A. Yes B. No	No
8	(figure) A. Yes B. No	Yes
9	(figure) A. Yes B. No	No
10	letter H	2
11	letter B	1
12	letter N	0
13	regular hexagon	6
14	Draw a shape with exactly one line of symmetry. Draw the line.	e.g. a heart or an isosceles triangle

WEEKLY CHECK · ANSWER KEY
After Lesson 8.4

About 15 minutes

Weekly Check 18: Lines and Shapes — Answer Key

Answers

#	Question	Answer
1	Name the figure. (figure)	ray
2	Parallel, perpendicular, or intersecting? (figure)	perpendicular
3	Name the shape. (figure)	trapezoid
4	How many lines of symmetry does a rectangle have?	2
5	Is every square a rhombus?	Yes
6	Draw a right triangle.	drawing

LESSON 8.5 · TEACHER PLAN
Week 2 · Lesson 1 of 4
Unit 8: Geometry, Angles, and Line Plots

Lesson 8.5: Angles: Right, Acute, and Obtuse

Student goal	I can identify right, acute, obtuse, and straight angles.
Objective	Students classify angles by comparing to a right angle.
Standard	4.G.1, 4.G.2 — Draw angles; classify figures by angles.
Pacing	Unit 8, week 2, lesson 1 of 4
Time	About 50 minutes
Materials	Index card, clock, workbook Lesson 8.5
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Review.
Launch	5	Read Let's Explore with a book.
Teach	15	Make each angle with arms or two pencils. Test with an index card corner.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Judging angles by the length of the rays.

SUPPORT

- Index card corner.

EXTEND

- Find each type of angle in the house.

STUDENT LESSON 8.5 · WITH ANSWERS

Lesson 8.5: Angles: Right, Acute, and Obtuse

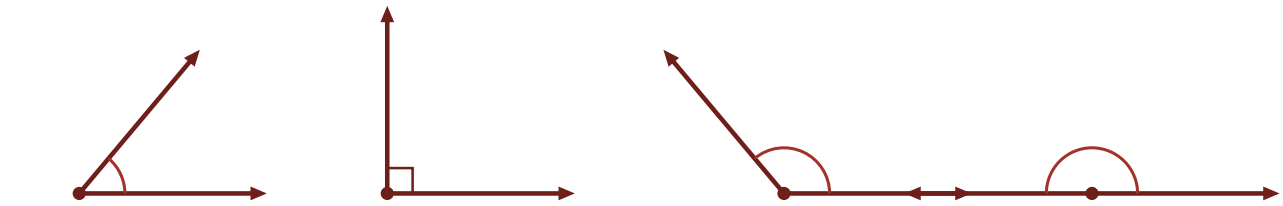
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.5 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.5, Part A.
- [ ] Complete the rest of Workbook Lesson 8.5 (Parts B–D).

LET'S EXPLORE

Open a book a little, then wider, then flat on the table. How does the corner between the covers change?

An angle is made of two rays with a shared endpoint called the vertex. We compare angles to a right angle (a square corner).



acute right obtuse straight

WORKED EXAMPLE · CLASSIFY

A clock shows 3:00. What kind of angle do the hands make?

1 The hands point to 12 and 3.

2 That is a square corner.

Answer: right angle

★ KEY IDEA

Use the corner of a paper as a right-angle tester: smaller is acute, bigger is obtuse.

✎ TRY IT

Answers are shown in color.

1. The hands of a clock at 1:00 make what kind of angle?

Answer: acute

2. At 6:00?

Answer: straight

3. At 4:00?

Answer: obtuse

Workbook Answer Key · Lesson 8.5

#	Question	Answer
1	The hands of a clock at 1:00 make what kind of angle?	acute
2	At 6:00?	straight
3	At 4:00?	obtuse
4	What do perpendicular lines make?	right angles
5	How many lines of symmetry does a square have?	4
6	(figure)	acute
7	(figure)	obtuse
8	(figure)	right
9	(figure)	obtuse
10	(figure)	acute
11	(figure)	straight
12	Draw an acute angle and an obtuse angle. Label the vertex of each.	drawings

LESSON 8.6 · TEACHER PLAN
Week 2 · Lesson 2 of 4
Unit 8: Geometry, Angles, and Line Plots

Lesson 8.6: Angles and Turns

Student goal	I can explain angles as turns and fractions of a full circle (360°).
Objective	Students relate angles to turns and degrees.
Standard	4.MD.5 — Recognize angles as geometric shapes formed wherever two rays share a common endpoint; understand angle measurement relative to a circle.
Pacing	Unit 8, week 2, lesson 2 of 4
Time	About 50 minutes
Materials	Paper plate with a brad and two strips, workbook Lesson 8.6
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	360 ÷ 4.
Launch	5	Read Let's Explore.
Teach	15	Turn your body a quarter, half, full turn. Build an angle maker with a paper plate.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking a full turn is 100°.

SUPPORT

- Body turns.

EXTEND

- How many degrees does the hour hand move in 1 hour? (30°)

STUDENT LESSON 8.6 · WITH ANSWERS

Lesson 8.6: Angles and Turns

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.6 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.6, Part A.
- [ ] Complete the rest of Workbook Lesson 8.6 (Parts B–E).

LET'S EXPLORE

A skateboarder does a "180" and a "360." What do those numbers mean?

An angle measures a turn. A full turn all the way around is 360 degrees (360°). A half turn is 180°. A quarter turn is 90°: a right angle.

Turn	Fraction of a circle	Degrees
quarter turn	$\frac{1}{4}$	90°
half turn	$\frac{1}{2}$	180°
three-quarter turn	$\frac{3}{4}$	270°
full turn	1	360°

WORKED EXAMPLE · FRACTION TO DEGREES

How many degrees is $\frac{1}{3}$ of a full turn?

1 360 ÷ 3 = 120.

Answer: 120°

★ KEY IDEA

A full circle is 360°. Find a fraction of 360 to find the angle.

✎ TRY IT

Answers are shown in color.

1. How many degrees in a half turn?

Answer: 180°

2. $\frac{1}{6}$ of a turn is how many degrees?

Answer: 60°

3. A clock's minute hand moves 15 minutes. How many degrees?

Answer: 90°

Workbook Answer Key · Lesson 8.6

#	Question	Answer
1	How many degrees in a half turn?	180°
2	$\frac{1}{6}$ of a turn is how many degrees?	60°
3	A clock's minute hand moves 15 minutes. How many degrees?	90°
4	360 ÷ 4	90
5	360 ÷ 2	180
6	$\frac{1}{4}$ turn	90°
7	$\frac{3}{4}$ turn	270°
8	$\frac{1}{8}$ turn	45°
9	$\frac{1}{10}$ turn	36°
10	$\frac{2}{3}$ turn	240°
11	$\frac{1}{12}$ turn	30°
12	A ceiling fan blade turns 2 full turns. How many degrees?	720°
13	A clock's minute hand moves from 12 to 2. How many degrees?	60°
14	Explain why a right angle is 90°.	It is a quarter of a full 360° turn: 360 ÷ 4 = 90.

LESSON 8.7 · TEACHER PLAN
Week 2 · Lesson 3 of 4
Unit 8: Geometry, Angles, and Line Plots

Lesson 8.7: Estimating Angle Measures

Student goal	I can estimate the measure of an angle using benchmark angles of 45°, 90°, 135°, and 180°.
Objective	Students estimate angle measures with benchmarks.
Standard	4.MD.5 — Understand angle measure in degrees.
Pacing	Unit 8, week 2, lesson 3 of 4
Time	About 50 minutes
Materials	Paper plate angle maker, workbook Lesson 8.7
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	90 ÷ 2.
Launch	5	Read Let's Explore.
Teach	15	Fold a paper corner in half to make 45°.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Confusing acute and obtuse estimates.

SUPPORT

- Fold benchmark angles from paper.

EXTEND

- Estimate angles in the letters of your name.

STUDENT LESSON 8.7 · WITH ANSWERS

Lesson 8.7: Estimating Angle Measures

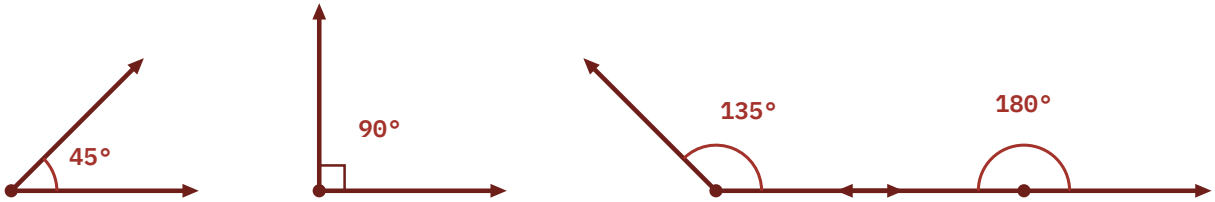
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.7 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.7, Part A.
- [ ] Complete the rest of Workbook Lesson 8.7 (Parts B–D).

LET'S EXPLORE

Without a tool, about how many degrees is the angle a door makes when it is half open?

Use benchmarks: a right angle is 90°. Half of a right angle is 45°. A right angle plus 45° is 135°. A straight line is 180°.



WORKED EXAMPLE · ESTIMATE

An angle is a bit bigger than a right angle but much smaller than a straight angle.

1 Between 90° and 180°.

2 Close to the 135° benchmark? It looks a little less.

Answer: about 120°

★ KEY IDEA

Compare to 45°, 90°, 135°, and 180° to estimate.

✎ TRY IT

Answers are shown in color.

1. Is a 40° angle closer to 45° or 90°?

Answer: 45°

2. Estimate an angle a little less than a straight angle.

Answer: about 170°

3. Which benchmark is closest to 100°?

Answer: 90°

Workbook Answer Key · Lesson 8.7

#	Question	Answer
1	Is a 40° angle closer to 45° or 90°?	45°
2	Estimate an angle a little less than a straight angle.	about 170°
3	Which benchmark is closest to 100°?	90°
4	90 ÷ 2	45
5	90 + 45	135
6	(figure)	about 45°
7	(figure)	about 150°
8	(figure)	about 30°
9	(figure)	about 100°
10	Draw an angle of about 135°. Explain how you made it.	A right angle plus half a right angle

LESSON 8.8 · TEACHER PLAN
Week 2 · Lesson 4 of 4
Unit 8: Geometry, Angles, and Line Plots

Lesson 8.8: Classifying Shapes by Angles

Student goal	I can sort shapes by their angles and tell how many right, acute, or obtuse angles they have.
Objective	Students classify shapes by angle types.
Standard	4.G.2 — Classify two-dimensional figures based on angles of a specified size.
Pacing	Unit 8, week 2, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 8.8
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Acute or obtuse?
Launch	5	Read Let's Explore.
Teach	15	Try to draw a triangle with two obtuse angles. Discuss why it fails.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Naming a triangle by its smallest angle.

SUPPORT

- Index card test for each angle.

EXTEND

- Sort 8 drawn shapes by angles.

STUDENT LESSON 8.8 · WITH ANSWERS

Lesson 8.8: Classifying Shapes by Angles

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.8 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.8, Part A.
- [ ] Complete the rest of Workbook Lesson 8.8 (Parts B–E).

LET'S EXPLORE

Can a triangle have two obtuse angles?

Look at each angle in a shape and classify it. Triangles are named by their largest angle.

Triangle	Angles
acute triangle	3 acute angles
right triangle	1 right angle, 2 acute
obtuse triangle	1 obtuse angle, 2 acute

WORKED EXAMPLE · TRY TO DRAW IT

Can a triangle have two obtuse angles?

1 If two sides spread wider than a square corner at both ends, they can never meet to close the triangle.

Answer: No

★ KEY IDEA

Every triangle has at least two acute angles.

✎ TRY IT

Answers are shown in color.

1. How many right angles does a rectangle have?

Answer: 4

2. A triangle has angles of 60°, 60°, 60°. What kind of triangle?

Answer: acute (and equilateral)

3. Can a triangle have 2 right angles?

Answer: No

Workbook Answer Key · Lesson 8.8

#	Question	Answer
1	How many right angles does a rectangle have?	4
2	A triangle has angles of 60°, 60°, 60°. What kind of triangle?	acute (and equilateral)
3	Can a triangle have 2 right angles?	No
4	Is 95° acute or obtuse?	obtuse
5	Is 89° acute or obtuse?	acute
6	(figure)	right
7	(figure)	acute
8	(figure)	obtuse
9	Angles 30°, 60°, 90°	right
10	How many obtuse angles does a regular hexagon have?	6 (each 120°)
11	How many acute angles does a parallelogram like the one in Lesson 8.3 have?	2
12	Draw a quadrilateral with exactly 2 right angles.	e.g. a right trapezoid

WEEKLY CHECK · ANSWER KEY
After Lesson 8.8

About 15 minutes

Weekly Check 19: Angles — Answer Key

Answers

#	Question	Answer
1	Classify the angle. (figure)	obtuse
2	Classify the angle. (figure)	acute
3	$\frac{1}{4}$ of a full turn = ___°	90
4	$\frac{1}{3}$ of a full turn = ___°	120
5	Estimate: about how many degrees? (figure)	about 45°
6	A straight angle = ___°	180

LESSON 8.9 · TEACHER PLAN
Week 3 · Lesson 1 of 4
Unit 8: Geometry, Angles, and Line Plots

Lesson 8.9: Measuring Angles with a Protractor

Student goal	I can use a protractor to measure angles in whole degrees.
Objective	Students measure angles with a protractor.
Standard	4.MD.6 — Measure angles in whole-number degrees using a protractor.
Pacing	Unit 8, week 3, lesson 1 of 4
Time	About 50 minutes
Materials	Protractor, workbook Lesson 8.9
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Estimates.
Launch	5	Read Let's Explore.
Teach	15	Model protractor placement slowly three times. Say: "Center, zero, read."
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Reading the wrong scale.
- Not putting the center on the vertex.

SUPPORT

- Estimate first.

EXTEND

- Measure angles in a picture.

STUDENT LESSON 8.9 · WITH ANSWERS

Lesson 8.9: Measuring Angles with a Protractor

STUDENT'S ASSIGNMENT

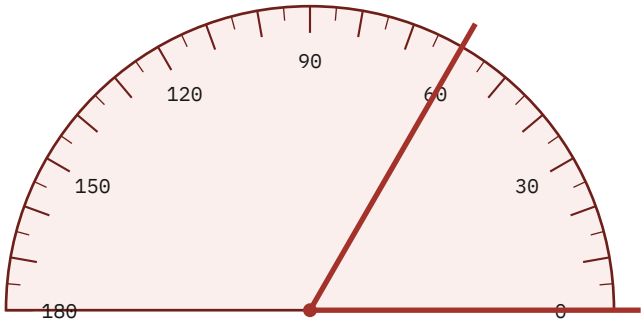
- [ ] Read Lesson 8.9 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.9, Part A.
- [ ] Complete the rest of Workbook Lesson 8.9 (Parts B–E).

LET'S EXPLORE

How can we find the exact size of an angle?

A protractor is a half circle marked from 0° to 180°.

- Put the center point of the protractor on the vertex.
- Line up the 0° line with one ray.
- Read where the other ray crosses the scale. Use the scale that starts at 0 on your first ray.



This angle measures 60°.

! WATCH OUT

Most protractors have two scales. If your angle is obviously acute but you read 120°, you used the wrong scale.

★ KEY IDEA

Center on the vertex, zero on one ray, read the other ray. Check with an estimate.

✎ TRY IT

Answers are shown in color.

1. An angle looks a little smaller than a right angle. The protractor shows 80° and 100°. Which is it?

Answer: 80°

2. Where does the protractor's center go?

Answer: on the vertex

3. A straight angle measures ___.

Answer: 180°

Workbook Answer Key · Lesson 8.9

#	Question	Answer
1	An angle looks a little smaller than a right angle. The protractor shows 80° and 100°. Which is it?	80°
2	Where does the protractor's center go?	on the vertex
3	A straight angle measures ___.	180°
4	Estimate a 'square corner' in degrees.	90°
5	Is 140° acute or obtuse?	obtuse
6	(figure)	40°
7	(figure)	110°
8	(figure)	75°
9	(figure)	150°
10	Draw any angle and measure it.	Answers vary
11	Measure the corner of this page.	90°
12	Explain how you know which scale to read.	Use the scale that starts at 0 on the first ray; check with an estimate.

LESSON 8.10 · TEACHER PLAN
Week 3 · Lesson 2 of 4
Unit 8: Geometry, Angles, and Line Plots

Lesson 8.10: Drawing Angles

Student goal	I can draw an angle of a given measure using a protractor.
Objective	Students draw angles of specified measures.
Standard	4.MD.6 — Sketch angles of specified measure.
Pacing	Unit 8, week 3, lesson 2 of 4
Time	About 50 minutes
Materials	Protractor, ruler, workbook Lesson 8.10
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Review measuring.
Launch	5	Read Let's Explore.
Teach	15	Model the four steps.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Drawing from the wrong side of 0.

SUPPORT

- Check: does it look acute or obtuse as expected?

EXTEND

- Draw a 'pie chart' with angles 90°, 120°, 150°.

STUDENT LESSON 8.10 · WITH ANSWERS

Lesson 8.10: Drawing Angles

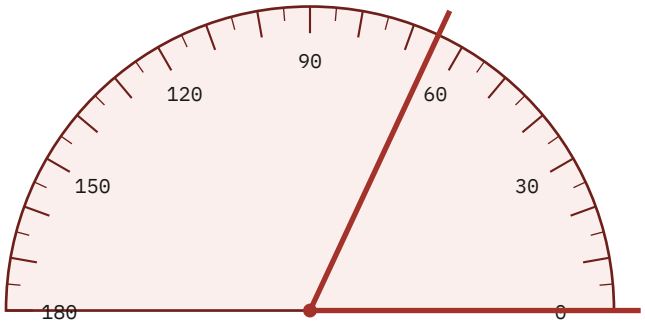
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.10 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.10, Part A.
- [ ] Complete the rest of Workbook Lesson 8.10 (Parts B–D).

LET'S EXPLORE

How would you draw an angle of exactly 65°?

- Draw a ray. Its endpoint is the vertex.
- Place the protractor's center on the vertex and line up 0° with the ray.
- Make a dot at 65° on the scale.
- Draw a ray from the vertex through the dot. Label the angle.



Drawing a 65° angle.

★ KEY IDEA

Draw a ray, place the protractor, mark the degree, connect.

✎ TRY IT

Answers are shown in color.

1. Draw a 30° angle.

Answer: drawing

2. Draw a 120° angle.

Answer: drawing

3. Draw a 90° angle. What is another name for it?

Answer: right angle

Workbook Answer Key · Lesson 8.10

#	Question	Answer
1	Draw a 30° angle.	drawing
2	Draw a 120° angle.	drawing
3	Draw a 90° angle. What is another name for it?	right angle
4	Read: center, zero, ___	read the other ray
5	Is 65° acute?	Yes
6	45°	45° angle drawn (acute)
7	70°	70° angle drawn (acute)
8	125°	125° angle drawn (obtuse)
9	160°	160° angle drawn (obtuse)
10	Draw a 100° angle. Is it acute, right, or obtuse?	obtuse

LESSON 8.11 · TEACHER PLAN
Week 3 · Lesson 3 of 4
Unit 8: Geometry, Angles, and Line Plots

Lesson 8.11: Adding Angle Measures

Student goal	I can find the measure of an angle made of smaller angles by adding.
Objective	Students add angle measures.
Standard	4.MD.7 — Recognize angle measure as additive.
Pacing	Unit 8, week 3, lesson 3 of 4
Time	About 50 minutes
Materials	Paper angles to cut and combine, workbook Lesson 8.11
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	45 + 45.
Launch	5	Read Let's Explore with paper pie slices.
Teach	15	Combine two paper angles and measure the total.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Overlapping the parts.

SUPPORT

- Paper wedges.

EXTEND

- Make 360° from 5 different angles.

STUDENT LESSON 8.11 · WITH ANSWERS

Lesson 8.11: Adding Angle Measures

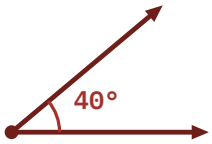
STUDENT'S ASSIGNMENT

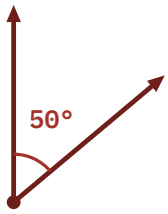
- [ ] Read Lesson 8.11 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.11, Part A.
- [ ] Complete the rest of Workbook Lesson 8.11 (Parts B–D).

LET'S EXPLORE

Two slices of pie meet at the center. One slice makes a 40° angle and the next makes 50°. What angle do they make together?

When an angle is split into parts that don't overlap, the whole angle equals the sum of the parts. 40° + 50° = 90°.





WORKED EXAMPLE · ADD PARTS

An angle is split into parts of 35°, 25°, and 60°. What is the whole angle?

1 35 + 25 + 60 = 120.

Answer: 120°

★ KEY IDEA

Angle measures are additive: whole = sum of the parts.

✎ TRY IT

Answers are shown in color.

1. 30° + 45°

Answer: 75°

2. Three 30° angles side by side make ___.

Answer: 90°

3. Two right angles together make ___.

Answer: 180° (a straight angle)

Workbook Answer Key · Lesson 8.11

#	Question	Answer
1	30° + 45°	75°
2	Three 30° angles side by side make ___.	90°
3	Two right angles together make ___.	180° (a straight angle)
4	45 + 45	90
5	180 − 90	90
6	20° + 70°	90°
7	55° + 35°	90°
8	40° + 40° + 40°	120°
9	90° + 45°	135°
10	15° + 60° + 30°	105°
11	100° + 80°	180°
12	Four equal angles fill a right angle. How big is each?	22.5° (accept 22 ½°)

LESSON 8.12 · TEACHER PLAN
Week 3 · Lesson 4 of 4
Unit 8: Geometry, Angles, and Line Plots

Lesson 8.12: Finding Unknown Angles

Student goal	I can find an unknown angle by adding or subtracting.
Objective	Students find unknown angles.
Standard	4.MD.7 — Solve addition and subtraction problems to find unknown angles on a diagram; use an equation with a symbol for the unknown.
Pacing	Unit 8, week 3, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 8.12
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	90 − 25.
Launch	5	Read Let's Explore.
Teach	15	Work both examples. Write equations.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Subtracting from 100 instead of 90.

SUPPORT

- Label 90° or 180° on the diagram first.

EXTEND

- Find angles in a clock face.

STUDENT LESSON 8.12 · WITH ANSWERS

Lesson 8.12: Finding Unknown Angles

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.12 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.12, Part A.
- [ ] Complete the rest of Workbook Lesson 8.12 (Parts B–E).

LET'S EXPLORE

A right angle is split into two parts. One part is 35°. How big is the other part?

WORKED EXAMPLE · SUBTRACT FROM 90°

Right angle: 35° + ? = 90°

1 90 − 35 = 55.

Answer: 55°

WORKED EXAMPLE · SUBTRACT FROM 180°

A straight angle is split into 110° and ?.

1 180 − 110 = 70.

Answer: 70°

★ KEY IDEA

Write an equation with a letter for the unknown: 35 + x = 90.

✎ TRY IT

Answers are shown in color.

1. Right angle split into 60° and x. Find x.

Answer: 30°

2. Straight angle split into 45° and x.

Answer: 135°

3. Full turn split into 100°, 120°, and x.

Answer: 140°

Workbook Answer Key · Lesson 8.12

#	Question	Answer
1	Right angle split into 60° and x. Find x.	30°
2	Straight angle split into 45° and x.	135°
3	Full turn split into 100°, 120°, and x.	140°
4	90 − 25	65
5	180 − 75	105
6	Right angle: 40° and x	50°
7	Right angle: 72° and x	18°
8	Straight angle: 65° and x	115°
9	Straight angle: 90°, 30°, and x	60°
10	360°: 90°, 150°, and x	120°
11	An angle of 125° is split in two. One part is 80°. The other?	45°
12	A ramp meets the ground at 25°. What angle does it make on the other side of the ground line (a straight angle)?	155°
13	Write and solve an equation: a straight angle is split into two parts, 48° and x.	48 + x = 180; x = 132°

WEEKLY CHECK · ANSWER KEY
After Lesson 8.12

About 20 minutes

Weekly Check 20: Measuring Angles — Answer Key

Answers

#	Question	Answer
1	Read the protractor. (figure)	130°
2	Draw a 70° angle.	drawing
3	35° + 55° = ?	90°
4	Right angle split into 28° and x. Find x.	62°
5	Straight angle split into 95° and x. Find x.	85°
6	360° split into 90°, 90°, 100°, and x.	80°

LESSON 8.13 · TEACHER PLAN
Week 4 · Lesson 1 of 4
Unit 8: Geometry, Angles, and Line Plots

Lesson 8.13: Line Plots with Fractions

Student goal	I can make a line plot to show measurements in fractions of a unit ($\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$).
Objective	Students create line plots with fractional measurements.
Standard	4.MD.4 — Make a line plot to display a data set of measurements in fractions of a unit (1/2, 1/4, 1/8).
Pacing	Unit 8, week 4, lesson 1 of 4
Time	About 50 minutes
Materials	Rulers, small objects to measure, workbook Lesson 8.13
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Equivalent fractions.
Launch	5	Read Let's Explore.
Teach	15	Measure 8 pencils or crayons to the nearest ¼ inch together and make a line plot.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Uneven spacing on the number line.

SUPPORT

- Label every tick.

EXTEND

- Make a line plot of family members' shoe lengths.

STUDENT LESSON 8.13 · WITH ANSWERS

Lesson 8.13: Line Plots with Fractions

STUDENT'S ASSIGNMENT

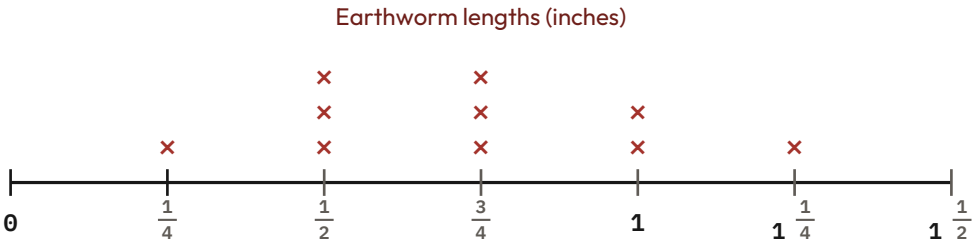
- [ ] Read Lesson 8.13 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.13, Part A.
- [ ] Complete the rest of Workbook Lesson 8.13 (Parts B–E).

LET'S EXPLORE

Ellie measured 10 earthworms after it rained. How can she show all the lengths in one picture?

A line plot shows data on a number line. Each X stands for one measurement.

Worm lengths (inches)
$\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{3}{4}, \frac{3}{4}, \frac{3}{4}, 1, \frac{5}{4}, \frac{1}{2}, 1$



- Draw a number line that covers the smallest to the largest value.
- Mark it in the right fraction (here, fourths).
- Put an X above the value for each measurement.
- Give it a title and units.

★ KEY IDEA

Each X is one data value. Tall stacks show the most common values.

✎ TRY IT

Answers are shown in color.

1. How many worms were $\frac{3}{4}$ inch?

Answer: 3

2. What was the longest worm?

Answer: $1\frac{1}{4}$ inches

3. How many worms were longer than $\frac{1}{2}$ inch?

Answer: 6

Workbook Answer Key · Lesson 8.13

#	Question	Answer
1	How many worms were $\frac{3}{4}$ inch?	3
2	What was the longest worm?	$1\frac{1}{4}$ inches
3	How many worms were longer than $\frac{1}{2}$ inch?	6
4	$\frac{2}{4} = \frac{1}{?}$	2
5	$\frac{4}{8} = \frac{?}{4}$	2
6	Pencil lengths (inches): 5, $5\frac{1}{2}$, $5\frac{1}{4}$, 6, $5\frac{1}{2}$, $5\frac{3}{4}$, $5\frac{1}{2}$, 6. Make a line plot from 5 to 6 in fourths.	X's: 5 (1), $5\frac{1}{4}$ (1), $5\frac{1}{2}$ (3), $5\frac{3}{4}$ (1), 6 (2)
7	Which length is most common?	$5\frac{1}{2}$ in.
8	How many pencils are there?	8
9	Why is a line plot useful for measurements?	It shows all values at once and which are most common.

LESSON 8.14 · TEACHER PLAN
Week 4 · Lesson 2 of 4
Unit 8: Geometry, Angles, and Line Plots

Lesson 8.14: Solving Problems with Line Plots

Student goal	I can add and subtract fractions to answer questions about a line plot.
Objective	Students solve problems using line plot data.
Standard	4.MD.4 — Solve problems involving addition and subtraction of fractions using information in line plots.
Pacing	Unit 8, week 4, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 8.14
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	3 × 1/4.
Launch	5	Read Let's Explore.
Teach	15	Work both examples.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Counting X's wrong.

SUPPORT

- Cross off X's as you use them.

EXTEND

- Make and answer 3 questions about your own data.

STUDENT LESSON 8.14 · WITH ANSWERS

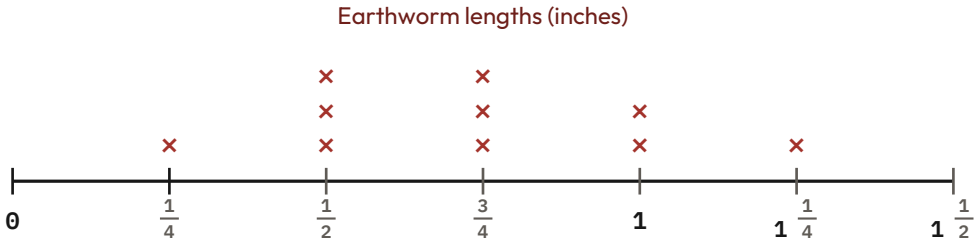
Lesson 8.14: Solving Problems with Line Plots

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.14 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.14, Part A.
- [ ] Complete the rest of Workbook Lesson 8.14 (Parts B–D).

LET'S EXPLORE

What is the difference between the longest and shortest earthworm? What is the total length of all the $3/4$-inch worms?



WORKED EXAMPLE · DIFFERENCE

Longest − shortest

1 Longest: $1\,1/4 = 5/4$. Shortest: $1/4$.

2 $5/4 - 1/4 = 4/4 = 1$.

Answer: 1 inch

WORKED EXAMPLE · TOTAL OF A GROUP

Total length of the $3/4$-inch worms

1 There are 3 of them: $3 \times 3/4 = 9/4 = 2\,1/4$.

Answer: $2\,1/4$ inches

★ KEY IDEA

Use the line plot to find the values, then add, subtract, or multiply fractions.

✎ TRY IT

Answers are shown in color.

1. Total length of the $^{1}/_{2}$-inch worms?

Answer: $1\,^{2}/_{4} = 1\,^{1}/_{2}$ inches

2. How many worms in all?

Answer: 10

3. How much longer is the longest worm than a $^{1}/_{2}$-inch worm?

Answer: $^{3}/_{4}$ inch

Workbook Answer Key · Lesson 8.14

#	Question	Answer
1	Total length of the $^{1}/_{2}$-inch worms?	$1\,^{2}/_{4} = 1\,^{1}/_{2}$ inches
2	How many worms in all?	10
3	How much longer is the longest worm than a $^{1}/_{2}$-inch worm?	$^{3}/_{4}$ inch
4	$3 \times {}^{1}/_{4}$	$^{3}/_{4}$
5	$^{7}/_{8} - {}^{3}/_{8}$	$^{4}/_{8}$
6	How many ribbon pieces are there? (figure)	9
7	What is the difference between the longest and shortest piece?	$^{7}/_{8}$ yard
8	What is the total length of the $^{5}/_{8}$-yard pieces?	$^{15}/_{8} = 1\,^{7}/_{8}$ yards
9	If all the $^{3}/_{8}$-yard pieces are joined, how long is the new piece?	$^{6}/_{8}$ yard
10	Write your own question about the ribbon line plot and answer it.	Answers vary.

LESSON 8.15 · TEACHER PLAN
Week 4 · Lesson 3 of 4
Unit 8: Geometry, Angles, and Line Plots

Lesson 8.15: Geometry Review: Attributes of Shapes

Student goal	I can describe shapes using lines, angles, and symmetry.
Objective	Students describe shapes using geometric attributes.
Standard	4.G.1, 4.G.2, 4.G.3 — 4.G.1–3 review.
Pacing	Unit 8, week 4, lesson 3 of 4
Time	About 50 minutes
Materials	Paper shapes, workbook Lesson 8.15
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Review.
Launch	5	Read Let's Explore.
Teach	15	Play Guess My Shape with paper shapes.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Using informal words like ‘slanty’.

SUPPORT

- Word bank.

EXTEND

- Write riddles for a sibling.

STUDENT LESSON 8.15 · WITH ANSWERS

Lesson 8.15: Geometry Review: Attributes of Shapes

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.15 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.15, Part A.
- [ ] Complete the rest of Workbook Lesson 8.15 (Parts B–D).

LET'S EXPLORE

Play "Guess My Shape": I have 4 sides, 2 pairs of parallel sides, no right angles, and 4 equal sides. What am I?

Describe shapes by their attributes: number of sides, parallel and perpendicular sides, types of angles, equal sides, and lines of symmetry.

WORKED EXAMPLE · GUESS MY SHAPE

4 sides, 2 pairs of parallel sides, no right angles, 4 equal sides.

1 Parallelogram with equal sides.

2 No right angles, so not a square.

Answer: rhombus

★ KEY IDEA

Use exact vocabulary: parallel, perpendicular, acute, right, obtuse, symmetric.

✎ TRY IT

Answers are shown in color.

1. I have 3 sides and one right angle. What am I?

Answer: right triangle

2. I have 4 right angles and 2 lines of symmetry, and my sides are not all equal.

Answer: rectangle

3. I have exactly one pair of parallel sides.

Answer: trapezoid

Workbook Answer Key · Lesson 8.15

#	Question	Answer
1	I have 3 sides and one right angle. What am I?	right triangle
2	I have 4 right angles and 2 lines of symmetry, and my sides are not all equal.	rectangle
3	I have exactly one pair of parallel sides.	trapezoid
4	90° + 45°	135°
5	Is a square a rhombus?	Yes
6	square	2 pairs parallel, 4 right angles, 4 lines
7	equilateral triangle	0 parallel, 0 right angles, 3 lines
8	rectangle (not a square)	2 pairs parallel, 4 right angles, 2 lines
9	parallelogram (not a rectangle or rhombus)	2 pairs parallel, 0 right angles, 0 lines
10	Write a ‘Guess My Shape’ riddle with at least 3 clues.	Answers vary.

LESSON 8.16 · TEACHER PLAN
Week 4 · Lesson 4 of 4
Unit 8: Geometry, Angles, and Line Plots

Lesson 8.16: Geometry and Data Problem Solving

Student goal	I can solve problems with angles, shapes, and line plots.
Objective	Students solve mixed geometry and data problems.
Standard	4.MD.4, 4.MD.7, 4.G.2 — 4.MD.4, 4.MD.7, 4.G review.
Pacing	Unit 8, week 4, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 8.16, Unit 8 Review
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	360 − 250.
Launch	5	Read Let's Explore. Sketch a kite.
Teach	15	Work the example.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook Lesson 8.16 and the Unit 8 Review.
Exit Ticket	3	Check. Tomorrow: Unit 8 Test.

COMMON MISCONCEPTIONS

- Forgetting angle totals.

SUPPORT

- Reference card: 90°, 180°, 360°.

EXTEND

- Design and build a real kite.

STUDENT LESSON 8.16 · WITH ANSWERS

Lesson 8.16: Geometry and Data Problem Solving

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.16 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.16, Part A.
- [ ] Complete the rest of Workbook Lesson 8.16 (Parts B–D).

LET'S EXPLORE

Design a kite: it must have one line of symmetry and two pairs of equal sides. What angles might it have?

Today you'll use everything from this unit to solve design and data problems.

WORKED EXAMPLE · KITE ANGLES

A kite has angles of 110°, 70°, 110°, and x°. The four angles of a quadrilateral add to 360°.

1 110 + 70 + 110 = 290.

2 360 − 290 = 70.

Answer: x = 70°

★ KEY IDEA

Use angle facts (90°, 180°, 360°), shape attributes, and line plot data together.

✎ TRY IT

Answers are shown in color.

1. Three angles of a quadrilateral are 90°, 90°, and 120°. Find the fourth.

Answer: 60°

2. How many lines of symmetry does a kite have?

Answer: 1

3. Can a trapezoid have a right angle?

Answer: Yes

Workbook Answer Key · Lesson 8.16

#	Question	Answer
1	Three angles of a quadrilateral are 90°, 90°, and 120°. Find the fourth.	60°
2	How many lines of symmetry does a kite have?	1
3	Can a trapezoid have a right angle?	Yes
4	360 − 250	110
5	Name an obtuse angle measure.	e.g. 120°
6	A pizza is cut into 8 equal slices. What angle does each slice make at the center?	45°
7	A triangle's angles add to 180°. Two angles are 50° and 60°. Find the third and classify the triangle.	70°; acute triangle
8	Seed growth (inches): $1/8$, $3/8$, $3/8$, $1/2$, $5/8$, $3/8$. Make a line plot. What is the difference between the most and least growth?	$4/8 = 1/2$ inch
9	Draw a shape with exactly one right angle and one line of symmetry.	e.g. isosceles right triangle

UNIT REVIEW · ANSWER KEY
Study before the unit test

About 35 minutes

Unit 8 Review — Answer Key

Homework or a study session before the test. Students need a protractor.

Answers

#	Question	Answer
1	How many endpoints does a line segment have?	2
2	Name a quadrilateral with exactly one pair of parallel sides.	trapezoid
3	How many lines of symmetry does a square have?	4
4	Can a triangle have two obtuse angles?	No
5	$\frac{1}{6}$ turn = ___°	60
6	Classify a 91° angle.	obtuse
7	Right angle: 47° + x. Find x.	43°
8	Read the protractor. (figure)	20°
9	Plant heights: $2\frac{1}{4}$, $2\frac{1}{2}$, $2\frac{1}{2}$, $2\frac{3}{4}$, 3. Make a line plot and find the difference between the tallest and shortest.	$\frac{3}{4}$ inch

UNIT TEST · ANSWER KEY
At the end of the unit
About 45 minutes

Unit 8 Test — Answer Key

Students need a protractor and ruler.

Answers

#	Question	Answer
1	Name the figure. (figure)	line segment
2	Name the lines. (figure)	parallel
3	Name the shape. (figure)	rhombus
4	Classify the triangle by angles. (figure)	obtuse
5	How many lines of symmetry does an equilateral triangle have?	3
6	True or false: every rectangle is a parallelogram.	True
7	Classify a 150° angle.	obtuse
8	$\frac{1}{2}$ turn = ___°	180
9	Read the protractor. (figure)	55°
10	40° + 75° = ?	115°
11	Right angle split into 64° and x.	26°
12	Straight angle split into 30°, 60°, and x.	90°
13	Use a protractor to draw a 115° angle.	drawing within 2°
14	Draw a quadrilateral with 2 pairs of parallel sides and no right angles.	parallelogram/rhombus
15	Bead lengths (inches): $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{3}{4}$, $\frac{1}{2}$, 1. Make a line plot. How much longer is the longest bead than the shortest? What is the total length of the $\frac{1}{2}$-inch beads?	$\frac{3}{4}$ inch longer; total $1\frac{2}{4} = 1\frac{1}{2}$ inches

Scoring

Score	What it means / next step
17–19 (85%+)	Mastered.
14–16	Proficient; correct missed items.
Below 14	Reteach and retest.

Measurement, Area, and Perimeter

9

Perimeter and area of rectangles, combined shapes, unit conversions, and elapsed time.

Measurement connects math to everyday life. You will find perimeter and area, solve for unknown sides, convert units of length, weight, capacity, and time, and solve real problems with measurements.

Big ideas

- Find perimeter and area with formulas.
- Find unknown side lengths.
- Find the area of combined rectangles.
- Convert customary, metric, and time units.
- Solve elapsed time and multi-step measurement problems.

Standards

Code	Standard
4.MD.1	Know relative sizes of measurement units within one system; express larger units in smaller units; record conversions in a table.
4.MD.2	Use the four operations to solve word problems involving distances, intervals of time, liquid volumes, masses, and money, including fractions and decimals.
4.MD.3	Apply the area and perimeter formulas for rectangles in real-world and mathematical problems.

What students should already know

- Multiplication and division (Units 2–3).
- Fractions and decimals (Units 5–7).

Materials for the unit

- Ruler, yardstick, meter stick

- Measuring cups and containers
- Kitchen scale
- Clock
- Grid paper and square tiles

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 4	9.1	Perimeter
Week 1	Lesson 2 of 4	9.2	Area
Week 1	Lesson 3 of 4	9.3	Area and Perimeter Formulas
Week 1	Lesson 4 of 4	9.4	Same Perimeter, Different Area
Week 1	End of week	Check	Weekly Check 21: Area and Perimeter
Week 2	Lesson 1 of 4	9.5	Finding an Unknown Side
Week 2	Lesson 2 of 4	9.6	Area of Combined Rectangles
Week 2	Lesson 3 of 4	9.7	Area and Perimeter Word Problems
Week 2	Lesson 4 of 4	9.8	Area or Perimeter?
Week 2	End of week	Check	Weekly Check 22: Area Problems
Week 3	Lesson 1 of 4	9.9	Customary Length
Week 3	Lesson 2 of 4	9.10	Customary Weight and Capacity
Week 3	Lesson 3 of 4	9.11	Metric Units
Week 3	Lesson 4 of 4	9.12	Time Conversions
Week 3	End of week	Check	Weekly Check 23: Converting Units
Week 4	Lesson 1 of 4	9.13	Elapsed Time
Week 4	Lesson 2 of 4	9.14	Measurement Problems with Fractions and Decimals
Week 4	Lesson 3 of 4	9.15	Multi-Step Measurement Problems
Week 4	Lesson 4 of 4	9.16	Measurement Problem Solving
Week 4	Before the test	Review	Unit 9 Review
Week 4	End of unit	Test	Unit 9 Test

TEACHING THIS UNIT AT HOME

Cook, build, and garden together this unit. Let your child do the measuring and converting.

UNIT VOCABULARY

perimeter	Distance around a figure.
area	Square units that cover a surface.
convert	Change to a different unit.
elapsed time	Time that passes between start and end.

LESSON 9.1 · TEACHER PLAN
Week 1 · Lesson 1 of 4

Unit 9: Measurement, Area, and Perimeter

Lesson 9.1: Perimeter

Student goal	I can find the perimeter of a shape by adding the lengths of its sides.
Objective	Students find perimeter.
Standard	4.MD.3 — Apply the area and perimeter formulas for rectangles in real-world and mathematical problems.
Pacing	Unit 9, week 1, lesson 1 of 4
Time	About 50 minutes
Materials	Ruler, string, workbook Lesson 9.1
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Addition warm-up.
Launch	5	Read Let's Explore.
Teach	15	Walk around a rug or table and count steps: that's perimeter. Work both examples.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Adding only two sides of a rectangle.

SUPPORT

- Label every side before adding.

EXTEND

- Find the perimeter of your bedroom.

STUDENT LESSON 9.1 · WITH ANSWERS

Lesson 9.1: Perimeter

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.1 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.1, Part A.
- [ ] Complete the rest of Workbook Lesson 9.1 (Parts B–E).

LET'S EXPLORE

Dad is putting a fence around a garden that is 12 feet long and 8 feet wide. How much fencing does he need?

Perimeter is the distance around a shape. Add the lengths of all the sides.



A garden 12 ft by 8 ft.

WORKED EXAMPLE · ADD THE SIDES

Find the perimeter of the garden.

1 A rectangle has two lengths and two widths.

2 12 + 8 + 12 + 8 = 40.

Answer: 40 feet of fencing

WORKED EXAMPLE · ANY POLYGON

A triangle has sides 7 cm, 9 cm, and 5 cm. Find the perimeter.

1 7 + 9 + 5 = 21.

Answer: 21 cm

★ KEY IDEA

Perimeter: add all the side lengths. The answer is in units (cm, ft), not square units.

✎ TRY IT

Answers are shown in color.

1. Rectangle 10 m by 4 m: perimeter?

Answer: 28 m

2. Square with 6-inch sides: perimeter?

Answer: 24 in.

3. Pentagon with all sides 3 ft: perimeter?

Answer: 15 ft

Workbook Answer Key · Lesson 9.1

#	Question	Answer
1	Rectangle 10 m by 4 m: perimeter?	28 m
2	Square with 6-inch sides: perimeter?	24 in.
3	Pentagon with all sides 3 ft: perimeter?	15 ft
4	12 + 8 + 12 + 8	40
5	4 × 9	36
6	(figure)	26 cm
7	(figure)	42 in.
8	(figure)	28 m
9	(figure)	62 ft
10	A picture frame is 14 inches by 10 inches. How much wood goes around the edge?	48 inches
11	A square tile has a perimeter of 32 cm. How long is each side?	8 cm
12	Draw a rectangle with a perimeter of 18 units.	e.g. 5 × 4 or 6 × 3 or 7 × 2

LESSON 9.2 · TEACHER PLAN
Week 1 · Lesson 2 of 4
Unit 9: Measurement, Area, and Perimeter

Lesson 9.2: Area

Student goal	I can find the area of a rectangle by counting square units and by multiplying.
Objective	Students find area of rectangles.
Standard	4.MD.3 — Apply the area formula for rectangles.
Pacing	Unit 9, week 1, lesson 2 of 4
Time	About 50 minutes
Materials	Square tiles or grid paper, workbook Lesson 9.2
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	6 × 4.
Launch	5	Read Let's Explore. Cover a rectangle with tiles.
Teach	15	Show rows × columns = area. Read the Watch Out.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Mixing up area and perimeter.
- Forgetting square units.

SUPPORT

- Tiles.

EXTEND

- Find the area of a book cover.

STUDENT LESSON 9.2 · WITH ANSWERS

Lesson 9.2: Area

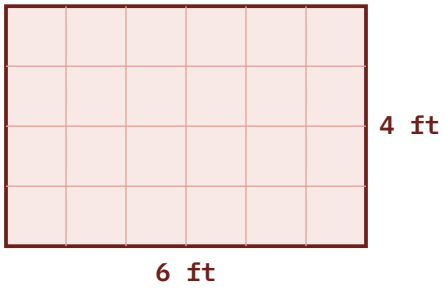
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.2 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.2, Part A.
- [ ] Complete the rest of Workbook Lesson 9.2 (Parts B–E).

LET'S EXPLORE

How many 1-foot square tiles cover a floor that is 6 feet long and 4 feet wide?

Area measures how much surface a shape covers. We count square units.



4 rows of 6 squares = 24 square feet.

The tiles form an array: 4 rows of 6. Multiplication is faster than counting: 6 × 4 = 24 square feet.

WORKED EXAMPLE · MULTIPLY

Find the area of a rectangle 9 m long and 7 m wide.

1 Area = length × width.

2 9 × 7 = 63.

Answer: 63 square meters

! WATCH OUT

Area is in square units (sq ft, sq m). Perimeter is in plain units (ft, m).

★ KEY IDEA

Area of a rectangle = length × width.

✎ TRY IT

Answers are shown in color.

1. Rectangle 8 in. by 5 in.: area?

Answer: 40 sq in.

2. Square with 9 cm sides: area?

Answer: 81 sq cm

3. A room 12 ft by 10 ft: area?

Answer: 120 sq ft

Workbook Answer Key · Lesson 9.2

#	Question	Answer
1	Rectangle 8 in. by 5 in.: area?	40 sq in.
2	Square with 9 cm sides: area?	81 sq cm
3	A room 12 ft by 10 ft: area?	120 sq ft
4	6 × 4	24
5	Perimeter of a 6 × 4 rectangle	20
6	(figure)	15 sq units
7	(figure)	14 sq units
8	(figure)	16 sq units
9	(figure)	18 sq units
10	10 m by 6 m	60 sq m
11	12 ft by 4 ft	48 sq ft
12	25 in. by 3 in.	75 sq in.
13	14 cm by 11 cm	154 sq cm
14	A rug is 7 feet by 5 feet. What is its area? What is its perimeter?	35 sq ft; 24 ft

LESSON 9.3 · TEACHER PLAN
Week 1 · Lesson 3 of 4
Unit 9: Measurement, Area, and Perimeter

Lesson 9.3: Area and Perimeter Formulas

Student goal	I can use the formulas P = 2 × l + 2 × w and A = l × w.
Objective	Students apply perimeter and area formulas.
Standard	4.MD.3 — Apply the area and perimeter formulas for rectangles.
Pacing	Unit 9, week 1, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 9.3
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Write the formulas on a card. Work the example.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Using A = 2 × l + 2 × w.

SUPPORT

- Formula card.

EXTEND

- Write a formula for the perimeter of a square (P = 4 × s).

STUDENT LESSON 9.3 · WITH ANSWERS

Lesson 9.3: Area and Perimeter Formulas

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.3 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.3, Part A.
- [ ] Complete the rest of Workbook Lesson 9.3 (Parts B–D).

LET'S EXPLORE

Is there a shortcut for adding 2 lengths and 2 widths?

Mathematicians write rules as formulas. For a rectangle with length l and width w:

Measure	Formula	Example: 9 × 5
Perimeter	P = 2 × l + 2 × w	2 × 9 + 2 × 5 = 28
Area	A = l × w	9 × 5 = 45

WORKED EXAMPLE · USE BOTH FORMULAS

A garden is 16 ft by 9 ft. Find P and A.

1 P = 2 × 16 + 2 × 9 = 32 + 18 = 50.

2 A = 16 × 9 = 144.

Answer: P = 50 ft, A = 144 sq ft

★ KEY IDEA

P = 2 × l + 2 × w. A = l × w.

✎ TRY IT

Answers are shown in color.

1. 13 cm by 6 cm

Answer: P = 38 cm, A = 78 sq cm

2. 20 m by 15 m

Answer: P = 70 m, A = 300 sq m

3. Square, 11 in. sides

Answer: P = 44 in., A = 121 sq in.

Workbook Answer Key · Lesson 9.3

#	Question	Answer
1	13 cm by 6 cm	P = 38 cm, A = 78 sq cm
2	20 m by 15 m	P = 70 m, A = 300 sq m
3	Square, 11 in. sides	P = 44 in., A = 121 sq in.
4	2 × 16 + 2 × 9	50
5	16 × 9	144
6	8 by 3 ft	P = 22 ft, A = 24 sq ft
7	12 by 7 cm	P = 38 cm, A = 84 sq cm
8	25 by 10 m	P = 70 m, A = 250 sq m
9	9 by 9 in.	P = 36 in., A = 81 sq in.
10	30 by 18 yd	P = 96 yd, A = 540 sq yd
11	45 by 20 ft	P = 130 ft, A = 900 sq ft
12	A basketball court is 94 ft by 50 ft. Find its perimeter and area.	P = 288 ft, A = 4700 sq ft

LESSON 9.4 · TEACHER PLAN
Week 1 · Lesson 4 of 4
Unit 9: Measurement, Area, and Perimeter

Lesson 9.4: Same Perimeter, Different Area

Student goal	I can show that rectangles with the same perimeter can have different areas.
Objective	Students explore how area changes with fixed perimeter.
Standard	4.MD.3 — Apply area and perimeter in real-world problems.
Pacing	Unit 9, week 1, lesson 4 of 4
Time	About 50 minutes
Materials	String loop of 24 inches, grid paper, workbook Lesson 9.4
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Perimeter and area of 5 × 3.
Launch	5	Read Let's Explore.
Teach	15	Make rectangles with a 24-inch string loop on grid paper; record each area.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking same perimeter means same area.

SUPPORT

- Use grid paper.

EXTEND

- Same area 24: find all rectangles and their perimeters.

STUDENT LESSON 9.4 · WITH ANSWERS

Lesson 9.4: Same Perimeter, Different Area

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.4 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.4, Part A.
- [ ] Complete the rest of Workbook Lesson 9.4 (Parts B–D).

LET'S EXPLORE

You have 24 feet of fence for a rabbit pen. Which rectangle gives the rabbit the most room?

Length	Width	Perimeter	Area
1	11	24 ft	11 sq ft
2	10	24 ft	20 sq ft
3	9	24 ft	27 sq ft
4	8	24 ft	32 sq ft
5	7	24 ft	35 sq ft
6	6	24 ft	36 sq ft

All these pens use 24 feet of fence, but the areas are very different. The 6 × 6 square gives the most room: 36 square feet.

★ KEY IDEA

For a fixed perimeter, the closer the rectangle is to a square, the bigger its area.

✎ TRY IT

Answers are shown in color.

1. Two rectangles have perimeter 16. One is 7 × 1, one is 4 × 4. Which has more area?

Answer: 4 × 4 (16 vs 7)

2. A rectangle has area 12. Name two possible perimeters.

Answer: e.g. 14 (3 × 4), 16 (2 × 6), 26 (1 × 12)

3. Same area 36: which has the smaller perimeter, 9 × 4 or 6 × 6?

Answer: 6 × 6 (24 vs 26)

Workbook Answer Key · Lesson 9.4

#	Question	Answer
1	Two rectangles have perimeter 16. One is 7 × 1, one is 4 × 4. Which has more area?	4 × 4 (16 vs 7)
2	A rectangle has area 12. Name two possible perimeters.	e.g. 14 (3 × 4), 16 (2 × 6), 26 (1 × 12)
3	Same area 36: which has the smaller perimeter, 9 × 4 or 6 × 6?	6 × 6 (24 vs 26)
4	Perimeter of 5 × 3	16
5	Area of 5 × 3	15
6	Length 1, width ___, area ___	width 9, area 9
7	Length 2, width ___, area ___	width 8, area 16
8	Length 3, width ___, area ___	width 7, area 21
9	Length 4, width ___, area ___	width 6, area 24
10	Length 5, width ___, area ___	width 5, area 25
11	Length 6, width ___, area ___	width 4, area 24
12	Which rectangle with perimeter 20 has the greatest area? Why?	5 × 5 = 25; closest to a square

WEEKLY CHECK · ANSWER KEY
After Lesson 9.4

About 15 minutes

Weekly Check 21: Area and Perimeter — Answer Key

Answers

#	Question	Answer
1	Perimeter? (figure)	26 cm
2	Area? (figure)	36 sq cm
3	Square with 7 ft sides: perimeter and area	28 ft; 49 sq ft
4	Rectangle 12 m × 5 m: P and A	P = 34 m, A = 60 sq m
5	Perimeter 16: which has more area, 7 × 1 or 4 × 4?	4 × 4
6	Area or perimeter: fence around a yard?	perimeter

LESSON 9.5 · TEACHER PLAN
Week 2 · Lesson 1 of 4
Unit 9: Measurement, Area, and Perimeter

Lesson 9.5: Finding an Unknown Side

Student goal	I can find a missing side length when I know the area or perimeter.
Objective	Students find unknown side lengths.
Standard	4.MD.3 — Apply the area and perimeter formulas, e.g., find the width given the area and length.
Pacing	Unit 9, week 2, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 9.5
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	48 ÷ 8.
Launch	5	Read Let's Explore.
Teach	15	Work both examples with equations.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Dividing the perimeter by the length.

SUPPORT

- Draw and label the rectangle.

EXTEND

- Garden design with a set area.

STUDENT LESSON 9.5 · WITH ANSWERS

Lesson 9.5: Finding an Unknown Side

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.5 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.5, Part A.
- [ ] Complete the rest of Workbook Lesson 9.5 (Parts B–E).

LET'S EXPLORE

A rectangle has an area of 48 square feet and a length of 8 feet. How wide is it?

WORKED EXAMPLE · FROM AREA

A = 48 sq ft, l = 8 ft. Find w.

1 $8 \times w = 48$.

2 $48 \div 8 = 6$.

Answer: 6 feet

WORKED EXAMPLE · FROM PERIMETER

P = 30 m, l = 9 m. Find w.

1 Two lengths: $2 \times 9 = 18$.

2 $30 - 18 = 12$ for the two widths.

3 $12 \div 2 = 6$.

Answer: 6 m

★ KEY IDEA

Write the formula with a letter for the unknown, then use the inverse operation.

✎ TRY IT

Answers are shown in color.

1. A = 56 sq in., l = 7 in. Find w.

Answer: 8 in.

2. P = 26 cm, l = 8 cm. Find w.

Answer: 5 cm

3. Square with A = 64 sq m. Side?

Answer: 8 m

Workbook Answer Key · Lesson 9.5

#	Question	Answer
1	A = 56 sq in., l = 7 in. Find w.	8 in.
2	P = 26 cm, l = 8 cm. Find w.	5 cm
3	Square with A = 64 sq m. Side?	8 m
4	48 ÷ 8	6
5	30 − 18	12
6	A = 72 sq ft, l = 9 ft	w = 8 ft
7	A = 90 sq cm, w = 6 cm	l = 15 cm
8	P = 40 m, l = 12 m	w = 8 m
9	P = 50 in., w = 10 in.	l = 15 in.
10	Square, P = 36 ft	side = 9 ft
11	A = 120 sq yd, l = 12 yd	w = 10 yd
12	A patio has an area of 96 square feet. It is 12 feet long. How wide is it? What is its perimeter?	8 ft wide; 40 ft
13	A rectangle has P = 34 cm and l = 11 cm. Find w and the area.	w = 6 cm; A = 66 sq cm

LESSON 9.6 · TEACHER PLAN
Week 2 · Lesson 2 of 4
Unit 9: Measurement, Area, and Perimeter

Lesson 9.6: Area of Combined Rectangles

Student goal	I can find the area of a shape made of rectangles by breaking it apart.
Objective	Students find the area of composite rectilinear figures.
Standard	4.MD.3 — Apply area formulas to real-world problems (rectilinear figures; supports 3.MD.7d).
Pacing	Unit 9, week 2, lesson 2 of 4
Time	About 50 minutes
Materials	Grid paper, workbook Lesson 9.6
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Cut an L-shape from grid paper into two rectangles. Say: “Break it, find each area, add.”
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Using the wrong side lengths for the smaller rectangle.

SUPPORT

- Shade each part a different color.

EXTEND

- Find the area of your home's floor plan.

STUDENT LESSON 9.6 · WITH ANSWERS

Lesson 9.6: Area of Combined Rectangles

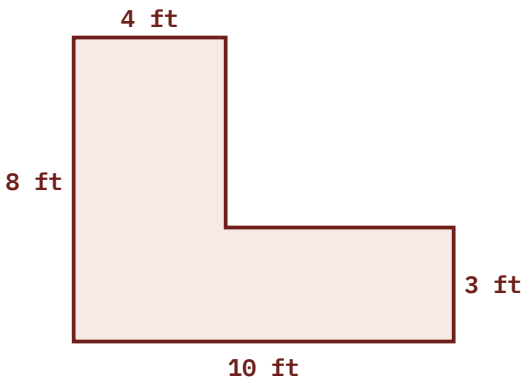
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.6 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.6, Part A.
- [ ] Complete the rest of Workbook Lesson 9.6 (Parts B–D).

LET'S EXPLORE

An L-shaped room needs new carpet. How can we find its area?

Break the shape into rectangles, find each area, and add.



An L-shaped room

WORKED EXAMPLE · BREAK IT APART

Find the area of the L-shape.

1 Bottom rectangle: 10 × 3 = 30.

2 Top rectangle: 4 × (8 − 3) = 4 × 5 = 20.

3 Add: 30 + 20 = 50.

Answer: 50 sq ft

★ KEY IDEA

Split into rectangles, find each area, add. Find missing side lengths by subtracting.

✎ TRY IT

Answers are shown in color.

1. Two rectangles: 5 × 4 and 3 × 2. Total area?

Answer: 26 sq units

2. A 10 × 10 square with a 2 × 2 corner cut out. Area?

Answer: 96 sq units

3. L-shape: bottom 8 × 2, top 3 × 4. Area?

Answer: 28 sq units

Workbook Answer Key · Lesson 9.6

#	Question	Answer
1	Two rectangles: 5 × 4 and 3 × 2. Total area?	26 sq units
2	A 10 × 10 square with a 2 × 2 corner cut out. Area?	96 sq units
3	L-shape: bottom 8 × 2, top 3 × 4. Area?	28 sq units
4	10 × 3 + 4 × 5	50
5	8 – 3	5
6	(figure)	48 sq m (12 × 3 + 3 × 4)
7	(figure)	34 sq ft (6 × 4 + 2 × 5)
8	(figure)	70 sq cm (10 × 5 + 4 × 5)
9	(figure)	45 sq yd (9 × 3 + 6 × 3)
10	Draw an L-shape, label its sides, and find its area.	Answers vary.

LESSON 9.7 · TEACHER PLAN
Week 2 · Lesson 3 of 4
Unit 9: Measurement, Area, and Perimeter

Lesson 9.7: Area and Perimeter Word Problems

Student goal	I can solve real-world problems with area and perimeter.
Objective	Students solve area and perimeter word problems.
Standard	4.MD.3 — Apply area and perimeter formulas in real-world problems.
Pacing	Unit 9, week 2, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 9.7
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work both examples. Ask: covering or going around?
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Using perimeter for covering problems.

SUPPORT

- Draw the shape.

EXTEND

- Plan and price a real room makeover.

STUDENT LESSON 9.7 · WITH ANSWERS

Lesson 9.7: Area and Perimeter Word Problems

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.7 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.7, Part A.
- [ ] Complete the rest of Workbook Lesson 9.7 (Parts B–D).

LET'S EXPLORE

Tile costs $3 per square foot. How much does it cost to tile a 9-foot by 6-foot bathroom?

WORKED EXAMPLE · AREA, THEN COST

Tile a 9 ft × 6 ft room at $3 per sq ft.

1 Area: 9 × 6 = 54 sq ft.

2 Cost: 54 × 3 = 162.

Answer: $162

WORKED EXAMPLE · PERIMETER, THEN COST

Border trim costs $2 per foot. Trim the same room.

1 Perimeter: 2 × 9 + 2 × 6 = 30 ft.

2 Cost: 30 × 2 = 60.

Answer: $60

★ KEY IDEA

Covering → area. Going around → perimeter.

✎ TRY IT

Answers are shown in color.

1. Carpet for a 12 ft × 10 ft room at $4 per sq ft?

Answer: $480

2. Fence around a 20 m × 15 m field at $5 per meter?

Answer: $350

3. How many 1-ft tiles for a 7 ft × 8 ft floor?

Answer: 56 tiles

Workbook Answer Key · Lesson 9.7

#	Question	Answer
1	Carpet for a 12 ft × 10 ft room at $4 per sq ft?	$480
2	Fence around a 20 m × 15 m field at $5 per meter?	$350
3	How many 1-ft tiles for a 7 ft × 8 ft floor?	56 tiles
4	54 × 3	162
5	30 × 2	60
6	Grass seed covers 100 square feet per bag. A yard is 30 ft by 20 ft. How many bags?	6 bags
7	Lillie puts ribbon around a 16 in. × 12 in. poster. Ribbon costs $1 per 4 inches. How much does it cost?	$14 (56 in.)
8	A garden 15 ft × 8 ft has a 3 ft × 4 ft shed in it. How much of the garden can be planted?	108 sq ft
9	Mom paints a wall 12 ft by 9 ft with a 3 ft by 4 ft window. How many square feet does she paint?	96 sq ft
10	Write your own area or perimeter word problem and solve it.	Answers vary.

LESSON 9.8 · TEACHER PLAN
Week 2 · Lesson 4 of 4
Unit 9: Measurement, Area, and Perimeter

Lesson 9.8: Area or Perimeter?

Student goal	I can decide whether a problem needs area or perimeter and explain why.
Objective	Students choose between area and perimeter.
Standard	4.MD.3 — Apply area and perimeter formulas.
Pacing	Unit 9, week 2, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 9.8
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Sort situation cards into area/perimeter.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Using key words without visualizing.

SUPPORT

- Act it out with a finger: trace (perimeter) or rub (area).

EXTEND

- Find 3 area and 3 perimeter jobs at home.

STUDENT LESSON 9.8 · WITH ANSWERS

Lesson 9.8: Area or Perimeter?

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.8 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.8, Part A.
- [ ] Complete the rest of Workbook Lesson 9.8 (Parts B–D).

LET'S EXPLORE

Are you finding area or perimeter when you put a frame on a picture? When you paint the picture?

Situation	Area or perimeter?
fence around a yard	perimeter
carpet for a room	area
trim around a door	perimeter
paint for a wall	area
lace around a pillow	perimeter
grass seed for a lawn	area

★ KEY IDEA

Covering a surface → area (square units). Edge or border → perimeter (units).

✎ TRY IT

Answers are shown in color.

1. A border of tape around a table top

Answer: perimeter

2. Wrapping paper to cover a box's top

Answer: area

3. A path around a pond

Answer: perimeter

Workbook Answer Key · Lesson 9.8

#	Question	Answer
1	A border of tape around a table top	perimeter
2	Wrapping paper to cover a box's top	area
3	A path around a pond	perimeter
4	Area of 8 × 7	56
5	Perimeter of 8 × 7	30
6	Tiles for a 5 ft × 5 ft floor	area; 25 sq ft
7	Fence for a 10 m × 6 m garden	perimeter; 32 m
8	Rug for a 9 ft × 12 ft room	area; 108 sq ft
9	Frame for a 8 in. × 10 in. photo	perimeter; 36 in.
10	Explain the difference between area and perimeter using your bedroom.	Area: floor covering; perimeter: baseboard around the room.

WEEKLY CHECK · ANSWER KEY
After Lesson 9.8

About 20 minutes

Weekly Check 22: Area Problems — Answer Key

Answers

#	Question	Answer
1	A = 54 sq ft, l = 9 ft. Find w.	6 ft
2	P = 28 cm, l = 9 cm. Find w.	5 cm
3	Area of the L-shape (figure)	39 sq m (9 × 3 + 3 × 4)
4	Carpet for 10 ft × 8 ft at $3 per sq ft	$240
5	Area or perimeter: paint for a wall?	area
6	Trim around all 4 sides of a 6 ft × 4 ft window	20 ft

LESSON 9.9 · TEACHER PLAN
Week 3 · Lesson 1 of 4
Unit 9: Measurement, Area, and Perimeter

Lesson 9.9: Customary Length

Student goal	I can convert between inches, feet, yards, and miles.
Objective	Students convert customary length units.
Standard	4.MD.1 — Know relative sizes of measurement units; express larger units in terms of smaller units; record in a two-column table.
Pacing	Unit 9, week 3, lesson 1 of 4
Time	About 50 minutes
Materials	Ruler, yardstick, tape measure, workbook Lesson 9.9
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	12 × 6.
Launch	5	Read Let's Explore. Measure each other.
Teach	15	Show 1 ft on a ruler, 1 yd on a yardstick. Say: “Big to small, multiply.”
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Dividing instead of multiplying.

SUPPORT

- Conversion card.

EXTEND

- Measure a room in feet and convert to inches.

STUDENT LESSON 9.9 · WITH ANSWERS

Lesson 9.9: Customary Length

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.9 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.9, Part A.
- [ ] Complete the rest of Workbook Lesson 9.9 (Parts B–E).

LET'S EXPLORE

Lillie is 4 feet 6 inches tall. How many inches tall is she?

Unit	Equals
1 foot (ft)	12 inches (in.)
1 yard (yd)	3 feet = 36 inches
1 mile (mi)	5,280 feet

To change a larger unit to a smaller unit, multiply. There are more of the small units.

WORKED EXAMPLE · FEET AND INCHES TO INCHES

4 ft 6 in. = ___ in.

1 4 × 12 = 48 inches.

2 48 + 6 = 54.

Answer: 54 inches

WORKED EXAMPLE · MAKE A TABLE

Yards to feet

1 Multiply by 3: 1 → 3, 2 → 6, 3 → 9, 4 → 12.

Answer: yards × 3 = feet

★ KEY IDEA

Big unit → small unit: multiply by how many small units are in one big unit.

✎ TRY IT

Answers are shown in color.

1. 5 ft = ___ in.

Answer: 60

2. 7 yd = ___ ft

Answer: 21

3. 2 mi = ___ ft

Answer: 10,560

Workbook Answer Key · Lesson 9.9

#	Question	Answer
1	5 ft = ___ in.	60
2	7 yd = ___ ft	21
3	2 mi = ___ ft	10,560
4	12 × 6	72
5	3 × 9	27
6	3 ft = ___ in.	36
7	8 ft = ___ in.	96
8	4 yd = ___ ft	12
9	10 yd = ___ ft	30
10	2 yd = ___ in.	72
11	3 ft 4 in. = ___ in.	40
12	6 ft 2 in. = ___ in.	74
13	3 mi = ___ ft	15,840
14	Feet: 1, 2, 3, 4, 5 → Inches	12, 24, 36, 48, 60
15	Which is longer: 2 yards or 70 inches? Explain.	2 yards (72 in.) > 70 in.

LESSON 9.10 · TEACHER PLAN
Week 3 · Lesson 2 of 4
Unit 9: Measurement, Area, and Perimeter

Lesson 9.10: Customary Weight and Capacity

Student goal	I can convert ounces and pounds, and cups, pints, quarts, and gallons.
Objective	Students convert customary weight and capacity.
Standard	4.MD.1 — Express larger units in terms of smaller units within one system.
Pacing	Unit 9, week 3, lesson 2 of 4
Time	About 50 minutes
Materials	Measuring cups, pint/quart/gallon containers, kitchen scale, workbook Lesson 9.10
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	16 × 3.
Launch	5	Read Let's Explore.
Teach	15	Pour water: 2 cups into a pint, 2 pints into a quart, 4 quarts into a gallon. Weigh a 1-lb item.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Confusing fluid ounces with ounces of weight.

SUPPORT

- Kitchen measuring.

EXTEND

- Gallon creature drawing (G with Q's, P's, C's).

STUDENT LESSON 9.10 · WITH ANSWERS

Lesson 9.10: Customary Weight and Capacity

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.10 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.10, Part A.
- [ ] Complete the rest of Workbook Lesson 9.10 (Parts B–E).

LET'S EXPLORE

A recipe needs 3 quarts of water. How many cups is that?

Weight	Capacity
1 pound (lb) = 16 ounces (oz)	1 pint (pt) = 2 cups (c)
	1 quart (qt) = 2 pints = 4 cups
	1 gallon (gal) = 4 quarts = 16 cups

WORKED EXAMPLE · QUARTS TO CUPS

3 qt = ___ c

1 1 qt = 4 c.

2 3 × 4 = 12.

Answer: 12 cups

WORKED EXAMPLE · POUNDS TO OUNCES

2 lb 5 oz = ___ oz

1 2 × 16 = 32.

2 32 + 5 = 37.

Answer: 37 oz

★ KEY IDEA

Learn the facts: 16 oz = 1 lb; 2 c = 1 pt; 2 pt = 1 qt; 4 qt = 1 gal.

✎ TRY IT

Answers are shown in color.

1. 4 lb = ___ oz

Answer: 64

2. 2 gal = ___ qt

Answer: 8

3. 5 pt = ___ c

Answer: 10

Workbook Answer Key · Lesson 9.10

#	Question	Answer
1	4 lb = ___ oz	64
2	2 gal = ___ qt	8
3	5 pt = ___ c	10
4	16 × 3	48
5	4 × 4	16
6	3 lb = ___ oz	48
7	1 lb 8 oz = ___ oz	24
8	6 qt = ___ c	24
9	3 gal = ___ qt	12
10	4 qt = ___ pt	8
11	1 gal = ___ c	16
12	7 pt = ___ c	14
13	5 lb = ___ oz	80
14	A jug holds 1 gallon of tea. How many 1-cup glasses can it fill?	16 glasses
15	A puppy weighs 4 lb 3 oz. How many ounces is that?	67 oz
16	Which is more: 3 quarts or 10 cups? Explain.	3 quarts (12 cups) > 10 cups

LESSON 9.11 · TEACHER PLAN
Week 3 · Lesson 3 of 4
Unit 9: Measurement, Area, and Perimeter

Lesson 9.11: Metric Units

Student goal	I can convert metric units of length, mass, and capacity.
Objective	Students convert metric units.
Standard	4.MD.1 — Express larger units in terms of smaller units within the metric system.
Pacing	Unit 9, week 3, lesson 3 of 4
Time	About 50 minutes
Materials	Meter stick, liter bottle, kitchen scale, workbook Lesson 9.11
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	6 × 1,000.
Launch	5	Read Let's Explore.
Teach	15	Show cm and m on a meter stick; a 1 L bottle; a 1 kg bag.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Using 12 or 16 for metric.

SUPPORT

- Place-value chart for metric.

EXTEND

- Measure your height in cm and m.

STUDENT LESSON 9.11 · WITH ANSWERS

Lesson 9.11: Metric Units

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.11 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.11, Part A.
- [ ] Complete the rest of Workbook Lesson 9.11 (Parts B–E).

LET'S EXPLORE

A race is 5 kilometers long. How many meters is that?

Length	Mass	Capacity
1 cm = 10 mm	1 kg = 1,000 g	1 L = 1,000 mL
1 m = 100 cm		
1 km = 1,000 m		

Metric units use place value: kilo- means 1,000, centi- means one hundredth, milli- means one thousandth.

WORKED EXAMPLE · KILOMETERS TO METERS

5 km = ___ m

1 1 km = 1,000 m.

2 5 × 1,000 = 5,000.

Answer: 5,000 m

★ KEY IDEA

Metric conversions multiply by 10, 100, or 1,000.

✎ TRY IT

Answers are shown in color.

1. 3 m = ___ cm

Answer: 300

2. 4 kg = ___ g

Answer: 4,000

3. 2 L = ___ mL

Answer: 2,000

Workbook Answer Key · Lesson 9.11

#	Question	Answer
1	3 m = ___ cm	300
2	4 kg = ___ g	4,000
3	2 L = ___ mL	2,000
4	6 × 1,000	6,000
5	8 × 100	800
6	7 m = ___ cm	700
7	2 km = ___ m	2,000
8	9 cm = ___ mm	90
9	6 kg = ___ g	6,000
10	5 L = ___ mL	5,000
11	3 m 25 cm = ___ cm	325
12	1 kg 500 g = ___ g	1,500
13	4 km 20 m = ___ m	4,020
14	A water bottle holds 750 mL. How many mL do 4 bottles hold? Is that more than 3 L?	3,000 mL; equal to 3 L
15	Which is longer: 2 m or 150 cm? Explain.	2 m (200 cm)

LESSON 9.12 · TEACHER PLAN
Week 3 · Lesson 4 of 4
Unit 9: Measurement, Area, and Perimeter

Lesson 9.12: Time Conversions

Student goal	I can convert hours, minutes, seconds, days, and weeks.
Objective	Students convert units of time.
Standard	4.MD.1 — Know relative sizes of units of time; express larger units in smaller units.
Pacing	Unit 9, week 3, lesson 4 of 4
Time	About 50 minutes
Materials	Clock, calendar, workbook Lesson 9.12
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	60 × 3.
Launch	5	Read Let's Explore.
Teach	15	Work the examples on a clock.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Using 100 minutes per hour.

SUPPORT

- Clock.

EXTEND

- How many seconds are in a day? (86,400)

STUDENT LESSON 9.12 · WITH ANSWERS

Lesson 9.12: Time Conversions

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.12 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.12, Part A.
- [ ] Complete the rest of Workbook Lesson 9.12 (Parts B–D).

LET'S EXPLORE

Ellie's soccer practice is 90 minutes. Is that more or less than 2 hours?

Unit	Equals
1 minute	60 seconds
1 hour	60 minutes
1 day	24 hours
1 week	7 days
1 year	12 months = 52 weeks (about) = 365 days

WORKED EXAMPLE · HOURS AND MINUTES

2 hours = ___ minutes

1 2 × 60 = 120.

Answer: 120 minutes, so 90 minutes is less

WORKED EXAMPLE · MIXED UNITS

1 hr 25 min = ___ min

1 60 + 25 = 85.

Answer: 85 minutes

★ KEY IDEA

60 seconds = 1 minute; 60 minutes = 1 hour; 24 hours = 1 day; 7 days = 1 week.

✎ TRY IT

Answers are shown in color.

1. 3 hr = ___ min

Answer: 180

2. 4 min = ___ sec

Answer: 240

3. 2 weeks = ___ days

Answer: 14

Workbook Answer Key · Lesson 9.12

#	Question	Answer
1	3 hr = ___ min	180
2	4 min = ___ sec	240
3	2 weeks = ___ days	14
4	60 × 3	180
5	24 × 2	48
6	5 min = ___ sec	300
7	4 hr = ___ min	240
8	3 days = ___ hr	72
9	6 weeks = ___ days	42
10	2 hr 15 min = ___ min	135
11	1 min 40 sec = ___ sec	100
12	1 day 6 hr = ___ hr	30
13	3 years = ___ months	36
14	Which is longer: 200 minutes or 3 hours? By how much?	200 min, by 20 min

WEEKLY CHECK · ANSWER KEY
After Lesson 9.12

About 15 minutes

Weekly Check 23: Converting Units — Answer Key

Answers

#	Question	Answer
1	6 ft = ___ in.	72
2	5 yd = ___ ft	15
3	3 lb = ___ oz	48
4	2 gal = ___ qt	8
5	4 m = ___ cm	400
6	3 L = ___ mL	3,000
7	2 hr 10 min = ___ min	130
8	3 weeks = ___ days	21

LESSON 9.13 · TEACHER PLAN
Week 4 · Lesson 1 of 4
Unit 9: Measurement, Area, and Perimeter

Lesson 9.13: Elapsed Time

Student goal	I can find how much time has passed and find start or end times.
Objective	Students solve elapsed time problems.
Standard	4.MD.2 — Solve word problems involving intervals of time using number line diagrams.
Pacing	Unit 9, week 4, lesson 1 of 4
Time	About 50 minutes
Materials	Clock, workbook Lesson 9.13
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	60 – 45.
Launch	5	Read Let's Explore.
Teach	15	Model counting up on a time number line.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Subtracting times like whole numbers (420 – 245).

SUPPORT

- Time number lines.

EXTEND

- Plan a day's schedule with elapsed times.

STUDENT LESSON 9.13 · WITH ANSWERS

Lesson 9.13: Elapsed Time

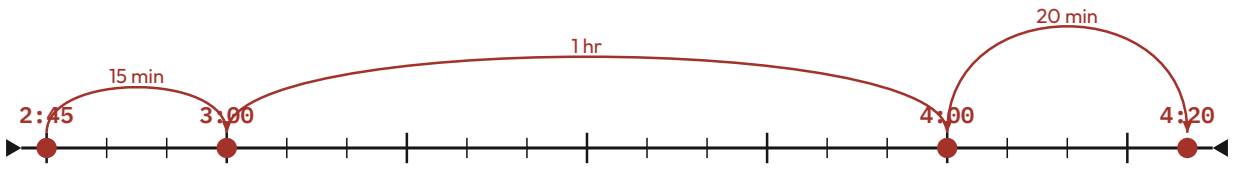
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.13 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.13, Part A.
- [ ] Complete the rest of Workbook Lesson 9.13 (Parts B–E).

LET'S EXPLORE

A movie starts at 2:45 p.m. and ends at 4:20 p.m. How long is the movie?

Count up from the start time in friendly jumps: to the next hour, then whole hours, then minutes.



WORKED EXAMPLE · COUNT UP

2:45 to 4:20

1 2:45 → 3:00 is 15 minutes.

2 3:00 → 4:00 is 1 hour.

3 4:00 → 4:20 is 20 minutes.

4 Total: 1 hr 35 min.

Answer: 1 hour 35 minutes

WORKED EXAMPLE · FIND THE END TIME

Start 9:30 a.m., lasts 2 hr 45 min.

1 9:30 + 2 hr = 11:30.

2 11:30 + 45 min = 12:15 p.m.

Answer: 12:15 p.m.

★ KEY IDEA

Jump to the next hour, then by hours, then by minutes. Watch for a.m. and p.m.

✎ TRY IT

Answers are shown in color.

1. 1:20 p.m. to 3:50 p.m.

Answer: 2 hr 30 min

2. Start 10:15, 1 hr 50 min later?

Answer: 12:05

3. Ends 5:00 p.m., lasted 2 hr 20 min. Start?

Answer: 2:40 p.m.

Workbook Answer Key · Lesson 9.13

#	Question	Answer
1	1:20 p.m. to 3:50 p.m.	2 hr 30 min
2	Start 10:15, 1 hr 50 min later?	12:05
3	Ends 5:00 p.m., lasted 2 hr 20 min. Start?	2:40 p.m.
4	60 − 45	15
5	1 hr 35 min = ___ min	95
6	8:15 a.m. to 11:00 a.m.	2 hr 45 min
7	12:40 p.m. to 2:10 p.m.	1 hr 30 min
8	3:25 p.m. to 6:05 p.m.	2 hr 40 min
9	10:50 a.m. to 1:15 p.m.	2 hr 25 min
10	Start 7:45 a.m., lasts 1 hr 30 min	9:15 a.m.
11	Start 4:20 p.m., lasts 2 hr 55 min	7:15 p.m.
12	End 3:30 p.m., lasted 45 min	2:45 p.m.
13	End 11:10 a.m., lasted 3 hr 25 min	7:45 a.m.
14	School starts at 8:30 a.m. and ends at 2:15 p.m. How long is the school day?	5 hr 45 min

LESSON 9.14 · TEACHER PLAN
Week 4 · Lesson 2 of 4
Unit 9: Measurement, Area, and Perimeter

Lesson 9.14: Measurement Problems with Fractions and Decimals

Student goal	I can solve measurement problems that use fractions and decimals.
Objective	Students solve measurement problems with fractions and decimals.
Standard	4.MD.2 — Solve word problems involving measurements with simple fractions or decimals.
Pacing	Unit 9, week 4, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 9.14
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	4 × 1/4.
Launch	5	Read Let's Explore.
Teach	15	Work both examples.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Converting fractions of hours using 100.

SUPPORT

- Clock face fractions.

EXTEND

- Double a recipe with fractions.

STUDENT LESSON 9.14 · WITH ANSWERS

Lesson 9.14: Measurement Problems with Fractions and Decimals

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.14 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.14, Part A.
- [ ] Complete the rest of Workbook Lesson 9.14 (Parts B–E).

LET'S EXPLORE

A recipe needs $3/4$ cup of milk per batch. How much for 3 batches? How many pints is that?

WORKED EXAMPLE · FRACTIONS

3 batches of $3/4$ cup

1 $3 \times 3/4 = 9/4 = 2\,1/4$ cups.

2 2 cups = 1 pint, so it is a little more than 1 pint.

Answer: $2\,1/4$ cups

WORKED EXAMPLE · DECIMALS

A bottle holds 1.5 L. How many mL?

1 1 L = 1,000 mL. 1.5 L = 1,000 + 500.

Answer: 1,500 mL

★ KEY IDEA

Convert to the same unit, then compute with fractions or decimals.

✎ TRY IT

Answers are shown in color.

1. $1/2$ ft = ___ in.

Answer: 6

2. $1/4$ hour = ___ minutes

Answer: 15

3. 2.5 m = ___ cm

Answer: 250

Workbook Answer Key · Lesson 9.14

#	Question	Answer
1	$\frac{1}{2}$ ft = ___ in.	6
2	$\frac{1}{4}$ hour = ___ minutes	15
3	2.5 m = ___ cm	250
4	$4 \times \frac{1}{4}$	1
5	$\frac{1}{2}$ of 60	30
6	$\frac{1}{2}$ hr = ___ min	30
7	$\frac{3}{4}$ ft = ___ in.	9
8	$\frac{1}{4}$ lb = ___ oz	4
9	0.5 kg = ___ g	500
10	Ellie runs $\frac{3}{4}$ mile each day for 6 days. How far does she run?	$4\frac{2}{4}$ miles
11	A ribbon 3.5 m long is cut into two pieces. One is 1.25 m. How long is the other?	2.25 m
12	Dad's workout is $1\frac{1}{2}$ hours. How many minutes is that?	90 minutes
13	Which is longer: $\frac{2}{3}$ hour or 35 minutes?	$\frac{2}{3}$ hour (40 min)

LESSON 9.15 · TEACHER PLAN
Week 4 · Lesson 3 of 4
Unit 9: Measurement, Area, and Perimeter

Lesson 9.15: Multi-Step Measurement Problems

Student goal	I can solve multi-step problems with measurement and conversions.
Objective	Students solve multi-step measurement problems.
Standard	4.MD.2 — Use the four operations to solve word problems involving distances, time, volume, mass, and money.
Pacing	Unit 9, week 4, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 9.15
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Conversions.
Launch	5	Read Let's Explore.
Teach	15	Work the example: convert first.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Mixing units in one computation.

SUPPORT

- Convert first.

EXTEND

- Plan a road trip with distances and times.

STUDENT LESSON 9.15 · WITH ANSWERS

Lesson 9.15: Multi-Step Measurement Problems

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.15 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.15, Part A.
- [ ] Complete the rest of Workbook Lesson 9.15 (Parts B–D).

LET'S EXPLORE

Lillie needs 3 pieces of rope, each 2 feet 4 inches long. She has a 9-foot rope. Is it enough?

WORKED EXAMPLE · CONVERT, MULTIPLY, COMPARE

3 pieces of 2 ft 4 in. from 9 ft

1 2 ft 4 in. = 28 in.

2 3 × 28 = 84 in.

3 9 ft = 108 in.

4 108 − 84 = 24 in. left.

Answer: Yes, with 2 feet left

★ KEY IDEA

Change to the smallest unit first. Then solve step by step.

✎ TRY IT

Answers are shown in color.

1. 4 boards of 1 ft 6 in. Total in feet?

Answer: 6 ft

2. A 2-gallon jug fills how many 1-cup glasses?

Answer: 32

3. 3 movies of 1 hr 40 min. Total time?

Answer: 5 hr

Workbook Answer Key · Lesson 9.15

#	Question	Answer
1	4 boards of 1 ft 6 in. Total in feet?	6 ft
2	A 2-gallon jug fills how many 1-cup glasses?	32
3	3 movies of 1 hr 40 min. Total time?	5 hr
4	2 ft 4 in. = ___ in.	28
5	9 ft = ___ in.	108
6	A baby weighed 7 lb 8 oz at birth and gained 1 lb 12 oz. How much does she weigh now?	9 lb 4 oz
7	Ellie practices piano 25 minutes a day for 2 weeks. How many hours and minutes is that?	5 hr 50 min (350 min)
8	A 2-liter bottle fills glasses of 250 mL. How many glasses?	8 glasses
9	A hike is 3 miles. Lillie has walked 9,240 feet. How many more feet? (1 mi = 5,280 ft)	6,600 feet
10	Write and solve a two-step measurement problem.	Answers vary.

LESSON 9.16 · TEACHER PLAN
Week 4 · Lesson 4 of 4
Unit 9: Measurement, Area, and Perimeter

Lesson 9.16: Measurement Problem Solving

Student goal	I can solve a variety of measurement, area, and perimeter problems.
Objective	Students apply measurement skills in problem solving.
Standard	4.MD.1, 4.MD.2, 4.MD.3 — 4.MD.1–3 review.
Pacing	Unit 9, week 4, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 9.16, Unit 9 Review
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work the garden example.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook Lesson 9.16 and the Unit 9 Review.
Exit Ticket	3	Check. Tomorrow: Unit 9 Test.

COMMON MISCONCEPTIONS

- Mixing up area and perimeter.

SUPPORT

- Formula and conversion cards.

EXTEND

- Measure and plan a real garden bed.

STUDENT LESSON 9.16 · WITH ANSWERS

Lesson 9.16: Measurement Problem Solving

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.16 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.16, Part A.
- [ ] Complete the rest of Workbook Lesson 9.16 (Parts B–E).

LET'S EXPLORE

Design a garden: 20 feet of fence, the biggest area possible, and a 2-hour planting job starting at 9:45 a.m. What do you need?

WORKED EXAMPLE · GARDEN DESIGN

20 ft of fence, biggest area.

1 Perimeter 20: try 5 × 5 = 25 sq ft (largest).

2 Planting: 9:45 + 2 hr = 11:45 a.m.

Answer: 5 ft × 5 ft garden; done at 11:45 a.m.

★ KEY IDEA

Use formulas, conversions, and elapsed time together.

✎ TRY IT

Answers are shown in color.

1. Area of a 15 ft × 4 ft garden

Answer: 60 sq ft

2. 6 qt = ___ gal ___ qt

Answer: 1 gal 2 qt

3. Start 1:50, end 4:05. Elapsed?

Answer: 2 hr 15 min

Workbook Answer Key · Lesson 9.16

#	Question	Answer
1	Area of a 15 ft × 4 ft garden	60 sq ft
2	6 qt = ___ gal ___ qt	1 gal 2 qt
3	Start 1:50, end 4:05. Elapsed?	2 hr 15 min
4	Perimeter of 7 × 3	20
5	Area of 7 × 3	21
6	A = 63 sq m, l = 9 m. Find w.	7 m
7	P of a square with 13 in. sides	52 in.
8	4 ft 9 in. = ___ in.	57
9	3 kg = ___ g	3,000
10	7:35 a.m. + 2 hr 40 min	10:15 a.m.
11	2 gal = ___ c	32
12	A rectangular pool is 25 m by 10 m. A rope goes around it. Tiles cover the bottom. How much rope? How many square meters of tile?	70 m of rope; 250 sq m of tile
13	Which measurement skill from this unit do you use most at home? Give an example.	Answers vary.

UNIT REVIEW · ANSWER KEY
Study before the unit test

About 35 minutes

Unit 9 Review — Answer Key

Homework or a study session before the test.

Answers

#	Question	Answer
1	Rectangle 14 in. × 6 in.: P and A	P = 40 in., A = 84 sq in.
2	A = 80 sq ft, w = 8 ft. Find l.	10 ft
3	Area of the L-shape (figure)	44 sq ft (8 × 4 + 3 × 4)
4	7 ft = ___ in.	84
5	4 qt = ___ c	16
6	5 kg = ___ g	5,000
7	9:40 a.m. to 12:15 p.m.	2 hr 35 min
8	A dog walk is $3/4$ mile, twice a day, for 7 days. How many miles in all?	$10\ 2/4$ miles

UNIT TEST · ANSWER KEY
At the end of the unit

About 45 minutes

Unit 9 Test — Answer Key

Answers

#	Question	Answer
1	Perimeter? (figure)	34 ft
2	Area? (figure)	66 sq ft
3	A = 42 sq cm, l = 7 cm. Find w.	6 cm
4	P = 30 m, l = 10 m. Find w.	5 m
5	Area or perimeter: grass for a lawn?	area
6	Area of the L-shape (figure)	48 sq in. (10 × 3 + 3 × 6)
7	9 ft = ___ in.	108
8	4 yd = ___ ft	12
9	2 lb 3 oz = ___ oz	35
10	3 gal = ___ qt	12
11	6 m = ___ cm	600
12	2 kg = ___ g	2,000
13	4 hr = ___ min	240
14	1 min 15 sec = ___ sec	75
15	10:35 a.m. to 1:10 p.m.	2 hr 35 min
16	Start 6:50 p.m., lasts 1 hr 45 min. End?	8:35 p.m.
17	A room is 12 ft by 11 ft. Carpet costs $5 per square foot. Baseboard costs $2 per foot. What is the total cost of carpet and baseboard?	$752 ($660 carpet + $92 baseboard)
18	Ellie needs 4 pieces of ribbon, each 1 ft 9 in. long. Ribbon comes in 2-yard rolls. Is one roll enough? Explain.	No. 4 × 21 in. = 84 in., but 2 yd = 72 in., so one roll is not enough (12 in. short).

Scoring

Score	What it means / next step
19–22 (85%+)	Mastered.

Score	What it means / next step
16–18	Proficient; correct missed items.
Below 16	Reteach and retest.

UNIT 10 · TEACHER OVERVIEW

Year in Review

Pulling the whole year together: numbers, operations, fractions, decimals, geometry, and measurement.

In the last four weeks you will review every big idea from Grade 4, practice the skills that need it most, and show what you have learned on the Year-End Test.

Big ideas

- Review place value and all four operations.
- Review factors, multiples, and patterns.
- Review fractions and decimals.
- Review geometry, measurement, and data.
- Show your learning on the Year-End Test.

Standards

Code	Standard
4.OA, 4.NBT	Operations and Algebraic Thinking; Number and Operations in Base Ten (review).
4.NF	Number and Operations—Fractions (review).
4.MD, 4.G	Measurement and Data; Geometry (review).

What students should already know

- Units 1–9.

Materials for the unit

- All materials from the year
- Unit Study Guides in the textbook

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 4	10.1	Review: Place Value and Rounding
Week 1	Lesson 2 of 4	10.2	Review: Adding and Subtracting
Week 1	Lesson 3 of 4	10.3	Review: Multiplying
Week 1	Lesson 4 of 4	10.4	Review: Dividing
Week 1	End of week	Check	Weekly Check 24: Whole Numbers
Week 2	Lesson 1 of 4	10.5	Review: Factors and Multiples
Week 2	Lesson 2 of 4	10.6	Review: Patterns
Week 2	Lesson 3 of 4	10.7	Review: Multi-Step Word Problems
Week 2	Lesson 4 of 4	10.8	Review: Whole-Number Challenge
Week 2	End of week	Check	Weekly Check 25: Factors, Patterns, and Problems
Week 3	Lesson 1 of 4	10.9	Review: Equivalent Fractions
Week 3	Lesson 2 of 4	10.10	Review: Comparing Fractions
Week 3	Lesson 3 of 4	10.11	Review: Fraction Operations
Week 3	Lesson 4 of 4	10.12	Review: Decimals
Week 3	End of week	Check	Weekly Check 26: Fractions and Decimals
Week 4	Lesson 1 of 4	10.13	Review: Geometry and Angles
Week 4	Lesson 2 of 4	10.14	Review: Area, Perimeter, and Conversions
Week 4	Lesson 3 of 4	10.15	Review: Data and Line Plots
Week 4	Lesson 4 of 4	10.16	Celebrating a Year of Math
Week 4	Before the test	Review	Year-End Review
Week 4	End of unit	Test	Grade 4 Year-End Test

TEACHING THIS UNIT AT HOME

Use each unit's test results to choose what to reteach during review weeks. Celebrate the year!

UNIT VOCABULARY

review	To study again so you remember.
strategy	A plan for solving a problem.

LESSON 10.1 · TEACHER PLAN
Week 1 · Lesson 1 of 4
Unit 10: Year in Review

Lesson 10.1: Review: Place Value and Rounding

Student goal	I can read, write, compare, and round numbers to one million.
Objective	Students review and strengthen review: place value and rounding.
Standard	4.NBT.1–3 — Cumulative review of Grade 4 standards.
Pacing	Unit 10, week 1, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook lesson pages; the unit's Study Guide in the textbook
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Quick mental-math warm-up from the workbook.
Launch	5	Read Let's Explore and ask the student to explain a strategy from memory.
Teach	15	Reread the Remember box together. Work the example. Ask the student to teach it back to you. Look back at the student's Unit Test: reteach any item that was missed.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Old mistakes coming back — check the misconceptions listed in the original unit.

SUPPORT

- Use the textbook lesson from the original unit.

EXTEND

- Have the student write and solve a challenge problem.

STUDENT LESSON 10.1 · WITH ANSWERS

Lesson 10.1: Review: Place Value and Rounding

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 10.1 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 10.1, Part A.
- [ ] Complete the rest of Workbook Lesson 10.1 (Parts B–D).

LET'S EXPLORE

What is the greatest six-digit number you can write? Round it to the nearest hundred thousand.

★ REMEMBER

Each place is 10 times the place to its right. To round, look one place to the right: 5 or more rounds up.

WORKED EXAMPLE · ROUND AND COMPARE

Round 638,452 to the nearest ten thousand. Is it greater than 638,524?

1 Ten thousands digit 3; look right: 8 → round up.

2 640,000.

3 638,452 < 638,524 (hundreds: 4 < 5).

Answer: 640,000; no

✎ TRY IT

Answers are shown in color.

1. Write 407,019 in words.

Answer: four hundred seven thousand nineteen

2. Round 85,349 to the nearest thousand.

Answer: 85,000

3. Value of the 6 in 961,000

Answer: 60,000

Workbook Answer Key · Lesson 10.1

#	Question	Answer
1	Write 407,019 in words.	four hundred seven thousand nineteen
2	Round 85,349 to the nearest thousand.	85,000
3	Value of the 6 in 961,000	60,000
4	10 × 4,000	40,000
5	Round 96 to the nearest ten.	100
6	Write 230,506 in expanded form.	200,000 + 30,000 + 500 + 6
7	Write 76,000 + 900 + 3 in standard form.	76,903
8	Compare 409,999 ◯ 410,001	<
9	Order: 52,110; 51,210; 52,101	51,210; 52,101; 52,110
10	Round 472,815 to the nearest hundred thousand.	500,000
11	Round 7,449 to the nearest hundred.	7,400
12	3,600 = ___ hundreds	36
13	Explain how 5,500 rounds to the nearest thousand.	Hundreds digit is 5, so round up to 6,000.

LESSON 10.2 · TEACHER PLAN
Week 1 · Lesson 2 of 4
Unit 10: Year in Review

Lesson 10.2: Review: Adding and Subtracting

Student goal	I can add and subtract multi-digit numbers fluently and check with estimation.
Objective	Students review and strengthen review: adding and subtracting.
Standard	4.NBT.4 — Cumulative review of Grade 4 standards.
Pacing	Unit 10, week 1, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook lesson pages; the unit's Study Guide in the textbook
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Quick mental-math warm-up from the workbook.
Launch	5	Read Let's Explore and ask the student to explain a strategy from memory.
Teach	15	Reread the Remember box together. Work the example. Ask the student to teach it back to you. Look back at the student's Unit Test: reteach any item that was missed.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Old mistakes coming back — check the misconceptions listed in the original unit.

SUPPORT

- Use the textbook lesson from the original unit.

EXTEND

- Have the student write and solve a challenge problem.

STUDENT LESSON 10.2 · WITH ANSWERS

Lesson 10.2: Review: Adding and Subtracting

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 10.2 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 10.2, Part A.
- [ ] Complete the rest of Workbook Lesson 10.2 (Parts B–E).

LET'S EXPLORE

A truck carried 12,486 pounds on Monday and 9,758 pounds on Tuesday. How much more on Monday? How much in all?

★ REMEMBER

Line up places. Add or subtract from right to left, regrouping as needed. Estimate to check.

WORKED EXAMPLE · BOTH OPERATIONS

12,486 and 9,758

1 Sum: 12,486 + 9,758 = 22,244.

2 Difference: 12,486 − 9,758 = 2,728.

3 Check: 12,000 + 10,000 = 22,000; 12,000 − 10,000 = 2,000.

Answer: 22,244 in all; 2,728 more

✎ TRY IT

Answers are shown in color.

1. 46,327 + 38,895

Answer: 85,222

2. 70,000 − 26,418

Answer: 43,582

3. Estimate 6,812 − 2,390.

Answer: 7,000 − 2,000 = 5,000

Workbook Answer Key · Lesson 10.2

#	Question	Answer
1	46,327 + 38,895	85,222
2	70,000 − 26,418	43,582
3	Estimate 6,812 − 2,390.	7,000 − 2,000 = 5,000
4	900 − 456	444
5	378 + 549	927
6	28,507 + 46,918 (figure)	75,425
7	63,205 - 17,846 (figure)	45,359
8	400,000 - 152,736 (figure)	247,264
9	19,999 + 80,001 (figure)	100,000
10	A school raised $14,250. It spent $6,875 on books and $3,940 on computers. How much is left?	$3,435
11	Find 50,003 − 28,765 and check by adding.	21,238; 21,238 + 28,765 = 50,003

LESSON 10.3 · TEACHER PLAN
Week 1 · Lesson 3 of 4
Unit 10: Year in Review

Lesson 10.3: Review: Multiplying

Student goal	I can multiply up to four digits by one digit and two digits by two digits.
Objective	Students review and strengthen review: multiplying.
Standard	4.NBT.5 — Cumulative review of Grade 4 standards.
Pacing	Unit 10, week 1, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook lesson pages; the unit's Study Guide in the textbook
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Quick mental-math warm-up from the workbook.
Launch	5	Read Let's Explore and ask the student to explain a strategy from memory.
Teach	15	Reread the Remember box together. Work the example. Ask the student to teach it back to you. Look back at the student's Unit Test: reteach any item that was missed.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Old mistakes coming back — check the misconceptions listed in the original unit.

SUPPORT

- Use the textbook lesson from the original unit.

EXTEND

- Have the student write and solve a challenge problem.

STUDENT LESSON 10.3 · WITH ANSWERS

Lesson 10.3: Review: Multiplying

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 10.3 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 10.3, Part A.
- [ ] Complete the rest of Workbook Lesson 10.3 (Parts B–E).

LET'S EXPLORE

A stadium has 48 sections with 50 seats each. How many seats is that? How could you find it quickly?

★ REMEMBER

Use area models, partial products, or the standard algorithm. Estimate to check.

WORKED EXAMPLE · TWO STRATEGIES

Find 6 × 2,347 and 34 × 27.

1 6 × 2,347: 6 × 2,000 + 6 × 300 + 6 × 40 + 6 × 7 = 12,000 + 1,800 + 240 + 42 = 14,082.

2 34 × 27: 600 + 80 + 210 + 28 = 918.

Answer: 14,082 and 918

✎ TRY IT

Answers are shown in color.

1. 7 × 3,058

Answer: 21,406

2. 45 × 23

Answer: 1,035

3. 80 × 60

Answer: 4,800

Workbook Answer Key · Lesson 10.3

#	Question	Answer
1	7 × 3,058	21,406
2	45 × 23	1,035
3	80 × 60	4,800
4	9 × 700	6,300
5	40 × 50	2,000
6	486 × 7	3,402
7	2,905 × 6	17,430
8	63 × 48	3,024
9	75 × 32	2,400
10	A bakery makes 36 trays of 24 muffins each day. How many muffins in 5 days?	4,320 muffins
11	Show 27 × 14 with an area model.	378

LESSON 10.4 · TEACHER PLAN
Week 1 · Lesson 4 of 4
Unit 10: Year in Review

Lesson 10.4: Review: Dividing

Student goal	I can divide up to four digits by one digit and interpret remainders.
Objective	Students review and strengthen review: dividing.
Standard	4.NBT.6, 4.OA.3 — Cumulative review of Grade 4 standards.
Pacing	Unit 10, week 1, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook lesson pages; the unit's Study Guide in the textbook
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Quick mental-math warm-up from the workbook.
Launch	5	Read Let's Explore and ask the student to explain a strategy from memory.
Teach	15	Reread the Remember box together. Work the example. Ask the student to teach it back to you. Look back at the student's Unit Test: reteach any item that was missed.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Old mistakes coming back — check the misconceptions listed in the original unit.

SUPPORT

- Use the textbook lesson from the original unit.

EXTEND

- Have the student write and solve a challenge problem.

STUDENT LESSON 10.4 · WITH ANSWERS

Lesson 10.4: Review: Dividing

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 10.4 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 10.4, Part A.
- [ ] Complete the rest of Workbook Lesson 10.4 (Parts B–E).

LET'S EXPLORE

A farmer packs 1,350 eggs into cartons of 6. How many full cartons? Any left over?

★ REMEMBER

Divide, multiply, subtract, bring down. Check: quotient × divisor + remainder = dividend. Decide what the remainder means.

WORKED EXAMPLE · LONG DIVISION

1,350 ÷ 6

1 13 ÷ 6 = 2 R1; 15 ÷ 6 = 2 R3; 30 ÷ 6 = 5.

2 Check: 225 × 6 = 1,350.

Answer: 225 cartons, none left

✎ TRY IT

Answers are shown in color.

1. 2,184 ÷ 7

Answer: 312

2. 605 ÷ 4

Answer: 151 R1

3. 50 people, 8 per table. How many tables?

Answer: 7 tables

Workbook Answer Key · Lesson 10.4

#	Question	Answer
1	2,184 ÷ 7	312
2	605 ÷ 4	151 R1
3	50 people, 8 per table. How many tables?	7 tables
4	4,200 ÷ 6	700
5	58 ÷ 7	8 R2
6	936 ÷ 4 (figure)	234
7	1,475 ÷ 5 (figure)	295
8	3,209 ÷ 8 (figure)	401 R1
9	6,012 ÷ 9 (figure)	668
10	A ribbon 250 inches long is cut into 9-inch pieces. How many full pieces? How much is left?	27 pieces; 7 inches left
11	Write a story for 100 ÷ 8 where the answer is 13.	Example: 100 people, 8 per van → 13 vans

WEEKLY CHECK · ANSWER KEY
After Lesson 10.4

About 20 minutes

Weekly Check 24: Whole Numbers — Answer Key

Answers

#	Question	Answer
1	Round 649,512 to the nearest ten thousand.	650,000
2	37,485 + 48,697	86,182
3	60,000 − 23,758	36,242
4	8 × 3,047	24,376
5	53 × 26	1,378
6	2,346 ÷ 6	391

LESSON 10.5 · TEACHER PLAN
Week 2 · Lesson 1 of 4
Unit 10: Year in Review

Lesson 10.5: Review: Factors and Multiples

Student goal	I can find factors, multiples, and tell prime from composite.
Objective	Students review and strengthen review: factors and multiples.
Standard	4.OA.4 — Cumulative review of Grade 4 standards.
Pacing	Unit 10, week 2, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook lesson pages; the unit's Study Guide in the textbook
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Quick mental-math warm-up from the workbook.
Launch	5	Read Let's Explore and ask the student to explain a strategy from memory.
Teach	15	Reread the Remember box together. Work the example. Ask the student to teach it back to you. Look back at the student's Unit Test: reteach any item that was missed.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Old mistakes coming back — check the misconceptions listed in the original unit.

SUPPORT

- Use the textbook lesson from the original unit.

EXTEND

- Have the student write and solve a challenge problem.

STUDENT LESSON 10.5 · WITH ANSWERS

Lesson 10.5: Review: Factors and Multiples

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 10.5 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 10.5, Part A.
- [ ] Complete the rest of Workbook Lesson 10.5 (Parts B–D).

LET'S EXPLORE

Is 91 prime? (Many people think so!)

★ REMEMBER

Factor pairs multiply to the number. Prime: exactly 2 factors. Multiples come from skip counting.

WORKED EXAMPLE · PRIME OR COMPOSITE

Is 91 prime?

1 Test small factors: 2 no, 3 no (9 + 1 = 10), 5 no, 7: 7 × 13 = 91.

Answer: Composite

✎ TRY IT

Answers are shown in color.

1. Factors of 36

Answer: 1, 2, 3, 4, 6, 9, 12, 18, 36

2. Is 53 prime?

Answer: Yes

3. First common multiple of 6 and 8

Answer: 24

Workbook Answer Key · Lesson 10.5

#	Question	Answer
1	Factors of 36	1, 2, 3, 4, 6, 9, 12, 18, 36
2	Is 53 prime?	Yes
3	First common multiple of 6 and 8	24
4	7 × 13	91
5	Is 2 prime?	Yes
6	Factor pairs of 60	1×60, 2×30, 3×20, 4×15, 5×12, 6×10
7	Prime or composite: 39	composite
8	Prime or composite: 61	prime
9	First 5 multiples of 11	11, 22, 33, 44, 55
10	Is 126 divisible by 9?	Yes (1 + 2 + 6 = 9)
11	Is 7 a factor of 98?	Yes
12	Name two numbers that have exactly 3 factors.	e.g. 4 (1, 2, 4), 9, 25, 49

LESSON 10.6 · TEACHER PLAN
Week 2 · Lesson 2 of 4
Unit 10: Year in Review

Lesson 10.6: Review: Patterns

Student goal	I can extend number and shape patterns and describe their rules and features.
Objective	Students review and strengthen review: patterns.
Standard	4.OA.5 — Cumulative review of Grade 4 standards.
Pacing	Unit 10, week 2, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook lesson pages; the unit's Study Guide in the textbook
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Quick mental-math warm-up from the workbook.
Launch	5	Read Let's Explore and ask the student to explain a strategy from memory.
Teach	15	Reread the Remember box together. Work the example. Ask the student to teach it back to you. Look back at the student's Unit Test: reteach any item that was missed.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Old mistakes coming back — check the misconceptions listed in the original unit.

SUPPORT

- Use the textbook lesson from the original unit.

EXTEND

- Have the student write and solve a challenge problem.

STUDENT LESSON 10.6 · WITH ANSWERS

Lesson 10.6: Review: Patterns

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 10.6 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 10.6, Part A.
- [ ] Complete the rest of Workbook Lesson 10.6 (Parts B–D).

LET'S EXPLORE

Rule: start at 5, add 5. Will 101 be in the pattern?

★ REMEMBER

Find the change between terms. Look for features like odd/even. For repeating patterns, use division.

WORKED EXAMPLE · FEATURES

Start at 5, add 5: 5, 10, 15, 20...

1 All terms are multiples of 5.

2 They end in 5 or 0, alternating.

Answer: 101 is not a term

✎ TRY IT

Answers are shown in color.

1. Rule: start at 3, multiply by 2. First 5 terms.

Answer: 3, 6, 12, 24, 48

2. Next: 99, 88, 77, ___

Answer: 66

3. Core: A B B C. 18th letter?

Answer: B (18 ÷ 4 = 4 R2)

Workbook Answer Key · Lesson 10.6

#	Question	Answer
1	Rule: start at 3, multiply by 2. First 5 terms.	3, 6, 12, 24, 48
2	Next: 99, 88, 77, ___	66
3	Core: A B B C. 18th letter?	B (18 ÷ 4 = 4 R2)
4	Count by 7s to 49	7, 14, 21, 28, 35, 42, 49
5	Find the rule: 4, 13, 22, 31	add 9
6	Input 3, 5, 8 → output 18, 30, 48. Rule?	× 6
7	Start at 2, add 4. Odd or even terms?	always even
8	Shape core: ● ▲ ■. 25th shape?	● (25 ÷ 3 = 8 R1)
9	Make a pattern whose terms are all odd. Explain why.	e.g. start 1, add 2: odd + even = odd

LESSON 10.7 · TEACHER PLAN
Week 2 · Lesson 3 of 4
Unit 10: Year in Review

Lesson 10.7: Review: Multi-Step Word Problems

Student goal	I can solve multi-step word problems with all four operations.
Objective	Students review and strengthen review: multi-step word problems.
Standard	4.OA.3 — Cumulative review of Grade 4 standards.
Pacing	Unit 10, week 2, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook lesson pages; the unit's Study Guide in the textbook
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Quick mental-math warm-up from the workbook.
Launch	5	Read Let's Explore and ask the student to explain a strategy from memory.
Teach	15	Reread the Remember box together. Work the example. Ask the student to teach it back to you. Look back at the student's Unit Test: reteach any item that was missed.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Old mistakes coming back — check the misconceptions listed in the original unit.

SUPPORT

- Use the textbook lesson from the original unit.

EXTEND

- Have the student write and solve a challenge problem.

STUDENT LESSON 10.7 · WITH ANSWERS

Lesson 10.7: Review: Multi-Step Word Problems

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 10.7 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 10.7, Part A.
- [ ] Complete the rest of Workbook Lesson 10.7 (Parts B–D).

LET'S EXPLORE

A class of 28 students goes on a trip. Tickets are $9 each. Buses hold 12 students. How many buses, and how much do tickets cost?

★ REMEMBER

Find the hidden steps. Write an equation for each, with a letter for the unknown. Check reasonableness.

WORKED EXAMPLE · TWO QUESTIONS

28 students, $9 tickets, buses of 12

1 Tickets: 28 × 9 = 252.

2 Buses: 28 ÷ 12 = 2 R4 → 3 buses.

Answer: $252 and 3 buses

✎ TRY IT

Answers are shown in color.

1. 4 packs of 18 pencils shared by 6 students. Each gets?

Answer: 12

2. $50 − 3 × $12

Answer: $14

3. A 6-week summer camp, 5 days a week, 3 hours a day. Hours?

Answer: 90 hours

Workbook Answer Key · Lesson 10.7

#	Question	Answer
1	4 packs of 18 pencils shared by 6 students. Each gets?	12
2	$50 − 3 × $12	$14
3	A 6-week summer camp, 5 days a week, 3 hours a day. Hours?	90 hours
4	28 × 9	252
5	28 ÷ 12	2 R4
6	Ellie earns $8 a week. After 9 weeks she buys a $45 game. How much is left?	$27
7	A library has 6 shelves of 125 books. It gives away 230 books. How many remain?	520 books
8	150 cupcakes go in boxes of 8. How many boxes are needed to hold all of them?	19 boxes
9	Write a two-step word problem using multiplication and subtraction. Solve it.	Answers vary.

LESSON 10.8 · TEACHER PLAN
Week 2 · Lesson 4 of 4
Unit 10: Year in Review

Lesson 10.8: Review: Whole-Number Challenge

Student goal	I can solve challenging whole-number problems and explain my strategies.
Objective	Students review and strengthen review: whole-number challenge.
Standard	4.NBT, 4.OA — Cumulative review of Grade 4 standards.
Pacing	Unit 10, week 2, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook lesson pages; the unit's Study Guide in the textbook
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Quick mental-math warm-up from the workbook.
Launch	5	Read Let's Explore and ask the student to explain a strategy from memory.
Teach	15	Reread the Remember box together. Work the example. Ask the student to teach it back to you. Look back at the student's Unit Test: reteach any item that was missed.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Old mistakes coming back — check the misconceptions listed in the original unit.

SUPPORT

- Use the textbook lesson from the original unit.

EXTEND

- Have the student write and solve a challenge problem.

STUDENT LESSON 10.8 · WITH ANSWERS

Lesson 10.8: Review: Whole-Number Challenge

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 10.8 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 10.8, Part A.
- [ ] Complete the rest of Workbook Lesson 10.8 (Parts B–D).

LET'S EXPLORE

Use the digits 2, 4, 6, 8 once each to make the greatest product of a 2-digit × 2-digit problem.

★ REMEMBER

Try, check, and improve. Organized guessing is a real strategy.

WORKED EXAMPLE · ORGANIZED GUESSING

Greatest product with 2, 4, 6, 8

1 Put the big digits in the tens places: 8_ × 6_.

2 Try 82 × 64 = 5,248 and 84 × 62 = 5,208.

3 82 × 64 is greater.

Answer: 82 × 64 = 5,248

✎ TRY IT

Answers are shown in color.

1. Least product with the same digits?

Answer: 26 × 48 = 1,248

2. Greatest 4-digit number divisible by 5 using 1, 3, 5, 7 once each

Answer: 7,315

3. Two numbers multiply to 96 and add to 20. What are they?

Answer: 8 and 12

Workbook Answer Key · Lesson 10.8

#	Question	Answer
1	Least product with the same digits?	26 × 48 = 1,248
2	Greatest 4-digit number divisible by 5 using 1, 3, 5, 7 once each	7,315
3	Two numbers multiply to 96 and add to 20. What are they?	8 and 12
4	82 × 64	5,248
5	84 × 62	5,208
6	Find a 3-digit number that is a multiple of 2, 3, 5, and 9.	e.g. 270, 360, 450
7	A number rounded to the nearest hundred is 4,500. What are the least and greatest whole numbers it could be?	4,450 and 4,549
8	Place +, −, ×, ÷ to make it true: 8 ◯ 4 ◯ 2 = 16	e.g. 8 × 4 ÷ 2 = 16 or 8 + 4 × 2 = 16
9	Which whole-number skill are you proudest of this year? Show it.	Answers vary.

WEEKLY CHECK · ANSWER KEY
After Lesson 10.8
About 20 minutes

Weekly Check 25: Factors, Patterns, and Problems — Answer Key

Answers

#	Question	Answer
1	Factors of 45	1, 3, 5, 9, 15, 45
2	Prime or composite: 87	composite (3 × 29)
3	Rule: start at 4, add 7. Fifth term?	32
4	Is 234 divisible by 9?	Yes
5	100 students ride buses of 24. How many buses?	5 buses
6	$100 − 4 × $18	$28

LESSON 10.9 · TEACHER PLAN
Week 3 · Lesson 1 of 4
Unit 10: Year in Review

Lesson 10.9: Review: Equivalent Fractions

Student goal	I can find equivalent fractions and simplify.
Objective	Students review and strengthen review: equivalent fractions.
Standard	4.NF.1 — Cumulative review of Grade 4 standards.
Pacing	Unit 10, week 3, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook lesson pages; the unit's Study Guide in the textbook
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Quick mental-math warm-up from the workbook.
Launch	5	Read Let's Explore and ask the student to explain a strategy from memory.
Teach	15	Reread the Remember box together. Work the example. Ask the student to teach it back to you. Look back at the student's Unit Test: reteach any item that was missed.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Old mistakes coming back — check the misconceptions listed in the original unit.

SUPPORT

- Use the textbook lesson from the original unit.

EXTEND

- Have the student write and solve a challenge problem.

STUDENT LESSON 10.9 · WITH ANSWERS

Lesson 10.9: Review: Equivalent Fractions

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 10.9 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 10.9, Part A.
- [ ] Complete the rest of Workbook Lesson 10.9 (Parts B–D).

LET'S EXPLORE

Is $\frac{6}{9}$ the same as $\frac{2}{3}$? How can you prove it?

★ REMEMBER

Multiply or divide the numerator and denominator by the same number: $\frac{a}{b} = \frac{n \times a}{n \times b}$.

WORKED EXAMPLE · PROVE IT

$\frac{6}{9}$ and $\frac{2}{3}$

1 $\frac{2}{3} \times \frac{3}{3} = \frac{6}{9}$.

2 A model shows the same amount.

Answer: They are equivalent.

✎ TRY IT

Answers are shown in color.

1. $\frac{3}{5} = \frac{?}{20}$

Answer: 12

2. Simplify $\frac{16}{20}$

Answer: $\frac{4}{5}$

3. Write two fractions equal to $\frac{1}{3}$.

Answer: e.g. $\frac{2}{6}$, $\frac{4}{12}$

Workbook Answer Key · Lesson 10.9

#	Question	Answer
1	$\frac{3}{5} = \frac{?}{20}$	12
2	Simplify $\frac{16}{20}$	$\frac{4}{5}$
3	Write two fractions equal to $\frac{1}{3}$.	e.g. $\frac{2}{6}$, $\frac{4}{12}$
4	Factors of 16	1, 2, 4, 8, 16
5	$\frac{1}{2} = \frac{?}{8}$	4
6	$\frac{2}{3} = \frac{?}{12}$	8
7	$\frac{5}{6} = \frac{?}{12}$	10
8	$\frac{3}{8} = \frac{?}{24}$	9
9	Simplify $\frac{10}{12}$	$\frac{5}{6}$
10	Simplify $\frac{9}{15}$	$\frac{3}{5}$
11	Simplify $\frac{12}{18}$	$\frac{2}{3}$
12	Draw a model to show $\frac{3}{4} = \frac{6}{8}$.	Two equal bars: 3 of 4 and 6 of 8 shaded

LESSON 10.10 · TEACHER PLAN
Week 3 · Lesson 2 of 4
Unit 10: Year in Review

Lesson 10.10: Review: Comparing Fractions

Student goal	I can compare and order fractions with different denominators.
Objective	Students review and strengthen review: comparing fractions.
Standard	4.NF.2 — Cumulative review of Grade 4 standards.
Pacing	Unit 10, week 3, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook lesson pages; the unit's Study Guide in the textbook
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Quick mental-math warm-up from the workbook.
Launch	5	Read Let's Explore and ask the student to explain a strategy from memory.
Teach	15	Reread the Remember box together. Work the example. Ask the student to teach it back to you. Look back at the student's Unit Test: reteach any item that was missed.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Old mistakes coming back — check the misconceptions listed in the original unit.

SUPPORT

- Use the textbook lesson from the original unit.

EXTEND

- Have the student write and solve a challenge problem.

STUDENT LESSON 10.10 · WITH ANSWERS

Lesson 10.10: Review: Comparing Fractions

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 10.10 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 10.10, Part A.
- [ ] Complete the rest of Workbook Lesson 10.10 (Parts B–D).

LET'S EXPLORE

Which is greater: $\frac{5}{8}$ or $\frac{3}{5}$?

★ REMEMBER

Use benchmarks, common denominators, or common numerators. Compare only when the wholes are the same.

WORKED EXAMPLE · COMMON DENOMINATOR

$\frac{5}{8}$ vs $\frac{3}{5}$

1 $\frac{25}{40}$ vs $\frac{24}{40}$.

Answer: $\frac{5}{8} > \frac{3}{5}$

✎ TRY IT

Answers are shown in color.

1. $\frac{4}{9}$ ◯ $\frac{1}{2}$

Answer: <

2. $\frac{7}{10}$ ◯ $\frac{2}{3}$

Answer: >

3. Order: $\frac{1}{2}$, $\frac{3}{8}$, $\frac{3}{4}$

Answer: $\frac{3}{8}$, $\frac{1}{2}$, $\frac{3}{4}$

Workbook Answer Key · Lesson 10.10

#	Question	Answer
1	$\frac{4}{9} \bigcirc \frac{1}{2}$	<
2	$\frac{7}{10} \bigcirc \frac{2}{3}$	>
3	Order: $\frac{1}{2}, \frac{3}{8}, \frac{3}{4}$	$\frac{3}{8}, \frac{1}{2}, \frac{3}{4}$
4	Common multiple of 3 and 4	12
5	$\frac{2}{3} = \frac{?}{12}$	8
6	$\frac{2}{5} \bigcirc \frac{3}{8}$	>
7	$\frac{5}{6} \bigcirc \frac{7}{8}$	<
8	$\frac{4}{12} \bigcirc \frac{1}{3}$	=
9	$\frac{3}{4} \bigcirc \frac{5}{7}$	>
10	$\frac{1}{6} \bigcirc \frac{2}{9}$	<
11	$\frac{9}{10} \bigcirc \frac{4}{5}$	>
12	Order from greatest to least: $\frac{2}{3}, \frac{5}{12}, \frac{3}{4}$	$\frac{3}{4}, \frac{2}{3}, \frac{5}{12}$

LESSON 10.11 · TEACHER PLAN
Week 3 · Lesson 3 of 4
Unit 10: Year in Review

Lesson 10.11: Review: Fraction Operations

Student goal	I can add and subtract fractions and mixed numbers and multiply fractions by whole numbers.
Objective	Students review and strengthen review: fraction operations.
Standard	4.NF.3, 4.NF.4 — Cumulative review of Grade 4 standards.
Pacing	Unit 10, week 3, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook lesson pages; the unit's Study Guide in the textbook
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Quick mental-math warm-up from the workbook.
Launch	5	Read Let's Explore and ask the student to explain a strategy from memory.
Teach	15	Reread the Remember box together. Work the example. Ask the student to teach it back to you. Look back at the student's Unit Test: reteach any item that was missed.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Old mistakes coming back — check the misconceptions listed in the original unit.

SUPPORT

- Use the textbook lesson from the original unit.

EXTEND

- Have the student write and solve a challenge problem.

STUDENT LESSON 10.11 · WITH ANSWERS

Lesson 10.11: Review: Fraction Operations

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 10.11 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 10.11, Part A.
- [ ] Complete the rest of Workbook Lesson 10.11 (Parts B–D).

LET'S EXPLORE

A recipe needs $1\frac{3}{4}$ cups of flour. You triple it. How much flour?

★ REMEMBER

Like denominators: add or subtract numerators. Mixed numbers: regroup when needed. $n \times \frac{a}{b} = \frac{n \times a}{b}$.

WORKED EXAMPLE · TRIPLE A MIXED NUMBER

$3 \times 1\frac{3}{4}$

1 $1\frac{3}{4} = \frac{7}{4}$.

2 $3 \times \frac{7}{4} = \frac{21}{4} = 5\frac{1}{4}$.

Answer: $5\frac{1}{4}$ cups

✎ TRY IT

Answers are shown in color.

1. $2\frac{5}{8} + 1\frac{6}{8}$

Answer: $4\frac{3}{8}$

2. $5\frac{1}{3} - 2\frac{2}{3}$

Answer: $2\frac{2}{3}$

3. $4 \times \frac{5}{6}$

Answer: $3\frac{2}{6}$

Workbook Answer Key · Lesson 10.11

#	Question	Answer
1	$2\frac{5}{8} + 1\frac{6}{8}$	$4\frac{3}{8}$
2	$5\frac{1}{3} - 2\frac{2}{3}$	$2\frac{2}{3}$
3	$4 \times \frac{5}{6}$	$3\frac{2}{6}$
4	$\frac{7}{4}$ as a mixed number	$1\frac{3}{4}$
5	$2\frac{1}{5}$ as a fraction	$\frac{11}{5}$
6	$\frac{5}{12} + \frac{9}{12}$	$1\frac{2}{12}$
7	$\frac{9}{10} - \frac{4}{10}$	$\frac{5}{10}$
8	$3\frac{2}{6} + 4\frac{5}{6}$	$8\frac{1}{6}$
9	$6\frac{1}{4} - 3\frac{3}{4}$	$2\frac{2}{4}$
10	$5 \times \frac{2}{3}$	$3\frac{1}{3}$
11	$7 \times \frac{3}{10}$	$2\frac{1}{10}$
12	Lillie runs $\frac{4}{5}$ mile a day for 6 days. Ellie runs $4\frac{3}{5}$ miles. Who runs farther?	Lillie ($4\frac{4}{5}$ miles)

LESSON 10.12 · TEACHER PLAN
Week 3 · Lesson 4 of 4
Unit 10: Year in Review

Lesson 10.12: Review: Decimals

Student goal	I can write, compare, and add with tenths and hundredths, including money.
Objective	Students review and strengthen review: decimals.
Standard	4.NF.5–7 — Cumulative review of Grade 4 standards.
Pacing	Unit 10, week 3, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook lesson pages; the unit's Study Guide in the textbook
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Quick mental-math warm-up from the workbook.
Launch	5	Read Let's Explore and ask the student to explain a strategy from memory.
Teach	15	Reread the Remember box together. Work the example. Ask the student to teach it back to you. Look back at the student's Unit Test: reteach any item that was missed.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Old mistakes coming back — check the misconceptions listed in the original unit.

SUPPORT

- Use the textbook lesson from the original unit.

EXTEND

- Have the student write and solve a challenge problem.

STUDENT LESSON 10.12 · WITH ANSWERS

Lesson 10.12: Review: Decimals

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 10.12 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 10.12, Part A.
- [ ] Complete the rest of Workbook Lesson 10.12 (Parts B–D).

LET'S EXPLORE

Which is more money: $0.75 or $0.8?

★ REMEMBER

Tenths and hundredths: 0.8 = 0.80. Line up decimal points to compare and to add money.

WORKED EXAMPLE · COMPARE

$0.75 vs $0.8

1 $0.8 = $0.80.

2 80 > 75.

Answer: $0.80 is more

✎ TRY IT

Answers are shown in color.

1. $^{7}/_{100}$ as a decimal

Answer: 0.07

2. $^{4}/_{10} + {}^{35}/_{100}$

Answer: $^{75}/_{100}$

3. Order: 0.5, 0.45, 0.54

Answer: 0.45, 0.5, 0.54

Workbook Answer Key · Lesson 10.12

#	Question	Answer
1	$\frac{7}{100}$ as a decimal	0.07
2	$\frac{4}{10} + \frac{35}{100}$	$\frac{75}{100}$
3	Order: 0.5, 0.45, 0.54	0.45, 0.5, 0.54
4	0.6 = 0.___	60
5	\$1.00 − \$0.35	\$0.65
6	$\frac{39}{100}$ as a decimal	0.39
7	2.07 in words	two and seven hundredths
8	3.4 ◯ 3.38	>
9	$\frac{1}{4}$ ◯ 0.3	<
10	\$12.48 + \$7.75	\$20.23
11	\$20.00 − \$13.86	\$6.14
12	Ellie has \$5. She buys 2 snacks at \$1.35 each. How much is left?	\$2.30

WEEKLY CHECK · ANSWER KEY
After Lesson 10.12

About 20 minutes

Weekly Check 26: Fractions and Decimals — Answer Key

Answers

#	Question	Answer
1	Simplify $\frac{12}{16}$	$\frac{3}{4}$
2	$\frac{3}{5} \bigcirc \frac{5}{8}$	<
3	$3\frac{1}{4} - 1\frac{3}{4}$	$1\frac{2}{4}$
4	$6 \times \frac{3}{4}$	$4\frac{2}{4}$
5	$\frac{6}{10} + \frac{25}{100}$	$\frac{85}{100}$
6	$0.7 \bigcirc 0.65$	>

LESSON 10.13 · TEACHER PLAN
Week 4 · Lesson 1 of 4
Unit 10: Year in Review

Lesson 10.13: Review: Geometry and Angles

Student goal	I can classify shapes and measure, draw, and find angles.
Objective	Students review and strengthen review: geometry and angles.
Standard	4.G.1–3, 4.MD.5–7 — Cumulative review of Grade 4 standards.
Pacing	Unit 10, week 4, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook lesson pages; the unit's Study Guide in the textbook
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Quick mental-math warm-up from the workbook.
Launch	5	Read Let's Explore and ask the student to explain a strategy from memory.
Teach	15	Reread the Remember box together. Work the example. Ask the student to teach it back to you. Look back at the student's Unit Test: reteach any item that was missed.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Old mistakes coming back — check the misconceptions listed in the original unit.

SUPPORT

- Use the textbook lesson from the original unit.

EXTEND

- Have the student write and solve a challenge problem.

STUDENT LESSON 10.13 · WITH ANSWERS

Lesson 10.13: Review: Geometry and Angles

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 10.13 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 10.13, Part A.
- [ ] Complete the rest of Workbook Lesson 10.13 (Parts B–D).

LET'S EXPLORE

A pizza is cut into 6 equal slices. What angle does each slice make at the center?

★ REMEMBER

A full turn is 360°. Angles are additive. Classify shapes by sides, angles, and symmetry.

WORKED EXAMPLE · ANGLES IN A CIRCLE

6 equal slices

1 360 ÷ 6 = 60.

Answer: 60° each

✎ TRY IT

Answers are shown in color.

1. Right angle split into 25° and x

Answer: 65°

2. Lines of symmetry in a rectangle (not a square)

Answer: 2

3. A triangle with one 90° angle is a ___ triangle.

Answer: right

Workbook Answer Key · Lesson 10.13

#	Question	Answer
1	Right angle split into 25° and x	65°
2	Lines of symmetry in a rectangle (not a square)	2

#	Question	Answer
3	A triangle with one 90° angle is a ___ triangle.	right
4	360 ÷ 4	90
5	180 − 35	145
6	Classify a 98° angle.	obtuse
7	Name a quadrilateral with 4 equal sides and no right angles.	rhombus
8	Straight angle split into 120° and x	60°
9	How many endpoints does a ray have?	1
10	Read the protractor. (figure)	105°
11	Draw two perpendicular line segments.	drawing
12	Draw a triangle with one obtuse angle. Label its angles with estimates.	Answers vary; one angle > 90°.

LESSON 10.14 · TEACHER PLAN
Week 4 · Lesson 2 of 4
Unit 10: Year in Review

Lesson 10.14: Review: Area, Perimeter, and Conversions

Student goal	I can solve area, perimeter, and measurement conversion problems.
Objective	Students review and strengthen review: area, perimeter, and conversions.
Standard	4.MD.1–3 — Cumulative review of Grade 4 standards.
Pacing	Unit 10, week 4, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook lesson pages; the unit's Study Guide in the textbook
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Quick mental-math warm-up from the workbook.
Launch	5	Read Let's Explore and ask the student to explain a strategy from memory.
Teach	15	Reread the Remember box together. Work the example. Ask the student to teach it back to you. Look back at the student's Unit Test: reteach any item that was missed.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Old mistakes coming back — check the misconceptions listed in the original unit.

SUPPORT

- Use the textbook lesson from the original unit.

EXTEND

- Have the student write and solve a challenge problem.

STUDENT LESSON 10.14 · WITH ANSWERS

Lesson 10.14: Review: Area, Perimeter, and Conversions

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 10.14 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 10.14, Part A.
- [ ] Complete the rest of Workbook Lesson 10.14 (Parts B–D).

LET'S EXPLORE

A rectangular garden has an area of 72 square feet and a width of 6 feet. How much fence goes around it?

★ REMEMBER

A = l × w; P = 2 × l + 2 × w. Big unit to small unit: multiply.

WORKED EXAMPLE · TWO STEPS

A = 72, w = 6. Find P.

1 l = 72 ÷ 6 = 12.

2 P = 2 × 12 + 2 × 6 = 36.

Answer: 36 feet of fence

✎ TRY IT

Answers are shown in color.

1. 5 yd = ___ in.

Answer: 180

2. Area of 14 m × 9 m

Answer: 126 sq m

3. 2 hr 45 min = ___ min

Answer: 165

Workbook Answer Key · Lesson 10.14

#	Question	Answer
1	5 yd = ___ in.	180
2	Area of 14 m × 9 m	126 sq m
3	2 hr 45 min = ___ min	165
4	72 ÷ 6	12
5	3 × 12	36
6	P and A of 15 cm × 8 cm	P = 46 cm, A = 120 sq cm
7	P = 44 ft, l = 14 ft. Find w.	8 ft
8	3 gal = ___ c	48
9	4 kg 250 g = ___ g	4,250
10	8:50 a.m. to 11:25 a.m.	2 hr 35 min
11	6 lb = ___ oz	96
12	A room is 11 ft × 9 ft. Tiles are 1 sq ft and cost $4 each. Total cost?	$396

LESSON 10.15 · TEACHER PLAN
Week 4 · Lesson 3 of 4
Unit 10: Year in Review

Lesson 10.15: Review: Data and Line Plots

Student goal	I can make and use line plots with fractions.
Objective	Students review and strengthen review: data and line plots.
Standard	4.MD.4 — Cumulative review of Grade 4 standards.
Pacing	Unit 10, week 4, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook lesson pages; the unit's Study Guide in the textbook
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Quick mental-math warm-up from the workbook.
Launch	5	Read Let's Explore and ask the student to explain a strategy from memory.
Teach	15	Reread the Remember box together. Work the example. Ask the student to teach it back to you. Look back at the student's Unit Test: reteach any item that was missed.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Old mistakes coming back — check the misconceptions listed in the original unit.

SUPPORT

- Use the textbook lesson from the original unit.

EXTEND

- Have the student write and solve a challenge problem.

STUDENT LESSON 10.15 · WITH ANSWERS

Lesson 10.15: Review: Data and Line Plots

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 10.15 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 10.15, Part A.
- [ ] Complete the rest of Workbook Lesson 10.15 (Parts B–D).

LET'S EXPLORE

Five friends' jump distances in feet: $4\frac{1}{2}$, 5, $4\frac{3}{4}$, 5, $4\frac{1}{2}$. Make a line plot. What is the difference between the longest and shortest jumps?

★ REMEMBER

Each X is one data value. Use fraction operations to answer questions.

WORKED EXAMPLE · READ A LINE PLOT

Longest − shortest

1 $5 - 4\frac{1}{2} = \frac{1}{2}$.

Answer: $\frac{1}{2}$ foot

✎ TRY IT

Answers are shown in color.

1. How many jumps were 5 feet?

Answer: 2

2. Total of the $4\frac{1}{2}$-foot jumps

Answer: 9 feet

3. Most common distance(s)?

Answer: $4\frac{1}{2}$ and 5 ft (tie)

Workbook Answer Key · Lesson 10.15

#	Question	Answer
1	How many jumps were 5 feet?	2
2	Total of the $4\frac{1}{2}$-foot jumps	9 feet
3	Most common distance(s)?	$4\frac{1}{2}$ and 5 ft (tie)
4	$2 \times 4\frac{1}{2}$	9
5	$5 - 4\frac{3}{4}$	$\frac{1}{4}$
6	How many plants were measured? (figure)	8
7	Difference between greatest and least growth?	$\frac{3}{4}$ inch
8	Total growth of the $\frac{3}{4}$-inch plants?	$2\frac{1}{4}$ inches
9	Collect 6 measurements at home (to the nearest $\frac{1}{4}$ inch) and make a line plot.	Answers vary.

LESSON 10.16 · TEACHER PLAN
Week 4 · Lesson 4 of 4
Unit 10: Year in Review

Lesson 10.16: Celebrating a Year of Math

Student goal	I can show what I learned in Grade 4 math and set goals for Grade 5.
Objective	Students reflect on the year and set goals.
Standard	Grade 4 review — Cumulative review.
Pacing	Unit 10, week 4, lesson 4 of 4
Time	About 50 minutes
Materials	Poster paper, markers, workbook Lesson 10.16
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Favorite math fact from the year.
Launch	5	Read Let's Explore. Look at Lesson 1.1 together.
Teach	15	Student chooses a lesson to teach back to you or a sibling.
Guided Practice	10	Fill out 'My year in math'.
Independent	12	Final challenge project.
Exit Ticket	3	Celebrate!

SUPPORT
- Offer a list of unit titles to choose from.

EXTEND
- Make a video teaching a lesson.

STUDENT LESSON 10.16 · WITH ANSWERS

Lesson 10.16: Celebrating a Year of Math

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 10.16 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 10.16, Part A.
- [ ] Complete the rest of Workbook Lesson 10.16 (Parts B–C).

LET'S EXPLORE

Look back at Lesson 1.1. What can you do now that you couldn't do then?

★ REMEMBER

You learned place value to one million, all four operations with big numbers, fractions, decimals, geometry, and measurement.

WORKED EXAMPLE · TEACH IT BACK

Choose your favorite lesson and teach it to someone at home.

1 Pick a lesson.

2 Explain the Key Idea.

3 Work one example together.

Answer: Answers vary

✎ TRY IT

Answers are shown in color.

1. Name one thing you learned in each unit.

Answer: Answers vary.

2. What was the hardest unit? What helped you?

Answer: Answers vary.

3. Set one math goal for Grade 5.

Answer: Answers vary.

Workbook Answer Key · Lesson 10.16

#	Question	Answer
1	Name one thing you learned in each unit.	Answers vary.
2	What was the hardest unit? What helped you?	Answers vary.
3	Set one math goal for Grade 5.	Answers vary.
4	The math I'm proudest of this year:	Answers vary.
5	Something I still want to practice:	Answers vary.
6	My math goal for Grade 5:	Answers vary.
7	Create a math poster or game that uses at least three skills from this year.	Project; use the rubric in the Teacher Edition.

YEAR-END REVIEW · ANSWER KEY
Study before the unit test

About 45 minutes

Year-End Review — Answer Key

Use over two days if needed. Any section with more than one mistake shows what to review before the Year-End Test.

Answers

#	Question	Answer
1	Write 504,060 in words.	five hundred four thousand sixty
2	Round 87,652 to the nearest thousand.	88,000
3	45,089 + 37,946	83,035
4	7 × 4,086	28,602
5	3,725 ÷ 5	745
6	$\frac{2}{3} = \frac{?}{15}$	10
7	$4\frac{2}{5} + 2\frac{4}{5}$	$7\frac{1}{5}$
8	$3 \times \frac{5}{8}$	$1\frac{7}{8}$
9	Order: 0.6, 0.06, 0.66	0.06, 0.6, 0.66
10	Right angle split into 38° and x	52°
11	P and A of 13 ft × 7 ft	P = 40 ft, A = 91 sq ft
12	4 ft 7 in. = ___ in.	55
13	9:25 a.m. to 1:10 p.m.	3 hr 45 min

YEAR-END TEST · ANSWER KEY
At the end of the unit
About 60 minutes

Grade 4 Year-End Test — Answer Key

Allow about 60 minutes (or split over two sittings). No calculator.

Answers

#	Question	Answer
1	Value of the 7 in 372,815	70,000
2	Round 465,382 to the nearest hundred thousand.	500,000
3	Compare: 81,409 ◯ 81,490	<
4	63,507 + 28,896	92,403
5	50,000 − 17,284	32,716
6	6 × 2,309	13,854
7	47 × 32	1,504
8	1,856 ÷ 8	232
9	703 ÷ 6	117 R1
10	Factors of 28	1, 2, 4, 7, 14, 28
11	$\frac{3}{4} = \frac{?}{12}$	9
12	Simplify $\frac{15}{18}$	$\frac{5}{6}$
13	$\frac{5}{6}$ ◯ $\frac{3}{4}$	>
14	$2\frac{3}{8} + 3\frac{7}{8}$	$6\frac{2}{8}$
15	$5\frac{1}{3} - 2\frac{2}{3}$	$2\frac{2}{3}$
16	$4 \times \frac{2}{5}$	$1\frac{3}{5}$
17	$\frac{43}{100}$ as a decimal	0.43
18	0.8 ◯ 0.79	>
19	Classify a 135° angle.	obtuse
20	Straight angle split into 72° and x	108°
21	Lines of symmetry in a square	4
22	P and A of 9 m × 6 m	P = 30 m, A = 54 sq m

#	Question	Answer
23	5 lb = ___ oz	80
24	3 m = ___ cm	300
25	A school orders 24 boxes of 36 pencils. It gives 150 pencils to each of 5 classes. How many pencils are left?	114 pencils (864 − 750)
26	Ellie walks $\frac{3}{4}$ mile each day for 5 days. Lillie walks $3\frac{1}{2}$ miles in all. Who walks farther, and by how much?	Ellie, by $\frac{1}{4}$ mile ($3\frac{3}{4}$ vs $3\frac{1}{2}$)
27	A rectangular pool is 18 ft by 12 ft. A fence goes around it 3 ft away on every side. How long is the fence?	84 ft (the fence makes a 24 ft × 18 ft rectangle)

Scoring

Score	What it means / next step
28–33 (85%+)	Ready for Grade 5 math.
23–27	Ready; review missed topics over the summer.
Below 23	Spend a few weeks reviewing weak units before Grade 5.

Grade 4 Mathematics

Teacher Edition

A full year of Grade 4 mathematics, taught one clear idea at a time. Every lesson opens with a question worth thinking about, walks through worked examples step by step, and builds from models and pictures to fluent, confident problem solving.

This Teacher Edition holds everything needed to teach: a week-by-week pacing guide, a guided lesson plan for every lesson with timing, discussion prompts, common misconceptions, support and extension ideas, and complete answer keys.

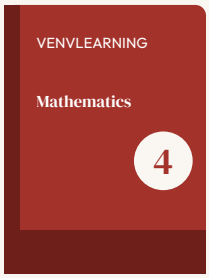
INSIDE THIS YEAR

01 Place Value, Rounding, and Whole Numbers
02 Multiplication
03 Division
04 Factors, Multiples, and Patterns
05 Fraction Equivalence and Comparison
06 Fraction Operations
07 Decimals
08 Geometry, Angles, and Line Plots
09 Measurement, Area, and Perimeter
10 Year in Review

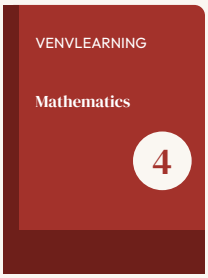
IN THIS BOOK

- Week-by-week pacing guide
- Guided lessons with timing and teaching moves
- Misconceptions, supports, and extensions
- Answer keys and scoring guides for every page

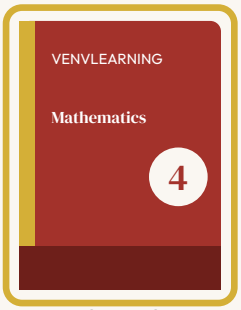
THE COMPLETE SET



Textbook



Workbook



Teacher Edition

You are holding this book

Each lesson has the same number in all three books, so the textbook, workbook, and teacher pages always match.